

IBM Z NetView
Version 6 Release 3

Security Reference



Note

Before using this information and the product it supports, read the information in [“Notices” on page 233](#).

This edition applies to version 6, release 3 of IBM Z NetView (product number 5697-NV6) and to all subsequent versions, releases, and modifications until otherwise indicated in new editions.

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About this publication

The IBM Z® NetView® product provides advanced capabilities that you can use to maintain the highest degree of availability of your complex, multi-platform, multi-vendor networks and systems from a single point of control. This publication, the *IBM Z NetView Security Reference*, describes the types of authorization checking that is available for the NetView environment and the definition statements that are required to implement authorization checking. This publication also includes information for debugging your authorization-checking specification.

Intended audience

This publication is for system programmers responsible for defining and maintaining authorization checking for the NetView environment.

Publications

This section lists publications in the IBM Z NetView library and related documents. It also describes how to access NetView publications online and how to order NetView publications.

IBM Z NetView library

The following documents are available in the IBM Z NetView library:

- *Administration Reference*, SC27-2869, describes the NetView program definition statements required for system administration.
- *Application Programmer's Guide*, SC27-2870, describes the NetView program-to-program interface (PPI) and how to use the NetView application programming interfaces (APIs).
- *Automation Guide*, SC27-2846, describes how to use automated operations to improve system and network efficiency and operator productivity.
- *Command Reference Volume 1 (A-N)*, SC27-2847, and *Command Reference Volume 2 (O-Z)*, SC27-2848, describe the NetView commands, which can be used for network and system operation and in command lists and command procedures.
- *Installation: Configuring Additional Components*, GC27-2851, describes how to configure NetView functions beyond the base functions.
- *Installation: Configuring the NetView Enterprise Management Agent*, GC27-2853, describes how to install and configure the IBM Z NetView Enterprise Management Agent.
- *Installation: Getting Started*, GI11-9443, describes how to install and configure the base NetView program.
- *Installation: Migration Guide*, GC27-2854, describes the new functions that are provided by the current release of the NetView product and the migration of the base functions from a previous release.
- *IP Management*, SC27-2855, describes how to use the NetView product to manage IP networks.
- *Messages and Codes Volume 1 (AAU-DSI)*, GC27-2856, and *Messages and Codes Volume 2 (DUI-IHS)*, GC27-2857, describe the messages for the NetView product, the NetView abend codes, the sense codes that are included in NetView messages, and generic alert code points.
- *Programming: Pipes*, SC27-2859, describes how to use the NetView pipelines to customize a NetView installation.
- *Programming: REXX and the NetView Command List Language*, SC27-2861, describes how to write command lists for the NetView product using the Restructured Extended Executor language (REXX) or the NetView command list language.

- *Security Reference*, SC27-2863, describes how to implement authorization checking for the NetView environment.
- *Troubleshooting Guide*, GC27-2865, provides information about documenting, diagnosing, and solving problems that occur in the NetView product.
- *Tuning Guide*, SC27-2874, provides tuning information to help achieve certain performance goals for the NetView product and the network environment.
- *User's Guide: Automated Operations Network*, SC27-2866, describes how to use the NetView Automated Operations Network (AON) component, which provides event-driven network automation, to improve system and network efficiency. It also describes how to tailor and extend the automated operations capabilities of the AON component.
- *User's Guide: NetView*, SC27-2867, describes how to use the NetView product to manage complex, multivendor networks and systems from a single point.
- *User's Guide: NetView Enterprise Management Agent*, SC27-2876, describes how to use the NetView Enterprise Management Agent.
- *Using Tivoli System Automation for GDPS/PPRC HyperSwap Manager with NetView*, GI11-4704, provides information about the Tivoli® System Automation for GDPS®/PPRC HyperSwap® Manager with NetView feature, which supports the GDPS and Peer-to-Peer Remote Copy (PPRC) HyperSwap Manager services offering.
- *Licensed Program Specifications*, GC31-8848, provides the license information for the NetView product.
- *Program Directory for IBM Z NetView US English*, GI11-9444, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Japanese*, GI11-9445, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Enterprise Management Agent*, GI11-9446, contains information about the material and procedures that are associated with installing the IBM Z NetView Enterprise Management Agent.

The following books are archived:

- *Customization Guide*, SC27-2849, describes how to customize the NetView product and points to sources of related information.
- *Data Model Reference*, SC27-2850, provides information about the Graphic Monitor Facility host subsystem (GMFHS), SNA topology manager, and MultiSystem Manager data models.
- *Installation: Configuring Graphical Components*, GC27-2852, describes how to install and configure the NetView graphics components.
- *Programming: Assembler*, SC27-2858, describes how to write exit routines, command processors, and subtasks for the NetView product using assembler language.
- *Programming: PL/I and C*, SC27-2860, describes how to write command processors and installation exit routines for the NetView product using PL/I or C.
- *Resource Object Data Manager and GMFHS Programmer's Guide*, SC27-2862, describes the NetView Resource Object Data Manager (RODM), including how to define your non-SNA network to RODM and use RODM for network automation and for application programming.
- *SNA Topology Manager Implementation Guide*, SC27-2864, describes planning for and implementing the NetView SNA topology manager, which can be used to manage subarea, Advanced Peer-to-Peer Networking, and TN3270 resources.
- *User's Guide: NetView Management Console*, SC27-2868, provides information about the NetView management console interface of the NetView product.

Related publications

You can find additional product information on the IBM Z NetView web site at <https://www.ibm.com/us-en/marketplace/ibm-tivoli-netview-for-zos>.

For information about the NetView Bridge function, see *Tivoli NetView for OS/390® Bridge Implementation*, SC31-8238-03 (available only in the V1R4 library).

Terminology in this Library

The following terms are used in this library:

CNMCMD

For the CNMCMD member and the members that are included in it using the %INCLUDE statement

CNMSTYLE

For the CNMSTYLE member and the members that are included in it using the %INCLUDE statement

DSIOPF

For the DSIOPF member and the members that are included in it using the %INCLUDE statement

IBM® Tivoli Netcool®/OMNIbus

For either of these products:

- IBM Tivoli Netcool/OMNIbus
- IBM Tivoli OMNIbus and Network Manager

MVS™

For z/OS® operating systems

MVS element

For the base control program (BCP) element of the z/OS operating system

NetView

For the following products:

- IBM Z NetView version 6 release 3
- IBM Tivoli NetView for z/OS version 6 release 2 modification 1
- NetView releases that are no longer supported

PARMLIB

For SYS1.PARMLIB and other data sets in the concatenation sequence

VTAM®

For Communications Server - SNA Services

Unless otherwise indicated, topics to programs indicate the latest version and release of the programs. If only a version is indicated, the topic is to all releases within that version.

When a topic is made about using a personal computer or workstation, any programmable workstation can be used.

Using IBM Z NetView online help

The following types of IBM Z NetView mainframe online help are available, depending on your installation and configuration:

- General help and component information
- Command help
- Message help
- Sense code information
- Recommended actions

Accessing publications online

IBM posts publications for this and all other products, as they become available and whenever they are updated, to the IBM Knowledge Center at <https://www.ibm.com/support/knowledgecenter>. You can find IBM Z NetView documentation on [IBM Z NetView Knowledge Center](#).

Note: If you print PDF documents on other than letter-sized paper, set the option in the **Print** window that enables Adobe Reader to print letter-sized pages on your local paper.

Ordering publications

You can order many Tivoli publications online at <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>

You can also order by telephone by calling one of these numbers:

- In the United States: 800-426-4968
- In Canada: 800-879-2755

In other countries, contact your software account representative to order Tivoli publications. To locate the telephone number of your local representative, perform the following steps:

1. Go to <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>.
2. Select your country from the list and click the grey arrow button beside the list.
3. Click **About this site** to see an information page that includes the telephone number of your local representative.

Accessibility

Accessibility features help users with a physical disability, such as restricted mobility or limited vision, to use software products successfully. Standard shortcut and accelerator keys are used by the product and are documented by the operating system. Refer to the documentation provided by your operating system for more information.

For additional information, see the Accessibility appendix in the *User's Guide: NetView*.

Tivoli user groups

Tivoli user groups are independent, user-run membership organizations that provide Tivoli users with information to assist them in the implementation of Tivoli Software solutions. Through these groups, members can share information and learn from the knowledge and experience of other Tivoli users.

Support information

If you have a problem with your IBM software, you want to resolve it quickly. IBM provides the following ways for you to obtain the support you need:

Online

Please follow the instructions located in the support guide entry: <https://www.ibm.com/support/home/pages/support-guide/?product=4429363>.

Troubleshooting information

For more information about resolving problems with the IBM Z NetView product, see the *IBM Z NetView Troubleshooting Guide*. You can also discuss technical issues about the IBM Z NetView product through the NetView user group located at <https://groups.io/g/NetView>. This user group is for IBM Z NetView customers only, and registration is required. This forum is also monitored by interested parties within IBM who answer questions and provide guidance about the NetView product. When a problem with the code is found, you are asked to open an official case to obtain resolution.

Conventions used in this publication

This section describes the conventions that are used in this publication.

Typeface conventions

This publication uses the following typeface conventions:

Bold

- Lowercase commands and mixed case commands that are otherwise difficult to distinguish from surrounding text
- Interface controls (check boxes, push buttons, radio buttons, spin buttons, fields, folders, icons, list boxes, items inside list boxes, multicolumn lists, containers, menu choices, menu names, tabs, property sheets), labels (such as **Tip:**, and **Operating system considerations:**)
- Keywords and parameters in text

Italic

- Citations (examples: titles of publications, diskettes, and CDs)
- Words defined in text (example: a nonswitched line is called a *point-to-point line*)
- Emphasis of words and letters (words as words example: "Use the word *that* to introduce a restrictive clause."; letters as letters example: "The LUN address must start with the letter *L*.")
- New terms in text (except in a definition list): a *view* is a frame in a workspace that contains data.
- Variables and values you must provide: ... where *myname* represents...

Monospace

- Examples and code examples
- File names, programming keywords, and other elements that are difficult to distinguish from surrounding text
- Message text and prompts addressed to the user
- Text that the user must type
- Values for arguments or command options

Operating system-dependent variables and paths

For workstation components, this publication uses the UNIX convention for specifying environment variables and for directory notation.

When using the Windows command line, replace *\$variable* with *%variable%* for environment variables and replace each forward slash (/) with a backslash (\) in directory paths. The names of environment variables are not always the same in the Windows and UNIX environments. For example, %TEMP% in Windows environments is equivalent to \$TMPDIR in UNIX environments.

Note: If you are using the bash shell on a Windows system, you can use the UNIX conventions.

Syntax diagrams

The following syntax elements are shown in syntax diagrams. Read syntax diagrams from left-to-right, top-to-bottom, following the horizontal line (the main path).

- [“Symbols” on page xv](#)
- [“Parameters” on page xvi](#)
- [“Punctuation and parentheses” on page xvi](#)
- [“Abbreviations” on page xvii](#)

For examples of syntax, see [“Syntax examples” on page xvii](#).

Symbols

The following symbols are used in syntax diagrams:



Marks the beginning of the command syntax.



Marks the end of the command syntax.



Indicates that the command syntax is continued on the next line.



Indicates that a statement is continued from the previous line.



Marks the beginning and end of a fragment or part of the command syntax.

Parameters

The following types of parameters are used in syntax diagrams:

Required

Required parameters are shown on the main path.

Optional

Optional parameters are shown below the main path.

Default

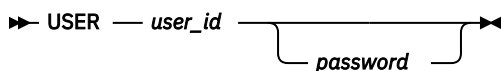
Default parameters are shown above the main path. In parameter descriptions, default parameters are underlined.

Syntax diagrams do not rely on highlighting, brackets, or braces. In syntax diagrams, the position of the elements relative to the main syntax line indicates whether an element is required, optional, or the default value.

When you issue a command, spaces are required between the parameters unless a different separator, such as a comma, is specified in the syntax.

Parameters are classified as keywords or variables. Keywords are shown in uppercase letters. Variables, which represent names or values that you supply, are shown in lowercase letters and are either italicized or, in NetView help, displayed in a differentiating color.

In the following example, the `USER` command is a keyword, the `user_id` parameter is a required variable, and the `password` parameter is an optional variable.



Punctuation and parentheses

You must include all punctuation that is shown in the syntax diagram, such as colons, semicolons, commas, minus signs, and both single and double quotation marks.

When an operand can have more than one value, the values are typically enclosed in parentheses and separated by commas. For a single value, the parentheses typically can be omitted. For more information, see [“Multiple operands or values” on page xviii](#).

If a command requires positional commas to separate keywords and variables, the commas are shown before the keywords or variables.

When examples of commands are shown, commas are also used to indicate the absence of a positional operand. For example, the second comma indicates that an optional operand is not being used:

```
COMMAND_NAME opt_variable_1,,opt_variable_3
```

You do not need to specify the trailing positional commas. Trailing positional and non-positional commas either are ignored or cause a command to be rejected. Restrictions for each command state whether trailing commas cause the command to be rejected.

Abbreviations

Command and keyword abbreviations are listed in synonym tables after each command description.

Syntax examples

The following examples show the different uses of syntax elements:

- “Required syntax elements” on page xvii
- “Optional syntax elements” on page xvii
- “Default keywords and values” on page xvii
- “Multiple operands or values” on page xviii
- “Syntax that is longer than one line” on page xviii
- “Syntax fragments” on page xviii

Required syntax elements

Required keywords and variables are shown on the main syntax line. You must code required keywords and variables.

➤ REQUIRED_KEYWORD — *required_variable* ➤

A required choice (two or more items) is shown in a vertical stack on the main path. The items are shown in alphanumeric order.

➤ REQUIRED_OPERAND_OR_VALUE_1
 REQUIRED_OPERAND_OR_VALUE_2 ➤

Optional syntax elements

Optional keywords and variables are shown below the main syntax line. You can choose not to code optional keywords and variables.

➤
 OPTIONAL_OPERAND ➤

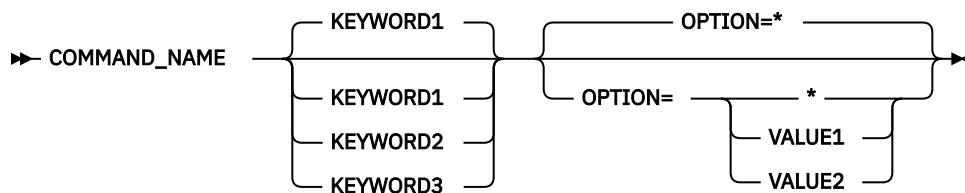
A required choice (two or more items) is shown in a vertical stack below the main path. The items are shown in alphanumeric order.

➤
 OPTIONAL_OPERAND_OR_VALUE_1
 OPTIONAL_OPERAND_OR_VALUE_2 ➤

Default keywords and values

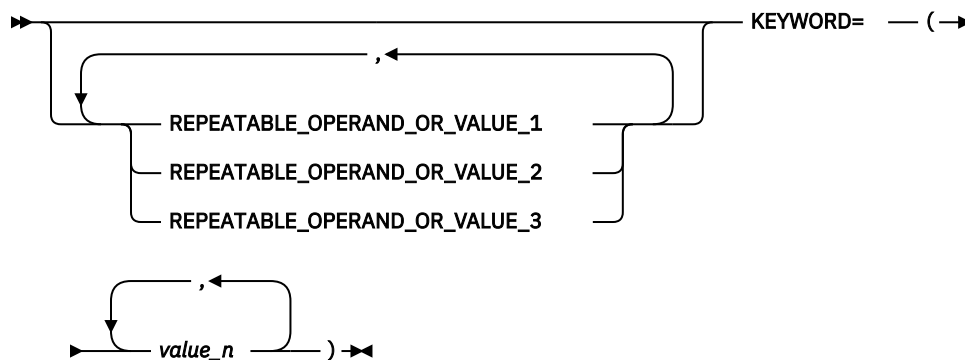
Default keywords and values are shown above the main syntax line in one of the following ways:

- A default keyword is shown only above the main syntax line. You can specify this keyword or allow it to default. The following syntax example shows the default keyword KEYWORD1 above the main syntax line and the rest of the optional keywords below the main syntax line.
- If an operand has a default value, the operand is shown both above and below the main syntax line. A value below the main syntax line indicates that if you specify the operand, you must also specify either the default value or another value shown. If you do not specify the operand, the default value above the main syntax line is used. The following syntax example shows the default values for operand OPTION=* above and below the main syntax line.



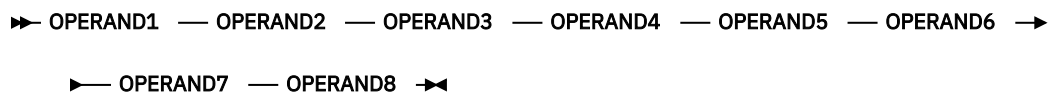
Multiple operands or values

An arrow returning to the left above a group of operands or values indicates that more than one can be selected or that a single one can be repeated.



Syntax that is longer than one line

If a diagram is longer than one line, each line that is to be continued ends with a single arrowhead and the following line begins with a single arrowhead.



Syntax fragments

Some syntax diagrams contain syntax fragments, which are used for lengthy, complex, or repeated sections of syntax. Syntax fragments follow the main diagram. Each syntax fragment name is mixed case and is shown in the main diagram and in the heading of the fragment. The following syntax example shows a syntax diagram with two fragments that are identified as `Fragment1` and `Fragment2`.



Fragment1



Fragment2



Chapter 1. Overview of NetView Security

This chapter defines security terminology and provides information to help security administrators and system programmers find information. More specific implementation details follow in subsequent chapters.

To minimize changes to your security while migrating from a previous release of the NetView product, see [Chapter 13, “Scenarios for Converting Types of Security,” on page 119.](#)

What Is Authorization Checking?

Authorization checking controls access to systems and networks. It restricts or enables users to view or change information, issue commands, and perform operator duties. Individuals are assigned the level of authorization necessary for their responsibilities.

The NetView program and System Authorization Facility (SAF) products, such as RACF®, enable you to have various levels and types of authorization. The following types of security can affect the NetView product:

- Operator passwords, password phrases, and logon attributes
- NetView command, keyword, and value authorization
- Span of control over VTAM and RODM resources
- Data set access
- Terminal access restrictions
- Protection of commands that are not NetView commands but that are issued from a NetView task
- Cross-domain logons

Why Limit Authorization?

Limiting authorization for tasks helps to prevent unauthorized system use and ensures that individuals are responsible for the actions taken by their operator task. Limiting authorization is also a way to help users avoid accidentally changing or destroying vital system information. For example:

- Password security prevents unauthorized personnel from logging on to the NetView program.
- Data set security keeps confidential data set members from being viewed or modified.
- Command authorization security ensures that only authorized operators can issue protected commands.

What Is the System Authorization Facility?

An SAF product, such as RACF, is an application that supports the RACROUTE interface and performs functions such as centralized auditing, resource authorization, and user identification and verification. For example, RACF lets you limit data set access. In addition, RACF allows you to:

- Define command authorization for the NetView program
- Maintain NetView operator passwords, password phrases, and logon attributes
- Define span names and the associated access levels

Using an SAF product also helps simplify and centralize your security. All the operator and command authorization syntax is created and managed within one product, rather than handled uniquely for each application. The NetView program provides sample CNMSAF2 which contains RACF definitions for NetView operators and commands. You can modify this sample to meet your installation requirements.

Ensure that the user ID that is associated with the NetView started task or job is defined to your SAF product with an OMVS segment. This is required so that the NetView program can use z/OS Unix System Services (USS) functions, which include TCP/IP functions.

Types of Security

The following table provides an overview of the types of security that you can use to ensure the integrity of your system:

Table 1. Overview of Types of Security

Type of Security	Using an SAF Product	Using the NetView Program	See
Defining operators and passwords	APPL class and NETVIEW segment	DSIPRF and DSIOPF	Chapter 2, “Defining Operators, Passwords, and Logon Attributes,” on page 7
NetView command authorization	NETCMDS class	NetView command authorization table	Chapter 3, “Controlling Access to Commands,” on page 29
RUNCMD	Not available	DSIEX19	“Defining RUNCMD Authorization” on page 54
RMTCMD	RMTOPS class	Remote security table (such as DSISECUR)	“Defining RMTCMD Authorization” on page 59
Defining resources to spans	Not available	Span table	Chapter 4, “Using Spans to Protect Resources and Views,” on page 67
Authorizing operators to spans	NETSPAN class	SPAN and ISPAN statements in DSIPRF	“Defining Operator Access to Spans” on page 80
Data set access	DATASET class	Not available	Chapter 5, “Controlling Access to Data Sets and Members,” on page 87
RODM	RODMMGR class	Not available	Chapter 12, “Defining NetView Security for RODM,” on page 113

Note: Commands running on a Virtual OST (VOST) are checked against the authority of the VOST owner.

To use an SAF product for security, ensure that the SAF product is running and the security classes used by the NetView program (such as the NETCMDS class) are active before starting the NetView program. For information about how to set up these types of security, see [Chapter 13, “Scenarios for Converting Types of Security,” on page 119](#).

Defining and Changing Types of Security

The initial NetView security settings are defined by the SECOPTS statements in the CNMSTYLE member. These statements define the security method that is used by the NetView product to specify the following items:

- The type of password checking
- Where the NetView operator logon attributes are defined
- Where span of control is defined
- The type of command authorization

After the NetView program is started, you can use the NetView REFRESH command to dynamically change the security settings. Refer to the NetView online help for more information about the REFRESH command.

You can use the NetView LIST SECOPTS command to display the current security settings. When used with the default security values, it produces output that is similar to the following output:

```

NetView V6R3 - 7E          NetView   NTV7E TOM      06/07/19 18:27:41
* NTV7E      list secopts
' NTV7E
BNH228I OPTION          VALUE          LAST UPDATED          UPDATE ID
BNH229I -----
BNH229I OPERSEC         NETVPW          06/07/19 09:59:30     INITIALIZATION
BNH229I OPSPAN          NETV           06/07/19 09:59:30     INITIALIZATION
BNH229I CMDAUTH         TABLE         06/07/19 09:59:30     INITIALIZATION
BNH229I TBLNAME         CNMSCAT2       06/07/19 09:59:30     INITIALIZATION
BNH229I AUTHCHK         SOURCEID       06/07/19 09:59:30     INITIALIZATION
BNH229I SPANAUTH        NONE           06/07/19 09:59:07     INITIALIZATION
BNH229I SPANCHK         SOURCEID       06/07/19 09:59:30     INITIALIZATION
BNH229I CATAUDIT        NONE           06/07/19 09:59:07     INITIALIZATION
BNH229I AUTOSEC         BYPASS         06/07/19 09:59:34     INITIALIZATION
BNH229I SURROGAT        NO             06/07/19 09:59:30     INITIALIZATION
BNH229I RMTSEC          NONE           06/07/19 09:59:40     INITIALIZATION
BNH229I RMTAUTH         SENDER         06/07/19 09:59:30     INITIALIZATION
BNH229I WEBAUTH         PASS           06/07/19 09:59:30     INITIALIZATION
BNH230I END OF LIST SECOPTS INFORMATION
-----

```

In the previous example, CATAUDIT and AUTOSEC are two additional security settings that can be set and altered using the NetView DEFAULTS command. Although RMTSEC can be altered using the REFRESH command, it is initially set by the RMTSECUR statement in the DSIUDST task initialization member.

For security problem diagnosis information see [Chapter 14, “Checklist for Debugging Security Problems,”](#) on page 139. For more security examples, see [Chapter 13, “Scenarios for Converting Types of Security,”](#) on page 119.

Centralized Security Definitions

You can use the NetView command authorization table and SAF command authorization to centralize your command security definitions. Centralization decreases the possibility of errors and makes it easier to change the definitions. Both of these types of command security are dynamic, which increases flexibility and reduces the system down time required to recycle the NetView program. Support for pattern-matching characters can reduce setup time and complexity.

Advantages of Dynamic Definitions

Using an SAF product, such as RACF, enables you to dynamically add, delete, or change security definitions for:

- Spans of control that an operator is allowed to start
- Operator passwords, password phrases, and logon attributes
- Command authorization

The NetView command authorization table also allows you to change command authorization without recycling the NetView program.

If you want to use an SAF product for command authorization, consider using an SAF product with the NetView command authorization table as a backup. See [Chapter 3, “Controlling Access to Commands,”](#) on page 29 for a description of command security.

Advantages of Pattern-Matching Characters

Using the NetView command authorization table or an SAF product for command authorization, you can use pattern-matching (wild card) characters, such as the asterisk (*), to globally protect command identifiers with names that match the pattern.

For example, the following NetView command authorization table statement prevents operators from loading code point tables (the CPTBL command) by protecting all members matching the pattern BNJ8*:

```
PROTECT *.*.CPTBL.MEMBER.BNJ8*
```

Flexibility of Authorization by Source or Target Task

You can also determine if the originators of commands are authorized to issue commands. For example, the NetView AT timer command can be used to issue commands to be run by the PPT task. The AT command can be protected; therefore, you can use source ID authorization checking to prevent the operator from using the AT command to cause the PPT to issue any restricted commands. This is important, because the PPT task is not normally subject to command security checking. See [“Authority Checking Commands against the Command Source” on page 34](#) for more information about types and definitions for source checking, and [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151](#).

Changing the Method of Command Authorization

Keep previous security definitions for backup and migration purposes. To improve performance, bypass unnecessary security checking.

The NetView program no longer supports *scope of command* authorization. If you specify `SECOPTS.COMDAUTH=SCOPE.table_name` in the CNMSTYLE member, the NetView program creates a command authorization table named *table_name*. The NetView program then uses the generated command authorization table for command security verification. If you choose to use this approach, copy your existing definitions because the samples supplied no longer include CMDCLASS, KEYCLASS, or VALCLASS statements. You can also use the SECMIGR tool with your existing statements to generate a NetView command authorization table, and then copy this generated table. If you choose to take either of these approaches, review the command authorization table for accuracy and change any statements as necessary. After your changes are complete, update the CNMSTUSR or CxxSTGEN member to specify `SECOPTS.COMDAUTH=TABLE.table_name`. As an alternative, you can migrate to an SAF product for command authorization.

If you plan to use an SAF product for command authorization, you can define the commands, keywords, and values as SAF resources and then authorize your operators to access these resources. You can use the SECMIGR tool to convert from either scope of command authorization or the NetView command authorization table to SAF command security definitions. Additionally, consider implementing a NetView command authorization table as backup. At a minimum, use the backup NetView command authorization table for immediate commands. You can also define the equivalent authorization checking in the backup NetView command authorization table as you did in the SAF product, in case the SAF product is not able to make a security decision. See [“Scenario 8: Converting from NetView Command Authorization Table to SAF Command Authorization” on page 136](#) for further information.

When the SAF command security is defined, you can use the REFRESH command with `CMDAUTH=SAF` to change the method of command authorization to SAF. Once again, you can initialize the NetView program with existing settings and use REFRESH to change them, if necessary.

Maintaining Existing Security Definitions

The following items are helpful in security migration:

- If you migrate from the NetView command authorization table to an SAF product, you might want to keep the NetView command authorization table command identifiers for reference while you ensure your conversion meets your expectations. This enables you to use the REFRESH command to change back in case your new security does not work.

- If your existing security allows all tasks to issue a command, you can use the SEC=BY keyword on that command. Using the SEC=BY keyword on the CMDDEF statement in CNMCMDU bypasses security checking and can improve system performance.
- If converting to SAF command security, consider leaving your existing definitions and profiles in DSIOPF and DSIPRF for reference at a later time. They are not used by the system when using an SAF product for NetView operator definitions and logon attributes.
- When converting to a command authorization table or SAF command security, consider leaving your scope definitions in CNMCMD for reference at a later time. They are ignored by the NetView program.

Protecting Immediate Commands

Immediate commands cannot be protected by an SAF product, but they can be protected using the NetView command authorization table.

If you are using an SAF product for command authorization, it is a good idea to use a backup NetView command authorization table. **If you are using an SAF product without a backup NetView command authorization table, all immediate commands are allowed to run.**

For a description of immediate commands and how to protect them, see [“Protecting Immediate Commands When CMDAUTH=SAF”](#) on page 52.

Chapter 2. Defining Operators, Passwords, and Logon Attributes

All operator information can be defined in an SAF product, such as RACF, eliminating the need for operator definitions in the DSIOPF member. Also, the method of defining operators, operator passwords and password phrases, and logon attributes can be dynamically changed using the REFRESH command, eliminating the recycling of the NetView program. Operator passwords, password phrases, and logon attributes can be defined in the following levels:

- Minimal checking, where operator passwords and password phrases are not checked, and logon attributes are ignored.
- The NetView program checks operator passwords and defines logon attributes.
- Operator passwords and password phrases are checked by an SAF product, with logon profiles that are specified in the NetView DSIOPF member (or by using the DEFAULTS command or the DEFAULTS.LogProf statement in the CNMSTYLE member) and attributes that are defined in members in the DSIPRF data set.
- All operator passwords, password phrases, and logon attributes are defined and checked by an SAF product.

The NetView program supports the use of IBM Multi-Factor Authentication for z/OS Version 1 Release 3 with an SAF product.

If you are migrating from a previous release of the NetView program and are not defining your operators to an SAF product, continue to use the existing operator definitions.

Some of the automated operator IDs (autotasks) that are defined in the NetView program are reserved; do not change these names. For information about reserved names, see the NetView operator definition sample (DSIOPF) or *IBM Z NetView Installation: Getting Started*.

Overview of Operator Security

Do not use the names of NetView commands, components, printers (hardcopy logs), terminals, or task identifiers for operator identifiers. Also, do not use the following reserved keywords:

ALL	NNT
DPR	OPT
DST	OST
HCL	PPT
HCT	SYSOP
LOG	TCT
MNT	

Additionally, if the operator identifier is the same as the LU name (terminal), some command lists assume that the operator is an autotask and do not run.

You can define passwords in either the NetView program or an SAF product, as described in [“Defining Operator Password Security”](#) on page 10.

The six types of operator security definitions are defined by the SECOPTS.OPERSEC statement in the CNMSTYLE member, as shown in [Table 2 on page 8](#). Each type specifies the combination of password and profile security:

Table 2. Operator Security Definition Types

OPERSEC value	Type of Operator Password and Logon Attributes
MINIMAL	Both operator passwords and logon attributes are ignored.
NETVPW	Operator passwords are specified in DSIOPF. Logon attributes are specified in the NetView DSIOPF member and defined in the DSIPRF data set. If the password value is *NONE*, the operator logon always fails. You can use the *NONE* value for operator IDs that are reserved for autotasks, where the operator ID is not used to log on (not even to reconnect by using takeover).
NETVPW and NOCHECK specified in DSIOPF	Passwords are not checked by the NetView program. Logon information is passed to NetView installation exit 12 (DSIEX12).
SAFPW	Operator passwords or password phrases are checked by an SAF product, with operator profiles that are specified in the NetView DSIOPF member and logon attribute values that are defined in the DSIPRF data set. A default profile can be specified by using the DEFAULTS command or the DEFAULTS.LogProf statement in the CNMSTYLE member, which eliminates the need for operator definitions in the DSIOPF member. Access to the data sets that are protected in the DATASET class and to MVS system commands that are protected in the OPERCMDS class of the SAF product are checked at the NetView product level.
SAFCHECK	Operator passwords or password phrases are checked by an SAF product, with operator profiles that are specified in the NetView DSIOPF member and logon attribute values that are defined in the DSIPRF data set. A default profile can be specified by using the DEFAULTS command or the DEFAULTS.LogProf statement in the CNMSTYLE member, which eliminates the need for operator definitions in the DSIOPF member. Access to the data sets that are protected in the DATASET class and to MVS system commands that are protected in the OPERCMDS class of the SAF product are checked at the individual task level.
SAFDEF	Operator passwords or password phrases are checked by an SAF product. Logon attributes are defined in the NETVIEW segment of an SAF product. Access to the data sets protected in the DATASET class and to MVS system commands protected in the OPERCMDS class of the SAF product are checked at the individual task level.

RACF supports the NGMFVSPN attribute for NetView management console view security. Support for this attribute might be available in other SAF products. Contact the product support group for your SAF product to find out.

By defining NetView operators exclusively to an SAF product and using the NETSPAN class, you can eliminate the need for the DSIOPF members and members in the DSIPRF data set. However, for migration and regression purposes, do not erase your operator profiles and definitions. Defining operator span of control is related to logon attributes, but is covered in [Chapter 4, “Using Spans to Protect Resources and Views,”](#) on page 67.

When operators are defined exclusively in an SAF product (OPERSEC=SAFDEF), they can be authorized to log on to a particular NetView host through a profile in the APPL class of an SAF product. You can use domain identifiers to define resources in the APPL class to represent instances of the NetView program.

If you are using an SAF security product to perform operator identification and password checking, you can log on to the NetView program using a PassTicket rather than a password. For example, the Network Security Program/Secure Logon Coordinator product (NetSP/SLC V1.2) supports PassTickets.

Programs that use UNIX System Services require an OMVS segment for the user ID under which they run. For this reason, NetView operator IDs that use UNIX System Services must have an OMVS segment

defined at either the user ID or group level using an SAF product. The NetView product has no specific requirements for OMVS segment attributes, unless stated for a specific NetView function, so the attribute values are determined by your security administrator in the same manner as other user IDs that require OMVS segments.

See [“Using an SAF Product Exclusively”](#) on page 19 for more information about defining operators and operator attributes using an SAF product. If you want to further limit access to the NetView program, see [“Using an SAF Product to Restrict Logon Access”](#) on page 23 and [“Protecting EMCS Console Names Using an SAF Product”](#) on page 27.

Defining Tivoli Enterprise Portal User IDs

You might need to define Tivoli Enterprise Portal user IDs to the NetView program for use by the following agents and products:

- Z NetView Enterprise Management Agent
- z/OS products that use the Tivoli Management services component and forward take action commands to the NetView program for authorization

Tivoli Enterprise Portal users can use the Z NetView Enterprise Management Agent to issue NetView commands and receive command responses in Tivoli Enterprise Portal workspaces.

The APSERV (CNMEAPCS) command allows authorized programs (clients) to run NetView or MVS commands under a properly established user name and to log the results. A Tivoli Enterprise Portal user ID can be passed to the APSERV command. For more information about the APSERV command, see the *IBM Z NetView Application Programmer's Guide*.

These commands are authorized using the Tivoli Enterprise Portal user ID. You must perform one or more of the following actions:

- Create a new Tivoli Enterprise Portal user ID that matches a new or existing NetView operator ID. This is the preferred method.
- Define the existing Tivoli Enterprise Portal user ID to the NetView program.
- Map the Tivoli Enterprise Portal user ID to a valid NetView operator ID.

Creating a New Tivoli Enterprise Portal User ID

The Tivoli Enterprise Portal user ID that you create must match the criteria for defining NetView operator IDs. See the *IBM Z NetView Administration Reference* for information about defining these IDs.

The user ID must also be authorized for take action commands. See the Tivoli Enterprise Portal online help and the *IBM Tivoli Monitoring Administrator's Guide* for more information.

Defining the Tivoli Enterprise Portal User ID to the NetView Program

You can define an existing Tivoli Enterprise Portal user ID to the NetView program if the Tivoli Enterprise Portal user ID meets NetView operator ID criteria.

Mapping the Tivoli Enterprise Portal User ID to a Valid NetView Operator ID

To map your Tivoli Enterprise Portal user IDs to NetView operator IDs, use the NACMD.OPID.TEPLoginid statement in the NetView CNMSTYLE member. This function is provided because Tivoli Enterprise Portal user IDs can have a maximum length of 10 characters, which is not supported by the NetView program. To use the mapping statement, the Tivoli Enterprise Portal user ID must meet the criteria for NetView operator IDs, except for the length criteria.

When you are mapping these user IDs, note the following considerations:

- Do not map multiple Tivoli Enterprise Portal user IDs to one NetView operator ID because multiple Tivoli Enterprise Portal users have the same command authorization. This might also cause throughput problems.
- Only the user ID is used for command authorization. Password validation is not performed.
- Be careful about mapping a Tivoli Enterprise Portal user ID to a NetView operator ID that is used to log on to a 3270 terminal. A take action command might be run on an active NetView operator with unwanted results. Mapping Tivoli Enterprise Portal user IDs to NetView autotasks is preferable.

Defining Operator Password Security

Using password security restricts unauthorized personnel from logging on to the NetView program. Define a unique operator identifier and password for each operator who logs on to the NetView product.

The NetView program provides the option of specifying whether password checking is performed by the NetView program or by SAF security, such as RACF. The method of checking is specified by the SECOPTS.OPERSEC setting, described in the *IBM Z NetView Administration Reference*. If you specified in that setting that password checking is to be performed by the NetView program, be aware that any password that is defined to the NetView program is automatically converted to uppercase and stored in uppercase. If you specified that password checking is to be performed by using an SAF security product, you might be able to use the mixed case password function.

If an SAF product, such as RACF, is installed on your system, use it for password protection rather than using static passwords in the DSIOPF member. You can also substitute a password phrase for a password if the SAF product supports password phrases.

Note: The RACF product allows password phrases to contain any valid EBCDIC characters. When a password phrase is set for a user ID that is to be accessed from a system that is not a z/OS system, the password is entered on that system as ASCII text. Not every EBCDIC character has an ASCII equivalent, so limit the EBCDIC password phrase to characters that have an ASCII equivalent.

Password phrases are not supported in the following environments:

- Password checking using the DSIOPF member.
- Logon validation using the DSIEX12 installation exit.
- Tivoli Enterprise Portal login. Because of this restriction, the password phrase cannot be passed to the Tivoli Enterprise Monitoring Server for RACF verification.
- Functions that use the NetView-NetView task (NNT).

Using an SAF product for password authorization

If you want an SAF product to perform password authorization, code OPERSEC with values of SAFPW, SAFCHECK, or SAFDEF. To change existing password security, see [“Scenario 2: Converting Operator Passwords”](#) on page 122.

Using an SAF product to define and maintain your operator passwords provides the following advantages:

- After the operator logs on for the first time with a system-defined password, new passwords are known only to the operator.
- Operators can change their own passwords from the NetView logon screens.
- Passwords and password phrases can be set to expire after a predetermined time period.
- Password phrases can be substituted for passwords. The password phrase can include phrases that are 9 to 100 characters in length, without character restrictions.
- Restrictions on the format of the password can be enforced.
- Passwords are not hardcoded and cannot be browsed.

- Because an operator can be uniquely defined to an SAF product by application, each operator can use the same operator identifier and password across multiple applications such as the NetView program and TSO.
- Additional logon restrictions are enabled, such as limiting the times, days, or terminal addresses which are valid.

The following example defines an operator password to RACF:

```
ADDUSER NEWOPER PASSWORD(PSWD1)
```

In this example, NEWOPER is the operator identifier and PSWD1 is the initial password. The first time NEWOPER logs on to an application, the password must be changed.

You can change a NetView operator password or password phrase from the logon panel if an SAF product is being used for NetView password authorization. If an operator tries to change a password or password phrase, but the logon attempt is not successful because of a bad parameter, and the password or password phrase is valid, then the password or password phrase is changed and the DSI757E message is sent to the NetView log. The operator is not logged on.

For example, if the operator specifies values for profile, HCL, or initial command that are not valid, even if the password or password phrase change is valid, the operator is not logged on and does not receive a message at the console. However, at the next logon attempt, the operator must use the new password or password phrase.

If you are using an SAF product for logon password or password phrase authorization, operator passwords in the DSIOPF member are ignored.

Changing an SAF Password or Password Phrase

If an SAF password or password phrase is to be changed, such as when it has expired, the NetView program provides the following methods for navigating to the password change panel:

- **Method 1:** On the logon panel for the NetView program, leave the **PASSWORD** field blank when entering credentials and other information. For more information about this method, see [“Details about Method 1” on page 11](#).
- **Method 2:** On the logon panel for the NetView program, when entering credentials and other information, provide a valid value in the **PASSWORD** field and type a non-blank character in the **Change password** field. For more information about this method, see [“Details about Method 2” on page 12](#).

Note: If the **PASSWORD** field of the logon panel is left blank, it is considered to be the first method for navigating to the password change panel, regardless of what is also specified in the **Change password** field.

Details about Method 1

When the first method for navigating to the password change panel is used, what the user specifies for the **OLD PASSWORD** field on the password change panel depends upon the following user profile configurations:

- Whether an SAF password or password phrase is used.
- Whether the profile is enabled for at least one authentication factor provided by the IBM Multi-Factor Authentication for z/OS product (IBM MFA).

If the user profile is configured to use an SAF password and is not enabled for any IBM MFA authentication factors, an SAF password should be specified in the **OLD PASSWORD** field. In addition to a user profile being considered not enabled for IBM MFA authentication factors when such factors are simply not configured for the profile, a user profile is considered not enabled for any IBM MFA authentication factors when these conditions are all true:

- IBM MFA authentication factors are configured for the user profile.

- The IBM MFA product is not active.
- The SAF product is configured to allow fallback to using SAF credentials when the IBM MFA product is not active.

If the user profile is configured to use an SAF password phrase and it is not enabled for any IBM MFA authentication factors, an SAF password phrase must be specified in the **OLD PASSWORD** field.

If the user profile is configured to use an SAF password and it is enabled for at least one IBM MFA authentication factor, valid IBM MFA credentials must be specified in the **OLD PASSWORD** field. If, however, the IBM MFA product is configured for compound in-band authentication, this method cannot be used to change an SAF password. The second method for navigating to the password change panel, described in more detail later, must be used.

If the user profile is configured to use an SAF password phrase and it is enabled for at least one IBM MFA authentication factor, valid IBM MFA credentials must be specified in the **OLD PASSWORD** field.

The user must then specify the same new SAF password or SAF password phrase, whichever the user profile is configured to use, in the **NEW PASSWORD** and **VERIFY NEW PASSWORD** fields.

Details about Method 2

The second method for navigating to the password change panel requires a valid value to be specified for the **PASSWORD** field of the logon panel, and a non-blank character to be specified in the **Change password** field of the logon panel. What comprises a valid value for the **PASSWORD** field depends upon the user profile and security product configurations.

If the user profile is configured to use an SAF password and it is not enabled for any IBM MFA authentication factors, specify the SAF password in the **PASSWORD** field of the logon panel.

If the user profile is configured to use an SAF password phrase and it is not enabled for any IBM MFA authentication factors, specify the SAF password phrase in the **PASSWORD** field of the logon panel.

If the user profile is configured to use an SAF password and it is enabled for at least one IBM MFA authentication factor, specify valid IBM MFA credentials in the **PASSWORD** field of the logon panel.

If the user profile is configured to use SAF password phrase and it is enabled for at least one IBM MFA authentication factor, specify valid IBM MFA credentials in the **PASSWORD** field of the logon panel.

On the password change panel, specify the old SAF password or password phrase, whichever the user profile is configured to use, in the **OLD PASSWORD** field. Specify the same new SAF password or SAF password phrase, whichever the user profile is configured to use, in the **NEW PASSWORD** and **VERIFY NEW PASSWORD** fields.

Using an SAF Product for PassTicket Authorization

To have an SAF product perform PassTicket authorization, code OPERSEC with a value of SAFPW, SAFCHECK, or SAFDEF. To change existing password security, see [“Scenario 2: Converting Operator Passwords”](#) on page 122. Using an SAF product to validate an operator PassTicket provides the following advantages over password usage:

- PassTickets are generated when needed, so the character string cannot be predicted.
- The password itself is not passed in the data stream, allowing for better network security.

To configure your SAF product for use with PassTickets, follow the procedures outlined in the appropriate SAF product documentation. When defining a PassTicket profile, the application name for NetView operator IDs depends on the value of OPERSEC:

- For an OPERSEC value of SAFPW or SAFCHECK, the NetView program does not provide an application name. Therefore, the SAF product uses the default name. For RACF, this is the name used for batch programs, which consists of the characters MVS followed by the SMF ID of the system.
- For an OPERSEC value of SAFDEF, the NetView program uses an application name of the NetView domain name.

You can change the actual password or password phrase for a NetView operator from the logon panel, even if a PassTicket is used to logon. See [“Using an SAF product for password authorization”](#) on page 10 for more information about changing passwords.

Using the NetView Program for Password Authorization

To use the NetView product for password authorization, specify SECOPTS.OPERSEC=NETVPW in the CNMSTUSR or CxxSTGEN member, or OPERSEC=NETVPW on the NetView REFRESH command. The password stored in the DSIOPF member is used to check logon password authorization, so you must update the DSIOPF member to change a password.

Note: To prevent unauthorized viewing or modification of the DSIOPF member and command lists which contain passwords, see [Chapter 5, “Controlling Access to Data Sets and Members,”](#) on page 87.

Define the operator identifier and password with the OPERATOR definition statement in the DSIOPF member.

To add an operator to the DSIOPF member, use a statement such as this:

```
NEWOPER OPERATOR PASSWORD=NEWOPER
        PROFILEN DSIPROFB
```

Where NEWOPER is the operator identifier and NEWOPER is the operator password. See [Chapter 13, “Scenarios for Converting Types of Security,”](#) on page 119 for information about converting logon password authorization from the NetView program to an SAF product, either manually or by using the SECMIGR command.

Using DSIEX12 for Password Authorization

To disable NetView password checking so that DSIEX12 is the only type of logon checking, take the following actions:

- Code NOCHECK on the OPERATOR statement in the operator definition in DSIOPF.
- Enable OPERSEC=NETVPW using either the SECOPTS statement in the CNMSTUSR or CxxSTGEN member, or by using the NetView REFRESH command.
- Uncomment the NetView LOGONPW command in the CNMCMD sample and run the command.

Refer to *IBM Z NetView Programming: Assembler* if you want to use NetView installation exit 12 (DSIEX12).

A password must be specified in the DSIOPF member to prevent definition errors, but it is ignored. The logon attributes in the NetView operator profile are used. The following example shows how to define operator NEWOPER using NOCHECK:

```
NEWOPER OPERATOR PASSWORD=USERPW,NOCHECK
```

Note that NOCHECK must follow the password and must have a comma before it.

In this case, the NetView program does not perform security checking on the password. The password is given to installation exit 12 (DSIEX12) for use by the customer-written assembler-language program. When an operator logs on using a definition that has NOCHECK coded, the DWO354I message is issued to the authorized receiver indicating that an operator has logged on with NOCHECK in effect.

Using the NetView Program without Password Authorization

When SECOPTS.OPERSEC=MINIMAL is coded in the CNMSTYLE member, the NetView program does not perform password checking. Unless you use other ways of keeping your system secure, such as physically restricting access to terminals, you must use password security.

Defining Operator Attributes

Operator logon attributes can be defined in the NetView product, in an SAF product, or in both. Although only one definition can be in effect at a time, you can dynamically change whether operator logon attributes are used from NetView operator profiles (the DSIPRF data set) or the NETVIEW segment of an SAF product.

Whether you define operator profiles in the DSIPRF data set or define operators in an SAF product, altering the logon attributes has no effect on the task until it is logged off and then logged back on. Before altering or migrating operator definitions, you must understand the operator attributes. In a NetView operator profile member, you can specify these logon attributes:

- CONSNAM keyword on a PROFILE statement
- HCL keyword on a PROFILE statement
- IC keyword on a PROFILE statement
- CTL keyword on an AUTH statement
- MSGRECV keyword on an AUTH statement
- NGMFADMN keyword on an AUTH statement
- NGMFCMDS keyword on an AUTH statement
- NGMFVSPN keyword on an AUTH statement
- DOMAINS statement
- ISPAN statement
- SPAN statement

Using an SAF product, these attributes can be defined in the NETVIEW segment:

- CONSNAM
- CTL
- DOMAINS
- IC
- MSGRECV
- NGMFADMN
- NGMFVSPN

Using an SAF product, the HCL and NGMFCMDS attributes cannot be defined in the NETVIEW segment, and span of control (in the NetView program, defined by SPAN and ISPAN statements) is defined in the NETSPAN class.

See [Chapter 13, “Scenarios for Converting Types of Security,” on page 119](#) for examples of migration techniques.

Using CONSNAM

The CONSNAM attribute can be used both in NetView operator profiles and in the NETVIEW segment of an SAF product. It is the identifier that is used for the default extended console name in the following situations:

- The operator does not specify a console name using the GETCONID or SETCONID command
- The ConsMask statement in the CNMSTYLE member either is not set or is set to an asterisk (*)

This attribute is also the console name used when you issue the MVS command and have not previously obtained an extended console. If you do not specify either a CONSNAM attribute or the ConsMask statement in the CNMSTYLE member, the default value that is supplied by the NetView program is the task name.

For more information about console names, refer to the z/OS library. For more information about using the CONSNAM keyword in a profile in the DSIPRF data set, refer to the *IBM Z NetView Administration Reference*. Refer to the NetView online help for a description of the GETCONID, SETCONID, and MVS commands.

Using CTL

The CTL attribute can be used in both NetView operator profiles and in the NETVIEW segment of an SAF product. It defines the operator's authority to access resources and views, and establish NNT sessions. The value of this operand must be one of the following values:

SPECIFIC

Indicates the operator can control only the resources and view names that are members of a span listed on ISPAN and SPAN statements in the operator's profile, or resources and view names that are members of the spans the operator has been granted access to using the NETSPAN class of an SAF product. The operator can establish NNT cross-domain sessions only with the NetView domains listed on the DOMAINS statement. SPECIFIC is the default value for the CTL operand.

GENERAL

Indicates the operator can control the resources and view names that are members of a span listed on ISPAN and SPAN statements in the operator's profile, or resources and view names that are members of the spans the operator has been granted access to using the NETSPAN class of an SAF product. The operator can also control resources that are not part of any span. The operator can establish NNT cross-domain sessions only with the NetView domains listed on the DOMAINS statement.

GLOBAL

Indicates span of control is not used. DOMAINS, ISPAN, and SPAN statements, as well as the span names in the NETSPAN class of an SAF product, are not used. An operator with global authority can establish NNT cross-domain sessions with domains specified in the resource routing definition (RRD) statements.

For more information about using the CTL keyword in a profile in the DSIPRF data set, refer to the *IBM Z NetView Administration Reference*.

Using DOMAINS

The DOMAINS attribute can be used both in NetView operator profiles and in the NETVIEW segment of an SAF product. This attribute enables the setup of NNT cross-domain communication for operators with CTL=SPECIFIC or CTL=GENERAL control. It lists which NNT cross-domain sessions this operator can start.

Cross-domain sessions started with the NetView RMTCMD command are not NNT sessions, and are not affected by the DOMAINS attribute.

The DOMAINS attribute does not apply to operators who have CTL=GLOBAL. For operators with CTL=GLOBAL, the valid domains are specified by the RRD statements in the CNMSTYLE member.

For more information about using the NetView DOMAINS statement in a profile in the DSIPRF data set, refer to the *IBM Z NetView Administration Reference*.

Using MSGRECVR

The MSGRECVR attribute can be used both in NetView operator profiles and in the NETVIEW segment of an SAF product. It specifies whether operators are eligible to receive unsolicited messages that are not routed to a particular operator using either the NetView ASSIGN command or NetView automation.

NO

The operator is not eligible to receive unsolicited messages. NO is the default.

YES

The operator is eligible to be the authorized message receiver.

For more information about using the MSGRECV keyword in a profile in the DSIPRF data set, refer to the *IBM Z NetView Administration Reference*.

Using NGMFADMN

The NGMFADMN attribute can be used in NetView operator profiles and in the NETVIEW segment of the SAF product. It specifies whether operators are allowed to perform administrative functions for the NetView management console (NMC). The functions controlled by this keyword are:

- Using the command profile editor
- Adjusting aggregation for individual resources
- Adjusting aggregation for classes of resources
- Adjusting SNA status mapping
- Adjusting unknown status
- Remapping views
- Deleting selected views from the NMC server databases
- Customizing views

NO

The operator does not have administrative authority for the NMC. NO is the default.

YES

The operator has administrative authority for the NMC.

Using NGMFCMDS

The NGMFCMDS attribute can be used in a NetView operator profile only. It cannot be specified using the NETVIEW segment of the SAF product. It specifies whether operators are allowed to issue commands from the pull-down menus of the NMC. NGMFCMDS does not prevent operators from typing commands in a NetView command line window.

YES

The operator is allowed to issue commands from the pull-down menus on the graphical display. YES is the default.

NO

The operator is not allowed to issue commands from the pull-down menus on the graphical display.

Using SPAN, ISPAN, and the NETSPAN class

The NetView SPAN and ISPAN statements are coded only in NetView operator profiles. The NETSPAN class, used when OPSPAN=SAF, is available starting with Version 2 of the RACF product, or in an SAF product with equivalent capabilities.

The SPAN and ISPAN statements are used in operator profiles when OPSPAN=NETV. Each SPAN statement identifies a VTAM or RODM span name which can be added to an operator's span of control using a NetView START SPAN command. An ISPAN statement specifies a span that is automatically activated when an operator logs on. The operator can deactivate a span in the span of control using the NetView STOP SPAN command.

You can use SPAN and ISPAN as often as necessary to define all the span names you want. Code these statements in the DSIPRF data set that is specified by a PROFILEN statement that is associated with the operator. Changes made to SPAN and ISPAN statements take effect the next time an operator logs on to the NetView program with the profile that contains the statement.

When OPSPAN=SAF, the NETSPAN class of an SAF product provides equivalent capabilities as SPAN statements in a NetView operator profile. The span names are defined as resources in the NETSPAN class, and can be protected from unauthorized use. To provide the same function as an ISPAN statement using an SAF product, take the following actions:

- Define the spans to the NETSPAN class of the SAF product.

- Permit the operator to the span resources.
- Add one START SPAN command to the operator's initial command list for each of the ISPAN statements in the NetView operator profile.

For more information about defining span of control in a profile in the DSIPRF data set, see [Chapter 4, “Using Spans to Protect Resources and Views,”](#) on page 67.

Using HCL

The HCL attribute is used only in NetView operator profiles. It defines the default name of the printer (hardcopy log). Define this name in the VTAM definition and in the NetView program HARDCOPY definition statement in the CNMSTUSR or CxxSTGEN member. HCL is an optional keyword.

Although each operator can be assigned to only one printer, several operators can share the same printer. However, if too many operators share the same printer, messages for that device can accumulate and messages might not be printed for some time after they are received.

For more information about using the HCL keyword in a profile in the DSIPRF data set, refer to the *IBM Z NetView Administration Reference*.

Using IC

The IC attribute can be used both in NetView operator profiles and in the NETVIEW segment of an SAF product. It specifies the command or command list that is run immediately after a successful logon. Any command lists are allowed, as are commands defined on a CMDDEF statement as Regular (R), Both (B), or High (H). For NetView profiles, all data between the IC keyword and column 71 is treated as initial command information.

No enclosing quotation marks are allowed around the IC value in NetView operator profiles, although it might be necessary to enclose values in quotation marks when using TSO to enter blank-delimited data into the NETVIEW segment of an SAF product.

If the IC keyword is specified, it must be the last keyword on the PROFILE statement. For more information about using the IC keyword in a profile in the DSIPRF data set, refer to the *IBM Z NetView Administration Reference*.

Using NGMFVSPN

The NGMFVSPN attribute can be used both in NetView operator profiles and in the NETVIEW segment of an SAF product. It defines the authority of the operator to display NetView management console views and resources within views. The NGMFVSPN attribute specifies whether each resource, each view name, or both are checked in the NetView span table when an operator asks to display a NetView management console view. The attribute also specifies whether views and lists indicate that view names or resources have been excluded if the operator is not authorized to see an entire view or some resources in a view.

The NGMFVSPN attribute is coded as a character string. Use each of the 4 characters in the string to specify a different option for operator authorization to display NetView management console views and resources.

If you are using RACF for RODM security, ensure that the NetView domain name is defined to RACF and has been permitted to a minimum of RODM security level 2.

span_level

Defines what level of span checking, if any, is to be enabled when this operator requests views and resources.

N

None. The span table is not checked for operator authority. Because access is not checked, the operator can see all views and resources displayed by the NMC. N is the default.

V

Views. Each view name is checked in the span table to see if the operator is authorized to display the view. This option avoids the overhead of span checking all resources in a view.

R

Resources. Each resource is checked in the span table to see if the operator is authorized to display the resource. View names are not span checked, but every resource in a view is checked.

A

All. Both view names and resources are checked in the span table to see if the operator is authorized to display them.

visible_objects

Specifies whether resources that are not in the operator's span of control are visible as null nodes and links in views displayed to the operator. **Null nodes** and **null links** are placeholders that do not indicate the type of node or link, or give any other information about a node or link, except its placement in the network hierarchy.

This option applies only if you specify R or A for *span_level*.

N

Not visible. Any resources not in an operator's span of control are not displayed in the operator's views. N is the default.

Y

Visible. Any resources not in an operator's span of control are displayed in the operator's views as null nodes and links.

restrict_view_info

Specifies whether an indication is given to the operator when objects not in the operator's span of control are excluded from a view the operator requests or when an entire view cannot be shown to the operator because it is not in the operator's span of control.

This option applies only if you specify V, R, or A for *span_level*.

N

Do not display restricted view information. The operator is not given an indication when resources are excluded from a view or a view is not displayed because the view or resources are not in the operator's span of control. **N** is the default.

Y

Display restricted view information. When the operator does not have authority to see either an entire view or some resources in a requested view, the operator is given an indication that either the entire view or certain resources have been restricted from the operator's display because they are not in the operator's span of control.

restrict_list_info

Specifies whether an indication is given to the operator when view names or resource names that are not in the operator's span of control are excluded from lists. The types of lists include the list of views on the Graphic Monitor Details screen and the results of a Locate Resource, More Detail, or List Suspended Resources request.

This option applies only if you specify V, R, or A for *span_level*.

N

Do not display restricted list information. When view names are excluded from the view list or from Locate Resource responses because the operator is not authorized to see those views, the operator is not given any indication that view names have been excluded. **N** is the default.

Y

Display restricted list information. The operator is given an indication when view names are excluded from the view list or from a Locate Resource response because the views are not in the operator's span of control.

For more information about using the NGMFVSPN attribute in a profile in the DSIPRF data set, refer to the *IBM Z NetView Administration Reference*.

Using an SAF Product Exclusively

If you specify SECOPTS.OPERSEC=SAFDEF in the CNMSTYLE member, or OPERSEC=SAFDEF on the NetView REFRESH command, operators that log on use the operator attributes contained in the NETVIEW segment.

By defining NetView operators exclusively in the SAF product and using the NETSPAN class, you can eliminate the DSIOPF member and the members from the DSIPRF data set. However, you can retain these definitions for migration and backup purposes. For example, if the SAF product is unavailable, you can use the REFRESH command to change to NetView operator definition and span of control while the NetView product is running. If you eliminate the DSIOPF member and the members from the DSIPRF data set, you must initialize with OPERSEC=SAFDEF and you cannot use REFRESH to return to another type of logon security.

When operators are defined exclusively in the SAF product, they can be authorized to log on to a particular NetView program through the APPL class of an SAF product. Define each instance of a NetView program to the APPL class using the domain identifiers. For example, if you are using RACF and your NetView *domainid* is CNM01, enter the following command:

```
RDEFINE APPL CNM01 UACC(NONE)
```

Usage Notes:

- Permit operators and other tasks to the NetView domain identifier in the APPL class with an access level of READ to allow tasks to log on to the NetView program.
- Permit operators to display operator information stored in the requested segment (except passwords) using the LIST SAFOP command. You must use RACF Version 2 Release 1 or later releases, or an SAF product with equivalent capabilities to issue the SAFOP parameter. Ensure that the SAF product is running and the security classes used by the NetView program (such as the NETCMD class) are active. For information about how to set up these types of security, see [Chapter 13, “Scenarios for Converting Types of Security,”](#) on page 119.
- Define operator attributes in the NETVIEW segment. Unlike in the NetView program, where many operators can share profiles, in an SAF product, operators must be individually defined with their own NETVIEW segments. For example, if you use RACF, use the ADDUSER and ALTUSER commands.

```
ADDUSER NEWOPER PASSWORD(PSWD1)
ALTUSER NEWOPER NETVIEW(IC(LOGPROF1) MSGRECV(YES) CTL(GLOBAL))
ALTUSER NEWOPER NETVIEW(NGMFADMN(YES))
```

- To enable a new operator to log on to the NetView domain CNM01:

```
PERMIT CNM01 CLASS(APPL) ID(NEWOPER) ACCESS(READ)
```

- If a resource exists in the APPL class of an SAF product for a NetView domain identifier such as CNM01, only operators who have been permitted to that domain can log on to that NetView program. If a resource does not exist for a particular NetView domain identifier, that NetView program is treated as though it had a resource in the APPL class with UACC(READ), and all operators can log on.
- If no NETVIEW segment exists for a user, no value exists for a field in a NETVIEW segment, or a value specified for a field in the NETVIEW segment is not valid, the following values are used:

CONSNAM

none

CTL

SPECIFIC

DOMAINS

none

IC

none

MSGRECV

NO

NGMFADMN

NO

NGMFVSPN

NNNN

- There is no method of specifying a default printer (HCL) or a value for the NGMFCMDS attribute in the NETVIEW segment. For a description of these keywords, their defaults, and their usage, see [“Defining Operator Attributes”](#) on page 14.
- The NetView program provides the CNMSAF2 sample, which contains RACF definitions for NetView operators. You can modify this sample to meet your installation requirements.

See Chapter 13, “Scenarios for Converting Types of Security,” on page 119 to modify new operator definitions or convert ones that defined command authorization in a previous release of the NetView program.

Defining Operator Attributes in NetView Profiles

You can code more than one profile for an operator. You can also use the same profile for more than one operator. For each operator profile, create a profile member in the DSIPRF data set with a PROFILE definition as the first statement in that file. Other definition statements, such as AUTH, follow this PROFILE statement.

This is an example showing how to add an operator definition to the DSIOPF member of DSIPARM:

```
NEWOPER    OPERATOR    PASSWORD=NEWOPER
           PROFILE     DSIPROFA
```

You can define profiles that:

- Specify a command or a command list to run automatically when an operator logs on
- Define the domains, resources, and commands available to an operator
- Specify whether an operator is eligible to be the authorized receiver of undeliverable messages or perform NMC administrative functions

For examples of profile definitions, browse profiles DSIPROFA and DSIPROFB.

This is an excerpt of sample profile DSIPROFA:

```
DSIPROFA  PROFILE  IC=LOGPROF1
           AUTH     MSGRECV=NO,CTL=GLOBAL
           END
```

The profile in the previous example specifies:

IC=LOGPROF1

A command list named LOGPROF1 (CNME1049) is run automatically when an operator logs on with this profile.

MSGRECV=NO

Operator is not eligible to be the authorized receiver.

CTL=GLOBAL

Span of control is not used.

Because no defaults are specified, they are set for the printer (HCL) or console name.

You can define other profiles as necessary by creating additional profile members in the DSIPRF data set. For more information about creating profile members, refer to the *IBM Z NetView Administration Reference*.

If you change the list of spans in an operator's profile, the operator must log off and then log on before the changes take effect. Spans defined in a NetView span table are dynamically enabled using the REFRESH command.

For information about...	See...
The REFRESH command	NetView online help
Defining span of control	“Defining Span of Control” on page 69

Dynamically Adding or Deleting Operators

If you are using NetView operator definitions in DSIOPF, you can use the NetView REFRESH OPERS command to dynamically add or delete operators while the NetView program is running. This command refreshes operator definitions in DSIOPF that were added since the last time the NetView program was stopped and restarted or since the last REFRESH OPERS command was issued. You can also use the REFRESH OPERS, TEST command to check the operator definitions that will change when you issue the REFRESH OPERS command.

NetView operators defined in an SAF product can be added or deleted dynamically. For more information about dynamic changes, refer to the *RACF General User's Guide* or to the documentation for your product. The REFRESH OPERS command has no effect for operators when the CNMSTYLE member includes a statement specifying SECOPTS.OPERSEC=SAFDEF or the REFRESH command is issued with the OPERSEC=SAFDEF option.

For both types of operator definitions (in either DSIOPF or an SAF product), if an operator's profile specifies CTL=GLOBAL and you dynamically delete the operator while the operator is logged on to the NetView program, the operator session continues until the operator logs off. However, the operator loses the ability to issue the VTAM DISPLAY, MODIFY, and VARY commands from the NetView program for resources that are defined in any span of control.

Dynamically Changing the Method of Defining Operators

You can use the NetView REFRESH command to change the value of the OPERSEC keyword, which allows you to dynamically change where logon attributes are defined.

To use operator definitions and attributes from the NETVIEW segment of an SAF product, use the OPERSEC setting of SAFDEF. If the OPERSEC setting is other than SAFDEF, operators are defined in DSIOPF and operator profiles are used from the DSIPRF data set.

Example of Migrating an Operator Password and Logon Attributes

Now that you have read about how to migrate operator passwords and logon attributes using an SAF product, here is a brief example to help you put it all together for a single operator.

On your system, all tasks must be migrated before you change the type of operator security. Define the PPT task, autotasks, and operators listed in DSIOPF before starting to use an SAF product for passwords and logon attributes.

The example that follows assumes:

- Your system has the NETSPAN class and NETVIEW segment, which require RACF Version 2 Release 1, or an SAF product with equivalent function.
- You want to code the NGMFVSPN attribute and the version of RACF that you are running supports this attribute.

- Use the SAF product with the functions equivalent to your NetView definitions.
- You have a task (such as a TSO operator task) that is authorized to issue these RACF commands.
- The NetView product was initialized with these values from the CNMSTYLE member:

```
SECOPTS.OPERSEC = NETVPW
SECOPTS.CMDAUTH = SCOPE.CNMSCOP1
SECOPTS.OPSPAN = NETV
SECOPTS.AUTHCHK = TARGETID
SECOPTS.SPANAUT = TABLE.span_table
```

- Because NGMFVSPN is set to V (see [Figure 1 on page 22](#)), you have defined view names to spans in the NetView span table.

With these assumptions and because SECOPTS.OPERSEC=NETVPW, both passwords and operator attributes are maintained by the NetView product. In this case, assume the following operator definition is in the DSIOPF member of DSIPARM:

```
NEWOPER    OPERATOR    PASSWORD=NEWOPER
           PROFILEN    OP1PROF
```

As specified on the previous PROFILEN statement, the operator uses this profile in the OP1PROF member of the DSIOPF data set:

```
OP1PROF    PROFILE    HCL=A01A705,CONSNAME=OP1CONS,IC=LOGPROF1
              AUTH      MSGRECVR=NO,CTL=SPECIFIC,NGMFADMN=NO,NGMFVSPN=VNNN
              DOMAINS    CNM02,CNM99
              ISPAN      SPAN1
              SPAN       SPAN2,SPAN3
              END
```

Figure 1. Operator Profile

The following scenario shows how to convert this operator's definition to use an SAF product for password, attribute, and span protection.

- To protect the NetView program from unauthorized logon, activate the APPL class by using this RACF command:

```
SETROPTS CLASSACT(APPL)
```

- Using domain name CNM01 as an example, protect the NetView domain from unauthorized access by defining the NetView domain name to RACF as a resource in the APPL class of the SAF product. Use the following RACF command:

```
RDEFINE APPL CNM01 UACC(NONE)
```

- To define NEWOPER to RACF and set an initial password for NEWOPER, use the following RACF commands:

```
ADDUSER NEWOPER PASSWORD(NEWOPER)
```

- To define all required operators to RACF, check DSIOPF for the existing task definitions which must be migrated, such as:

```
ADDUSER CNM01PPT
```

- To allow operator NEWOPER to log on to NetView domain CNM01, permit NEWOPER and the other operators and autotasks to the domain name using the following RACF command:

```
PERMIT CNM01 CLASS(APPL) ID(NEWOPER) ACCESS(READ)
```

- You can define operator attributes for NEWOPER in the NETVIEW segment of the user profile that are similar to the definitions for NEWOPER in the OP1PROF profile.

- This example uses a TSO job continuation character, the dash (-), which is required to enter such a long command when using a batch job. From an authorized TSO task, the text automatically wraps, and does not require a continuation character.

```
ALTUSER NEWOPER NETVIEW(CONSNAME(OP1CONS) IC(LOGPROF1) MSGRECVR(NO) -
CTL(SPECIFIC) NGMFADMN(NO) NGMFVSPN(VNNN) DOMAINS(CNM02,CNM99))
```

- Span definitions are defined in the NETSPAN class, rather than the NETVIEW segment. To protect spans named SPAN1, SPAN2, and SPAN3 from unauthorized access, define the spans to the NETSPAN class in RACF using the following commands:

```
RDEFINE NETSPAN SPAN1 UACC(NONE)
RDEFINE NETSPAN SPAN2 UACC(NONE)
RDEFINE NETSPAN SPAN3 UACC(NONE)
```

- To allow NEWOPER access to these spans, use the following RACF commands:

```
PERMIT SPAN1 CLASS(NETSPAN) ID(NEWOPER) ACCESS(UPDATE)
PERMIT SPAN2 CLASS(NETSPAN) ID(NEWOPER) ACCESS(UPDATE)
PERMIT SPAN3 CLASS(NETSPAN) ID(NEWOPER) ACCESS(UPDATE)
```

The access level of UPDATE enables the operators to issue VTAM commands (DISPLAY, VARY, and MODIFY) against resources in the span. If you set the access level to READ instead of UPDATE, the operators are limited to using the VTAM DISPLAY command against resources in the span.

- The NGMFVSPN attribute setting of NGMFVSPN=VNNN allows the operator to display NMC views to which the operator has span authorization. The access level must be READ or higher for the operator to see these views.
- To provide a functional equivalent to the ISPAN statement, add the following command to the operator's initial command list, LOGPROF1, as defined on the IC keyword:

```
START SPAN=SPAN1
```

- To allow RACF to protect spans defined in the NETSPAN class, ensure that the NETSPAN class is active:

```
SETROPTS CLASSACT(NETSPAN)
```

- You can change all your operator settings to use an SAF product by entering this NetView command:

```
REFRESH OPERSEC=SAFDEF,OPSPAN=SAF
```

- If this setting causes problems, use the REFRESH command to change your settings back to definitions that were working. If it works as you expect, you can change the SECOPTS statements in the CNMSTUSR or CxxSTGEN member to the following statements:

```
SECOPTS.OPERSEC = SAFDEF
SECOPTS.OPSPAN  = SAF
```

After successfully completing these steps, the operator password, logon attributes, and span-of-control definitions in an SAF product provide equivalent function as they did in the NetView product.

These steps are similar to a fast path through the scenarios in [Chapter 13, “Scenarios for Converting Types of Security,”](#) on page 119.

Using an SAF Product to Restrict Logon Access

You can restrict operator access to NetView systems by limiting where or when they can log on.

If you use an SAF product such as RACF to security check passwords and password phrases, you can restrict NetView operator access by limiting the times or terminal addresses that are valid for logging on. For example, you can use the RACF keywords WHEN and TERMINAL on RACF commands such as ADDUSER, ALTUSER, RDEFINE, or RALTER.

You can use an SAF product for passwords and password phrases by specifying OPERSEC with a value of SAFPW, SAFCHECK or SAFDEF. For details about using the OPERSEC keyword, see the *IBM Z NetView Administration Reference* or the REFRESH command in the NetView online help.

Using a RACF WHEN Keyword to Restrict Logon Times

To restrict a user from entering the system on certain days or during certain times, use the WHEN keyword on the ADDUSER or ALTUSER commands. For example, use this RACF command to specify that NEWOPER can enter the system only on weekdays between the hours of 7:00 am. and 5:00 PM.:

```
ALTUSER NEWOPER WHEN(DAYS(WEEKDAYS) TIME(0700:1700))
```

Using a RACF TERMINAL Keyword to Restrict Terminal Addresses

To control when users can access the system from a specific terminal, specify the TERMINAL keyword on the RDEFINE or RALTER commands for the appropriate terminal. When accessing the NetView program from a 3270 terminal, the TERMINAL keyword specifies the LU name of the terminal. When accessing the NetView program from a TCP/IP terminal, the TERMINAL keyword specifies the network byte representation of the IP address. For example, use these RACF commands to specify that user NETOP1 can access the NetView program from 3270 terminal A01A441 any time during the week, but not at all during the weekend:

```
RDEFINE TERMINAL A01A441 WHEN(DAYS(WEEKDAYS))
PERMIT A01A441 CLASS(TERMINAL) ID(NETOP1)
```

For a complete description of defining users to RACF, refer to the RACF library.

Determining Attributes for Extended Multiple Console Support (EMCS) Consoles

NetView operators use extended EMCS consoles to receive MVS system messages. Each operator issuing MVS system commands obtains an extended EMCS console. The attributes of extended EMCS consoles control the message delivery from the MVS system to the NetView program. This section describes the ways that these attributes can be set and changed.

The EMCS consoles that are obtained using the NetView program usually have attributes that are specified in the NetView system. Although there are MVS default attributes for EMCS consoles, the MVS defaults take effect only in certain cases.

For information about...	See...
The MVS default values for EMCS console attributes	Table 3 on page 25 and the z/OS library

When the NetView program obtains EMCS consoles, a set of NetView defaults are used for EMCS console attributes. You can change some of these NetView defaults using the MVSPARM definitions in the CNMSTUSR or CxxSTGEN member. You can also override some of the EMCS console attributes with the NetView GETCONID command.

Optionally, you can use an SAF product such as RACF to specify the EMCS console attributes in the OPERPARM segment. The EMCSPARM keyword of NetView DEFAULTS and OVERRIDE command can be used to specify whether the OPERPARM segment is used for console attributes. If EMCSPARM=SAF, the values from the OPERPARM segment are used in preference to the values supplied with the NetView product. For the values provided by the NetView product to take precedence, specify EMCSPARM=NETVIEW on the NetView DEFAULTS or OVERRIDE command.

The console attributes specified in the OPERPARM segment can only be used by operators that are:

- Permitted to use a generic MVS.MCSOPER.* profile in the OPERCMDS class.

- Permitted to use a discrete MVS.MCSOPER.consname profile in the OPERCMDS class, where *consname* is the name of the EMCS console to be obtained.

If you are using the OPERPARM segment, you define the OPERPARM under the user ID which is equal to the console name. If the console name is not also a user ID, issue an ADDUSER command for the console name. Your security administrator must use password protection to ensure that this profile is not misused as a user ID for logging on to the system.

```
ADDUSER NEWOPER OPERPARM( AUTH(SYS) )
ADDUSER CNM01PPT OPERPARM( AUTH(MASTER) )
ADDUSER TAPE1 OPERPARM( AUTH(SYS) )
```

These console attributes apply to the EMCS console names NEWOPER, CNM01PPT, and TAPE1. See “Protecting EMCS Console Names Using an SAF Product” on page 27 for additional RDEFINE and PERMIT commands.

When the EMCS console is active, you can change some of the console attributes with MVS system commands. Table 3 on page 25 describes the origins of the dynamically defined MVS EMCS console attributes.

Table 3. Origin of Attributes for Dynamically Defined EMCS Consoles

OPERPARM Value	NetView Default	Specify in CNMSTYLE (CNMSTUSR or CxxSTGEN member)	Specify on GETCONID	Modify with MVS Command
ALTGRP	N/A	No	No	VARY
AUTH	Master	MVSPARM. DEFAUTH	AUTH	VARY
AUTO	Do not receive automatable messages	No	No	No
CMDSYS	My system	No	No	CONTROL
DOM	Normal	No	No	No
KEY	<i>domainname</i>	No	No	No
LEVEL	ALL	No	No	CONTROL
LEVEL(NB)	Receive broadcast messages (except for the PPT task)	MVSPARM. OperRecvBrdcst	BRDCST	CONTROL
LOGCMDRESP	MVS default	No	No	No
MFORM	J S T	No	No	CONTROL
MIGID	No migration ID requested	MVSPARM. MIGRATE	MIGRATE	No
MONITOR	MVS Default	No	No	MONITOR
MSCOPE	All systems	No	No	VARY
ROUTCODE	NONE	No	No	VARY
STORAGE	2000	No	STORAGE	No
UD	MVS default	No	No	VARY
N/A	2147483647	No	QLIMIT	No

Table 3. Origin of Attributes for Dynamically Defined EMCS Consoles (continued)

OPERPARM Value	NetView Default	Specify in CNMSTYLE (CNMSTUSR or CxxSTGEN member)	Specify on GETCONID	Modify with MVS Command
N/A	80%	No	ALERTPCT	No
N/A	1%	No	QRESUME	No

Usage notes:

1. If EMCSPARM=SAF is in effect for the operator and an OPERPARM segment exists in an SAF product for a particular console name, all the OPERPARM segment values are used. The NetView program default values are not used. Therefore, if you specify any of the OPERPARM values with an SAF product, such as RACF, specify all of the OPERPARM values. The MVS default values are used for the attributes you do not specify in the OPERPARM segment.
2. Although the CNMCSSIR task is set up to receive all AUTO(YES) and AUTO(TOKEN) messages, it discards messages that MVS also delivered due to console name to another EMCS console in the same NetView program. This filtering helps prevent duplicate message processing by the CNMCSSIR task for the messages that were sent through a WTO (write to operator) to a particular console. Messages delivered to NetView consoles by other routing criteria, such as route codes, will not be automated at the receiving task unless Canzlog records indicate that it was not delivered to the CNMCSSIR task.
3. Messages that can be automated are marked AUTO(YES) or AUTO(TOKEN) in the MPF table or in a message revision table, or NETVONLY in the message revision table.
4. You can display the consoles that are in use in your NetView system or sysplex by entering the following MVS command:

```
D C,KEY=domainname
```

The default value for the console key is the NetView domain name. You can change this KEY using the OPERPARM segment in RACF (or a compatible security product).

5. MFORM does not control your NetView screen. You can use a screen format definition to tailor your NetView screen. The NetView program provides sample CNMSCNFT for screen formatting.
6. Setting up any of your EMCS consoles to receive messages by ROUTCODE can create duplicate NetView automation and message processing. If you choose to solicit messages using ROUTCODE, consider also that these same messages can be delivered to another task in your NetView program by other message routing criteria. For example, a message you solicited by route code can also be delivered to the EMCS console that is set up to receive automatable messages marked automatable in the MPF table or in a message revision table.

Additionally, if you have more than one EMCS console active in your NetView system that is set up to receive the same route code, NetView automation is driven twice for the same message. Also, consider that some messages have more than one route code.

7. You can use an MVS command to change certain EMCS console attributes (such as ROUTCODE, UD, and so on) before the delete operator message (DOM) for an outstanding WTOR (write to operator with reply) has arrived. However, when you change the attribute, the EMCS console does not receive DOMs for action messages that are received because of the original settings on these attributes. These action messages can be deleted manually using the cursor.
8. The first EMCS console to become active in NetView determines the maximum STORAGE size for the message data space.
9. This table shows settings for EMCS consoles as they are defined to the MVS program. Note that the decision to send a message to a particular console by the MVS program is not based completely on any one attribute in the table.

For information about...**See...**

Screen format definition statements

IBM Z NetView Administration Reference

For information about...**See...**

Customizing the NetView command facility screen *IBM Z NetView Customization Guide*

Protecting EMCS Console Names Using an SAF Product

You can protect console names by completing the following steps:

- Have the security administrator ensure that dynamic parsing is active. For more information about dynamic parsing, refer to the SAF product library.
- Specify `SECOPTS.OPERSEC=SAFCHECK` or `SECOPTS.OPERSEC=SAFDEF` in the `CNMSTUSR` or `CxxSTGEN` member to enable task-level checking.

Ensure you have defined profiles for the PPT task and any other operator tasks which require access to protected resources. To see which tasks you currently have defined in the NetView product, review `DSIOPF`.

The following examples are basic RACF commands entered in TSO with a user ID that is authorized for RACF administration.

- Use the `ADDUSER` function of the RACF product to define `TAPEOPER` and `CNM01PPT` as valid users.

```
ADDUSER TAPEOPER
ADDUSER CNM01PPT
```

- Define consoles to the SAF product using the `RDEFINE` command. Name these consoles using the form:

```
MVS.MCSOPER.consname
```

Where *consname* is the console name for the console you want to define. Define the *consname* in the class `OPERCMD5`, as follows:

```
RDEFINE OPERCMD5 MVS.MCSOPER.TAPE1 UACC(NONE)
RDEFINE OPERCMD5 MVS.MCSOPER.CNM01PPT UACC(NONE)
```

You can create a generic profile using an asterisk for the *consname* value, such as specifying a `MVS.MCSOPER.*` profile value. The asterisk is a pattern-matching character which allows you to protect all console names that are not specifically protected by an individual profile.

- Permit operators to use the preferred EMCS console name using the `PERMIT` function. In this example, operator `TAPEOPER` is allowed to obtain EMCS console `TAPE1`. Specify the console, operator ID, and class as follows:

```
PERMIT MVS.MCSOPER.TAPE1 CLASS(OPERCMD5) ID(TAPEOPER) ACCESS(READ)
PERMIT MVS.MCSOPER.CNM01PPT CLASS(OPERCMD5) ID(CNM01PPT) ACCESS(READ)
```

For information about...**See...**

SAF checking at the task level

[“Authority Checking Commands against the Command Source” on page 34](#)

Generic profiles

RACF library

Chapter 3. Controlling Access to Commands

Command authorization is the process of protecting commands from unauthorized use and selectively granting access to them. To protect commands in the NetView environment, set up command authorization using the NetView command authorization table or the NETCMDS class of an SAF product, such as RACF.

Types of Command Authorization

With the NetView program, you can use the command authorization table or SAF command authorization to restrict access to commands, certain keywords, and values. For more information about which commands, keywords, and values are eligible for protection, see [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,”](#) on page 151.

A *NetView command authorization table* enables you to restrict access to commands, keywords, and values. It then allows you to permit operators and groups of operators to access these restricted commands, keywords, and values. You can also specify commands, keywords, and values that pass authorization checking. Refer to the CNMSCAT2 sample for an example command authorization table. For the GDPS Continuous Availability solution, refer to the AQNCAT sample (%INCLUDED in the CNMSCAT2 sample). The NetView command authorization table is stored as a member of the DSIPARM data set. You can use the REFRESH command to dynamically update your table.

Note: The CNMSCAT2 sample includes the CNMSCATU file, which is intended for user modification.

SAF command authorization enables you to use RACF Version 2 or later, or a comparable SAF product, to restrict access to commands, keywords, and values, and to grant operator access to them. You do this by defining the commands, keywords, and values as resources in the NETCMDS class of the SAF product, and then selectively granting operator access. You can also specify commands, keywords, and values that are accessible universally. Refer to the CNMSAF2 sample for a RACF example. For the GDPS Continuous Availability solution, refer to the AQNSAF sample. When you make changes, you can have them take effect by requesting the SAF product to refresh the NETCMDS class definitions. You do not have to issue NetView commands to include the changes.

Note: The NetView program no longer supports *scope of command* authorization. In previous releases of the NetView program, scope of command authorization enabled you to assign classes of authority (scope classes) to operators, and to place commands, keywords, and values into these classes. Scope classes (OPCLASSES) for an operator were defined in the operator profile (defined in the DSIPRF data set or in the NETVIEW segment of an SAF product). Commands, keywords, and values were placed into scope classes using the CMDCLASS, KEYCLASS, and VALCLASS statements in the CNMCMD member of the DSIPARM data set. If you used scope of command authorization, migrate to either the command authorization table or to SAF command authorization.

If you want information on...	See...
Migration from scope of command	“Changing the Method of Command Authorization” on page 4
Command authorization table	“Using the NetView Command Authorization Table” on page 39
SAF command authorization	“Using the NETCMDS Class in an SAF Product for Command Authorization” on page 49
Command authorization for the GDPS Continuous Availability solution	<i>IBM Z NetView for Continuous Availability Configuring and Using the GDPS Continuous Availability Solution</i>
REFRESH command	NetView online help

If you want information on...	See...
Protecting keywords and values on user-written commands	DSIKVS macro in <i>IBM Z NetView Programming: Assembler</i> and CNMSCOP service in <i>IBM Z NetView Programming: PL/I and C</i>
Protecting keywords and values on user-written command lists	"Controlling Access to Command Lists" in <i>IBM Z NetView Programming: REXX and the NetView Command List Language</i>

Tivoli Enterprise Portal Take Action Command Authorization Checking

If you create new take action commands to be used with the Z NetView Enterprise Management Agent in the Tivoli Enterprise Portal, the command must be prefixed with the characters NA: (NA followed by a colon) to be security checked by the NetView program when it is issued. When a command has this prefix, the command is treated as an agent command rather than as a system command. For information about creating take action commands, see the *IBM Tivoli Monitoring: Tivoli Enterprise Portal User's Guide*.

Starting with Tivoli Management Services Version 6.3.0, all automation commands, such as take action commands, situation actions, tacmd executeCommand commands, and policy workflow commands, operate in a more secure environment. Specifically, Tivoli Management Services captures detailed audit information about every command request and rejects any command from an untrusted source. The mechanism for rejecting untrusted commands relies on an encrypted security token that accompanies each command request as it is passed between IBM Tivoli Monitoring components. The encryption requirement means that all Tivoli Monitoring servers and agents must have access to local cryptographic functions. On distributed systems, GSKit provides the cryptographic support. On z/OS systems, the Integrated Cryptographic Services Facility (ICSF) must be available. Agents running with a framework before Tivoli Management Services Version 6.3.0 are not affected, and encryption and auditing are not enforced.

Tivoli Management Services Version 6.3.0 Fix Pack 1 (FP1) provides remediation measures to help migrate to the enhanced command security. These measures depend on the Tivoli Enterprise Monitoring Server KMS_SECURITY_COMPATIBILITY_MODE environment variable, which has two possible values: Y or N. Y enables compatibility mode, which offers more lenient command security than N. If security compatibility mode is enabled (KMS_SECURITY_COMPATIBILITY_MODE=Y), commands can be processed even if cryptographic facilities are not available at the Tivoli Enterprise Monitoring Server or at the agent. This mode gives you time to ensure that your z/OS Tivoli Enterprise Monitoring Server has ICSF available. It is also compatible with earlier versions when commands originate from a source that is running with a framework before Tivoli Management Services Version 6.3.0.

When you are ready to implement stricter command security, reconfigure your Tivoli Enterprise Monitoring Server to disable security compatibility mode, which changes the value of KMS_SECURITY_COMPATIBILITY_MODE from Y to N. For more information about configuring security settings for a monitoring server that is running in a z/OS environment, see *IBM Tivoli OMEGAMON® XE and Tivoli Management Services: Configuring the Tivoli Enterprise Monitoring Server on z/OS*.

Defining Command Authorization Checking

You can define commands and command lists using the CMDDEF statement in CNMCMU or by using the ADDCMD command. You can specify whether authorization checking is performed by using the SEC keyword, as described in ["Bypassing and Requiring Security Checking Using the SEC Keyword"](#) on page 31. There are several exceptions to normal command security.

Several commands are restricted in the sample command authorization definitions CNMSAF2 and CNMSCAT2. You might also want to restrict other commands in your network that can affect the NetView environment or access to it. Some additional commands to consider restricting are:

- AFTER (using PPT keyword)
- AT (using PPT keyword)
- CHANGEFP (protected as CNME7009)

- CHRON (Use of the ROUTE keyword)
- EVERY (using PPT keyword)
- GETCONID
- GLOBALV
- RMTCMD
- RUNCMD
- SETCONID
- STARTDOM (protected as CNME7001)

If you want information on...	See...
CMDDEF statement	<i>IBM Z NetView Administration Reference</i>
DEFAULTS and ADDCMD commands	NetView online help

Exceptions to Command Authorization Checking

Major exceptions to command authorization checking include:

- Commands that are entered as replies to the NetView WTOR (message DSI802A) are not authority checked. To prevent the NetView program from issuing the WTOR, specify TASK.DSIWTOMT.INIT=NO in the CNMSTUSR or CxxSTGEN member.
- Command authority checks are not made against the PPT or DST tasks, therefore, you do not need to authorize these tasks to access your protected commands. You can protect commands queued to the PPT by using AUTHCHK=SOURCEID, which causes authority to be checked against the user ID that sent the command. For more information about AUTHCHK=SOURCEID, see [“Authority Checking Commands against the Command Source”](#) on page 34.
- Commands issued from a source ID of *BYPASS* are not checked for command authorization by:
 - The NetView command authorization table
 - The SAF product OPERCMDS class
 - The SAF product NETCMDS class

The SOURCEID will default to *BYPASS* if the command was entered at an extended multiple console support (EMCS) console and the operator was not logged on to the EMCS console.

Bypassing and Requiring Security Checking Using the SEC Keyword

Using the SEC keyword on the CMDDEF statement or on the ADDCMD command, along with the environment where the command is issued, determines authorization checking in the following manner:

- If SEC=CH is specified, authorization checking is always performed regardless of the environment.
- If SEC=BY is specified, authorization checking is bypassed regardless of the environment. When you specify SEC=BY, you maximize system performance by avoiding unnecessary security checking.
- If SEC=DE is specified (either explicitly or by default), authorization checking for commands that do not originate in the automation table is performed, and authorization checking for commands that originate in the automation table is controlled by the value of AUTOSEC that can be specified using the DEFAULTS command.

Inherited Command Security in Authorized Command Lists

Primary commands are commands that are directly initiated by an operator, and are always checked for security when the command definition requires it, as described in [“Bypassing and Requiring Security Checking Using the SEC Keyword”](#) on page 31. When a primary command is a command list, it often carries out its function by invoking other commands within the particular context of the initial primary command. These subordinate commands are commands that are invoked by a command list. Subordinate

commands can also be command lists, which invoke other subordinate commands within the context of their operation.

When a command list has passed the audits for a specific environment, the command list can have privileged status to invoke subordinate commands without any further authorization checks. This allows the privileged command to carry out its function even if a subordinate command, when run as a primary command or within a less secure command list, would introduce an unacceptable security risk.

A command list becomes privileged to run subordinate commands without further authorization checks by being listed in the Authorization Bypass Enablement member, DSIAUTB, or in its included members. The privileged command utilizes the Authorization Bypass function, **AUTBYPAS**, to invoke the privilege. Note that **AUTBYPAS** cannot be used to bypass authorization checking for primary commands.

Syntax

AUTBYPAS

```
➔ AUTBYPAS — ( — ON — ) ➔
                |
                | SUSPEND
                |
                | RESUME
                |
                | OFF
                |
                | RESET
```

Purpose of Function

The **AUTBYPAS** function enables writers of command lists to bypass command authorization checks for NetView commands that are issued from command lists. This ensures that operators are allowed to run selected sensitive commands only within the context of specified command lists.

Operand Descriptions

Using **ON**, **SUSPEND**, **RESUME**, **OFF**, and **RESET** enables you to control which commands bypass security checks and which do not.

ON

Specifies that command authorization checks are not performed on NetView commands that are subsequently issued by the command list.

SUSPEND

Specifies that command authorization checks are performed on subsequent NetView commands that are issued by the command list, regardless of the **AUTBYPAS** setting prior to this call. Authorization bypass can be suspended a maximum of 254 times without specifying a **RESUME** in each command procedure.

RESUME

Specifies that the **AUTBYPAS** settings are returned to the state that was in effect before the previous **SUSPEND**. Specifying **RESUME** without a previous **SUSPEND** has no effect.

OFF

Specifies that command authorization checks are performed on NetView commands that are subsequently issued by the command list.

RESET

Specifies that any subsequent NetView commands issued by the command list will have command authorization checks performed, and the **SUSPEND** and **RESUME** statistics will be reset.

Function Results

Code

Meaning

0

Successful completion (the previous state of **AUTBYPAS** was **OFF**)

- 1** Successful completion (the previous state of AUTOBYPAS was ON)
- 4** Required parameter missing
- 8** SUSPEND or RESUME limit reached
- 12** Function not allowed in this environment
- 16** Unexpected error
- 20** Storage was not obtained

Usage notes

Note: Use great care when deciding which commands are to bypass command authorization checking. Controlling who is allowed to write command lists is very important.

Consider the following items when using AUTOBYPAS:

- Command lists that are supplied by IBM that issue the AUTOBYPAS command are listed in the DSIAUTB member in the DSIPARM data set. Review the DSIAUTB member for the list of NetView commands that are capable of bypassing authorization checking for subordinate commands.
- Add any command lists that you want to bypass command authorization checking to the DSIAUTBU member in the DSIPARM data set. The DSIAUTBU member is included by the DSIAUTB member. Make sure that you have a CMDDEF statement defined for any commands that you add. If the AUTOBYPAS command is issued without the proper definition, a return code of 12 is received. After you have completed your updates, issue the AUTOBNABL command against the DSIAUTB member to enable the AUTOBYPAS function. See the *IBM Z NetView Command Reference Volume 1 (A-N)* for more information about the AUTOBNABL command.
- Any command list that is run from a private data set or is loaded into memory using INSTORE LOCAL will not be authorized to issue AUTOBYPAS. A return code of 12 is received and no changes to security are made.
- Any command for which SEC=CH is specified on its command definition will be authority-checked, regardless of any AUTOBYPAS specification.
- Any command for which SEC=BY is specified on its command definition will not be authority-checked, regardless of any AUTOBYPAS specification.
- Any command for which SEC=DE is specified on its command definition which originates at an automation table when DEFAULTS AUTOSEC=BYPASS is in effect will not be authority-checked, regardless of any AUTOBYPAS specification.
- Any command for which SEC=DE is specified on its command definition which originates at an automation table when DEFAULTS AUTOSEC=CHECK is in effect will be authority-checked, regardless of any AUTOBYPAS specification.
- Any command for which SEC=DE is specified on its command definition which originates at an automation table when DEFAULTS AUTOSEC=DEFER is in effect will respect the current AUTOBYPAS specification.
- Any commands or command procedures that are called from a command list inherit the AUTOBYPAS setting. When control is returned to the caller, the security setting is the same as it was when the call was made. No called command list can change the AUTOBYPAS setting in the calling command list.
- The NetView CMD command does not recognize the AUTOBYPAS settings. When CMD is issued, authorization checks are always made.
- If a REXX function is called as an external subroutine or function, command checking is performed on the function and not on the caller.

Examples

REXX example:

```
ABPstate = AUTOBYPAS('ON')      /* Set auth bypass, save current setting */
'PIPE NETVIEW cmd'              /* Issue NetView command                */
If ABPstate = 0 Then             /* If preserved bypass setting was off, */
  Call AUTOBYPAS 'OFF'          /* restore to prior setting            */
```

NetView command list language example:

```
&ABPSTATE = &AUTOBYPAS ON
NetView cmd
&IF .&ABPSTATE = .0 &THEN +
&TRASH = &AUTOBYPAS OFF
```

Authority Checking Commands against the Command Source

Authority checking for commands can be done using either the USERID of the task that issued the command or the USERID of the task under which the command is run.

Determining the Target for Authority Checking

If SECOPTS.AUTHCHK=TARGETID is specified in the CNMSTYLE member or AUTHCHK=TARGETID is specified on the REFRESH command, authority checking is performed using the USERID of the task under which the command is run. For example, as shown in Figure 2 on page 34, when the OPER1 ID is authorized to use the EXCMD command to queue the PURGE command to the NETOP1 ID, the options on the PURGE command cannot be limited based on the authority of OPER1. Any operator who is authorized to use the EXCMD command to queue the PURGE command to NETOP1 has the same authority as NETOP1, because the authority checking is performed against the NETOP1 ID and not against the original issuer of the command.

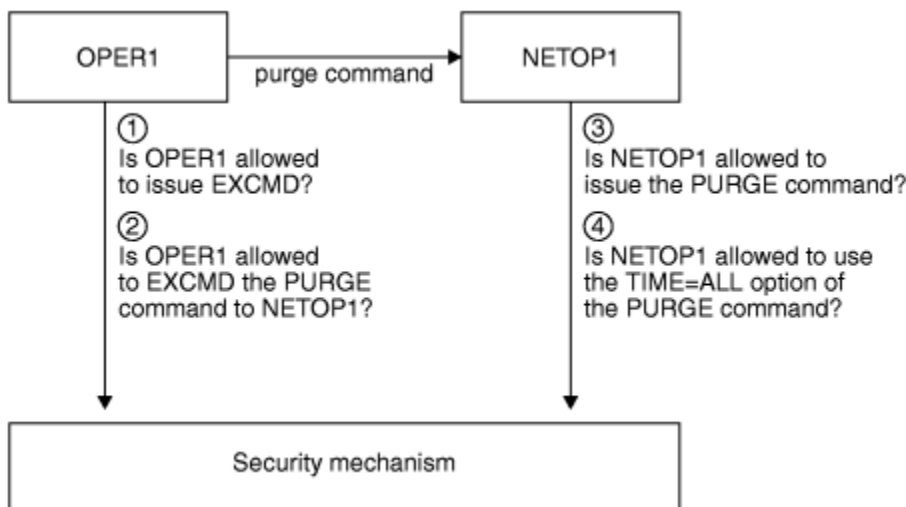


Figure 2. Example of AUTHCHK=TARGETID

NetView commands issued through batch (the JCL COMMAND command) or through TSO SUBMIT function run under the task that is assigned console 0. If console 0 is not assigned to any autotask, it defaults to the primary program operator interface task (PPT). Since command authority checking is not performed against the PPT, any command sent to the PPT will run successfully. Consider assigning console 0 to an autotask so NetView commands can be protected against unauthorized users.

Determining the Source for Authority Checking

You can specify that authority checking is performed against the original issuer of the command by specifying SECOPTS.AUTHCHK=SOURCEID in the CNMSTUSR or CxxSTGEN member, or

AUTHCHK=SOURCEID on the REFRESH command. Specifying authority checking against the original issuer of the command enables you to authorize multiple operators to issue EXCMD to NETOP1 and still control individual authority based on the issuer of the command.

For example, Figure 3 on page 35 shows that although OPER1 has been granted authority to issue EXCMD to NETOP1 with the PURGE command, the actual authority for running the PURGE command with the TIMER=ALL option is determined by the authority of OPER1 and not NETOP1.

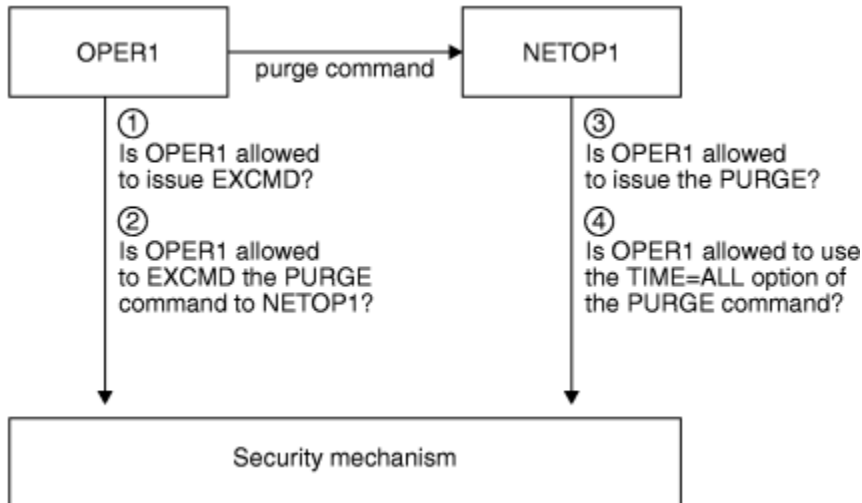


Figure 3. Example of AUTHCHK=SOURCEID

Source checking is also performed for span authorization for VTAM commands. If SPANCHK=SOURCEID is specified, span checking for VTAM commands (such as VERIFY and DISPLAY) is performed against the operator who originates the VTAM command.

Using Command Source ID Authority Checking

Unless you restrict command authorization by source ID, it is difficult to protect commands that are routed. For example, if you protect the CLOSE command as a keyword of the EXCMD command, then the CLOSE command is not protected at the target ID if an operator routes the CLOSE command using EXCMD OPER1, CMD CLOSE, unless CMD is also protected as a keyword of EXCMD.

Using the source ID, the command is checked for authorization against the ID closest to the command source, as defined in Table 4 on page 36. Using target ID, the command is checked for authorization against the ID running the command, checking the routed command as a keyword.

The SOURCEID of a command is determined by the command and the environment under which it is issued. In some cases, a command flowing through a single NetView program can pass through more than two tasks. In those cases, any intermediate tasks pass along the originating user ID as the SOURCEID. For example, OPER1 issues:

```
EXCMD AUTO2 AT 12:00:00,PPT,AUTOTBL MEMBER=DSITBL01
```

As the AUTO2 task processes the AT command, it does not become the original issuer but recognizes that OPER1 is the existing SOURCEID. Therefore, when the AUTOTBL command runs under the PPT, the SOURCEID used for authority checking is OPER1.

The NetView timer commands (AFTER, AT, CHRON, and EVERY) can be used to issue commands that are run by the PPT task. These commands are examples of authorization checking using the command source ID. Unless you restrict command authorization by source ID, or restrict the PPT keyword on timer commands, you cannot protect specific commands routed to the PPT task, because the PPT task is not subject to command security.

As you consider migration from a prior release of the NetView program, be aware that the NetView initialization default has changed to use SOURCEID authority checking. You might want to consider using

TARGETID authority checking until you are sure that the various sources of commands are authorized. Remember that command lists can contain timer commands and other embedded commands such as EXCMD commands. When you switch to SOURCEID, all sources must then be authorized before commands can process successfully.

Determining SOURCEID Values for Authority Checking

Table 4 on page 36 shows the various methods for determining the SOURCEID value based on command and environment:

Table 4. SOURCEID Determination

Command and Environment	SOURCEID Determination
EXCMD command or a same-domain LABEL command prefix used to queue an imbedded command to another task.	The SOURCEID is the task that issued the EXCMD command, or the existing SOURCEID at the time the EXCMD command was issued.
TIMER commands that are scheduled to run under the PPT.	The SOURCEID is the task that issued the AT, EVERY, CHRON, or AFTER command, or the existing SOURCEID at the time the AT, EVERY, CHRON, or AFTER command was issued. Note: The SOURCEID is not erased by saving and restoring timer commands.
NetView SUBMIT command for jobs submitted to the operating system from NetView program.	For OPERSEC=SAFCHECK or OPERSEC=SAFDEF, the identity that is checked by the operating system is the issuer of the SUBMIT command, or the existing SOURCEID at the time the SUBMIT command was issued. For other values of OPERSEC, NetView's authority is used for submitting the job.
NetView commands that are received over the subsystem interface (SSI) that were entered at an MVS operator console.	When an MVS console has been associated with an autotask using the AUTOTASK command with the CONSOLE= parameter, NetView commands can be entered from that MVS console. This is done by prefixing the NetView command with the NetView designator character, which by default is %. If the MVS operator has logged on to the MVS console with a user ID, the SOURCEID is the user ID of the MVS operator. If an operator has not logged on that MVS console, the SOURCEID of that task defaults to *BYPASS*. Commands issued from a source ID of *BYPASS* are not checked for command authorization by: • The NetView command authorization table • The SAF product OPERCMDS class • The SAF product NETCMDS class

Table 4. SOURCEID Determination (continued)

Command and Environment	SOURCEID Determination
NetView commands that are entered using the MVS MODIFY command.	When an MVS console has been associated with an autotask using the AUTOTASK command with the CONSOLE= parameter, NetView commands can be entered from that MVS console by issuing an MVS MODIFY or STOP command against the NetView task. The NetView command is entered as text following the MODIFY command. The first parameter on the MODIFY command is the application ID that is being modified. If the MVS operator has logged on to the MVS console with a user ID, the SOURCEID is the user ID of the MVS operator. If an operator has not logged on that MVS console, the SOURCEID of that task defaults to *BYPASS*. Commands issued from a source ID of *BYPASS* are not checked for command authorization by: • The NetView command authorization table • The SAF product OPERCMDS class • The SAF product NETCMDS class
NetView commands that are received over the subsystem interface (SSI) that were entered by TSO users.	When a TSO user ID has been associated with an autotask using the AUTOTASK command with the CONSOLE= parameter, NetView commands can be entered from that TSO user ID when the user is acting as an MVS operator by using an EMCS console session, or when using SDSF. The SOURCEID is the TSO user's user ID.
Commands issued from JCL.	When a job that issues a NetView command is submitted by a TSO user ID, the SOURCEID is the TSO user ID. If the ID of the submitter is unknown, a default user ID is inserted. The value of the default user ID is defined by the system installation.
MVS ROUTE command issued from the NetView program.	If the MVS command ROUTE is issued from a NetView task, the originating source ID is always passed to the SAF product for authorization checks in the OPERCMDS class. This occurs for all settings of AUTHCHK and CMDAUTH.
Generic commands that are selected in NMC.	The NMC operator ID is the SOURCEID that is associated with the NetView commands issued from a NMC workstation.
Commands that are routed to an operator from the automation table.	The SOURCEID is the operator ID to which the command is routed. Note: Commands from the automation table are subject to authority checking unless SEC=BY was specified on the CMDDEF statement or SEC=DE was specified (or SEC was not specified) and AUTOSEC=BYPASS is in effect. For more information, refer to the DEFAULTS command in the NetView online help.
Commands that are routed to another domain with the RMTCMD command or using a cross-domain LABEL command prefix.	The receiving autotask becomes the SOURCEID on the receiving domain.
Commands that are routed to another domain over OST-NNT sessions.	The ID of the NNT becomes the SOURCEID on the remote domain.
User written command processors that call DSIMQS macro.	When the DSIMQS macro is used to queue a command to another task, the SOURCEID is the task name (TVBOPID) of the DSIMQS caller, or the existing SOURCEID at the time DSIMQS was called.

Table 4. SOURCEID Determination (continued)

Command and Environment	SOURCEID Determination
CNMSMSG service (PL/I and C).	If CNMSMSG is called to queue a command from one task to another, the SOURCEID is the task name (TVBOPID) of the CNMSMSG issuer, or the existing SOURCEID at the time the CNMSMSG service was called.
Commands that are received over the PPI.	Commands received over the PPI by the command receiver in DSIQTSK must be in SNA GDS buffer format. The major GDS variable key is X'1043'. The SOURCEID is determined by the contents of subfield X'01' of subvector X'07' contained in GDS vector X'0001'. Note that the SOURCEID can potentially be the same as an operator ID for which command security is defined in your domain. In this case, the SOURCEID inherits the command authority of that operator in your domain. If the source of the command is a RODM method, the SOURCEID is the name of the RODM.
Commands that are entered from the NetView command line window on NMC.	The NMC operator ID is the SOURCEID that is associated with the NetView commands issued from an NMC workstation.

Protecting Commands Containing Special Characters

There are some special characters that cannot be included in the command identifier or SAF resource name. For this reason, the NetView program translates these special characters to other characters before passing them to either the NetView command authorization table or the SAF product. The special characters that are translated along with their translated results are:

Reserved Character	Translated Result
.	/
*	+
%	?
&	:
- (dash)	_ (underscore)
' (blank)	_ (underscore)

As an example, the following NetView command can be entered by a NetView operator:

```
FOCALPT DELETE FPCAT=*,TARGET=NETA.REMOTE.
```

To restrict access to the FPCAT keyword and its value in the NetView command authorization table, include the following statement:

```
PROTECT NETA.CNM01.FOCALPT.FPCAT.+
```

To restrict access to this keyword and value using RACF, include the following RACF profile:

```
RDEFINE NETCMDS NETA.CNM01.FOCALPT.FPCAT.+ UACC(NONE).
```

Note that, in both cases, the asterisk was translated to a plus.

When you develop command processors for the NetView program, take this translation into consideration when designing keywords and values. Avoid using both a reserved character and its passed character to differentiate between two keywords or two values; they cannot be protected by unique NetView command authorization table load statements or RACF profiles.

For example, if you write a command processor TESTCMD with a keyword of TEST that allowed a value of I/O and I . O, the table statement for both values is:

```
PROTECT NETA.CNM01.TESTCMD.TEST.I/O
```


And the RACF profile for both values is:

```
RDEFINE NETCMDS NETA.CNM01.TESTCMD.TEST.I/O UACC(NONE)
```

Using the NetView Command Authorization Table

The NetView command authorization table restricts the use of commands and operands to specific operators or groups of operators. The table consists of a member in DSIPARM containing the authorization statements. This table can include statements to embed other members from DSIPARM. The command authorization table also protects command lists that do not have a CMDDEF statement in CNMCMD. The sample command authorization table supplied with the NetView program is CNMSCAT2.

Command Authorization Table Syntax

Table statements consist of freeform text that specify a table statement type followed by its operands. You can enter the text in upper or lowercase, with the exception of %INCLUDE statements, which must be in uppercase. For all other statements, the text is converted to uppercase when the table is processed. The table statements must be coded between columns 1 and 72. If a statement is too long to fit between columns 1 and 72, you can use the <BEGIN> and <END> statements when multiple lines must be treated as a single statement. You can include a sequence number in columns 73 through 80 for problem determination purposes. If the NetView program encounters any errors while processing the table statements, the error messages issued include the sequence number of the line in error. An asterisk in column 1 denotes a comment and causes the rest of the NetView command authorization table line to be ignored.

Command Identifiers

You can protect commands, as well as certain keywords and values, with the NetView command authorization table.

You identify which commands, keywords, and values are protected using command identifiers. The format of the command identifiers for the NetView command authorization table is the same as the format of the resource names used in the NETCMDS class in an SAF product. In its full form, a command identifier uses these fields: *netid.luname.command.keyword.value*.

You can determine the *netid* and *luname* values for your systems using the NetView LISTVAR command, or the REXX functions NETID() and DOMAIN() in a command list. In the LISTVAR example shown in [Figure 4 on page 39](#), the current values are NETA for *netid* and the NetView domain, CNM01, for *luname*.

```
LISTVAR
CNM353I LISTVAR : OPSYSTEM = MVS/ESA
CNM353I LISTVAR : MVSLEVEL = SP7.2.2
CNM353I LISTVAR : CURSYS   = NMP177
CNM353I LISTVAR : VTAMLVL  = V622
CNM353I LISTVAR : VTCOMPID = 5695-11701-220
CNM353I LISTVAR : NetView  = IBM Z NetView V6R3
CNM353I LISTVAR : NETID    = NETA
CNM353I LISTVAR : DOMAIN  = CNM01
CNM353I LISTVAR : APPLID   = NTVAA007
CNM353I LISTVAR : OPID     = OPER3
CNM353I LISTVAR : LU       = NTAAL703
CNM353I LISTVAR : TASK     = OST
CNM353I LISTVAR : NCCFCNT  = 4
CNM353I LISTVAR : HCOPIY   =
CNM353I LISTVAR : TYPE     = ENT
CNM353I LISTVAR : CURCONID =
```

Figure 4. Example of LISTVAR Command Output

Note that some characters are reserved if you are using the NetView command authorization table or an SAF security product for command authorization checking. See [“Protecting Commands Containing Special Characters” on page 38](#) for more information.

The command identifier can be up to 246 characters in length, including the periods that serve as field delimiters. The individual fields of the command identifier have no maximum length as long as the entire command identifier length does not exceed 246 characters.

You can use generic characters in command identifiers. An asterisk (*) can be used to indicate that all possible values of a field are protected or permitted, except those that are more explicitly specified. You can use the asterisk either as a replacement for a field or as a trailing character to indicate that all items that begin with the specified characters are to be protected. The percent sign (%) can be used as a single character generic anywhere within the command identifier. Generic characters are useful to specify a level of protection for commands for which there is not a match in the table. You can do this because the most specific command identifier determines the level of protection for a command.

Note: The generic character combination %* is not valid.

Commands are checked separately from keywords and values. When designing command identifiers, keep in mind that the command is checked first, in addition to the subsequent security checking for the command, keyword, and value combinations. Keywords and their associated values are checked as a pair. To protect a keyword that has a value associated with it, there must be an entry in the value position of the command identifier. The command identifiers can be in these formats:

```
netid.luname.command  
netid.luname.command.keyword (used only for keywords without values)  
netid.luname.command.keyword.value
```

Where:

netid

Indicates the VTAM network identifier. You can specify a generic character (*) for this field.

The *netid* specification is syntax checked for format (*netid* cannot begin with a left parenthesis) but no checking is done to verify that the *netid* specified matches the current *netid*. This field is treated as a place holder and is supported so that the format of the command identifier in the NetView command authorization table is the same as the format of a resource name in the NETCMDS class of an SAF product.

luname

Indicates the domain identifier for an instance of a NetView program.

command

Indicates the command name on the CMDDEF statement in the CNMCMD member of DSIPARM, or a command list name. This must be the actual command name and not a synonym defined by the CMDSYN statement. No checking is performed to validate that *command* is a valid command or command list name.

Some commands have duplicate names within the components of the NetView program, and those are protected by a command identifier with a *command* field that is different from the command name.

For example, the command facility and the session monitor both have a LIST command. The command identifier for the command facility command is *netid.luname.LIST*, and the command identifier for the session monitor LIST is *netid.luname.NLDM.LIST*. See Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151 to determine the command identifiers that are supplied with the NetView product that you can use.

keyword

Indicates the keyword identifier that is protected.

value

Indicates the value identifier that is protected when used with the keyword on the command.

The keyword or value used with the command might not match the keyword or value being protected because of synonyms, defaults, and substitutions of command identifier values. The only commands, keywords, or values that are provided by the NetView program and that can be protected are identified in Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151. For user-

written commands, the keywords and values that can be protected are the values that are passed to the DSIKVS macro or the CNMSCOP service in the command processor.

Table Statements

This section describes the format and function of the following statement types:

- <BEGIN> and <END>
- %INCLUDE
- PROTECT
- EXEMPT
- GROUP
- PERMIT
- SETVAR

<BEGIN> and <END> Statements

The <BEGIN> and <END> statements specify the beginning and end of a NetView command authorization table statement that spans multiple input lines. The total length of any individual table statement must not exceed 4096 characters, including blanks, which provides a maximum of 56 input lines.

The syntax for the <BEGIN> and <END> statements are:

<BEGIN>

➤ <BEGIN> ➤

<END>

➤ <END> ➤

The <BEGIN> and <END> statements must occur on lines by themselves. Command identifiers can continue onto more than one line, but statement types, group names, and each user ID in a *userid_list* must not span more than one line. As you enter multiple input lines, be careful not to accidentally put an asterisk in column 1, because the remainder of that line will be treated as a comment.

Example:

To enter a command authorization statement that spans two input lines, specify it in the following way:

```
<BEGIN>
GROUP  ALLOPS  OPER1,OPER2,OPER3,OPER4,OPER5,OPER6,OPER7,OPER8,OPER9,
              OPER10,OPER11
<END>
```

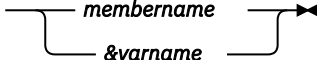
Note: Blank characters between input lines alignment are valid.

%INCLUDE

The %INCLUDE statement enables you to keep portions of your NetView command authorization table in separate DSIPARM members. Both the %INCLUDE statement and its values (either the *membername* or *&varname*) must be capitalized.

The syntax for the %INCLUDE statement is:

%INCLUDE Statement

➤ % — INCLUDE —  ➤

Where:

%INCLUDE

Indicates the keyword coded at the beginning of each %INCLUDE statement.

membername

Indicates the name of the DSIPARM member to be included.

&varname

Indicates the name of an existing local or global variable, preceded by the ampersand (&) character.

Usage Notes:

1. Each %INCLUDE statement can be no longer than one line.
2. A member that has been included can contain %INCLUDE statements as well as other NetView command authorization table statements.
3. A member that has been included cannot include itself either directly or indirectly.
4. If you specify a variable name for the value of the %INCLUDE, the NetView program includes the designated member when you issue the REFRESH command with CMDAUTH=TABLE. The NetView program searches for the variables in the following order:
 - If the REFRESH command is issued from a command procedure, the NetView program searches first for a local variable of the name *varname*, then for a task global variable, and finally for a common global variable.
 - If the REFRESH command is not issued from a command procedure, the NetView program searches for a task global variable of the name *varname* and then for a common global variable.

If you change the value of the variable after activating the NetView command authorization table, the member that is included does not change, unless you reissue the REFRESH command.

Note: You cannot use a variable name in a command authorization table that is specified on the SECOPTS.CMDAUTH statement in the CNMSTYLE member for NetView initialization.

For a full description of this statement, see the *IBM Z NetView Administration Reference*.

Example:

To include member TBL02 from DSIPARM, include the following statement in your NetView command authorization table:

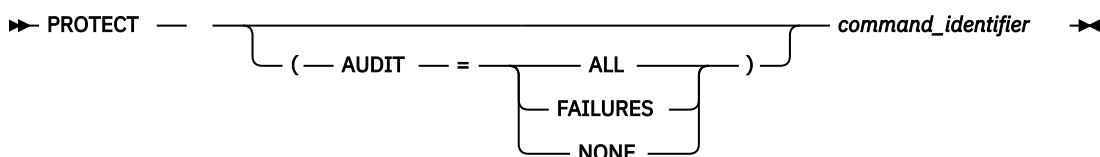
```
%INCLUDE TBL02
```

PROTECT Statement

The PROTECT statement identifies a command identifier to be protected.

The syntax for the PROTECT statement is:

PROTECT



Where:

command_identifier

Specifies the *netid*, *luname*, command, keyword, and value to be protected. See [“Command Identifiers”](#) on page 39 for information about specifying command identifiers.

AUDIT

Specifies whether an audit record is created when a command authority check yields a match on the command identifier. The audit records can be SMF type 38 records, or the DSIXITXL exit can write the records to an external log. The AUDIT keyword is optional. If not specified, auditing is determined by the value of CATAUDIT on the DEFAULTS command. When specified, the value overrides the value specified for CATAUDIT on the DEFAULTS command. Valid values for AUDIT are:

ALL

Specifies that an audit record is to be created when a match occurs on the command identifier.

FAILURES

Specifies that an audit record is to be created when a match occurs on the command identifier and the command authority decision is *fail*.

NONE

Specifies that no audit record is to be created when a match occurs on the command identifier.

Example:

To define a command identifier to protect the AUTH keyword and MASTER value of the GETCONID command in domain CNM01, use the following statement:

```
PROTECT *.CNM01.GETCONID.AUTH.MASTER
```

To define a command identifier to protect the AUTH keyword and MASTER value of the GETCONID command in domain CNM01, and to create audit records for all attempts to get a console with master authority, use the following statement:

```
PROTECT (AUDIT=ALL) *.CNM01.GETCONID.AUTH.MASTER
```

Usage Notes:

- Create one PROTECT statement for each command that you want to protect. For example, to protect the STOP command for *luname* CNM01, create a table entry as follows:

```
PROTECT *.CNM01.STOP
```

- Create one PROTECT statement for each command and keyword that does not have an associated value that you want to protect. For example, to protect the OFF keyword on the AUTOTBL command for *luname* CNM01, create a table entry as follows:

```
PROTECT *.CNM01.AUTOTBL.OFF
```

- Create one PROTECT statement for each command, keyword, and value combination that you want to protect. For example, to protect the OPERID keyword with a value of AUTO1 on the ENDTASK command for *luname* CNM01, create a table entry as follows:

```
PROTECT *.CNM01.ENDTASK.OPERID.AUTO1
```

- To protect all values of OPERID in the previous example, create a table entry as follows:

```
PROTECT *.CNM01.ENDTASK.OPERID.*
```

- To protect all values of OPERID that begin with "TEST" and end with "0", create a table entry as follows:

```
PROTECT *.CNM01.ENDTASK.OPERID.TEST%0
```

- To allow a NetView operator to issue a NetView command that is protected with a PROTECT statement, you must use a PERMIT statement for each operator ID or group of operators that must be authorized.
- If you have more than one statement that describes the same command, keyword, and value, the first is used and all others are ignored. The *netid* and *luname* values are ignored when the NetView command authorization table is loaded. The following example shows how generic characters cause the second command identifier to be ignored. If the following statements are included in the NetView command authorization table for domain CNM01, only the first is used:

```
PROTECT *.*.AUTOTBL.MEMBER.DSITBL01
PROTECT *.CNM01.AUTOTBL.MEMBER.DSITBL01
```

EXEMPT Statement

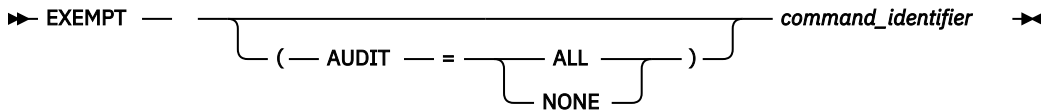
The EXEMPT statement identifies a command and optionally a keyword and value to be exempted from command authorization.

It enables all users to issue a command, keyword, or value, which is similar to defining a resource in the NETCMDS class with a universal access of read (UACC(READ)).

Using specific EXEMPT statements can reduce the amount of processing required for command authorization checking, and can improve performance. Refer to the *IBM Z NetView Tuning Guide* for specific performance suggestions.

The syntax for the EXEMPT statement is:

EXEMPT



Where:

command_identifier

Is the identifier specifying the *netid*, *luname*, command, keyword, and value to be exempted. See [“Command Identifiers”](#) on page 39 for information on specifying command identifiers.

AUDIT

Specifies whether an audit record is created when a command authority check yields a match on the command identifier. The audit records can be SMF type 38 records, or the DSIXITXL exit can write the records to an external log. The AUDIT keyword is optional. If not specified, auditing is determined by the value of CATAUDIT on the DEFAULTS command. When specified, the value overrides the value specified for CATAUDIT on the DEFAULTS command. The values allowed for AUDIT are:

ALL

Specifies that an audit record is to be created when a match occurs on the command identifier.

NONE

Specifies that no audit record is to be created when a match occurs on the command identifier.

Example:

To define a command identifier to exempt the LIST command in any domain, you must use the following statements:

```
EXEMPT *.*.LIST
EXEMPT *.*.LIST.*
```

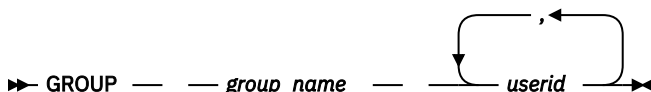
The first statement applies only to the LIST command. The trailing asterisk in the second statement causes this command identifier to apply to all keywords and values of the LIST command that are not more explicitly specified.

GROUP Statement

The GROUP statement defines a list of operators to be associated with a specific group name for command security purposes. The group name is unrelated to other groups of operators, such as the groups used to route messages using the NetView ASSIGN command.

The syntax for the GROUP statement is:

GROUP



Where:

group_name

Is the 1–8 character name of the group you are defining. The *group_name* cannot contain an ampersand (&), asterisk (*), or percent sign (%). The group name cannot be the same as any of your user IDs that are defined in the NetView command authorization table.

userid

Is the 1–8 character identifier of a user to be included in the group. The *userid* cannot contain an ampersand (&), asterisk (*), or percent sign (%). This must be an individual user ID and not the name of a group.

Example:

To define a group named NIGHTOPS containing operators FELIX, MORRIS, and TOM, use the following command:

```
GROUP NIGHTOPS FELIX,MORRIS,TOM
```

To define a large number of operators to a group, you can either repeat the same group name on multiple group statements or create a multiple-line group statement using the NetView <BEGIN> and <END> statements.

PERMIT Statement

The PERMIT statement authorizes a user ID or group to issue a command and optionally a keyword and value. The command identifier must have been previously protected with a PROTECT statement. You can include more than one PERMIT statement for the same command identifier.

The syntax for the PERMIT statement is:

PERMIT

```
➤ PERMIT — — authorized_name — — command_identifier ➤
```

Where:***authorized_name***

Is the 1–8 character name of a user ID or a group that is authorized to issue the command, keyword, and value identified by the *command_identifier*. The *authorized_name* cannot contain an ampersand (&), asterisk (*), or percent sign (%). No checking is done to verify that a user ID is a valid NetView operator ID.

Note: User IDs used in your table statements are independent of DSIOPF operator definitions and SAF product definitions. Even if an operator has been deleted from DSIOPF or the SAF product, the operator will continue to have the same command authority with respect to the active NetView command authorization table as long as the operator remains logged on.

command_identifier

Is the identifier specifying the *netid*, *luname*, command, keyword, and value to be authorized. See [“Command Identifiers” on page 39](#) for information about specifying command identifiers.

Examples of Generic Characters in PERMIT and PROTECT Statements

The following examples assume you are using the NetView command authorization table statements to define command authorization and that your NetView domain name (*luname*) is CNM01. To authorize only NETOP1 to issue the GETCONID command with the AUTH keyword and a value of MASTER, include the following statements:

```
PROTECT *.CNM01.GETCONID.AUTH.MASTER
PERMIT NETOP1 *.CNM01.GETCONID.AUTH.MASTER
```

- To protect all other keywords on the GETCONID command, include the following statement:

```
PROTECT *.CNM01.GETCONID.*
```

- It is not required to use a generic character in the value position, but because all of the GETCONID keywords have corresponding values, a command identifier of *.CNM01.GETCONID.*.* is functionally equivalent.
- Some commands have keywords that are issued without a corresponding value. For example, the SAVE and PPT keywords of the EVERY command do not have a value. To authorize NETOP1 in domain CNM01 to issue the EVERY command with both the SAVE and PPT keywords, include the following statements:

```
* PROTECT KEYWORDS ON "EVERY" COMMAND
PROTECT *.CNM01.EVERY.SAVE
PROTECT *.CNM01.EVERY.PPT
PERMIT NETOP1 *.CNM01.EVERY.SAVE
PERMIT NETOP1 *.CNM01.EVERY.PPT
```

- To protect both the SAVE and PPT keywords and all other keywords on the EVERY command, include the following statement:

```
PROTECT *.CNM01.EVERY.*
```

- Notice that there is no generic character used for value. The command identifier *.CNM01.EVERY.*.* does not protect the SAVE and PPT keywords but protects only keywords that are specified with a corresponding value. The command identifier *.CNM01.EVERY.* protects keywords that have corresponding values as well as keywords that do not have corresponding values.
- Using an asterisk (*) as a trailing generic character at the end of a command identifier allows you match on subsequent values in that field and subsequent fields. Using a trailing asterisk in the *command* field protects the command, and all its keywords and values. For example, if you use this statement:

```
PROTECT *.*.STOP*
```

It protects the NetView STOP command, and all its keywords and values. Note that this is equivalent to coding all three of the following statements:

```
PROTECT *.*.STOP
PROTECT *.*.STOP.*
PROTECT *.*.STOP*.*
```

- Using a trailing asterisk in the *keyword* field will protect the keyword for that command, with all the values on that keyword. For example, to protect all the values on all the REXX keywords for the NetView DEFAULTS command, use this statement:

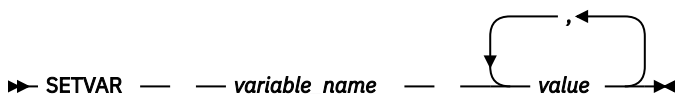
```
PROTECT *.*.DEFAULTS.REXX*
```

SETVAR Statement

The SETVAR statement defines a table variable to represent multiple values that can be used in command identifiers. Table variables must represent an entire field value and must be defined before being used.

The syntax for the SETVAR statement is:

SETVAR



Where:

variable_name

Is the 1–32 character name of the variable you are defining. The variable name cannot contain an ampersand (&), dash (-), period (.), asterisk (*), or percent sign (%).

value

Is the 1–242 character value to be included in the command identifier. The value cannot contain an ampersand (&), dash (-), or period (.).

Examples:

To define a variable EURODOM to represent domains CNM01, CNM02, and CNM99, use the following command:

```
SETVAR EURODOM CNM01,CNM02,CNM99
```

To subsequently use the variable &EURODOM in a PROTECT statement, include the following command:

```
PROTECT *.&EURODOM.STOP
```

When processed, this generates the equivalent of the following table statements:

```
PROTECT *.CNM01.STOP
PROTECT *.CNM02.STOP
PROTECT *.CNM99.STOP
```

Note that the table variable EURODOM represented the entire field value. A specification such as the following specification is *not* valid:

```
SETVAR EURODOM 01,02,99
PROTECT *.CNM&EURODOM.STOP
```

To define a variable XDOM to represent commands ROUTE and RMTCMD, use the following command:

```
SETVAR XDOM ROUTE,DSIUSNDM
```

To subsequently use the variable &XDOM in a PROTECT statement, include the following command:

```
PROTECT *.CNM01.&XDOM
```

This generates the equivalent of the following table statements:

```
PROTECT *.CNM01.ROUTE
PROTECT *.CNM01.DSIUSNDM
```

Loading the NetView Command Authorization Table

The NetView command authorization table can be loaded during NetView initialization as specified by the SECOPTS.COMDAUTH statement in the CNMSTYLE member. During initialization, if syntax errors are encountered, messages are issued, no table is loaded, and the system operator is asked with a WTOR message if they want initialization to continue. No command security will be in effect. After NetView initialization is complete, errors can be corrected and the table reloaded using the REFRESH command. If there are syntax errors in the table processed by the REFRESH command, the table is not loaded. There is a TEST keyword on the REFRESH command that you can use to check for syntax errors before attempting to load the table.

Restricting Keywords and Values of a VTAM Command using the NetView Command Authorization Table

You can restrict any keywords and values of a VTAM command using the NetView command authorization table. The value entered with the VTAM keyword ID, SLU, PLU, LU1, LU2, IPADDR (excluding the START command), or DVIPA is a VTAM resource or IP address. If the VTAM resource is qualified with a network ID, access to the network ID and resource name is checked separately. Therefore, they must be defined in separate PROTECT statements. The VTAM resource name and the network ID can be up to 8 characters long.

If IDTYPE=IPADDR is entered with the VTAM DISPLAY command, the value entered with the ID keyword is an IP address. The value entered with the IPADDR or DVIPA keyword is also an IP address. The IP address can be longer than 8 characters. See [Appendix F, "How to Protect IP Addresses for Command Security and Span of Control,"](#) on page 231 for examples of protecting IP addresses.

For example, the following statements can be defined in the NetView command authorization table:

```
PROTECT *.*.DISPLAY.ID.NETA
PROTECT *.*.DISPLAY.ID.DSIAMLUT
PROTECT *.*.DISPLAY.ID.87/123/136/121
```

A NetView operator cannot run the following VTAM commands, even though the resource, NTVB5LUC, DSIAMLUT and 87.123.136.121 are in the operator's span of control:

```
D NET,ID=NETA.NTVB5LUC
D NET,ID=NETA.DSIAMLUT
NET,ID=NETB.DSIAMLUT
D NET,ID=87.123.136.121,IDTYPE=IPADDR
```

However, the operator can run the following commands:

```
D NET,ID=NTVB5LUC
D NET,ID=NETB.NTVB5LUC
```

Command Authorization Table Usage Notes

Some command identifiers are more specific than others. For example, the following table statements are ordered from most specific to least specific, as you can determine by comparing the character strings from left to right:

```
PROTECT *.CNM01.STOP.FORCE.CNM01PPT
PROTECT *.CNM01.STOP.FORCE.*
PROTECT *.CNM01.STOP.*
```

The most specific PROTECT statement in your NetView command authorization table is the statement with the generic character latest in the sequence of fields, after the *netid* and *luname* fields. Only the most specific statement that matches the command being issued is used for command authorization.

The type of generic character is also used to determine which command identifier is most specific. Because the percent sign (%) generic character replaces just a single character, the percent sign is considered more specific than the asterisk (*) generic character. For example, ABC% is more specific than ABC* when evaluating the value ABCD.

For example, the value SYS1 matches both the SY%1 and the SYS* identifiers. In this case, SYS* is considered to be more specific because the generic character is in the fourth position, rather than SY%1 which has a generic character in the third position.

If both a PROTECT and an EXEMPT statement are coded for the same command identifier, message BNH184E will be issued indicating a syntax error in the NetView command authorization table.

If this message is issued because of a REFRESH command, the NetView command authorization table is not loaded. If the message is issued during initialization, the NetView command authorization table is loaded, but only the first (PROTECT or EXEMPT) statement is used. Use message BNH184E to find the problem.

Command Authorization Table Example

The following steps provide an example of defining operator authority using a NetView command authorization table:

1. Define groups of operators.

```
GROUP GRP1 NETOP1,NETOP2,AUT01,AUT02
<BEGIN>
GROUP GRP2 OPER1,OPER2,OPER3,OPER4,OPER5,OPER6,NETOP1,NETOP2,
AUT01,AUT02
<END>
```

Note that these operators are grouped into two classes of authorization.

2. Define the commands, keywords, and values to be protected.

- In the example that follows, the statement protects the LOGAUTO (CNME7016) command.

```
PROTECT      NETA.CNM01.CNME7016
```

- The following statements define the OVERRIDE command as unprotected except for the REXXSTRF keyword. This keyword can only be used by operators in group GRP1.

```
EXEMPT      NETA.CNM01.OVERRIDE
PROTECT     NETA.CNM01.OVERRIDE.REXXSTRF.*
PERMIT GRP1 NETA.CNM01.OVERRIDE.REXXSTRF.*
```

- Note that these statements protect the CHANGEFP (CNME7009) command and authorize operators in group GRP1 to issue the command.

```
PROTECT     NETA.CNM01.CNME7009
PERMIT GRP1 NETA.CNM01.CNME7009
```

- All of the following statements are comments. If you remove the asterisks from these statements, they protect the GLOBALV command and restrict its use to operators in groups GRP1 and GRP2. The statements also protect the SAVEC and RESTOREC keywords, and restrict their use to operators in groups GRP1 and GRP2. Finally, the statements protect the asterisk (*) and PURGEC keywords, and restrict their use to operators in group GRP1.

```
* PROTECT      NETA.CNM01.GLOBALV
* PERMIT GRP1  NETA.CNM01.GLOBALV
* PERMIT GRP2  NETA.CNM01.GLOBALV
* PROTECT     NETA.CNM01.GLOBALV.SAVEC
* PERMIT GRP1 NETA.CNM01.GLOBALV.SAVEC
* PERMIT GRP2 NETA.CNM01.GLOBALV.SAVEC
* PROTECT     NETA.CNM01.GLOBALV.RESTOREC
* PERMIT GRP1 NETA.CNM01.GLOBALV.RESTOREC
* PERMIT GRP2 NETA.CNM01.GLOBALV.RESTOREC
* PROTECT     NETA.CNM01.GLOBALV.ASTERISK
* PERMIT GRP1 NETA.CNM01.GLOBALV.ASTERISK
* PROTECT     NETA.CNM01.GLOBALV.PURGEC
* PERMIT GRP1 NETA.CNM01.GLOBALV.PURGEC
```

Using the NETCMDS Class in an SAF Product for Command Authorization

You can define NetView commands as resources in the NETCMDS class of your Systems Authorization Facility (SAF) product. In this way, you can use RACF Version 2 Release 1 or later to restrict access to the commands and some of their operands.

Defining NetView Commands as NETCMDS Resources

To define NetView commands as resources in the NETCMDS commands class, use resource names described in the following topics. The format of the resource names used in the NETCMDS class in an SAF product is the same as the format of the command identifiers for the NetView command authorization table. For more information about keywords and values that are eligible for protection, see [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,”](#) on page 151.

Commands are checked separately from keywords and values. When designing resource names, remember that the command is checked first. Commands, keywords, and value combinations are checked in the following order:

```
netid.luname.command
netid.luname.command.keyword
netid.luname.command.keyword.value
```

Where:

netid

Indicates the VTAM network identifier. You can specify a generic character (*) for this field.

This value is compared with the VTAM network identifier from the last activation of VTAM or +NONE+ if VTAM has not been activated. If you do not need to differentiate between *netids* and are not concerned about whether VTAM has been active, specify a generic character (*) for this field.

luname

Indicates the domain identifier for an instance of a NetView program.

command

Indicates the command name on the CMDDEF statement in the CNMCMD member of DSIPARM, or a command list name. This must be the command name and not a synonym defined by the CMDSYN statement. No checking is done to validate that *command* is a valid command or command list name.

Some commands have duplicate names within the components of the NetView program, and those are protected by a resource name with a *command* field that is different from the command name.

For example, the command facility and the session monitor both have a LIST command. The resource name for the command facility command is *netid.luname.LIST*, and the resource name for the session monitor LIST is *netid.luname.NLDM.LIST*. See [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151](#) to determine the resource names that must be used.

keyword

Indicates the keyword identifier that is protected.

value

Indicates the value identifier that is protected when used with the keyword on the command.

The keyword or value that is used with the command might not match the keyword or value being protected because of synonyms, defaults, and substitutions of values in the resource name. The only commands, keywords, or values that are provided by the NetView program and that can be protected are identified in [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151](#). For user-written commands, the keywords and values that can be protected are the values that are passed to DSIKVS or CNMSCOP in the command processor.

Examples of NETCMDS Resource Definitions

The following examples assume you are using the RDEFINE command (or its abbreviation, RDEF) of RACF to define your NETCMDS resources, that your *netid* is NETA, and your *luname* is CNM01.

- Create one resource in the NETCMDS class for each command that you want to protect. In the following example, RACF is used to protect the STOP command:

```
RDEFINE NETCMDS NETA.CNM01.STOP UACC(NONE)
```

- Create one resource in the NETCMDS class for each command and keyword that does not have an associated value that you want to protect. For example, protect the OFF keyword on the AUTOTBL command by issuing:

```
RDEFINE NETCMDS NETA.CNM01.AUTOTBL.OFF UACC(NONE)
```

- Create one resource in the NETCMDS class for each command, keyword, and value combination that you want to protect. For example, to protect the OPERID keyword with a value of AUTO1 on the ENDTASK command, create a protected resource by issuing:

```
RDEFINE NETCMDS NETA.CNM01.ENDTASK.OPERID.AUTO1 UACC(NONE)
```

- To protect all values of OPERID in the previous example, create a protected resource by issuing:

```
RDEFINE NETCMDS NETA.CNM01.ENDTASK.OPERID.* UACC(NONE)
```

- To allow a NetView operator to issue a NetView command protected in the NETCMDS class, you must grant a level of access of at least READ.
- To minimize the number of occasions when SAF can reach no command authorization decision, you can universally grant or deny access to the remaining commands, keywords, and values by defining a

generic resource name for your NetView program. Using RACF, you can do this for a *netid* of NETA and an *luname* of CNM01 by issuing one of the following commands:

```
RDEFINE NETCMDS NETA.CNM01.* UACC(READ)
```

```
RDEFINE NETCMDS NETA.CNM01.* UACC(NONE)
```

SAF Command Authorization Example

The following steps show an example of defining operator authority to RACF, assuming a *netid* of NETA, a NetView domain name (*luname*) of CNM01, and assuming the operators are already defined to RACF.

1. Activate the NETCMDS class, if not already active:

```
SETROPTS CLASSACT(NETCMDS) GRPLIST
```

2. Define the NETCMDS class as a GENERIC class to allow the use of generic characters, if generic characters will be used:

```
SETROPTS GENERIC(NETCMDS)
```

3. Define groups of operators.

```
ADDGROUP GRP1
CONNECT NETOP1 GROUP(GRP1) UACC(READ)
CONNECT NETOP2 GROUP(GRP1) UACC(READ)
CONNECT AUTO1 GROUP(GRP1) UACC(READ)
CONNECT AUTO2 GROUP(GRP1) UACC(READ)
ADDGROUP GRP2
CONNECT OPER1 GROUP(GRP2) UACC(READ)
CONNECT OPER2 GROUP(GRP2) UACC(READ)
CONNECT OPER3 GROUP(GRP2) UACC(READ)
CONNECT OPER4 GROUP(GRP2) UACC(READ)
CONNECT OPER5 GROUP(GRP2) UACC(READ)
CONNECT OPER6 GROUP(GRP2) UACC(READ)
CONNECT NETOP1 GROUP(GRP2) UACC(READ)
CONNECT NETOP2 GROUP(GRP2) UACC(READ)
CONNECT AUTO1 GROUP(GRP2) UACC(READ)
CONNECT AUTO2 GROUP(GRP2) UACC(READ)
```

Note that these operators have been grouped into two classes of authorization, and that RACF group definitions are not dynamic. Operators must log off, then log on again, before changes to groups become effective.

4. Define the commands, keywords and values to be protected.

- In the following example, this statement protects the LOGAUTO (CNME7016) command.

```
RDEFINE NETCMDS NETA.CNM01.CNME7016 UACC(NONE)
```

Note that no operators are authorized to issue the LOGAUTO command.

- The following statements define the OVERRIDE command as unprotected except for the REXSTRF keyword. This keyword can only be used by operators in group GRP1.

```
RDEFINE NETCMDS NETA.CNM01.OVERRIDE UACC(READ)
RDEFINE NETCMDS NETA.CNM01.OVERRIDE.REXXSTRF.* UACC(NONE)
PERMIT NETA.CNM01.OVERRIDE.REXXSTRF.* CLASS(NETCMDS) ID(GRP1) ACCESS(READ)
```

- These statements protect the CHANGEFP (CNME7009) command and authorize operators in group GRP1 to issue the command:

```
RDEFINE NETCMDS NETA.CNM01.CNME7009 UACC(NONE)
PERMIT NETA.CNM01.CNME7009 CLASS(NETCMDS) ID(GRP1) ACCESS(READ)
```

5. Put your definitions into effect by refreshing the NETCMDS class.

```
SETROPTS RACLIST(NETCMDS) REFRESH
```

Protecting Immediate Commands When CMDAUTH=SAF

Immediate commands are host NetView commands that are defined with TYPE=I on the CMDDEF statement in CNMCMD, or defined with TYPE=B on the CMDDEF statement in CNMCMD, and are entered from the command line. These commands are run under the control of an IRB exit. This environment prohibits the use of an SAF RACROUTE macro to call a security product. When using command authorization with the NetView command authorization table, immediate commands are protected the same way as other commands.

To restrict the use of immediate commands while using an SAF product for command authorization, you can specify a backup NetView command authorization table. Specify the name of the backup NetView command authorization table by using the SECOPTS.CMDAUTH = SAF.*backtbl_name* statement in the CNMSTYLE member, or the BACKTBL keyword on the REFRESH command. For an example backup table, refer to sample CNMSBAK1.

Using SAF with a NetView Command Authorization Table for Backup

When using an SAF product for command authorization, you can specify a NetView command authorization table to be used for immediate commands and for command authorization when the call to the SAF product does not give a security decision. The SAF product cannot make a security decision when:

- There is no resource name in the NETCMDS class that can protect or authorize the command.
- The NETCMDS class is not active.
- The SAF product is not active.
- The data space for the NETCMDS class was deleted.

You can activate a backup NetView command authorization table by specifying SECOPTS.CMDAUTH=SAF.*backtbl_name* in the CNMSTYLE member, or by using the BACKTBL keyword on the REFRESH command.

Decide how you will use a backup table. One approach is to use the backup table to protect immediate commands and to give access to a subset of commands. For example, give authorized operators access to the NetView REFRESH command in case you need to specify another form of command authorization.

Another approach is to create a backup table that duplicates the authority checking being done by the SAF product. If you choose this approach, keep both the SAF product and the backup table at the same level of protection.

A third approach is to use the backup table to protect immediate commands only, and use generics so that all other commands either pass or fail.

Using SAF without a Backup Command Authorization Table

You can specify the action taken for authorization when the SAF product gives no decision on passing or failing access to a command, and a backup NetView command authorization table has not been specified. You specify this action using the SECOPTS.CMDAUTH=SAF.PASS or SECOPTS.CMDAUTH=SAF.FAIL statement in the CNMSTYLE member, or by using the SAFNODEC keyword on the REFRESH command. You can specify that all such command access requests are passed or failed. However, use the BACKTBL keyword instead.

Specifying FAIL causes you to lose the ability to issue any non-immediate NetView commands (if SEC=BY is not specified) if the SAF product becomes unavailable. Specifying PASS provides no command authority checking in the event that the SAF product or NETCMDS class becomes unavailable.

Note: The SAFNODEC and BACKTBL keywords are mutually exclusive and cannot be used at the same time.

Defining TSO Stage Authorization

When a TSO command is issued using the TSO pipe stage, NetView command authorization for the TSO pipe stage is checked. Subsequently, when the command arrives in TSO, it is checked by TSO command authorization. When the command arrives in TSO, it is checked using the same rules that apply when the TSO username is directly logged on.

Protecting the TSO Pipe Stage

To prevent operators from using the TSO pipe stage, add security definitions for DSIPITSO. For example, in the NetView command authorization table, you can use:

```
PROTECT *.*.DSIPITSO
```

Protecting Specific TSO Commands Using the TSO Pipe Stage

To prevent operators from sending specific TSO commands using the TSO pipe stage, specify VERB in the keyword position of the command identifier, and the TSO command itself in the value position of the command identifier. For example, to prevent operators from issuing “PIPE TSO SEND”, you can use the following statement in a NetView command authorization table:

```
PROTECT *.*.DSIPITSO.VERB.SEND
```

For NetView command authorization purposes, the TSO command is considered to be the first blank delimited token after “PIPE TSO”.

Note: The TSO stage cannot resolve synonyms for the TSO command that is specified. All command synonyms for the imbedded TSO command must be individually protected.

Protecting TSO Servers from Unauthorized Use

To control which TSO servers are available to individual NetView operators, specify TSOSERV in the keyword position of the command identifier, and the name of the TSO server in the value position of the command identifier. For example, to prevent unauthorized NetView operators from sending commands to the TSO server that is identified by the *userid/jobname* USER1/SERVJOB1, you can code the following command in a NetView command authorization table:

```
PROTECT *.*.DSIPITSO.TSOSERV.USER1/SERVJOB1
```

In the *userid/jobname* specification, the *userid* is the TSO user ID and the *jobname* is what was specified using the MEM keyword on the START command when the TSO server was started. Both are specified for granularity since multiple TSO servers can be running with the same TSO user ID.

Note:

1. There is also a TSOSERV keyword on the NetView START command that can be protected. For the START command, the protection of the TSOSERV keyword controls whether operators can start the TSO server. On the TSO pipe stage, the protection provided by specifying "TSOSERV" in the keyword position of the command identifier controls whether or not an operator can send commands to that server. Starting the server is controlled separately from sending commands to the server.
2. For information about using SAF surrogate authority to provide control over who is allowed to submit TSO commands from the NetView program, see [“Surrogate Support for TSO Command Servers” on page 95](#).

Defining EXCMD Authorization

The NetView EXCMD command is used to send commands to another task.

Two operands are used with the EXCMD command. One is the *operator_id* where the command is being sent, and the other is the *command* that is being sent. These two operands are checked as a keyword-value pair.

Note: When protecting the target verb of EXCMD, specify the command verb, and not a synonym. Unless otherwise documented, the verb is either the command name used on the CMDDEF statement or the value of the NAME keyword of your ADDCMD command. The verb for labeled commands beginning with a slash is EXCMD.

For example, these are the command identifiers to protect EXCMD OPER1 LOGOFF:

```
PROTECT *.*.EXCMD.OPER1.LOGOFF
```

Defining RUNCMD Authorization

The NetView RUNCMD command is used to send a command from the NetView program to a service point application. You can protect RUNCMD, its keywords, and values using standard NetView command security: the NetView command authorization table or an SAF product. However, standard command security cannot protect the actual service point command that you send using the RUNCMD command.

To enable security for service point commands issued by RUNCMD, you must define a CMDDEF statement in CNMCMDU and write code in NetView installation exit DSIEX19. For each service point command you want to protect, add a CMDDEF statement with MOD=DSISPCMD. This is an example statement:

```
CMDDEF .servpcmd.MOD=DSISPCMD
```

Where *servpcmd* is the command you are checking. Then write DSIEX19 installation exit code to examine the command being sent to a service point application and either authorize or reject the command. CNMS4307 is an example of DSIEX19 in assembler language. The exit can parse the service point command and use NetView services to make a security decision based on the:

- Application name
- Command to be sent to the service point
- Network ID
- Service point name
- User ID against which the RUNCMD was checked

When the service point command is parsed, DSIEX19 can make the security decision or invoke NetView security services to make the decision. If your exit 19 returns a non-zero return code, BNH192E is issued and the command is not sent to the service point. Examples of security services include:

DSICES

Assembler macro for command authorization

DSIKVS

Assembler macro for keyword and value authorization

CNMSCOP

PL/I or C service for command authorization

NetView security services DSICES, DSIKVS, and CNMSCOP treat all data as uppercase. If you want the DSIEX19 installation exit to use these services to protect mixed-case commands, keywords, or values, you must translate the lowercase characters so they are uniquely represented in uppercase. For example, to protect p differently than P, you can use DSIEX19 to make a security call for PLOWCASE and PUPCASE.

DSICES, DSIKVS, and CNMSCOP also have a length restriction of only being able to protect the first eight characters. To protect longer values using these services, use DSIEX19 to substitute a shorter character string. You can solve special character, mixed case, and greater than 8-character problems with a single substitution. For example, to protect Service_Point_Command, use DSIEX19 to make a security call for SERVPAMD. When using DSIEX19 to change the value of parameters, ensure the new values you assign

(such as PLOWCASE, PUPCASE, or SERVPCMD) are unique character strings that do not conflict with existing CMDDEF statements or command security.

If the RUNCMD command is embedded in another command, such as EXCMD, the user ID that is passed to the DSIEX19 installation exit is either the source or target based on the setting of the SECOPTS.AUTHCHK statement in the CNMSTYLE member or the most recently issued REFRESH command.

If you want information on...	Refer to...
DSIEX19	<i>IBM Z NetView Programming: Assembler or IBM Z NetView Programming: PL/I and C</i>
Using NetView security services	DSIKVS and DSICES macros in <i>IBM Z NetView Programming: Assembler</i> and CNMSCOP service in <i>IBM Z NetView Programming: PL/I and C</i>

Defining Security for the CHRON Command

The CHRON command has syntax that is more complex than most commands. CHRON uses multiple levels of keywords, items in lists, and quoted strings.

Command security for the CHRON command is checked so that operands within parentheses can be uniquely defined in the command authorization table or an SAF product.

The following rules describe CHRON commands and what command identifiers are checked:

RULE 1: Each keyword that does not take a value (NOSAVE, SAVE, LOCAL, GMT, REFRESH, TEST, and DEBUG) is checked in the form:

```
netid.luname.CHRON.keyword
```

RULE 2: Each keyword with a value is checked in the form:

```
netid.luname.CHRON.keyword.value
```

With the CHRON command, the value can be a list or quoted string.

Command Example:

```
CHRON AT=(),RECOVERY=IGNORE,NOSAVE,LOCAL,ROUTE=OPER1,ID=TEST1,COMMAND='MSG ALL
HELLO'
```

The following command identifiers are checked:

```
netid.luname.CHRON
netid.luname.CHRON.AT.()
netid.luname.CHRON.RECOVERY.IGNORE
netid.luname.CHRON.NOSAVE
netid.luname.CHRON.LOCAL
netid.luname.CHRON.ROUTE.OPER1
netid.luname.CHRON.ID.TEST1
netid.luname.CHRON.COMMAND.'MSG_ALL_HELLO'
```

Rule 3A: Keywords that are within parenthesized lists of other keywords are checked using the hierarchy of keywords with a "(" between so that the keyword hierarchy can be uniquely identified. The compound keyword that is generated is tested with the value of the innermost keyword. This checking is done at each level of the nesting of the lists. When a keyword is within a list that is the value of another keyword, the notation uses both keywords with a "(" between them.

Rule 3B: From the outermost to innermost, if a "keyword=(list)" is used, if any values in the list are without keywords, the "keyword=value" check is done for that value. The keyword that is checked is the keyword hierarchy defined by Rule 3A.

Command Example:

```
CHRON EVERY=(INTERVAL=(000-01.00.00 FOR=08.00.00))
```

The following command identifiers are checked:

```
netid.luname.CHRON
netid.luname.CHRON.EVERY.(INTERVAL=(000_01/00/00_FOR=08/00/00))
netid.luname.CHRON.EVERY(INTERVAL.(000_01/00/00_FOR=08/00/00)
netid.luname.CHRON.EVERY(INTERVAL.000_01/00/00
netid.luname.CHRON.EVERY(INTERVAL(FOR.08/00/00
```

Substitution of certain special characters is performed as described in “Protecting Commands Containing Special Characters” on page 38. For example, a dash becomes an underscore in the command identifier.

Rule 4: Quoted string values are checked as a single value, including the apostrophes and all text within the apostrophes.

Command Example:

```
netid.luname.CHRON.REM.'ISN''T THIS A REMARK STRING?'
```

The following command identifier is checked:

```
CHRON REM='ISN''T THIS A REMARK STRING?'
```

Rule 5: For the DAYSWEET keyword, days of the week can be followed by a sublist identifying particular weeks of the month. The day name and each item in the sublist are treated as a unit.

Command Example:

```
CHRON EVERY=(DAYSWEET=(NOT MON(1ST 2nd)))
```

The following command identifiers are checked:

```
netid.luname.CHRON
netid.luname.CHRON.EVERY.(DAYSWEET=(NOT_MON(1ST_2ND)))
netid.luname.CHRON.EVERY(DAYSWEET.(NOT_MON(1ST_2ND))
netid.luname.CHRON.EVERY(DAYSWEET.NOT
netid.luname.EVERY(DAYSWEET.MON(1ST)
netid.luname.EVERY(DAYSWEET.MON(2ND)
```

This lets you check the sublist values without concern for the order of the items within the sublist. Notice that the value MON(1st 2nd) is not checked since the values MON(1st) and MON(2nd) are checked.

The following table illustrates a detailed list of possible command identifiers that can be defined for the CHRON command. This table is more detailed than what is provided in [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,”](#) on page 151. The rule that causes the command identifier to be checked is shown in the second column.

Table 5. NetView Command Identifiers for the CHRON Command		
Commands and Keywords Identifier	RULE	SAF Resource or Command Authorization Table Identifier
CHRON	Command Name	netid.luname.CHRON
AT=	2 2 3B 3B 2	netid.luname.CHRON.AT() netid.luname.CHRON.AT.(timespec datespec) ¹ netid.luname.CHRON.AT.timespec netid.luname.CHRON.AT.datespec ¹ netid.luname.CHRON.AT.yyy_mm_dd_hh/mm/ss/micros ¹

Table 5. NetView Command Identifiers for the CHRON Command (continued)		
Commands and Keywords Identifier	RULE	SAF Resource or Command Authorization Table Identifier
AFTER=	2 2	netid.luname.CHRON.AFTER.timespec ¹ netid.luname.CHRON.AFTER.ddd_hh/mm/ss/micros ¹
EVERY=	2 2 2	netid.luname.CHRON.EVERY.NONE netid.luname.CHRON.EVERY.() netid.luname.CHRON.EVERY.(everyoptions) ¹
EVERY=(INTERVAL=	3A 3B 3B 3A	netid.luname.CHRON.EVERY(INTERVAL.() netid.luname.CHRON.EVERY(INTERVAL.(intervaloptions) ¹ netid.luname.CHRON.EVERY(INTERVAL.timespec ¹ netid.luname.CHRON.EVERY(INTERVAL. ddd_hh/mm/ss/ micros ¹
EVERY=(INTERVAL= (FOR=	3A 3A	netid.luname.CHRON.EVERY(INTERVAL(FOR.timespec netid.luname.CHRON.EVERY(INTERVAL(FOR. hh/mm/ss/ micros ¹
EVERY=(INTERVAL= (MXREPEAT=	3A 3A	netid.luname.CHRON.EVERY(INTERVAL(MXREPEAT. NOLIMIT netid.luname.CHRON.EVERY(INTERVAL(MXREPEAT. repeat_count
EVERY=(INTERVAL= (OFF=	3A 3A	netid.luname.CHRON.EVERY(INTERVAL(OFF.timespec netid.luname.CHRON.EVERY(INTERVAL(OFF. hh/mm/ss/ micros ¹
EVERY=(REMOVE=	3A 3A, 3B 3B 3B 3A	netid.luname.CHRON.EVERY(REMOVE.MANUALLY netid.luname.CHRON.EVERY(REMOVE.(removeoptions) ¹ netid.luname.CHRON.EVERY(REMOVE.datespec ¹ netid.luname.CHRON.EVERY(REMOVE.timespec ¹ netid.luname.CHRON.EVERY(REMOVE. yyyy_mm_dd_hh/mm/ss/micros ¹
EVERY= (REMAFTER=	3A 3A	netid.luname.CHRON.EVERY(REMAFTER.timespec ¹ netid.luname.CHRON.EVERY(REMAFTER. ddd_hh/mm/ss/ micros ¹
EVERY= (DAYSWEEK=	3A 3B 3B 3B 5	netid.luname.CHRON.EVERY(DAYSWEEK.ALL netid.luname.CHRON.EVERY(DAYSWEEK.(daysweeklist) ¹ netid.luname.CHRON.EVERY(DAYSWEEK.NOT netid.luname.CHRON.EVERY(DAYSWEEK.dayname netid.luname.CHRON.EVERY(DAYSWEEK. dayname (sublist_element) ¹
EVERY=(DAYSMON=	3A 3B 3B 3B	netid.luname.CHRON.EVERY(DAYSMON.ALL netid.luname.CHRON.EVERY(DAYSMON.(dayslist) ¹ netid.luname.CHRON.EVERY(DAYSMON.NOT netid.luname.CHRON.EVERY(DAYSMON.dayofmonth ¹

Table 5. NetView Command Identifiers for the CHRON Command (continued)

Commands and Keywords Identifier	RULE	SAF Resource or Command Authorization Table Identifier
EVERY=(CALENDAR=	3A 3B 3B 3B	netid.luname.CHRON.EVERY(CALENDAR.ALL netid.luname.CHRON.EVERY(CALENDAR.(calendarlist) ¹ netid.luname.CHRON.EVERY(CALENDAR.NOT netid.luname.CHRON.EVERY(CALENDAR.keyname ¹
RECOVERY=	2 2 2	netid.luname.CHRON.RECOVERY.IGNORE netid.luname.CHRON.RECOVERY.AUTOLGN netid.luname.CHRON.RECOVERY.PURGE
SAVE	1	netid.luname.CHRON.SAVE
NOSAVE	1	netid.luname.CHRON.NOSAVE
LOCAL	1	netid.luname.CHRON.LOCAL
ID=	2	netid.luname.CHRON.ID.idname
NOTIFY=	2	netid.luname.CHRON.NOTIFY.(notifylists)
NOTIFY=(PURGE=	3B 3B	netid.luname.CHRON.NOTIFY(PURGE.(purgelist) netid.luname.CHRON.NOTIFY(PURGE.taskname
NOTIFY=(REMOVE=	3B 3B	netid.luname.CHRON.NOTIFY(REMOVE.(removelist) netid.luname.CHRON.NOTIFY(REMOVE.taskname
NOTIFY=(IGNORE=	3B 3B	netid.luname.CHRON.NOTIFY(IGNORE.(ignorelist) netid.luname.CHRON.NOTIFY(IGNORE.taskname
NOTIFY=(RUN=	3B 3B	netid.luname.CHRON.NOTIFY(RUN.(runlist) netid.luname.CHRON.NOTIFY(RUN.taskname
REFRESH	1	netid.luname.CHRON.REFRESH
TEST	1	netid.luname.CHRON.TEST
DEBUG	1	netid.luname.CHRON.DEBUG
COMMAND=	4	netid.luname.CHRON.COMMAND.'quoted string' ¹
REM=	4	netid.luname.REM.'quoted string' ¹

Note:

1. This value can have a special character, such as "." or "-", for example in the programmer time notation. You substitute the character "/" for "." and "_" for "-" when making the security definition.

Defining RMTCMD Authorization

The RMTCMD command sends system, subsystem, and network commands to one or more remote NetView systems for processing. The commands are invoked at a task specified by the RMTCMD issuer and the responses are returned across an LU 6.2 or IP session. You can also use the RMTCMD QUERY and RMTSESS command to display information about cross-domain sessions.

Note:

1. The verb for RMTCMD and for remote labeled commands is DSIUSNDM. The verb for labeled commands beginning with a slash is EXCMD
2. CNMCMD contains the following command definition statements for remote operations:

```
CMDDEF.CNME1092.MOD=DSICCP
CMDDEF.CNME1092.ECHO=Y
CMDDEF.CNME1092.CMDSYN=RMTSESS

CMDDEF.DSIUSNDM.MOD=DSIUSNDM
CMDDEF.DSIUSNDM.PARSE=Y
CMDDEF.DSIUSNDM.TYPE=RD
CMDDEF.DSIUSNDM.RES=Y
CMDDEF.DSIUSNDM.CMDSYN=RMTCMD

CMDDEF.ENDTASK.MOD=DSIUDISM
CMDDEF.ENDTASK.PARSE=N
CMDDEF.ENDTASK.TYPE=R
CMDDEF.ENDTASK.RES=Y

CMDDEF.REFRESH.MOD=DSISECP
CMDDEF.REFRESH.PARSE=Y
CMDDEF.REFRESH.TYPE=R
CMDDEF.REFRESH.RES=N
```

You can protect these commands through one of the available command authorization methods. Additional security through the RMTOPS class of an SAF product or a RMTCMD security table is also available for the RMTCMD and ENDTASK commands.

3. The RMTSESS command uses RMTCMD QUERY to gather its data; therefore, all security setup and provisions, such as the RMTOPS class or the RMTCMD security table made for RMTCMD, apply to RMTSESS.

Note: The RMTSESS command is available for LU6.2 sessions only.

Setup for RMTCMD Authorization

The RMTINIT.SECOPT and SECOPTS.RMTAUTH statements in the CNMSTYLE member define the method of RMTCMD security. This method can be changed using the RMTSEC or RMTAUTH keyword on the REFRESH command.

Using the RMTOPS Class in an SAF Product for RMTCMD Authorization

If you specified RMTINIT.SECOPT=*SAF* in the CNMSTYLE member, or RMTSEC=SAF on the REFRESH command, you can use the following steps to define your authorization checking.

1. Ensure that the RMTOPS security class is available in your SAF product.
2. Create profiles in the generic RACF RMTOPS class for remote operators that have the following characteristics:
 - Use the RMTCMD command
 - Are blocked from using the RMTCMD command
 - Start a remote autotask
 - Use the ENDTASK command to stop a remote autotask in your domain

The profiles are in this format:

```
RDEFINE RMTOPS netid.domname.operatorid.RMTCMD
RDEFINE RMTOPS netid.domname.operatorid.ENDTASK
```

Where:

netid

Is the ID of the remote network in which the operator resides.

domname

Is the name of the remote NetView domain in which the operator resides.

operatorid

Is the ID of the remote operator.

Note: The operator used for comparing with the *netid*, *domname*, and *operatorid* values is determined based on the current value of the RMTAUTH security setting, which is set by either the SECOPTS.RMTAUTH statement in the CNMSTYLE member or the REFRESH RMTAUTH command:

- If the RMTAUTH value is SENDER, the operator that issued the incoming RMTCMD or ENDTASK request is used to determine the operator identifiers.
- If the RMTAUTH value is ORIGIN, the operator that originated the request, which eventually resulted in the RMTCMD or ENDTASK request that was received at this domain, is used to determine the operator identifiers.

The remote domain sends information to identify the origin only if it is a NetView V5R3 program or a NetView V5R2 program with PTF UA25897. When the information is not supplied, authority checking proceeds as if SECOPTS.RMTAUTH = SENDER was active.

See the *IBM Z NetView Administration Reference* for more information about the SECOPTS.RMTAUTH statement.

3. Define the local operator IDs that the remote operator can use to start autotasks. Use the universal access parameter (UACC) in the profile to allow or block the remote operator from starting autotasks with any *userid*. For example, to allow all operators from domain CNM99 to start or stop any local autotask, use the following profile:

```
RDEFINE RMTOPS NET1.CNM99.*.* UACC(UPDATE)
```

To block all operators from network NET1 from starting any autotask, use the following profile:

```
RDEFINE RMTOPS NET1.*.*.RMTCMD UACC(NONE)
```

To globally allow RMTCMD autotasks to start when the origin operator name matches the requested RMTCMD autotask name, you can use the global access method:

```
SETOPTS GENCMD (RMTOPS)
SETOPTS GLOBAL (RMTOPS)
RDEFINE GLOBAL RMTOPS
RALTER GLOBAL RMTOPS ADDMEM (*.*.&RACUID.*/UPDATE)
SETOPTS GLOBAL (RMTOPS) REFRESH
```

4. To allow an operator to start remote autotasks, use the PERMIT statement with ACCESS(UPDATE) for the RMTOPS profile of the autotask. For example, to allow operator OPER01 from network NET1 to start and end an autotask with a local operator ID of NCCF1, use the following profile:

```
RDEFINE RMTOPS NET1.CNM99.OPER01.* UACC(NONE)
PERMIT NET1.CNM99.OPER01.* CLASS(RMTOPS) ACCESS(UPDATE) ID(NCCF1)
```

5. Activate generic profile checking for the RMTOPS class.

```
SETOPTS GENERIC(RMTOPS)
```

6. Activate the RMTOPS class.

```
SETOPTS CLASSACT(RMTOPS)
```

Using a Dynamic RMTCMD Security Table for RMTCMD Authorization

If you specified `RMTINIT.SECOPT=tbl_name` in the `CNMSTYLE` member, or `RMTSEC=TABLE` on the `REFRESH` command, you can use the following steps to define your authorization checking. Using the sample `DSISECUR` in `DSIPARM` as a model, add `RMTSEC` statements to the table in `DSIPARM` to define which remote operators you want to authorize or block. A statement can be up to 80 characters long. `RMTSEC` statements that begin with an asterisk (*) are considered comments and statements that begin with `RMTSEC` are processed as `RMTCMD` security entries. Statements that begin with anything else are ignored.

Note: The `RMTSEC` statement must be contained on one line.

Use this example of an `RMTSEC` statement provided in `DSISECUR` when defining your `RMTSEC` statements:

```
* RMTSEC PASS,TARGOP=DAUTO1,NET=NETX,DOMAIN=CNM02,RMTOP=MASTEROP,CMD=*
```

Where:

PASS

Specifies that remote operator `MASTEROP` in network `NETX` and domain `CNM02` can start or stop the distributed autotask with the target ID `DAUTO1`.

TARGOP=DAUTO1

Specifies that the operator ID `DAUTO1` is the distributed autotask.

NET=NETX

Is the network from which the operator is making the request.

DOMAIN=CNM02

Is the domain from which the operator is making the request.

RMTOP=MASTEROP

Is the operator making the request.

CMD=*

Specifies that the operator can issue both the `RMTCMD` and `ENDTASK` command.

Note: The operator ID, network identifier, and domain identifier of the operator making the request is determined based on the current value of the `RMTAUTH` security setting, which is set by either the `SECOPTS.RMTAUTH` statement in the `CNMSTYLE` member or the `REFRESH RMTAUTH` command:

- If the `RMTAUTH` value is `SENDER`, the operator that issued the incoming `RMTCMD` or `ENDTASK` request is used to compare with the `NET`, `LU`, and `RMTOP` values.
- If the `RMTAUTH` value is `ORIGIN`, the operator that originated the request, which eventually resulted in the `RMTCMD` or `ENDTASK` request that was received at this domain, is used to compare with the `NET`, `LU`, and `RMTOP` values.

The remote domain sends information to identify the origin only if it is a NetView V5R3 program or a NetView V5R2 program with PTF UA25897. When the information is not supplied, authority checking proceeds as if `SECOPTS.RMTAUTH = SENDER` was active.

See the *IBM Z NetView Administration Reference* for more information about the `SECOPTS.RMTAUTH` statement.

To enable operators from domain `CNM99` to start or stop any local operator ID, code the following statement:

```
RMTSEC PASS,TARGOP=*,NET=*,LU=CNM99,RMTOP=*,CMD=*
```

To indicate that a request passes when the local *operatorid* and the remote *operatorid* match, code `TARGOP=\` and `RMTOP=\` or omit the keywords from the `RMTSEC` statement.

For example, to enable operators in network NETA to issue the RMTCMD command without specifying an operator ID on the command, code one of the following statements:

```
RMTSEC PASS,TARGOP=\\,NET=NETA,LU=*,RMTOP=\\,CMD=RMTCMD

RMTSEC PASS,NET=NETA,LU=*,CMD=RMTCMD
```

Note: The \ is X'E0'.

RMTCMD Authorization Usage Notes

If RMTINIT.SECOPT=*tbl_name* is specified in the CNMSTYLE member, the table is scanned from top to bottom and any match results in the authorization being set to PASS or BLOCK as specified. The first match ends the search. Therefore, code specific authorizations or blocks before coding generic statements that contain asterisks (*). If the NetView program scans the entire table without finding a match, the RMTCMD or ENDTASK request is rejected.

If you specify RMTINIT.SECOPT=*SAF* in the CNMSTUSR or CxxSTGEN member, and a resource name does not exist in the RMTOPS class or the RMTOPS class is not active, RMTCMD and ENDTASK requests are rejected because the default response to a security request is BLOCK.

If you specify either *SAF* or *tbl_name*, consider the following items:

- Ensure that the DSIUDST task is active before you issue a RMTCMD command. You can switch from one method of security checking to another while the data services task (DSIUDST) is running by using the REFRESH command.
- If an operator is allowed to start an autotask using the RMTCMD command, the operator is automatically authorized to use the ENDTASK command to end the session.
- You can display the status of your RMTCMD authorization checking with the LIST DSIUDST command. You can also display the settings of your RMTCMD security options with the LIST SECOPTS command.

Note:

1. Include authorization blocking for all task names that are defined, even if the tasks are always active. If authorization allows, then RMTCMD can be used to send commands to any active task including the PPT, regular autotasks, and optional tasks.
2. Specifying ENDTASK as the command to block or pass only pertains to the ENDTASK FORCE option. Although security checking is performed on the originating operator when ORIGIN checking is used and the operator issued ENDTASK STOP and the operator was not the owner of the distributed operator, the command passes authority checking if the RMTCMD command is allowed for the operator even if ENDTASK is not. This is because ENDTASK STOP is equivalent to the LOGOFF command, which is sent using RMTCMD. Therefore, if authority checking is required for ENDTASK STOP, the command is allowed if the originator is allowed to perform either ENDTASK or RMTCMD command processing.

If you want information on...	See...
LIST and REFRESH commands	NetView online help
Defining additional operators	Chapter 2, “Defining Operators, Passwords, and Logon Attributes,” on page 7

Protecting MVS Command Revision Processing

The NetView autotask for the MVS Command Revision function is defined in the following CNMSTYLE statement:

```
function.autotask.MVSCmdRevision = MVSCMDS
```

Ensure that this is the only task that is permitted to issue the REISSUE command.

MVSCMDS or the defined autotask that is used by MVS Command Revision processing must be protected so that other NetView operators cannot send commands to the autotask for execution if AUTHCHK=TARGETID is used.

Protecting MVS System Commands Using an SAF Product

You can protect individual MVS system commands from unauthorized use with the OPERCMDS class of an SAF product, such as RACF. This is additional authorization checking done at the MVS level, after the command security checking done by the NetView command authorization table or NETCMDS class of an SAF product.

To protect MVS commands:

1. Ensure your OPERSEC setting has a value of SAFCHECK or SAFDEF.
2. Define command profiles to restrict specific commands from operators. For example, to restrict all operators from being able to issue an MVS QUIESCE command, enter:

```
RDEFINE OPERCMDS MVS.QUIESCE UACC(NONE)
```

3. Ensure that the OPERCMDS class is active and enabled for processing. The following RACF commands can be used to do this:

```
SETOPTS CLASSACT(OPERCMDS)
SETOPTS RACLIST(OPERCMDS)
```

4. When the OPERCMDS class is active, use the RACF REFRESH function when you change a definition:

```
SETOPTS RACLIST(OPERCMDS) REFRESH
```

Depending on whether AUTHCHK is set to SOURCEID or TARGETID, either the task which runs the command or the source ID is checked for command authorization in the OPERCMDS class. However, if an MVS ROUTE command is issued from a NetView task, the originating source ID is always passed to the SAF product for authorization checks in the OPERCMDS class, for all settings of AUTHCHK and CMDAUTH.

In order to use SECOPTS.OPERSEC=SAFDEF in the CNMSTYLE member, or OPERSEC=SAFDEF on the REFRESH command, you must have RACF or an equivalent SAF product that supports the OPERCMDS class.

If you want information on...	Go to...
The profile names of the system commands	z/OS library
RACF	RACF library

Protecting Jobs Submitted from the NetView Program using the SUBMIT Command

When the NetView SUBMIT command is issued, three layers of protection can be used:

1. The SUBMIT command can be protected using NetView command authorization. This is your first layer of protection. By protecting at this level, you can stop the processing for unauthorized users before the job is ever submitted to the system. See [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151](#) for proper specification of command identifiers for the NetView command authorization table and for SAF products.

Note: Individual members or data sets can be protected or permitted. For example, the following entries enable operator JEFF to submit anything, and operator TOM to submit any DSIPARM member and one other job:

```
PROTECT *.*.SUBMIT.*.*
PROTECT *.*.SUBMIT.DSIPARM.*
PROTECT *.*.SUBMIT.USER/INIT.JOB1
PERMIT JEFF *.*.SUBMIT.*.*
PERMIT JEFF *.*.SUBMIT.DSIPARM.*
PERMIT JEFF *.*.SUBMIT.USER/INIT.JOB1
PERMIT TOM *.*.SUBMIT.DSIPARM.*
PERMIT TOM *.*.SUBMIT.USER/INIT.JOB1
```

2. For jobs that reside in data sets that are *not* part of the DSIPARM data set concatenation, you can use the SAF DATASET class to prevent users from accessing those data sets. Using the SAF DATASET class prevents users from submitting jobs that are members of those data sets. This is the second layer of protection. An attempt to access the data set is made before the job is submitted. This layer of protection is available only when SECOPTS.OPERSEC=SAFCHECK or SECOPTS.OPERSEC=SAFDEF is specified in the CNMSTYLE member, or when OPERSEC=SAFCHECK or OPERSEC=SAFDEF is specified on the REFRESH command.
3. The SAF JESJOBS class can be used to prevent users from submitting specific jobs. This is effective for DSIPARM and non-DSIPARM data sets. This is the third layer of protection. This check happens after the job has been submitted to JES, (not synchronously with the NetView SUBMIT command). This layer of protection is available only when SECTOPS.OPERSEC=SAFCHECK or SECTOPS.OPERSEC=SAFDEF is specified in the CNMSTYLE member, or when OPERSEC=SAFCHECK or OPERSEC=SAFDEF is specified on the REFRESH command.

Note: A failure at this level is not reported to the NetView console. The JESJOBS class failure is only reported to a system console/syslog.

Auditing Command Authority Checking

If your command authority checking is performed through a NetView command authorization table or an SAF product, you can audit accesses to protected commands, keywords, and values. This auditing can be done on an individual command, keyword, or value basis.

For command authorization using a NetView command authorization table, you can specify the CATAUDIT keyword on the DEFAULTS command to determine the level of auditing performed. With the option not to audit, you can choose to audit unsuccessful or failed attempts to access protected command identifiers, or to audit all matches on command identifiers in your table. For more information about the DEFAULTS command, refer to the NetView online help. You can also specify specific auditing levels on specific command identifiers using PROTECT and EXEMPT statements with the AUDIT keyword, as described in “Table Statements” on page 41.

If auditing is specified, the records are written to SMF as record type 38, or else can be written to an external log using installation exit DSIXITXL. For more information about the DSIXITXL installation exit, refer to the DST external logging information in *IBM Z NetView Programming: PL/I and C*.

For command authorization using an SAF product, you can audit accesses to SAF-defined resources. You can control this auditing on a resource basis. For each resource, you can specify whether to perform no auditing, to audit authorization failures, to audit authorization successes, or to audit all access attempts whether successful or not. For RACF, the auditing level is specified using the RDEFINE or RALTER commands when you define the resource name. Additionally, to allow NetView commands in general to be audited, you must ensure that the RACF SETROPTS statement specifies AUDIT(NETCMDS). RACF generates SMF records that contain details at the audit level you specify for commands. You can then use the RACF report writer to create reports that describe attempts to access RACF-protected resources. For more information about the RACF report writer, refer to the RACF library.

The more auditing you request the SAF product to perform, the more system resources are required by the SAF product. You need to determine the value of the audit level you choose versus the expense in system overhead, both processor and DASD.

Logging Suppressed Operator Commands

NetView commands that are entered on the NetView command line and are preceded by one or two suppression characters are not displayed or logged. This normal operation can be overridden by adding the IGNRLSUP parameter on the CMDDEF statement or on the ADDCMD command. For more information, refer to the *IBM Z NetView Administration Reference* or the NetView online help.

Logging Command Responses for the NetView Management Console

The NetView management console topology server provides an audit log for commands and command responses. For more information, refer to the *IBM Z NetView User's Guide: NetView Management Console*.

Chapter 4. Using Spans to Protect Resources and Views

When a resource is accessed, the NetView program verifies the operator's authority to access the span that contains the resource. This resource protection is in addition to the normal command security checking.

Span of control provides a means to control access to particular resources and views. Operators access resources in the following ways:

- Issuing commands (for example, VTAM DISPLAY or VARY commands issued from the NetView program)
- Opening a NetView management console view
- Selecting a resource in a NetView management console view and performing an action against that resource

Implementing span of control is a multistep process:

- Group your resources and views into logical groups called **spans**.

This is accomplished by using the NetView span table.

- Authorize operators to the defined spans as desired.

This is accomplished with the SPAN and ISPAN statements in the operator profile in the DSIPRF data set, or in the NETSPAN class of the SAF product.

- Ensure that the AUTH statement CTL keyword setting for the operator is appropriate.

The CTL setting is in the profile in the DSIPRF data set for the operator or in the NETVIEW segment for the operator in the SAF product. It is used to determine the level of resource control for this operator.

- Ensure that the AUTH statement NGMFVSPN keyword setting for the operator is appropriate.

The NGMFVSPN setting is in the profile for the operator in the DSIPRF data set or in the operator's NETVIEW segment in the SAF product. It is used to determine the level of span checking that occurs for this operator when viewing resources and views on a NetView management console.

- Have the operator issue the START SPAN=*spanname* command for each span that must be within the operator's access.

Access is allowed to resources in a span only if the span has been started by the operator.

- It is not necessary to issue the start command for spans identified by ISPAN statements in the operator's profile.

Activating Span Checking on VTAM Commands entered with the MVS Prefix

For the MODIFY (F) command, the VTAM *procname*, or the task identifier or *procname.identifier*, is used instead of NET. For the NetView program to perform span checking on the command, a CMDDEF PARMSYN statement must be added in CNMCMDU for the command MVS so the VTAM *procname* or *task identifier* is defined as a synonym of NET. Without the CMDDEF PARMSYN statement in CNMCMD, MODIFY (or F) command is not span checked.

The VTAM command processing in the NetView program does not support *procname.identifier*. If it is used in the MODIFY command, the NetView program sends a syntax error message.

If span check is successful, the command is sent to MVS for processing. If span check fails, processing of the command stops.

Examples:

Example 1:

VTAM is started by the following command:

```
S V43VTAM.VTAM43,,,(LIST=01)
```

V43VTAM is the VTAM procname and VTAM43 is the task identifier. In CNMCMDSU, the following PARMSYN statement was added for MVS:

```
CMDDEF.MVS.PARMSYN.VTAM43=NET
```

Suppose that a NetView operator enters the following command:

```
MVS F VTAM43,CDRM=new_cdim,ID=cdimname,TYPE=NORM
```

The command is span checked, and the command fails if span checking is not successful.

Suppose that a NetView operator enters the following command:

```
MVS D NET,ID=ABC
```

The command is span checked, and the command fails if the ABC resource is not in the operator's span of control.

Suppose that a NetView operator enters the following command:

```
MVS F V43VTAM.VTAM43,CDRM=new_cdim,ID=cdimname,TYPE=NORM
```

The command fails with a NetView syntax error even though the same MVS command is processed by MVS.

Example 2:

If the following VTAM command is specified, V43VTAM is the VTAM procname and there is no task identifier:

```
S V43VTAM,,,(LIST=01)
```

In CNMCMDSU, the following PARMSYN statement was added for MVS:

```
CMDDEF.MVS.PARMSYN.V43VTAM=NET
```

Suppose that a NetView operator enters the following command. The command is span checked, and the command fails if the ABC resource is not in the operator's span of control.

```
MVS D NET,ID=ABC
```

Suppose that a NetView operator enters the following command. The command is span checked and the command fails if span checking is not successful:

```
MVS F V43VTAM,CDRM=new_cdim,ID=cdimname,ACTION=REP
```

Example 3:

If VTAM is started by the following command, then V43VTAM is the VTAM procname and there is no task identifier. No PARMSYN statement is added to CNMCMDSU for the MVS command.

```
S V43VTAM,,,(LIST=01)
```

If a NetView operator enters the following command, the command is span checked and the command fails if the resource ABC is not in the operator's span of control

```
MVS D NET, ID=ABC
```

If a NetView operator enters, the following command, the command is *not* span checked:

```
MVS F V43VTAM, CDRM=new_cdrn, ID=cdname, ACTION=REP
```

Example 4:

If VTAM is started by the following command, then V43VTAM is the VTAM proname and VTAM43 is the task identifier. No PARMSYN statement is added to CNMCMDSU for the MVS command.

```
S V43VTAM.VTAM43,,, (LIST=01)
```

If a NetView operator enters the following command, the command is span checked and the command fails if the resource ABC is not in the operator's span of control:

```
MVS D NET, ID=ABC
```

If a NetView operator enters the following command, the command is *not* span checked:

```
MVS F VTAM43, CDRM=new_cdrn, ID=cdname, ACTION=REP
```

If a NetView operator enters, the following command, the command is *not* span checked:

```
MVS F VT43VTAM.VTAM43, CDRM=new_cdrn, ID=cdname, ACTION=REP
```

Reference information for span of control is contained in this chapter. Also see the statement descriptions in *IBM Z NetView Administration Reference*.

Defining Span of Control

You can define which resources belong to the span-of-control logical groupings using the NetView span table. Using the NetView span table, you can perform the following tasks:

- Control access not only to resources but also to views containing those resources.
- Change these groupings without having to recycle the NetView program.
- Define generic resource and view names for ease of maintenance.
- Define a resource name that is longer than 8 characters.

The VTAMLST method of defining which resources belong to a particular span of control group is supported for migration purposes only. See the CNMSTYLE member for more information.

Defining Span of Control Using NetView Span Table

To use the NetView span table, specify SECOPTS.SPANAUT=span_tbl in the CNMSTUSR or CxxSTGEN member, or SPANAUT=TABLE on the REFRESH command. The span_table is the name of a member in the DSIPARM data set containing a series of span table statements. Each statement identifies one or more resources or views that are contained in one or more span names. You can use wild card characters in both the resource and view names. The NetView program also provides a migration tool (SECMIGR) to help you create the NetView span table from any existing VTAMLST and DSISPN definitions.

Note: You can use the SECMIGR command to migrate any existing span-of-control definitions in DSISPN and VTAMLST to a NetView span table. DSISPN and VTAMLST are supported for migration purposes only. For more information, see [“Scenario 5: Converting from DSISPN and VTAMLST to the NetView Span Table”](#) on page 127.

As you are defining your NetView span table, keep in mind that a resource or view is considered authorized if it can be matched against any resource or view identifier in at least one span name that is

active for the operator. The exception is when resource access has been stopped using the STOP RESOURCE command. If a resource that has been stopped is matched by a specific resource identifier in the NetView span table, the search ends and access is denied. A stopped resource can only be accessed by operators with CTL=GLOBAL.

The CTL keyword setting determines whether:

- Explicit authorization is required (CTL=SPECIFIC, the default value)
- Explicit authorization is required unless there is no match for a resource or view in any span definition that implies authorization (CTL=GENERAL)
- The operator is authorized to work with all resources (CTL=GLOBAL)

Applying Span of Control to the NetView Management Console

Spans can be used to restrict operators from seeing views and the resources within views. To apply span of control to NetView management console views and resources in views:

- Use the NGMFVSPN attribute.

The NGMFVSPN attribute specifies whether an operator's span of control is checked for authority to display views, resources, or both. The NGMFVSPN attribute is defined in either the NetView operator profiles in the DSIPRF data set or the NETVIEW segment of the USER profiles in an SAF product, such as RACF. If you use SAF definitions, you must use a release of the SAF product that supports the NGMFVSPN attribute. See [“Using an SAF Product Exclusively” on page 19](#) for more information about SAF releases.

- Use the NetView span table.

SPANDEF statements in the NetView span table specifies which views and resources an operator can display on the NetView management console. The SPANDEF statements define the resources and views to spans. Each operator is then given authority to the appropriate spans.

- Define the NetView domain name to RACF.

If you are using RACF for RODM security, ensure that the NetView domain name is defined to RACF and has been permitted to a minimum of RODM security level 2 so that the NetView program can issue queries to RODM to determine span authorization.

Resource and View Identifiers

For this discussion, a resource can be a hardware device such as a terminal, an application program, or anything in the network that can be identified by name.

Specific resource and view identifiers can be 1–255 characters in length. The identifiers can be in any form, such as:

- netid.sscp.resource
- netid.resource
- resource
- view name

NetView management console restricts view names to a length less than 33 characters. For information about the maximum length allowed for view names and to determine the names of the resources and views you want to define to spans, refer to the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*.

The previous reference includes information about how to define generic resource and view identifiers for resource and view names defined and stored in RODM. It shows how GMFHS determines which resource names and view names are used to check span authorization when building span restricted views. It also has examples and information about defining views and resources to spans.

Resource Names Used for Span Checking

The process that determines the resource names used for span checking resources in NetView management console views is described in the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*. The description includes information about the UserSpanName which can be specified to override span authorization checking on the resource name of a resource.

Wild Card Characters for NetView Span Table

An identifier can contain wild card characters such as a question mark (?), a single asterisk (*), or double asterisk (**).

Note: Wild cards used in NetView span tables are not equivalent to the generic characters (wild cards) used in the NetView command authorization table.

Table 6 on page 71 shows examples of identifiers that begin with wild cards and names that match those identifiers.

*Table 6. Names Matched by ?, *, or ** at the Beginning of an Identifier*

Identifier in SPANDEF Statement	Names Matching the Identifier	Names Not Matching the Identifier
NET?.A01APPLS	NETB.A01APPLS	NETAB.A01APPLS
	NETA.A01APPLS	NETABC.A01APPLS
*.A01APPLS	.A01APPLS	NETAB
	NETA.A01APPLS	NETAB.A01M
	NETAB.A01APPLS	A01APPLS
	NETABC.A01APPLS	
	NETAB.SSCP1.A01APPLS	
**.A01APPLS	A01APPLS	NETAB
	NETA.A01APPLS	NETABC.A01APPLS.RES1
	NETAB.A01APPLS	
	NETA.SSCP1.A01APPLS	

Table 7 on page 71 shows examples of identifiers that end with wild cards and names that match those identifiers.

*Table 7. Names Matched by * or ** at the End of an Identifier*

Identifier in SPANDEF Statement	Names Matching the Identifier	Names Not Matching the Identifier
NETAB.SSCP1*	NETAB.SSCP1	NETABC.RES1
	NETAB.SSCP1.RES1	
	NETAB.SSCP1.RES1.C1	
	NETAB.SSCP1.RES2	
	NETAB.SSCP12	

Table 7. Names Matched by * or ** at the End of an Identifier (continued)

Identifier in SPANDEF Statement	Names Matching the Identifier	Names Not Matching the Identifier
NETAB.SSCP1.*	NETAB.SSCP1.	NETABC.RES1
	NETAB.SSCP1.RES1	NETAB.SSCP1
	NETAB.SSCP1.RES1.C1	NETAB.SSCP12
	NETAB.SSCP1.RES2	
NETAB.SSCP1.**	NETAB.SSCP1	NETABC.RES1
	NETAB.SSCP1.RES1	NETAB.SSCP12
	NETAB.SSCP1.RES1.C1	NETAB.SSCP12.RES3
	NETAB.SSCP1.RES2	
NETAB.SSCP1*.*	NETAB.SSCP1	NETABC.RES1
	NETAB.SSCP1.RES1	
	NETAB.SSCP1.RES1.C1	
	NETAB.SSCP1.RES2	
	NETAB.SSCP1.	
	NETAB.SSCP12	
	NETAB.SSCP12.RES1	
NETAB.SSCP1*.*	NETAB.SSCP1.	NETABC.RES1
	NETAB.SSCP1.RES1	NETAB.SSCP1
	NETAB.SSCP1.RES1.C1	NETAB.SSCP12
	NETAB.SSCP1.RES2.	NETAB.SSCP12.RES3
		NETABC.SSCP3.RES4.C1

Question Mark

A question mark (?) can be used to represent a single character of any value. For example, specifying a pattern such as: PU?0, includes identifiers PU10, PU20, and so on.

Single Asterisk

A single asterisk (*) matches 0 (zero) or more characters, ignoring qualifiers. The period (.) of a qualifier is treated as a character. A resource or view name of A . B . C . D matches identifiers: *.D, *D, A*D, or A*.

Specify a single asterisk as follows:

- As a character at the beginning of an identifier to match zero or more characters preceding the specified characters (for example, *NETA)
- As a character at the end of an identifier to match zero or more characters following the specified characters (for example, NET*)
- As a character within an identifier to match zero or more characters following the specified characters (for example, NETA.N*PU)

- Followed by a period, as a qualifier at the beginning of an identifier to match one qualifier at the beginning of an identifier (for example, *.NET)
- Preceded by a period, as a qualifier at the end of an identifier to match one qualifier at the end of an identifier (for example, NET.*).

For example:

- Specifying an identifier such as: NET*, includes resources or view names such as NETA, NETB1, NETA.NCP1, and so on.
- Specifying an identifier such as: NETA . *, includes resources or view names such as NETA.NCP1 and NETA.NCP1.RES1, but excludes NETA as a match.

Double Asterisk

A double asterisk with a period matches 0 (zero) or more qualifiers. A double asterisk can be leading: **. or it can be trailing: **. For example, a resource or view name of A . B . C . D matches identifiers: **. C . D (leading double asterisk) or A . B . ** (trailing double asterisk).

Specifying a double asterisk is useful for defining an identifier that matches both resource names on a VTAM command as well as fully qualified resource names in NetView management console views. The double asterisk is only valid at the beginning or end of an identifier, and cannot be used more than once in an identifier.

Specifying an identifier such as NETA . ** includes any resources and view names such as NETA, NETA.RES1, or NETA.CTLR1.RES1, for example.

Omit Strings

Identifiers can be further defined to exclude a subset of another identifier. For example, specifying the following identifier includes all resources and view names beginning with the characters NR, except for those that begin with NRU:

```
NR*<NRU*>
```

The omit string must be a subset of the base identifier and is only allowed when the base identifier includes at least one wild card character.

NetView Span Table Statements

To apply span of control to operator authority, you must use the NetView span table that is stored as a member in the DSIPARM data set. Each statement can use multiple lines, with the last line of a statement ending in a semicolon.

The NetView span table can contain the following statements:

- Optionally, one or more SPANSYN statements that define span synonyms to be used in SPANDEF statements.
- One or more SPANDEF statements that map span names to resource and view names.
- Optionally, one or more %INCLUDE statements. Note that the NetView span table does not support the use of END statements. If an END statement is found in a member specified on a %INCLUDE statement, results are unpredictable. For more information about the %INCLUDE statement, see the *IBM Z NetView Administration Reference*.

Each statement can include a sequence number in columns 73 through 80. This is useful when errors occur as the statement in error is displayed and the sequence number can be used to quickly locate the statement. Statements with an asterisk in column 1 are considered comments.

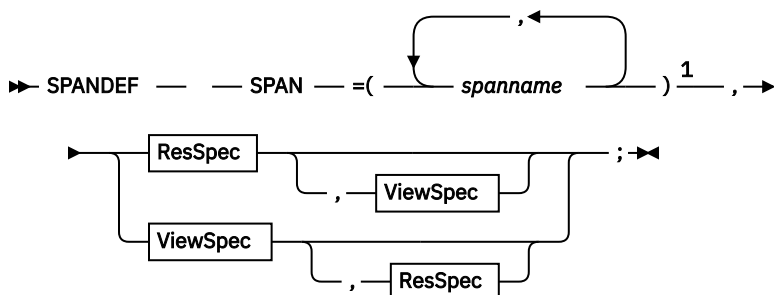
You can use system symbolics in span table statements. For example, you can use &DOMAIN when it has been assigned a value equal to the NetView domain name.

SPANDEF Statement

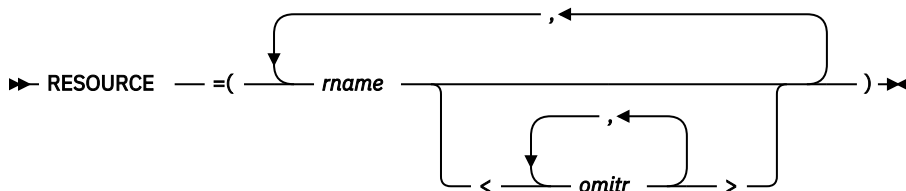
You use the SPANDEF statement in a NetView span table to define resource and view names to a span name. This statement can be used in any DSIPARM data set member that contains NetView span definitions. The SPANDEF statement can be defined on multiple lines, up to a maximum of 455 lines per statement. The span names in SPANDEF statements can be assigned to operators using either a SPAN or ISPAN statement in an operator's profile or in the NETSPAN class of an SAF product.

The syntax for the SPANDEF statement is:

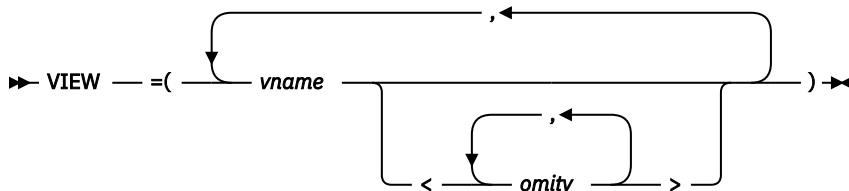
SPANDEF



ResSpec



ViewSpec



Notes:

¹ If a single name is specified for *spanname*, *rname*, or *vname*, parentheses are not required.

Where:

SPANDEF, SPAN, RESOURCE, and VIEW must be in uppercase.

omit

Indicates a list of resource names to be omitted from the previous set of resource names specified using a wild card character. Each name is case sensitive, can be 1–255 characters in length, and can include wild card characters. For more information about wild card characters, see [“Resource and View Identifiers” on page 70](#).

This exclusion only eliminates resources from this definition of the span. It does not keep an operator from accessing the excluded resources. The resource can be included by another SPANDEF statement.

omitv

Indicates a list of view names to be omitted from the previous set of view names specified using a wild card character. Each name is case sensitive, can be 1–32 characters in length, and can include wild card characters. For more information, see [“Resource and View Identifiers” on page 70](#).

This exclusion only eliminates views from the span. It does not keep an operator from accessing the excluded views. The view can be included by another SPANDEF statement.

rname

Indicates the name of a resource to be included in *spanname*. The name is case sensitive, can be 1–255 characters in length, and can contain DBCS characters. The resource name can also include wild card characters, unless the name is a DBCS string. In this case, wild card characters are not allowed. To determine the names of the resources you want to define to spans, refer to the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*. This book describes how resource and view names are defined in RODM.

Some examples of resource name formats include:

- resource
- netid.resource
- netid.sscp.resource
- netid.*
- IP address

If a single resource name (including an optional *omitr* string) is specified, parentheses are not required. For more information, see [“Resource and View Identifiers”](#) on page 70.

Because resource names are defined by users, you must use a naming scheme to define classes of resources if you want to use wild card characters to define generic resource names in SPANDEF statements. As you are creating entries for SNA resources in the NetView span table, use network-qualified resource names. Special rules apply when you are applying span of control to non-SNA resources defined in RODM. For more information, refer to the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*.

Note: If the resource name is specified in a VTAM DISPLAY EE command, no span checking is done on the resource name. This means that the resource name is accessible to all NetView operators who have the authority to issue the DISPLAY EE command, regardless of their span-of-control authorization settings. The keywords and values of the DISPLAY EE command can be protected by the NetView command authorization table if they are 8 characters or less in length.

spanname

Indicates the name of a span. The name is case sensitive and can be 1–8 characters in length. The SPAN keyword must be the first keyword specified on the SPANDEF statement. Wild card and escape characters are not allowed in span names. The *spanname* can also be a span synonym as defined by a SPANSYN statement. If a single span name is specified, parentheses are not required. For more information, see [“SPANSYN Statement”](#) on page 76.

vname

Indicates the name of a view to be included in *spanname*. The name is case sensitive, can be 1–32 characters in length, and can contain DBCS characters. NetView management console restricts view names to a length less than 33 characters. For information about the maximum length allowed for view names and how predefined and dynamically defined view names are defined and stored in RODM, refer to the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*.

The view name can also include wild card characters, unless the name is a DBCS string. In this case, wild card characters are not allowed. If a single view name (including an optional *omitv* string) is specified, parentheses are not required. For more information, see [“Resource and View Identifiers”](#) on page 70.

Restrictions:

The following reserved characters must be preceded by the escape character (“) when used in resource and view identifiers:

Character	Hexadecimal Value	Description
*	X'5C'	Asterisk
	X'40'	Blank
,	X'6B'	Comma

Character	Hexadecimal Value	Description
>	X'6E'	Greater than
(	X'4D'	Left Parenthesis
<	X'4C'	Less than
%	X'6C'	Percent sign
?	X'6F'	Question mark
"	X'7F'	Quotation mark
)	X'5D'	Right Parenthesis
;	X'5E'	Semicolon

Usage Notes:

- Resource and view names can contain DBCS characters. However, in this case, the names cannot contain wild card characters. If a DBCS name is specified, the statement line containing the SO must have an SI as the last character of the line. Span names can only contain SBCS characters. Note that you cannot use an escape character to alter the effect of a field delimiter (.) or DBCS shift-out and shift-in characters.
- NetView management console restricts view names to a length less than 33 characters. Refer to the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for information about the maximum length allowed for view names.
- The following reserved characters cannot be used in span names:

Character	Hexadecimal value	Description
	X'0E'	DBCS Shift-in
	X'0F'	DBCS Shift-out
(	X'4D'	Left Parenthesis
*	X'5C'	Asterisk
)	X'5D'	Right Parenthesis
;	X'5E'	Semicolon
,	X'6B'	Comma
%	X'6C'	Single percent sign (synonyms are supported)
?	X'6F'	Question mark
"	X'7F'	Quotation mark

- For examples of how to protect IP addresses, refer to [Appendix F, “How to Protect IP Addresses for Command Security and Span of Control,”](#) on page 231.

Related Statements: AUTH, ISPAN, OPERATOR, PROFILEN, SPANLIST, SPANSYN

SPANSYN Statement

Use the SPANSYN statement in a NetView span table to define synonyms for use later in the NetView span table. This statement can be used in any DSIPARM data set member that contains NetView span table definitions. The SPANSYN statement can be defined on multiple lines, up to a maximum of 455 lines per statement. A synonym has a name and a value. After defining a synonym, you can use the name of the synonym on a SPANDEF statement elsewhere in the table. When you activate the table, the NetView program substitutes the synonym value for the name.

Synonyms enable you to provide a shorthand notation for long, repetitive strings. Synonyms can also help you modify and maintain a NetView span table, because you can change a value throughout a table by changing it in one place.

The syntax for the SPANSYN statement is:

SPANSYN Statement

```
➤ SPANSYN — — % — synname — % — — = — — synvalue — — ; ➤
```

Where:

synname

Indicates the name of the synonym. This name can be from 1 to 32 characters in length.

synvalue

Indicates the value of the synonym. This value consists of all characters following the equal sign up to but not including the semicolon. This value can contain DBCS characters.

Restrictions:

- A synonym definition must precede the use of the synonym in the NetView span table.
- Define a synonym's value only once in the table; thereafter, you can use the synonym freely.
- Consider defining all synonyms at the beginning of the table.

The following segment of a NetView span table contains an example of a SPANSYN statement:

```
SPANSYN %NET_SPANS% = (SPAN1,SPAN2,SPAN4);
SPANDEF SPAN = %NET_SPANS%,RESOURCE = (NRR*,NT*);
```

This combination of SPANSYN and SPANDEF is equivalent to:

```
SPANDEF SPAN = (SPAN1,SPAN2,SPAN4),RESOURCE = (NRR*,NT*);
```

Because the SPANSYN value is a character string to be substituted in a SPANDEF statement, the SPANSYN and SPANDEF statements can also have been specified as follows to produce the same result:

```
SPANSYN %NET_SPANS% = SPAN1,SPAN2,SPAN4;
SPANDEF SPAN = (%NET_SPANS%),RESOURCE = (NRR*,NT*);
```

Do not use a synonym when specifying the value of another synonym. For example, do not use statements such as:

```
SPANSYN %NSPANS% = SPAN1,SPAN2;
SPANSYN %XSPANS% = %NSPANS%,SPAN4;
```

Do not use synonyms in the place of SPANDEF keywords. Use only synonyms in place of keyword values. For example, do not use statements such as:

```
SPANSYN %X% = SPAN;
SPANDEF %X% = (SPAN1,SPAN2),RESOURCE = (NRR*)
```

Use alphanumeric characters and other characters in synonym names and synonym values with the following exceptions:

- Synonym names cannot contain a percent sign (%) or a semicolon (;)
- Synonym values cannot contain a semicolon (;)

When you use system symbolics and span synonym names in the same span table statement, the system symbolics are evaluated first, followed by the span synonym names.

Consider using a naming convention for synonyms.

Related Statements: SPANDEF

Creating a NetView Span Table

You can use the SECMIGR command to migrate any existing span-of-control definitions in DSISPN and VTAMLST to a NetView span table.

You can also create the table as a new definition. By creating a new definition, you can choose to structure your span definitions in different ways, because the RESOURCE and VIEW keywords can be specified in any order. Some examples are:

- By *netid* of the resources within a span
- Alphabetically by resource name within a span
- Alphabetically by span name
- Alphabetically by major node name within a span

In all cases, the result is the same. You can choose any method to make maintaining the table easiest for you.

Loading a NetView Span Table

During NetView initialization, you can load a NetView span table by specifying its name on the SECOPTS.SPANAUTHT statement in the CNMSTYLE member. You can also load a NetView span table using the SPANTBL keyword on the REFRESH command. You can use the TEST option on the REFRESH command to check a NetView span table for syntax errors without actually loading the table. The LIST SECOPTS command can be used to determine which span table is active and when it was loaded.

When part or all of a NetView span table is loaded, the following processes occur.

1. Entries for specific (containing no wild cards) resource or view names are searched first. If the view or resource is found in a span that is active for the operator, access is allowed regardless of the value of CTL in the operator profile.
2. View and resource entries containing wild cards are searched next. When the operator's profile specifies a CTL keyword value of:

- SPECIFIC (the default)

The resource or view access fails if there is no match for the view or resource name in the NetView span table for a span that is active for the operator.

- GENERAL

The resource or view access is only granted if the resource or view is not defined either specifically or with wild card entries in the NetView span table.

- GLOBAL

No span checking is performed; access is granted.

NetView Span Table Example

The following example shows a NetView span table:

```
SPANSYN %ALLSPANS% = (SPANT, SPANA, SPANB, SPANC);
SPANSYN %NETBOPER% = SPANB;
SPANSYN %OMITNETB% = NETB.NCP*;
SPANDEF SPAN=%ALLSPANS%,
        RESOURCE=NETTEST.*;
SPANDEF SPAN=SPANA,
        RESOURCE=NETA.*;
SPANDEF SPAN=%NETBOPER%,
        RESOURCE=(NETB.*<%OMITNETB%>);
SPANDEF SPAN=SPANC,
        RESOURCE=NETC.*,
        VIEW=NETC*;
```

In this example, the enterprise is divided into four *netids* — one test *netid* and three production *netids*. Assume the following conditions for this example:

- All resources in the test *netid* (NETTEST) have NETTEST. as the first 8 characters of their resource names.
- All resources in the NETA *netid* have NETA. as the first 5 characters of their resource names.
- All resources in the NETB *netid* have NETB. as the first 5 characters of their resource names.
- All resources in the NETC *netid* have NETC. as the first 5 characters of their resource names and all views in the NETC *netid* have NETC as the first 4 characters of their view names.

A span is defined for each *netid*. The following list describes the resources and views authorized for an operator with each of these spans started.

SPANT

Because new operators are only allowed to affect resources on the test network, they are only assigned to SPANT. Thus, they are assigned to a span that allows them to work with resources that have been named to indicate they are part of the NETTEST *netid*. The resource statement is coded as:

```
NETTEST.*
```

Note that the SPANDEF for this resource statement is coded for synonym %ALLSPANS%. Thus, an operator who has SPANA, SPANB, or SPANC started can also access resources in the NETTEST *netid*.

SPANA

Operators assigned to SPANA are allowed to affect all resources in the NETA *netid*. The resource statement is coded as:

```
NETA.*
```

As mentioned previously, operators with SPANA started are also allowed to affect resources in the NETTEST *netid*. The resource statement is coded as:

```
NETTEST.*
```

SPANB

Operators assigned to SPANB are allowed to affect non-NCP resources in the NETB *netid*. The resource statement is coded as:

```
NETB.*<%OMITNETB%>
```

As mentioned previously, operators with SPANB started are also allowed to affect resources in the NETTEST *netid*. The resource statement is coded as:

```
NETTEST.*
```

SPANC

Operators assigned to SPANC are allowed to affect resources in the NETC *netid* as well as certain NetView management console views. Thus, they are assigned to a span that allows them to work with all NETC resources and any NetView management console view that matches the view statement. The resource statement is coded as:

```
NETC.*
```

The view statement is coded as:

```
NETC*.
```

As mentioned previously, operators with SPANC started are also allowed to affect resources in the NETTEST *netid*. The resource statement is coded as:

```
NETTEST.*
```

Note that span synonyms are used to simplify the SPANDEF statements.

Migrating Span of Control Using the SECMIGR Command

You can use the SECMIGR command to migrate any existing span-of-control definitions in DSISPN and VTAMLST to a NetView span table. DSISPN and VTAMLST are supported for migration purposes only. For more information, see [“Scenario 5: Converting from DSISPN and VTAMLST to the NetView Span Table” on page 127](#).

Note: The SECMIGR command does not migrate usage of the CommandSpanName attribute of RODM objects. The CommandSpanName attribute is supported for migration purposes.

Defining Operator Access to Spans

You can define an operator's access to spans using a profile in the DSIPRF data set or by using an SAF product such as RACF. The method that you use depends on the SECOPTS.OPSPAN statement in the CNMSTYLE member or the OPSPAN keyword on the REFRESH command.

Span of control is meaningful for an operator only when both of the following options are specified:

- CTL=SPECIFIC or CTL=GENERAL in the operator's profile or in the NETVIEW segment of an SAF product
- The following OPERSEC keyword values (as specified on the SECOPTS.OPERSEC statement in the CNMSTYLE member, or the REFRESH command) are valid:
 - NETVPW
 - SAFPW
 - SAFCHECK
 - SAFDEF

To apply span of control to NetView management console views, the following items must also be specified:

- The first position of the NGMFVSPN attribute must be set to R, V, or A in the operator's profile or in the NETVIEW segment of an SAF product.
- SECOPTS.SPANAUTH = *span_table* must be specified in the CNMSTYLE member, or SPANAUTH=TABLE on the REFRESH command.

The CTL and NGMFVSPN attributes are specified for an operator in the operator's profile (when OPERSEC=NETVPW, SAFPW, or SAFCHECK) or in the NETVIEW segment of an SAF product (when OPERSEC=SAFDEF).

When you are using the NetView span table, it is recommended you use CTL=SPECIFIC and include wild card characters in your span definitions rather than use CTL=GENERAL. This improves performance and ensures that your authorization is predictable.

When you define operator access to a span of control, consider the setting of SECOPTS.AUTHCHK in the CNMSTYLE member. See [“Authority Checking Commands against the Command Source” on page 34](#) and the *IBM Z NetView Administration Reference* for more information.

When you initialize the NetView program, the access level is defined by the level specified on either the PERMIT command when using the NETSPAN class of an SAF product or by the level specified in the operator profile when not using the NETSPAN class.

Note: If you switch between using the DSIPRF definition and the NETSPAN class of an SAF product for span of control, the level of access can change as a result of issuing the REFRESH command.

Defining Operator Access to Spans Using an SAF Product

When SECOPTS.OPSPAN=SAF is specified in the CNMSTYLE member, or OPSPAN=SPAN is specified on the REFRESH command, the current definitions in the NETSPAN class of the SAF product determine which spans of control can be specified by the SPAN keyword on the START command. The authority level that a user is given to the span defined in the NETSPAN class determines which commands the operator can

issue to the resources in that span. The authority level also determines the operator's ability to display NetView management console views and resources contained in that span.

When using operator profiles in the DSIPRF data set, the ISPAN statements in the profiles set up the spans of control that an operator has when logging on. Including START SPAN commands in the initial command list, which is processed when the operator logs on, provides a functional equivalent of ISPAN.

Defining Span to the NETSPAN Class

The NETSPAN class must contain a definition for any span an operator starts. A span does not start unless it matches a specific name in the NETSPAN class. The following example shows using a RACF command:

```
RDEFINE NETSPAN SPAN1 UACC(NONE)
```

The resource name in the NETSPAN class is *SPAN1*, which is a span name from the NetView span table.

Authorizing Operators to Access Spans of Control

An operator must be authorized to access spans of control in the NETSPAN class. To perform this authorization in RACF, use:

```
PERMIT SPAN1 CLASS(NETSPAN) ID(OPER1) ACCESS(access_level)
```

Operators are given an access level, such as READ, UPDATE, CONTROL, or ALTER, to a span of control that is defined in the NETSPAN class. This level is used to determine which commands the operator can issue to resources in that span of control and which NetView management console views and resources the operator can display. The following kinds of access are required:

For VTAM commands that are issued from the NetView program:

- READ access is required to issue a DISPLAY command.
- UPDATE access is required to issue the VARY, MODIFY, and REPLY commands.

For NetView management console views:

- READ access is required to display a particular view or resources in the view.

For commands that are issued from NetView management console for resources:

- READ access is required for DUIFSDIS (display current status).
- UPDATE access is required for DUIFSACT (activate).
- UPDATE access is required for DUIFSINA (inactivate).
- UPDATE access is required for DUIFSRCY (recycle).
- No access is required for DUIFSNTV because native commands do not check spans.
- UPDATE access is required for DUIFSRSC (resource-specific commands).
- UPDATE access is required for DUIFSSET (set status).
- No access is required for DUIFCSGW because COS gateway does not check spans.

For marker and suspend actions from NetView management console for resources:

- UPDATE access is required to issue a marker or suspend action.

When you change your OPSPAN setting from SAF to NETV, currently active spans of control that were defined by logging on with a NetView operator profile (OPSPAN=NETV) have their access level reset to the access level defined in the operator profile. Active spans that are known only to the SAF product retain only the access level permitted by the SAF product, and if these spans are stopped, they cannot be restarted.

Note: When you change from NETV to SAF, currently active spans have their access level reset to the authority permitted by the SAF product. You might lose some currently active spans unless they are defined in SAF and the operator is permitted to the span.

Activating the NETSPAN Class

Finally, activate the NETSPAN class to enable span-of-control resource protection in the SAF product.

```
SETRPTS CLASSACT(NETSPAN)
```

If OPSPAN=SAF is set, and the NETSPAN class becomes unavailable, an operator cannot start spans.

Defining Operator Access to Spans Using the DSIPRF Data Set

To enable NetView span-of-control authorization checking, specify SECOPTS.OPSPAN=NETV in the CNMSTUSR or CxxSTGEN member, or OPSPAN=NETV on the REFRESH command. When an operator issues a NetView START SPAN command, the authority to start the span is checked against the SPAN and ISPAN statements in the operator profile unless the operator has CTL=GLOBAL authority.

The NetView ISPAN statements define the spans of control to which an operator has access at logon.

The NetView SPAN statements define additional spans that an operator can start using the START command with the SPAN keyword. An operator can use the STOP command with the SPAN keyword to remove a span of control.

The NGMFVSPN attribute on the AUTH statement specifies whether span checking is performed for NetView management console view names and resources.

The following example illustrates a profile using the SPAN and ISPAN statements:

```
SPANPROF  PROFILE
          AUTH      CTL=SPECIFIC NGMFVSPN=RNNN
          DOMAINS   CNM01
          ISPAN     SPAN1(R) , SPAN2(R)
          SPAN      SPAN3(A) , SPAN4(A)
          END
```

In this example, SPAN1, SPAN2, SPAN3, and SPAN4 are the names of specific spans. Although span names are not required to start with the character string span, it is recommended you use an easily recognizable value.

Operators are given an access level, such as READ(R), UPDATE(U), CONTROL(C), or ALTER(A), to a span of control. This level is used to determine which commands the operator can issue to resources in that span of control and which views and resources the operator can display.

The NGMFVSPN=RNNN setting in the example means that each resource in a NetView management console view is span checked before the view is displayed. If the operator is not authorized to display some of the resources, those resources are excluded from the view when it is displayed. Operators are not given any indication when resources are excluded from a view or a view list.

Code as many span names, and as many ISPAN and SPAN statements, as you need. The names coded in SPAN and ISPAN statements are associated with span names defined in the NetView span table.

You can change the spans specified on the ISPAN and SPAN statements in an operator profile at any time. The changes take effect the next time the operator logs on with the changed profile.

An operator using profile SPANPROF can control resources and display views contained in SPAN1 and SPAN2. In addition, that operator can enter the commands START SPAN=SPAN3 and START SPAN=SPAN4 to access resources and views in those two spans.

Span Checking against the VTAM Command Source

When you are using SPANAUTH=TABLE, you can specify that span checking is performed against the original issuer of the VTAM command, or against the operator where the VTAM command is running. A value of SPANCHK=SOURCEID indicates that the original issuer of the command is used to span check the resources.

If SPANCHK=SOURCEID is specified, the task that originates the VTAM command must be active when the VTAM command runs, except for a VTAM command that is issued from a system console where no

logon has been performed (the source ID is *BYPASS*). This VTAM command is assumed to be fully authorized and no span checking is done.

Auditing Your Span-of-Control Checking

You can specify that auditing is to be performed for span-of-control authorization checking. You can audit successful and unsuccessful access attempts for both commands and NetView management console accesses to resources and views. The type of auditing that is performed is controlled by the DEFAULTS command and can be changed with the OVERRIDE command. For more information about the DEFAULTS and OVERRIDE commands, refer to the NetView online help.

If auditing is specified, the records are written to SMF as record type 38 or can be written to an external log using installation exit DSIXITXL.

For information about...	Refer to...
DSIXITXL installation exit	<i>IBM Z NetView Programming: PL/I and C</i>
Span authorization auditing record format	<i>IBM Z NetView Application Programmer's Guide</i>

Span-of-control Examples

To summarize the NetView span-of-control functions, consider the following examples. First, define the span names and their contents; then, authorize operators to access the spans of control.

Defining Span

Define the contents of spans of control using the NetView span table.

In the following example, three resources and a view (an NCP NCP1, two SNA resources RSC1 and RSC2, and view VW1) are defined in spans of control SPAN1 and SPAN4.

Define the contents of SPAN1 and SPAN4 as follows:

Create the MYSPANTB member in DSIPARM:

```
SPANDEF SPAN=(SPAN1,SPAN4),RESOURCE=NCP1;
SPANDEF SPAN=SPAN1,RESOURCE=RSC1;
SPANDEF SPAN=SPAN4,RESOURCE=RSC2;
SPANDEF SPAN=SPAN4,VIEW=VW1;
```

Note: SPANDEF SPAN=SPAN4,VIEW=VW1; allows only the operator to see the one view named VW1. To enable the operator to see more views, you can either code all the individual view names or use wild cards. The use of wild cards is recommended.

The spans of control can be used when the NetView program is restarted or when you issue the following REFRESH command:

```
REFRESH SPANAUTH=TABLE,SPANTBL=MYSPANTB
```

Defining Access to a Span of Control

You can choose to define access to spans of control using either:

- An SAF product such as RACF
- The DSIOPF member and the DSIPRF data set

In the following examples, one operator, OPER1, is defined, and is granted access to spans of control SPAN1 and SPAN2. The SPAN2 is referenced but not defined in the previous example, and SPAN4 is defined in the previous example but OPER1 is not granted authority to access it.

Defining Access to a Span of Control Using an SAF Product

Define the operators that are authorized to access the spans of control. To use RACF:

1. Create an operator ID with access to NetView domain CNM01:

```
ADDUSER OPER1 PASSWORD(PWORD)
PERMIT CNM01 CLASS(APPL) ID(OPER1) ACCESS(READ)
```

2. Specify NetView attributes for user OPER1:

```
ALTUSER OPER1 NETVIEW(IC(OP1INIT) CTL(SPECIFIC) NGMFVSPN(ANNN))
```

3. Grant OPER1 access authority to SPAN1 and SPAN2:

```
RDEFINE NETSPAN SPAN1 UACC(NONE)
RDEFINE NETSPAN SPAN2 UACC(NONE)
PERMIT SPAN1 CLASS(NETSPAN) ID(OPER1) ACCESS(READ)
PERMIT SPAN2 CLASS(NETSPAN) ID(OPER1) ACCESS(READ)
```

4. Define initial command list OP1INIT in DSICLD to start initial spans of control:

```
/* OPER1 Initial Command List */
'START SPAN=SPAN1'
'START SPAN=SPAN2'
```

Now, OPER1 can use the spans of control. For NetView management console views, view names and resources are span checked for OPER1.

Defining Access to a Span of Control Using the DSIOPF Member and the DSIPRF Data Set

Define the operators that are authorized to access the spans of control.

1. Create an entry in the DSIOPF member for OPER1:

```
OPER1      OPERATOR PASSWORD=PWORD
           PROFILEN DSIPROFX
```

2. Create an entry in the DSIPRF data set that is named DSIPROFX:

```
DSIPROFX  PROFILE  IC=OP1INIT
          AUTH      CTL=SPECIFIC,NGMFVSPN=ANNN
          SPAN       SPAN1(R),SPAN2(R)
          ISPAN      SPAN3(R),SPAN4(R)
          END
```

The spans of control can be used by OPER1 by recycling the NetView program or, if the NetView program is already active, by issuing the following REFRESH command:

```
REFRESH OPERS
```

Determining the Contents of a Span of Control

To display the resources and views within a span of control, use the NetView LIST command. For example, if you have used the LIST command to determine that you have SPAN1 as an active span, and you are uncertain which resources and views this span controls, enter:

```
LIST SPAN=SPAN1
```

Output that is similar to the following output is displayed:

```
SPAN NAME: SPAN1
SPECIFIC RESOURCES:  NCP1, RSC1
GENERIC RESOURCES:   NONE
SPECIFIC VIEWS:      NONE
```

GENERIC VIEWS:	NONE
----------------	------

For more examples on how to define span of control to limit displays of view names and resources, refer to *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*.

Chapter 5. Controlling Access to Data Sets and Members

To prevent unauthorized alteration of data, you can protect data sets with an SAF product, such as RACF, and with NetView command authorization. To prevent unauthorized viewing of passwords and other restricted information, protect them with NetView commands such as READSEC, WRITESEC, and PWDSEC. See [“NetView READSEC, WRITESEC, and PWDSEC Commands”](#) on page 88 for more information.

Data Set Security

You can restrict unauthorized alteration of data sets from the NetView environment using the DATASET class of the security product. These are some considerations when using the DATASET class of the security product:

- When the EXECIO command is used from a REXX command list, choose one of the following options:
 - DISKR**
READ
 - DISKRU**
UPDATE
 - DISKW**
UPDATE
- The NetView program requires CONTROL access to the DSILOG data set to write to the NetView log.
- The NetView program requires READ access to the first data set identified by the DSILIST DD statement.
- The NetView program requires READ access to non-DSIPARM data sets that are specified on the NetView SUBMIT command.
- Each of the following NetView commands requires UPDATE access to the first data set that is identified by the DSILIST DD statement.
 - AUTOTBL (with the LISTING keyword)
 - AUTOTEST (with the LISTING keyword)
 - AUTOCNT (with the FILE keyword)
 - QRYGLOBL (with the FILE keyword)
 - SECMIGR (with output to DSILIST)
- For the AUTOTEST command with the SOURCE keyword, the NetView program requires READ access to the data sets identified by the DSIASRC DD statement.
- For the AUTOTEST command with the MEMBER keyword and the DD=DSIASRC keyword, the NetView program requires READ access to the data sets identified by the DSIASRC DD statement.
- For the AUTOTEST command with the REPORT keyword, the NetView program requires UPDATE access to the first data set identified by the DSIARPT DD statement.
- For the AUTOTEST command with the RECORD keyword, the NetView program requires UPDATE access to the first data set identified by the DSIASRC DD statement.
- If you use SECMIGR to convert from the NetView command authorization table to RACF (TBL2RACF), the operator running SECMIGR requires READ authority to the data set containing the NetView command authorization table being converted.
- If you use SECMIGR to convert from scope of command authorization to RACF (SCP2RACF), you require access to the intermediate NetView command authorization table data set.

- If you use the default temporary DD name, the operator running SECMIGR requires UPDATE authority to the first data set in DSIPARM.
- If you specify a temporary DD name of DSILIST, the operator running SECMIGR requires UPDATE authority to the first data set in DSILIST.
- If you specify your own output data set, the operator running SECMIGR requires UPDATE authority to your output data set.

Note: NetView trace records are not made for calls to the DATASET class, because the calls are made by MVS for the NetView tasks.

Data Set Access on MVS Systems

If you have RACF, or an equivalent system authorization facility product, you can protect data sets. To activate the data set protection described in the preceding section, do the following:

1. To enable task-level authorization checking, specify SECOPTS.OPERSEC=SAFCHK or SECOPTS.OPERSEC=SAFDEF in the CNMSTUSR or CxxSTGEN member. You can also specify OPERSEC=SAFCHK or OPERSEC=SAFDEF on the REFRESH command.

Note that the SAFDEF option requires the NETVIEW segment, which is available using RACF version 2 or an SAF product with equivalent capabilities.

2. If you are using an SAF product, add profiles for the data sets you want to protect. The RACF product requires that the highest-level qualifier of the data set name be either a task or group name.

For example, use the RACF ADDSD command to add data set profiles. From an authorized TSO user, enter the following command to protect the OPER1.STATS data set:

```
ADDSD 'OPER1.STATS'
```

3. If you are using an SAF product, authorize the operator tasks so they can access the data set. For example, use the RACF PERMIT command to authorize operator tasks to the data set. To authorize NETOP1 to have update access to OPER1.STATS, enter the following command from an authorized TSO user:

```
PERMIT 'OPER1.STATS' CLASS(DATASET) ID(NETOP1) ACCESS(UPDATE)
```

NetView READSEC, WRITESEC, and PWDSEC Commands

Use the NetView READSEC and WRITESEC commands to restrict access to data sets and members by NetView commands. When you specify security for the READSEC command, it affects all of the NetView commands that can display sensitive information, such as:

- BROWSE with a member name
- NCCF LIST with the CLIST or PROFILE keywords
- PIPE stages
 - < (From disk)
 - QSAM
- VSAM command DSIVSMX
- REXX EXECIO (if PRCIOSEC=ENABLE). PRCIOSEC can be set using the NetView DEFAULTS command. Please refer to NetView online help for more information about the DEFAULTS command.

Using READSEC and WRITESEC is the only way to prevent operators from viewing data sets and members using these NetView commands. In the NetView program, security is defined so that operators have access to DSIOPEN and NetView online help. DSIOPEN is a DD name designed to hold information that must not be secured, such as NEWS data and PF key definitions. Anything other than DSIOPEN and NetView online help can be considered sensitive information.

Starting with NetView Version 3, access to data sets and members is no longer controlled by protecting individual commands. Because attempts to define security for these NetView commands is considered a severe error, message BNH115A is generated every time an operator logs on. The error text for this message is “SPECIAL SECURITY IN EFFECT FOR BROWSE AND READSEC”, which indicates that the NetView program has defined default protection for sensitive data sets and members, and the NetView commands that display data sets or members will fail. You must delete any security definitions for the commands and reinitialize the NetView program to clear the error condition.

If you use command authorization without specifying values for READSEC and WRITESEC, operators have access to all data sets and members. Because unlimited access can only be prevented with READSEC and WRITESEC, the lack of READSEC and WRITESEC CMDDEF statements in CNMCMD is considered a severe error. This condition generates error message BNH115A and indicates the NetView program has defined default protection for sensitive data sets and members. For specifics about message BNH115A, refer to the NetView online help.

Do not protect DD name CNMPNL1, because operators must be able to access online help that is contained there.

You can use the PWDSEC command to protect password information.

For more information about how to use the NetView READSEC, WRITESEC, and PWDSEC commands, refer to the NetView online help.

VSAM Data Set Access Security

VSAM data sets are accessed by using the DSIVSMX and DSIVSAM commands. This section provides information about how to protect VSAM data sets by controlling who is allowed to access VSAM data sets with these commands.

DSIVSMX Command

You can control who is allowed to access data sets with the DSIVSMX command by using:

- Command security directly on the DSIVSMX command

The DSIVSMX command and keywords can be checked with your command security as specified by the CMDAUTH setting. See Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151 for the list of keyword and value combinations that can be protected, such as:

```
netid.luname.DSIVSMX.PUT.ddname
```

- The DATASET class of the SAF product

When the DSIVSMX request causes an MVS OPEN macro to be issued, MVS checks the authorization in the DATASET class. If you are running with OPERSEC=SAFCHECK or OPERSEC=SAFDEF, the access request is verified against the operator who made the request.

- The READSEC and WRITESEC commands

DSIVSMX requests are checked against the security definitions for READSEC and WRITESEC. DSIVSMX GET requests are checked against the appropriate READSEC definitions, and DSIVSMX PUT requests are checked against the appropriate WRITESEC definitions.

Protect the VSAM files used by NetView data services tasks to prevent operators from interfering with the operations of data services tasks (for example, DSILOG). Any unprotected VSAM file to which a DDNAME can be allocated, using the ALLOC command, can be accessed by the DSIVSMX command.

DSIVSAM Command

The DSIVSAM command is used to initiate a VSAM request to a data services task. The data services task runs the VSAM requests, not the initiating operator. Because of this, there is less of a security concern than there is with the DSIVSMX command, which is run directly by an operator task.

For DSIVSAM, your point-of-control is to limit which commands can be sent to which data services task by operators. You can accomplish this by using:

- Command security directly on the DSIVSAM command

The DSIVSAM command and keywords can be checked with your command security as specified by the CMDAUTH setting. See Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151 for the list of keyword and value combinations that can be protected, such as:

```
netid.luname.DSIVSAM.PUT.taskname
```

- The DATASET class of the SAF product

When the DSIVSAM request causes an MVS OPEN macro to be issued, MVS checks the authorization in the DATASET class.

Note: Data services tasks do not have an identity in the SAF product. Therefore, the NetView program must be authorized to the proper data sets, if they are protected in the DATASET class of the SAF product.

The READSEC and WRITESEC commands do not apply to DSIVSAM requests.

Security for Access to DB2 Data using the SQL Pipe Stage

When the SQL pipe stage is used to access DB2® data, security is determined by the DB2SEC statement in the CNMSTYLE member.

If DB2SEC=CAF is specified, the user ID that is authority checked by MVS is the user ID associated with the NetView address space. No task-level checking is available, even if you have SECOPTS.OPERSEC=SAFCHECK or SECOPTS.OPERSEC=SAFDEF specified in the CNMSTYLE member, or when OPERSEC=SAFCHECK or OPERSEC=SAFDEF is specified on the REFRESH command. This limitation is caused by the DB2 product recognizing only one identity for each address space with the CAF interface to the DB2 product.

If DB2SEC=RRS is specified, and the RRS interface is available, each task in the NetView program has its own user ID associated with the SQL pipe stage requests.

Chapter 6. Controlling Access to the NetView Program from TCP/IP Hosts

To prevent unauthorized connections to the NetView program from a TCP/IP host, you can restrict access by using the NetView command authorization table, the NETCMDS class of an SAF product, and through the use of sample definition members that are shipped with the NetView program.

Note: If you use an SAF product for command authorization, autotask DSIIPCHK must be active and defined to that SAF product. This autotask is used for authorization checks that are made from certain DSTs and OPTs.

Application Transparent Transport Layer Security for TCP/IP Connections with the NetView Program

You can protect the data that flows through TCP/IP connections by using the Application Transparent Transport Layer Security (AT-TLS) function, which was new in z/OS V1R7 Communications Server. For details about using AT-TLS, see the z/OS Communications Server library.

To enable and configure AT-TLS for secure communications between the NetView program and the NetView management console, the NetView web application, or the MultiSystem Manager IBM Tivoli Network Manager agent, or for secure communications between remote clients and the NetView Web Services Gateway, refer to the CNMSJTLS sample.

NetView Web Access and TCP/IP Alert Receiver

You can prevent unauthorized TCP/IP hosts with a Web browser from connecting to the NetView program and the TCP/IP alert receiver task by using the WEBACC command.

- To enable a connection to the NetView program from a Web browser, the DSIIPCHK task must be permitted to issue the WEBACC command for the IP address of the TCP/IP host requesting a connection.

Note: To enable the connection, authorize the Web application server to access the NetView program.

- To enable a connection to the TCP/IP alert receiver, the DSIIPCHK task must be permitted to issue the WEBACC command for the IP address of the TCP/IP host requesting a connection.

Note: The WEBACC command is for security authorization purposes only.

The syntax for the WEBACC command is:

WEBACC

➤ WEBACC — — TCPADDR — = — *tcphostaddr* ➤

Where:

tcphostaddr

Defines the TCP/IP host address.

Command authorization table example:

To control access to the NetView program by using the WEB server from a TCP/IP host at address 9.68.55.45, code the following command authorization table:

```
PROTECT netid.luname.WEBACC.TCPADDR.*
PROTECT netid.luname.WEBACC.TCPADDR.9/68/55/45
```

To allow this TCP/IP host access to the NetView program, code the following statement in the command authorization table:

```
PERMIT DSIIPCHK netid.luname.WEBACC.TCPADDR.9/68/55/45
```

REXEC Server

You can prevent unauthorized TCP/IP hosts from connecting to the NetView program by using the REXEC server by using the JAVAACC command. To allow a connection that uses REXEC, the DSIIPCHK task must be permitted to issue the JAVAACC command for the IP address of the TCP/IP host requesting a connection.

Note: The JAVAACC command is for security authorization purposes only.

The syntax for the JAVAACC command is:

JAVAACC

➤ JAVAACC — TCPADDR — = — *tcphostaddr* ➤

Where:

tcphostaddr

Defines the TCP/IP host address.

Command authorization table example:

To control access to the NetView program by using the REXEC server from a TCP/IP host at address 9.68.55.45 or at address 2002:92A:111:513:9:42:5:133, code the following command authorization table:

```
PROTECT netid.luname.JAVAACC.TCPADDR.*  
PROTECT netid.luname.JAVAACC.TCPADDR.9/68/55/45  
PROTECT netid.luname.JAVAACC.TCPADDR.2002:92A:111:513:9:42:5:133
```

To allow this TCP/IP host access to the NetView program, code the following statements in the command authorization table:

```
PERMIT DSIIPCHK netid.luname.JAVAACC.TCPADDR.9/68/55/45  
PERMIT DSIIPCHK netid.luname.JAVAACC.TCPADDR.2002:92A:111:513:9:42:5:133
```

NetView Correlation Service

You can prevent unauthorized TCP/IP hosts from connecting to the NetView program by using the correlation service by using the JAVAACC command. To allow a connection that uses the correlation service, the DSICORSV task must be permitted to issue the JAVAACC command for the IP address of the TCP/IP host requesting a connection.

Note:

1. The JAVAACC command is for security authorization purposes only.
2. Implementing security for the NetView correlation service also affects the security for the REXEC server because this task also uses the JAVAACC command.

The syntax for the JAVAACC command is:

JAVAACC

➤ JAVAACC — TCPADDR — = — *tcphostaddr* ➤

Where:

tcphostaddr

Defines the TCP/IP host address.

Command authorization table example:

To control access to the NetView program by using the correlation service from a TCP/IP host at address 9.68.55.45 or at address 2002:92A:111:513:9:42:5:133, code the following command authorization table:

```
PROTECT netid.luname.JAVAACC.TCPADDR.*
PROTECT netid.luname.JAVAACC.TCPADDR.9/68/55/45
PROTECT netid.luname.JAVAACC.TCPADDR.2002:92A:111:513:9:42:5:133
```

To allow this TCP/IP host access to the NetView program, code the following statements in the command authorization table:

```
PERMIT DSIIPCHK netid.luname.JAVAACC.TCPADDR.9/68/55/45
PERMIT DSIIPCHK netid.luname.JAVAACC.TCPADDR.2002:92A:111:513:9:42:5:133
```

NetView SOCKET Interface

You can prevent unauthorized TCP/IP hosts from connecting to the NetView program by using the NetView socket interface by using the SOCACC command. To allow a connection that uses the NetView socket interface, the task that issued the socket command must be permitted to issue the SOCACC command for the IP address of the TCP/IP host requesting a connection.

Note: The SOCACC command is for security authorization purposes only.

The NetView socket interface includes support for Application Transparent Transport Layer Security (AT-TLS) encrypted communications by using the new TYPE=IOCTL OPTNAME=SIIOCTLCTL call. The OPTNAME=SIIOCTLCTL operand returns the status of the AT-TLS function, the AT-TLS policy governing the connection, and the status of the current connection. The following status is included:

- If the handshake has completed
- If the connection is encrypted
- The protocol with which the connection is encrypted, for example SSLv2, SSLv3, TLSv1.

The syntax for the SOCACC command is:

SOCACC

➡ SOCACC — TCPADDR — = — *tcphostaddr* →

Where:***tcphostaddr***

Defines the TCP/IP host address.

Command authorization table example:

To control access to the NetView program by using the SOCKET interface from a TCP/IP host at address 9.68.55.45, code the following command authorization table:

```
PROTECT netid.luname.SOCACC.TCPADDR.*
PROTECT netid.luname.SOCACC.TCPADDR.9/68/55/45
```

To allow this TCP/IP host access to the NetView program by using sockets opened by operator AUTOSOC, code the following statements in the command authorization table:

```
PERMIT AUTOSOC netid.luname.SOCACC.TCPADDR.9/68/55/45
```

RMTCMD over IP

You can use the RMTACC command to prevent unauthorized TCP/IP hosts or NetView domains from connecting to the NetView program with RMTCMD over IP. To allow a connection that uses RMTCMD over IP, the DSIIPCHK task must be permitted to issue the RMTACC command for the IP address of the TCP/IP host requesting a connection.

Note: The RMTACC command is for security authorization purposes only.

When you code the syntax for the RMTACC command at least one of the keywords TCPADDR or NETIDDOM must be coded. The syntax for the RMTACC command is:

RMTACC

```

➤ RMTACC  ┌── TCPADDR  ── = ── tcphostaddr ──┐
           │      ────┘
           └── NETIDDOM ── = ── netid.domain ──┘

```

Where:

tcphostaddr

Defines the TCP/IP host address.

netid.domain

Defines the originating network id and domain.

Command authorization table example 1 - using IPv4:

To control general access to your NetView domain from TCP/IP hosts or other NetView domains that use the RMTCMD command over IP (IPv4) and also allow commands to be sent from a specific host at address 10.10.90.5 with domain CNM01 on network NETA, code the following statements in your command authorization table:

```

PROTECT netid.luname.RMTACC.TCPADDR.*
PROTECT netid.luname.RMTACC.TCPADDR.10/10/90/5
PROTECT netid.luname.RMTACC.NETIDDOM.ETA/CNM01
PROTECT netid.luname.RMTACC.NETIDDOM.*
PERMIT  DSIIPCHK netid.luname.RMTACC.NETIDDOM.ETA/CNM01
PERMIT  DSIIPCHK netid.luname.RMTACC.TCPADDR.10/10/90/5

```

Command authorization table example 2 - using IPv6:

To control general access to your NetView domain from TCP/IP hosts or other NetView domains that use the RMTCMD command over IP (IPv6) and also allow commands to be sent from a specific host at address 2002:92A:111:501:192:168:1:1 with domain CNM02 on network NETB, code the following statements in your command authorization table:

```

PROTECT netid.luname.RMTACC.TCPADDR.*
PROTECT netid.luname.RMTACC.TCPADDR.2002:92A:111:501:192:168:1:1
PROTECT netid.luname.RMTACC.NETIDDOM.NETB/CNM02
PROTECT netid.luname.RMTACC.NETIDDOM.*
PERMIT  DSIIPCHK netid.luname.RMTACC.NETIDDOM.NETB/CNM02
PERMIT  DSIIPCHK netid.luname.RMTACC.TCPADDR.2002:92A:111:501:192:168:1:1

```

RSH Server

To allow a connection to the RSH server, modify the DSIRHOST sample. The sample describes the syntax. For example, to allow TCP/IP host at address 9.68.55.45 access to the NetView program, the following statement can be added to DSIRHOST:

```
+9.68.55.45
```


Chapter 7. Security for TSO and UNIX Command Servers

Any authorized NetView user can issue the START TSOSERV and START UNIXSERV commands. The authority to start and stop these servers is defined by NetView command security on the START command. It is strongly suggested that the START command, DSIPITSO, and DSIPUNX be secured. Because these servers do not run in the NetView address space, it is suggested that SAF security be used to secure these commands. The level of security in the CNMSTYLE member must be at least SECOPTS.OPERSEC=SAFCHECK and SECOPTS.CMDAUTH=SAF.

The CNMSJTZO, CNMSJUNX, CNMSSTZO, CNMSJUNX, CNMSTZOS, and CNMSJUNXS members in the DSIPARM data set determine how the TSO and UNIX command servers are started in MVS. These members must therefore be protected so that only authorized users can change these members, or create members with the same name. For additional information, refer to the *IBM Z NetView Programming: Pipes*.

For more information about defining authorization to the TSO Pipe stage, see [“Protecting the TSO Pipe Stage”](#) on page 53.

TSO Command Server

If NetView operator security (SECOPTS.OPERSEC) is set to NETVPW or minimal, the TSO server JCL, CNMSJTZO, requires PASSWORD to be explicitly coded on the JOB statement; or, if the SAF SURROGATE class is active, specify USER=&userid; (without the password) to ensure that the correct SAF user ID is associated with the TSO command server job.

If NetView operator security (SECOPTS.OPERSEC) is set to SAFPW, SAFCHECK, or SAFDEF, the TSO server JCL (CNMSJTZO) does not require PASSWORD explicitly coded on the JOB statement. However, the explicit values for USER and PASSWORD can be overridden. In addition, the TSO user ID specified on the TSOSERV keyword (or the issuing NetView operator ID if this value is not specified) must be defined to SAF as a valid TSO user ID.

Surrogate Support for TSO Command Servers

An SAF-defined surrogate is an SAF-defined user who has been authorized to act on behalf of another user without having to specify that user's password. NetView users can use SAF surrogate support for NetView TSO servers.

To start, stop, or access the TSO server (using the NetView TSO PIPE command stage) you must define the NetView user ID as a surrogate of the TSO user ID under whose authority the TSO server is running. To define the server, perform the following steps:

1. For RACF SAF environments, define the NetView user as a surrogate of the TSO user.
 - a. Define the TSO user IDs in the RACF surrogate class as follows:

```
RDEFINE SURROGAT tso-userid.DSITSOSV UACC(NONE)
```

- b. Add NetView users (of the NetView TSO server) to the RACF database.

For RACF, use the ADDUSER command. Refer to [“Using an SAF Product to Restrict Logon Access”](#) on page 23 for more information.

- c. Authorize the appropriate NetView users to act as surrogates of the TSO user IDs who can run NetView TSO servers. Use the following RACF PERMIT command:

```
PERMIT tso-userid.DSITSOSV CLASS(SURROGAT) ID(NetView-userid) ACCESS(READ)
```

- d. Activate the RACF surrogate class by using the following command:

```
SETROPTS CLASSACT(SURROGAT)
```

Refer to CNMSAF2 for sample SAF definitions. For more information about the RDEFINE, PERMIT and SETROPTS commands, refer to the z/OS SecureWay Security Server library.

2. Set up the NetView program to use the RACF surrogate security checking in one of the following ways:

- Specifying SECOPTS.SURROGAT=YES *and* one of the following operands in the CNMSTYLE member:
 - SECOPTS.OPERSEC=SAFDEF
 - SECOPTS.OPERSEC=SAFCHECK
- Issuing the REFRESH command specifying SURROGAT=YES *and* one of the following operands:
 - OPERSEC=SAFDEF,
 - OPERSEC=SAFCHECK

After the NetView program is set up using one of these two methods, it uses the RACF surrogate definitions activated for the SURROGAT class.

See the NetView online help for more information about the REFRESH command.

UNIX Command Server

The UNIX command server runs under the authority of the user ID that is associated with the job or procedure that is used to start it. If SAF-based security is used, the user ID can be provided with the authority to enable execution of the UNIX command server. If the Resource Access Control Facility (RACF) program is used to implement SAF-based security on the system, the security for the UNIX command server may be implemented by performing the following tasks:

- Activate the SAF FACILITY class and define the BPX.DAEMON to it:

```
REDEFINE FACILITY BPX.DAEMON UACC(NONE)  
SETROPTS CLASSACT(FACILITY)  
SETROPTS RACLIST(FACILITY)
```

- Ensure that the user ID from which the UNIX command server gets its authority has at least READ access to the BPX.DAEMON resource of the FACILITY class as follows, where *userid* is the user ID:

```
PERMIT BPX.DAEMON CLASS(FACILITY) ID(userid)  
ACCESS(READ)  
SETROPTS RACLIST(FACILITY) REFRESH
```

- Ensure that the user ID that is associated with the UNIX command server job or procedure has the necessary authority by implementing at least one of the following options.
 - Give the user ID permission to execute UNIX command server code and write output files:
 1. Ensure that the user ID is an owner of UNIX command server executable file cnmeunix with execute permission for that file; Or ensure that the user ID is a member of the owner group assigned to UNIX command server executable file cnmeunix and the group has execute permission for that file.
 2. For the UNIX command server executable files cnmechld and cnmework, ensure that the user ID is an owner of the files with execute permission for them; Or ensure that the user ID is a member of the owner group assigned to those files, and the group has execute permission for them; Or ensure that the other permissions for those files include execute permission.
 3. Ensure that the user ID is an owner of the directory and output files, for example, cnmeunix.stdout and cnmeunix.stderr in the directory that are defined in the UNIX command server job or procedure, and has write permission for those items; Or ensure that the user ID is a member of the owner group that is assigned to the directory and output files, and the group has write permission for those items.

For example, if an owner group name has been set for the applicable z/OS UNIX files and directories, a command to add the user ID to the owner group may be similar to the following:

```
CONNECT user_id GROUP(group_name)
```

See *z/OS UNIX System Services Command Reference* for commands **CHMOD**, **CHOWN**, and **CHGRP** that are used to set permissions and an owner or group owner of files and directories in the z/OS UNIX file system.

- Give the user ID superuser authority overall or within the context of the UNIX command server job or procedure.

One method of ensuring that a user ID has superuser authority overall is to define the user ID and assign a zero UID value in the OMVS segment of the user ID profile as follows, where *userid* is the user ID and *group_name* is the name of a group that has a group ID (GID) set in the OMVS segment of the profile:

```
ADDUSER userid DFLTGRP(group_name) OMVS(UID(0) HOME('/')) NOPASSWORD
```

A user ID with a non-zero UID in the OMVS segment of the user ID profile can also be given superuser authority in just the context of the running UNIX command server job or procedure. To do that, the TRUSTED attribute must be set for the UNIX command server job or started procedure, as follows:

1. Ensure that the STARTED class is active:

```
SETROPTS CLASSACT(STARTED)
SETROPTS RACLIST(STARTED)
SETROPTS GENERIC(STARTED)
```

2. Define a profile for the UNIX command server job or started procedure, associate a user ID with that job or started procedure, and set the TRUSTED attribute. As an example, for a STARTED class profile, CNMSSUNX.*, which might represent a procedure for starting the UNIX command server, execute a command as follows:

```
RDEFINE STARTED CNMSSUNX.* STDATA(USER(userid) TRUSTED(YES))
```

3. Refresh the STARTED class information:

```
SETROPTS RACLIST(STARTED) REFRESH
SETROPTS GENERIC(STARTED) REFRESH
```

- Turn on PROGRAM CONTROL to provide security for privileged functions the UNIX command server issues (for example, setuid):

```
SETROPTS WHEN(PROGRAM)
```

- Define the following libraries to PROGRAM CONTROL:

```
RALT PROGRAM * ADDMEM('SYS1.LINKLIB' /*****/NOPADCHK)
ACC(READ)
RALT PROGRAM * ADDMEM('hlq.SCEERUN' /*****/NOPADCHK)
UACC(READ)
RALT PROGRAM * ADDMEM('hlq.SCNMLNK1' /*****/NOPADCHK)
ACC(READ)
RALT PROGRAM * ADDMEM('hlq.SCNMLNKN' /*****/NOPADCHK)
ACC(READ)
RALT PROGRAM * ADDMEM('hlq.SEAGALT' /*****/NOPADCHK)
UACC(READ)
SETROPTS WHEN(PROGRAM) REFRESH
```

Note:

1. If you are running with the REXX alternate library, the SEAGALT data set must be defined to PROGRAM CONTROL. However, if your installation is using the REXX/370 runtime library, define the SEAGLPA data set. For more information about SEAGALT or SEAGLPA usage, refer to *IBM Z NetView Installation: Getting Started*.

2. If the PROGRAM class * already exists, use the RALT command. If the PROGRAM class * does not exist, use the RDEF command for the first command and use the RALT command for subsequent commands.
 3. Also define to program control any library that provides modules to support any commands that might be run by the UNIX command server. An example is a library that contains a REXX function package module that supports a REXX EXEC that is driven by a command that is sent to UNIX command server for processing.
- Define each NetView operator ID that issues (when using an AUTHCHK setting of TARGETID) or originates (when using an AUTHCHK setting of SOURCEID) the PIPE UNIX command to RACF with a valid OMVS segment. The OMVS segment must include a valid HOME directory.

This includes all operators associated with the AON AUTOOPS TCPOPER statement in FKXCFG01 and the MultiSystem Manager autotask that is associated with the STRTDISC command. For more information about the STRTDISC command and the autotask it runs under, refer to the NetView online help.

This also includes the primary program operator interface task (PPT) if an AUTHCHK setting of SOURCEID is used and a PIPE UNIX command originates in the PPT (such as when an auxInitCmd statement in the CNMSTYLE member is used).

It also includes the IDSAUTO2 operator that is used by Intrusion Detection Services (IDS) for capturing syslog probes.

Chapter 8. Security Considerations for Automation

Security in an automated environment is similar to security in an environment without automation.

The major security considerations for automation include:

- Data set access to the automation table and command lists
- Access to automated operator tasks
- Autotask definitions
- Command authorization from the automation table

The NetView defaults for these security settings cannot be optimized for your automation security. This chapter is designed to help you maximize your automation security while minimizing automation performance impacts.

Restricting Data Set Access to the Automation Table and Command Lists

If you have an SAF product, such as RACF, installed on your system, restrict data set access to DSIPARM data sets for the automation tables, and DSICLD data sets for command lists.

Without restricted access, someone with TSO editing capabilities can alter automation processing and, either intentionally or accidentally, disturb normal processing. See [Chapter 5, “Controlling Access to Data Sets and Members,” on page 87](#) for more information about restricting access to these data sets.

Restricting Access to Automated Operator Tasks

Automated operator tasks (autotasks) typically are given a broad range of command authorization, because they perform a wide range of system and network functions.

If you do not specify passwords for autotasks, the system defines a default password value. Specify non-expiring passwords for autotasks to prevent operators from logging on to autotask operator IDs, and to prevent automation from being affected by an expired password. The security administrator must periodically change the passwords.

If passwords are contained in a command list, the command list must have data set protection as described in [Chapter 5, “Controlling Access to Data Sets and Members,” on page 87](#). See [Chapter 2, “Defining Operators, Passwords, and Logon Attributes,” on page 7](#) for a full description of password protection.

Use command authorization to restrict the type of commands which operators can send to autotasks. For example, restrict the NetView EXCMD command so that operators cannot send unauthorized commands to autotasks to use the autotask's level of authorization. See [Chapter 3, “Controlling Access to Commands,” on page 29](#) for examples of command authorization.

Another way of preventing the use of the authority of some other task is by changing your type of command authorization. When you specify AUTHCHK=SOURCEID, command security checks the authorization of the originator for some commands, such as EXCMD. For examples of command authorization based on TARGETID or SOURCEID, see [“Authority Checking Commands against the Command Source” on page 34](#).

Autotask Definitions

Within the NetView program, all operator station tasks (OSTs), including autotasks and NetView-NetView tasks (NNTs), must be defined either in the NetView member DSIOPF or in an SAF product such as RACF.

If you have an SAF product, either the NetView product or the SAF product can contain the operator IDs, password, and logon attributes. You can dynamically add and delete operators using both security methods. See [Chapter 2, “Defining Operators, Passwords, and Logon Attributes,”](#) on page 7 for more information.

You can use autotasks in the following ways:

- As an operator for running commands that are issued as automation responses to messages and management services units (MSUs). No operator needs to log on to this operator identifier. Instead, the AUTOTASK command is used to log the operator identifier on without a terminal.
- As an operator for running NetView commands issued at an MVS operator console and returning command responses and other messages to that console. The AUTOTASK command is used with the CONSOLE operand to log on the operator identifier and associate it with an MVS console. The DISCONID command displays the information that corresponds to each MVS console.
- As an operator for running commands from remote systems using RMTCMD. The RMTCMD command causes distributed autotasks to be automatically logged on to the target NetView system to run remote commands and return responses back to the originating system. Therefore, ensure that all operators who are used on a target NetView system are defined for that NetView system.

The NetView samples provide AUTO1 as an example of an operator identifier used for NetView automation. AUTO2 is provided as an example of an operator identifier used as an MVS system console interface task. The samples also provide operator profiles DSIPROFC and DSIPROFD so that you can assign different characteristics to the different types of autotasks. Their corresponding logon initial command lists, LOGPROF2 (CNME1032) and LOGPROF3 (CNME1033), are processed when their tasks are started as a result of an AUTOTASK command.

Defining Operator IDs for Autotasks

NetView autotask operator IDs can be defined in three ways, depending on the value of the SECOPTS.OPERSEC statement in the CNMSTYLE member, or the OPERSEC keyword on the REFRESH command:

- Entirely within the NetView program, by using OPERSEC=NETVPW
- With passwords or password phrases in an SAF product and operator profiles in the NetView product, by using OPERSEC=SAFPW or OPERSEC=SAFCHECK
- Entirely within an SAF product, by using OPERSEC=SAFDEF

Defining Autotasks Using the NetView Program

The following examples show how autotasks are defined in the NetView operator profiles. The values are used when the NetView program checks passwords (SECOPTS.OPERSEC=NETVPW in the CNMSTYLE member or OPERSEC=NETVPW on the REFRESH command). Operator task names and passwords are defined in the NetView DSIOPF member, and NetView operator profiles are defined in the DSIPRF data set.

In member DSIOPF, the autotask name is in the first column, and the corresponding password is defined by the PASSWORD statement.

AUTO1	OPERATOR PROFILEN	PASSWORD=AUTO1 DSIPROFC
AUTO2	OPERATOR PROFILEN	PASSWORD=AUTO2 DSIPROFD

Figure 5. Defining Operator IDs and Passwords in the NetView DSIOPF Member

To prevent operators from logging on using an operator ID that is reserved for an autotask, specify PASSWORD=★NONE★

To see the operator logon attributes specified in DSIPROFC and DSIPROFD, use the NetView BROWSE command.

Defining Autotask Passwords Using SAF and Profiles Using the NetView Program

Using SECOPTS.OPERSEC=SAFPW or SECOPTS.OPERSEC=SAFCHECK (or the equivalent on the REFRESH command), NetView profiles define the autotask names and logon attributes but the passwords or password phrases are now defined in an SAF product, such as RACF. The NetView profiles are defined in the same way as shown in Figure 6 on page 101. The following example defines the autotasks to RACF; in this example, the AUTO1 autotask is defined with a password, and the AUTO2 autotask is defined with a password phrase. See “Defining Operator IDs for Autotasks Using an SAF Product” on page 101 for an explanation of the RACF commands in this example.

```
ADDUSER AUTO1 PASSWORD(XYZZY)
ADDUSER AUTO2 PHRASE('my_password_phrase_is_xyzzy')
```

Figure 6. Defining Autotasks in RACF

Note:

1. For OPERSEC=SAFPW, autotasks do not need PERMIT statements.
2. For OPERSEC=SAFCHECK, appropriate PERMIT statements are required.

Defining Operator IDs for Autotasks Using an SAF Product

“Defining Operator IDs for Autotasks Using an SAF Product” on page 101 shows examples of the commands that define autotask profiles in an SAF product and that are used when SAF verifies passwords or password phrases and logon attributes (when SECOPTS.OPERSEC=SAFDEF in the CNMSTYLE member, or OPERSEC=SAFDEF on the REFRESH command). The autotask is defined by the ADDUSER statement, the ALTUSER defines the logon attributes, and the PERMIT statement enables the operator access to the domain. In this example, the AUTO1 autotask is defined with a password, the AUTO2 autotask is defined with a password phrase, and CNM01 is the name used to define the NetView program. Note that these values match the function of the DSIPROFC and DSIPROFD NetView operator profiles, and that the password does not change if the autotasks are defined.

```
ADDUSER AUTO1 PASSWORD(XYZZY)
ALTUSER AUTO1 NETVIEW(IC(LOGPROF2) MSGRECVR(NO) CTL(GLOBAL))
PERMIT CNM01 CLASS(APPL) ID(AUTO1) ACCESS(READ)
ADDUSER AUTO2 PHRASE('my_password_phrase_is_xyzzy')
ALTUSER AUTO2 NETVIEW(IC(LOGPROF3) MSGRECVR(NO) CTL(GLOBAL))
PERMIT CNM01 CLASS(APPL) ID(AUTO2) ACCESS(READ)
```

Figure 7. Defining Operator IDs and Passwords in an SAF Product

If the autotask already exists, the entire ADDUSER command fails, so define the logon attributes for AUTO1 on a separate ALTUSER statement which will not fail. Even though AUTO1 is defined, the ALTUSER changes take effect. If the ADDUSER fails, the task retains its old password, and the password is not reset to a default system value.

For information about...	See...
Using an autotask and the syntax for the AUTOTASK command	NetView online help
Using the NetView RMTCMD command	NetView online help

Restricting Authorization for Commands Issued from the Automation Table

If you use the NetView command authorization table or an SAF product for command authorization checking, you can bypass command security for commands issued from NetView automation table statements.

You can choose not to use command security for your automation table under the following conditions:

- Your automation table is secure enough to run without command authorization checking.

- Using the NetView command authorization table or an SAF product has an impact on your system performance.

If you need to reduce system resource usage caused by command authorization for NetView automation processing, use the DEFAULTS command to set AUTOSEC=BYPASS. To view your current settings, use the NetView LIST SECOPTS command.

If you know any individual commands that can cause interruptions in your system, using SEC=CH on the CMDDEF statement unconditionally checks access for those commands. This ensures that an authorization check is always made and prevents a task from issuing a command to which it is not authorized, even if the command originated from the automation table and BYPASS or DEFER is in effect for the DEFAULTS AUTOSEC setting. Depending on how much processing is being used by command security for automation tasks, using AUTOSEC=BYPASS or AUTOSEC=DEFER can noticeably improve system performance. For more information, see the NetView online help or *IBM Z NetView Command Reference Volume 1 (A-N)* for the DEFAULTS and LIST commands.

Chapter 9. Security for Automated Operations Network

This section provides information about security for Automated Operations Network (AON) policy and gateway commands.

Security for the NetView Policy Command

To prevent an operator from issuing POLICY requests, protect the EZLEPOLY CLIST. You can use the POLICY command to load policy definitions and to add, modify, and delete policy definitions. Permit access to EZLEPOLY to only those operators whose jobs require them to perform policy functions. These operators also require READSEC authority to the POLICY data set.

When considering security for the Policy functions, remember to authorize the autotask, which loads the policy definitions into the Policy Repository. By default that autotask is AUTOAON.

Security for AON Gateway Sessions

To prevent an operator from accessing the AON Manage Cross Domain Gateway Sessions panel, protect command list EZLE5200. Only the operators whose jobs require them to issue commands through the gateway should have access to this panel. Additional layers of security are provided for gateway sessions.

The EZLE1REQ command list provides the ability to perform authorization checks for parameters passed to it. This is an example of the gateway command issued from the command line:

```
EZLE1REQ todom-fromdom,request,targoper,origoper,acknowl,command
```

Where:

todom

The domain where the command is to be run.

fromdom

The domain originating the command request.

request

The type of request. Only types of CMD are authority checked.

targoper

The operator for whom the command is to be run.

origoper

The operator who originated the command.

acknowl

The acknowledgment setting. Valid values are YES, NO, and ACK.

command

The command to be issued. Only the command is checked and none of the keywords.

This is an example of a gateway command. This request is being sent to domain NTV6D from domain NTV74. The command is to be routed to OPER4 on domain NTV6D. OPER1 originated the request on domain NTV74. An acknowledgment has been requested. The command to be run is DISCONID.

```
EZLE1REQ NTV6D-NTV74,CMD,OPER4,OPER1,YES,DISCONID
```

The following example shows NetView command authorization table entries that provide security for the gateway command, which prevents anyone from sending or receiving commands from domain NTV70 (a

& b). It also prevents sending commands to any task that begins with AUTO (c) or sending any command from OPER3 (d). The command DISCONID (e) cannot be issued or received from any operator or domain.

```
(a) Protect *.*EZLE1REQ.TODOM.NTV70
(b) Protect *.*EZLE1REQ.FROMDOM.NTV70
(c) Protect *.*EZLE1REQ.TARGOPER.AUTO*
(d) Protect *.*EZLE1REQ.ORIGOPER.OPER3
(e) Protect *.*EZLE1REQ.COMMAND.DISCONID
```

Security for SNMP Commands

To prevent an operator from issuing SNMP commands, you must control access to the command from both the AON operator panels as well as from the NVSNMP commands.

Security for SNMP Commands from AON Operator Panels

To control access to SNMP requests issued from AON panels, protect the CNMESNMP command.

To control access to which stack requests can be sent, or which host names can be the targets of stack requests, protect the FKXESCMD command.

For information about protecting these commands, see [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,”](#) on page 151.

Security for SNMP Commands from the NVSNMP Command

The NVSNMP command can issue SNMP requests directly to the Communications Server using OSNMP or it can use NetView SNMP services.

To control access to SNMP requests through NetView SNMP services, protect the CNMESNMP command.

To control access to SNMP requests to the Communications Server using OSNMP, protect the OSNMP command.

For information about protecting these commands, see [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,”](#) on page 151

Usage Notes

Consider the following items:

- When considering security for the SNMP functions, remember to authorize the AON/TCP autotasks, which might use SNMP functions for their processing. By default, those autotasks are AUTTCP1 through AUTTCP10; they can be customized differently for your environment. To determine your AON/TCP autotasks, browse your policy definitions and search for the AUTOOPS TCPOPER policy.
- Ensure that the NetView program has the appropriate directory accesses for the SNMP command.

For each directory in the list of directories that are defined on the COMMON.CNMSNMP.MIBPATH CNMSTYLE statement in the NetView style sheet (sample CNMSTYLE and its included members), give read access to the user ID that is associated with the NetView job or started task and any user ID that might execute or schedule the **SNMP (CNMESNMP)** command by use of the **NetView ATTACH** command.

The NetView program ships its own set of MIB definition files, which are located by default in the /usr/lpp/netview/v6r3/mibs directory. The MIB definition files in this directory are only updated by NetView maintenance.

Other MIB definition files, which could exist in a directory similar to /etc/netview/mibs, can be updated by using a job similar to sample CNMSJ032.

Note: Configuring read or write access to directories is done by using permission bits in addition to your SAF product.

Give write access to the `/var/snmp/mib_indexes` directory to the user ID associated with the NetView job or started task and any user ID that can schedule or run the **SNMP (CNMESNMP)** command via the NetView **ATTACH** command. The **SNMP** command can write a configuration file in that directory and can create a `mib_indexes` subdirectory into which one or more index files may be written. A directory other than `/var/snmp`, but subject to the same conditions for accesses, can be used for configuration and index files. For this directory, you need to define a CEEOPTS statement in the NetView start procedure to point to the file that defines the `SNMP_PERSISTENT_DIR` environment variable that specifies the path to replace `/var/snmp`. For example, `//CEEOPTS DD DISP=SHR,DSN=USER.PARMLIB(SNMPNVxx)`. In an `SNMPNVxx` member of `USER.PARMLIB`, code a statement like `ENVAR('SNMP_PERSISTENT_DIR=/pathname')`.

If you are using RACF for security, an alternative method for giving the user ID that is associated with the NetView job or started task write-access to directories is to give the user ID superuser authority, which can be done using one of the following methods:

- Set UID(0) in the OMVS segment of the user ID (profile) that is associated with the NetView job or started task.
- Set either the TRUSTED attribute to YES in the STDATA information section of a STARTED class profile that matches the NetView job or started task.

For more information, see the *z/OS Security Server RACF Security Administrator's Guide*.

- To protect a particular MIB variable from a GET request, restrict the MIB variable or its SNMP table. Be aware that the MIB variable can also be retrieved using a GETBULK, GETNEXT, or WALK request.
- If you are using NetView source ID checking for your security, functions such as starting monitoring by IPMAN can be restricted. The requests are checked against the operator starting the monitoring.
- When routing SNMP requests to remote domains, access to the FKXESCMD and STACK is checked in the source domain. If the operator has authority, the command is routed to the target domain and the remaining parameters are checked in that domain. For more information, refer to NVSNMP in the *IBM Z NetView User's Guide: Automated Operations Network*.
- Restrict access to the NVSNMP command (FKXE251A) and FKXESCMD command list. For more information, refer to NVSNMP in the *IBM Z NetView User's Guide: Automated Operations Network*.
- For NVSNMP requests that use the NetView SNMP service (the default option), see [Appendix A, "NetView Commands, Keywords, and Values that Can Be Protected,"](#) on page 151.

Examples

Example 1: Call to FKXESCMD to Issue OSNMP SET

This is an example of a call to FKXESCMD to issue an OSNMP SET of MIB variable `SysContact.0` to a value of *Cinderella* for IP host `nmpipl10.raleigh.ibm.com` from NetView operator OPER4:

```
NVSNMP -st NMPIPL10 -osnmp -h nmpipl10.raleigh.ibm.com
      SET SysContact.0 cinderella
(results in the following FKXESCMD invocation)
FKXESCMD NMPIPL10 OSNMP -c Master -h nmpipl10.raleigh.ibm.com
      SET SysContact.0 Cinderella
```

Using the following command authorization table definitions, all tasks are protected from issuing requests to all stacks (a) with the exception of OPER4 who can use LOCAL stack (b). Not all tasks can issue SNMP SET (c). In this case, the request fails because OPER4 is attempting to use NMPIPL10 and is also attempting to issue a SET.

- (a) Protect `*.FKXESCMD.STACK.*`
- (b) Protect `*.FKXESCMD.STACK.LOCAL`
- (c) Permit OPER4 `*.FKXESCMD.STACK.LOCAL`
- (d) Protect `*.FKXESCMD.SET.*`

Example 2: Call to FKXESCMD to Issue OSNMP GET

This is an example of a call to FKXESCMD to issue an OSNMP GET of MIB variable SysContact.0 for IP host 146.84.158.78 from NetView operator OPER4 using the LOCAL TCP/IP Stack:

```
NETVASIS NVSNMP -st LOCAL -osnmp -h 146.84.158.78 GET SysContact.0
(results in the following FKXESCMD invocation)
FKXESCMD LOCAL OSNMP -h 146.84.158.78
GET SysContact.0
```

Using the following command authorization table definitions, all tasks are prevented from issuing requests to all stacks (a) with the exception of OPER4 who can use LOCAL stack (b). Not all tasks can issue any SNMP GET (c) with the exception of OPER4. Not all tasks can issue requests for any device in subnet 146.84.158, except OPER4. In this case, OPER4 can issue a GET SysContact.0.

- (a) Protect *.*FKXESCMD.STACK.*
- (b) Protect *.*FKXESCMD.STACK.LOCAL
- (c) Permit OPER4 *.*FKXESCMD.STACK.LOCAL
- (d) Protect *.*FKXESCMD.GET.*
- (e) Permit OPER4 *.*FKXESCMD.GET.*
- (f) Protect *.*FKXESCMD._h.146/84/158/*
- (g) Permit OPER4 *.*FKXESCMD._h.146/84/158/*

Security for AON SNMP Processes

Table 8 on page 106 documents SNMP requests, MIB variables, and related AON functions. Any operator needing to use a function within this table must be granted access to the appropriate SNMP request and MIB variable.

Table 8. AON Command Identifiers. SNMP Functions Used by AON/TCP		
SNMP Command	SNMP MIB Variable	NetView Function
WALK	IpAddrTable	IPMAN - Resource Details
WALK	IfTable	SNMP Resource Information • IPMAN • Monitoring • Autoview
GET	defined in policy	SNMP MIB Thresholding
GET	ibmMvsRPing ResponseTime	SNMP Remote Ping
WALK	ibmTcpipMvsTcpConnTable	• IPSTAT • SNMPVIEW • DVIPA functions • FKXEDROP
WALK	tcpConnTable	IPSTAT
SET	tcpConnState	FKXEDROP
GET	ibmMvsTcpipSubagentVersion	Stack Monitoring
WALK	ibmTcpipMvsTcpListenerTable	• Port Monitoring • DVIPA functions
WALK	ibmMvsDvipaTable	DVIPA functions
WALK	ibmMvsDvipaConnRoutingTable	DVIPA functions

<i>Table 8. AON Command Identifiers. SNMP Functions Used by AON/TCP (continued)</i>		
SNMP Command	SNMP MIB Variable	NetView Function
WALK	ibmMvsDvipaDistPortTable	DVIPA functions
GET	ibmMvsTcpipProcname	DVIPA functions
WALK	ibmTcpipMvsUdpTable	Port Monitoring
GET	sysUpTime	Stack Monitoring
WALK	ipAddrTable	DVIPA functions
WALK	ibmMvsTcpConnResourceId	IPSTAT

Security for SNMP Community Name

You can define a COMMUNITYNAME= keyword on each of your TCP390 stack definitions. You must also protect access to your TCP390 definitions. To accomplish security for your NetView community names, take the following actions:

- Restrict access to operators who can read DSIPARM members where your TCP390 definitions reside. By default, this is CNMPOLCY.
- Restrict access to operators who can issue commands to display and update your TCP390 definitions. You must restrict the following commands:
 - DSPCFG
 - POLICY

Chapter 10. Security for the NetView Management Console

This section provides information about security for the NetView management console.

Controlling Access to the NetView Management Console

You can prevent unauthorized users from logging on to the NetView management console by restricting access to the DUILOGON NMC command that is used to connect to the topology server. You can restrict access using either an SAF product or the NetView command authorization table. The sample command authorization table CNMSCAT2 includes an example of the necessary statements. The RACF definition sample CNMSAF2 includes a commented-out example of DUILOGON protection. For more information about restricting the DUILOGON command, see [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,”](#) on page 151.

Security for Commands Issued from the NetView Management Console

There are several ways to restrict commands from being issued from NetView management console.

NGMFCMDS

Using the NetView operator profile on the host, you can specify whether this operator is allowed to issue commands from pull-down menus on the graphical display. See [“Using NGMFCMDS”](#) on page 16.

Note: This specification does not prevent operators from entering commands on a NetView command line.

Command Profile Editor

With the command profile editor, you can limit the commands that are displayed for a particular operator. To prevent an operator from seeing an action in the pop-up menu for a particular type of resource, remove the command that corresponds to that action and resource combination from the command profile.

Note: This does not prevent operators from entering commands on a NetView command line. The ihsscpe.xxx.rsp and flccpe.xxx.rsp samples that are supplied with the NetView product, where xxx is a country code indicator such as en_us, show Command Profile Editor examples of NetView, VTAM, SNA Topology Manager, and MSM commands.

Using host task authorization

NetView commands issued from NetView management console are subject to normal NetView command authorization as described in [Chapter 3, “Controlling Access to Commands,”](#) on page 29.

Using DSIEX19

If a NetView RUNCMD is issued as a result of the command issued by the graphic display, NetView installation exit DSIEX19 can be used to protect the imbedded service point command.

The NetView management console topology server provides an audit log for commands and command responses. For more information, refer to the *IBM Z NetView Administration Reference*.

Applying Policy to Views

This provides information about security for the DUIFPOLI autotask.

CHRON Timers Created by the DUIFPOLI Autotask

When NMCSTATUS policy definitions are processed, the DUIFPOLI autotask creates CHRON timers to specify the beginning and end of a policy window. Each TIME keyword specified for the NMCSTATUS policy definition results in the creation of two CHRON timers; one timer to begin the scheduled window and one timer to end it. Timers that are created by the DUIFPOLI autotask are prefixed with the characters NMC, for example, NMC1.

It is possible, but not recommended, to change the schedule of a policy with the TIMER command. You can set up security on these timers so they are not deleted or updated. To see all timers created by the DUIFPOLI autotask, enter:

```
TIMER NMC
```

where NMC is the filter.

For additional information about defining NMCSTATUS policy definitions refer to NMCSTATUS in the *IBM Z NetView Administration Reference*.

Commands Issued by the DUIFPOLI Autotask

NetView commands are issued by the DUIFPOLI autotask during processing of NMCSTATUS policy definitions. Make sure that the autotask has authorization for the following NetView commands:

- EZLEDAPI
- EZLEPOLY
- EZLETAPI
- NMCPINIT
- NMCPTTEST
- EXCMD

Note: If you issue NetView command NMCPINIT or NMCPTTEST from a NetView OST, the command is sent to autotask DUIFPOLI to run.

Chapter 11. Security for the NetView REST Server

Security Configuration of the NetView REST Server

About this task

To complete the security configuration of the NetView REST Server, perform the following steps:

Procedure

1. Make a keystore.

From Server Configuration folder:

```
mkdir keystore
```

2. Generate keystore.

```
keytool -genkey -alias NetView -keystore keystore/localhost.keystore.p12 -storetype PKCS12 -  
keyalg RSA \ -storepass <unique-password> -validity 730 -keysize 2048
```

3. Export certificate.

```
keytool -export -alias NetView -storetype PKCS12 -storepass <password> -file keystore/  
localhost.cer \ -keystore keystore/localhost.keystore.p12
```

4. Generate truststore and store certificate.

```
keytool -import -storetype PKCS12 -keystore keystore/localhost.truststore.p12 \ -storepass <password> -alias NetView -file keystore/localhost.cer -noprompt
```

5. Update the application.yml file with the passwords for the keystore and truststore.

Note: The application.yml file is encoded in ASCII.

Optional: Configuration for Using Zowe™ Authentication With Single Sign-On

About this task

This task is optional. If you want to use Zowe™ authentication with single sign-on, complete the following steps:

Procedure

1. Add Zowe™ certificate to truststore.

From Zowe™ home directory:

```
keytool -import -alias localca -storetype PKCS12 -file <zowe_home_directory>/api-mediation/  
keystore/localca/localca.cer \ -keystore keystore/localhost.truststore.p12
```

2. Specify Zowe™ location in the yaml File.

The server directory contains an `application.yml` file for updating the properties of the application. One common use case for updating the `application.yml` file is if the default URLs for Zowe™ need to be changed.

The following URLs are the default URLs for Zowe™ in the `application.yml` file. These are used by the NetView REST Server to communicate with Zowe™ and they should match your Zowe™ configuration.

```
zowe:
  urls:
    apiCatalog: https://localhost:7554/api/v1/apicatalog/
    query: https://localhost:7554/api/v1/apicatalog/auth/query/
    login: https://localhost:7554/api/v1/apicatalog/auth/login/
```

If you want to update the `yml` file, simply change the text of the entry that needs to be updated and save the file, for example,

```
zowe:
  urls:
    apiCatalog: https://localhost:7554/api/v1/apicatalog/
    query: https://localhost:7554/api/v1/apicatalog/auth/query/
    login: https://myRemoteServer.company.com:7554/api/v1/apicatalog/auth/login/
```

Note: 7554 in the default URLs is a default port for Zowe™, and may be changed to another value.

Chapter 12. Defining NetView Security for RODM

If you are using an SAF product, such as RACF, on your system, you can use any of the three ways to defining RODM security:

- Bypass system security with *TSTR0DM.
- Define the RODM task and authority level to the RODMMGR class of your SAF product, if it is available.
- Define the RODM task and resources that represent authority levels to a user-defined class in your SAF product.

Bypassing RODM Security

To bypass RODM security, initialize RODM with *TSTR0DM in the SEC_CLASS field in EKG CUST when:

- Your system uses an SAF product, such as RACF, but you do not want to define RODM and operator tasks to the SAF product for security.
- Your system does not use an SAF product.
- The SAF product is not active on your system.

Note: If you are referring to this section from the *IBM Z NetView Installation: Getting Started* during the installation process and your system does not have an SAF product, you can now IPL the target system with the CLPA Option.

Defining RODM Security with the RODMMGR Class

If you are using the RACF product for RODM security, you need to have RACF at this level or higher to use the SAF RODMMGR class:

- RACF Version 1 Release 9 with PTF UW00497
- RACF Version 1 Release 9.2 with PTF UW00498
- RACF Version 2 Release 1 with PTF UW90113

Note: If you are referring to this section from *IBM Z NetView Installation: Configuring Graphical Components* during the installation process and your security product provides the RODMMGR class, you can now IPL the target system with the CLPA Option.

If you are using an SAF product which provides a RODMMGR class, you still need to define security resource names to that product, and you need to authorize users to the correct SAF resources, as described in [“Defining RACF Resource Names”](#) on page 115.

Defining RODM Security with a User-Defined Class

If you are using an SAF product for RODM security, and the SAF product does not provide the SAF RODMMGR class, these steps must be completed before RODM can initialize:

- Define a security class in the SAF product for RODM. For RACF, you also need to create a RACF router table for this security class, as described in [“Creating a Sample RACF Router Table”](#) on page 114.
- Define security resource names for the class you define. See the *IBM Z NetView Administration Reference* for more information.

The commands you issue to define RODM and the operators to the security class can vary depending on whether you use RACF or another SAF product.

Defining the Resource Class to the RACF Class Descriptor Table

The SEC_CLASS operand in EKGCUST in CNMSAMP can be used to specify the security class definition for your installed security system. If you do not define the class name in the EKGCUST customization file, or if you do not include the EKGCUST DD statement in the JCL, the default security name is RODMMGR.

Note: Prior to NetView Version 3, the default RODM security class was DATAMGR.

To use another class name as the default RACF security name, define the *class_name* to the RACF class descriptor table and the RACF router table. Locate the RFTABLE job in the RACINSTL member in SYS1.SAMPLIB. The RACINSTL member contains sample RACF installation jobs.

The following example shows the assembler and link-edit statements that you need to use to modify the RFTABLE job to create a sample class descriptor table, ICHRRCDE, using *class_name* as the security class.

```
//ASM.SYSIN DD *
class_name  ICHERCDE CLASS=class_name          *
              ID=130,                          *
              MAXLNTH=44,                      *
              FIRST=ALPHANUM,                  *
              OTHER=ANY,                       *
              POSIT=21,                        *
              OPER=NO                          *
//LKED.SYSMOD DD DSN=SYS1.LINKLIB,DISP=SHR
//LKED.SYSIN DD *
ORDER class_name
ORDER ICHRRCDE
NAME ICHRRCDE(R)
//
```

Note:

1. You might receive a warning note that indicates that the class name does not contain a national character or digit in the first four positions. You can ignore this message.
2. You receive a return code of 4 from the assembler steps. In this case, return code 4 is not an error.
3. The valid values for the ID number are in the range of 128–255.
4. MAXLNTH specifies the maximum length of names of resources defined by the CLASS operand. RODM further restricts this length by requiring that the resource name be one less than the number you specify for MAXLNTH.

Creating a Sample RACF Router Table

The following example shows the assembler and link-edit statements that you need to use to modify the RFTABLE job to create a sample RACF router table, ICHRRFR01, for your user-defined security class, *class_name*. RODMMGR is the default security class for RODM since RACF Version 2 Release 1. See the *IBM Z NetView Administration Reference* for more information.

```
//ASM.SYSIN DD *
ICHRRFR01 CSECT
class_name  ICHRRFR01 CLASS=class_name,ACTION=RACF
ENDTAB     ICHRRFR01 TYPE=END
              END ICHRRFR01
//LKED.SYSMOD DD DSN=SYS1.LINKLIB,DISP=SHR
//LKED.SYSIN DD *
NAME ICHRRFR01(R)
//
```

Note: You might receive a warning note that indicates that the class name does not contain a national character or digit in the first four positions. You can ignore this message. Also, you receive a return code of 4 from the assembler steps. This is not a reason for concern.

Re-IPL MVS for the RACF resource class and the RACF router table to become effective.

If you are referring to this section from the *IBM Z NetView Installation: Configuring Graphical Components* during the installation process, as you proceed through your installation, [“Using RACF for RODM Security”](#) on page 115 contains additional steps that you must complete to define RODM security.

If you want information about...	Refer to...
Defining a security class	RACF library
ICHRFRTB macros	RACF library

Using RACF for RODM Security

If you use “Defining RODM Security with the RODMMGR Class” on page 113 or “Defining RODM Security with a User-Defined Class” on page 113, you must also perform the following operations:

- Define six RACF resource names under RODMMGR or your user-defined security class for the six user authority levels.
- Define user IDs for users connecting to RODM. If the user ID is already defined to RACF (for example, for normal login), no additional registration is required.
- Authorize user IDs to the appropriate RACF resource names.

Note: RODM only verifies security levels for API calls into RODM, and not on the MODIFY command interface. To implement security for the MODIFY command interface, refer to the RACF library and the MVS/ESA library.

Defining RACF Resource Names

To define the RACF resource names under RODMMGR for the six user authority levels, complete the following steps from your RACF-authorized TSO ID.

1. To define the RODM resource names, if SEC_RNAME is RODM, enter:

```
RDEFINE RODMMGR RODM1 UACC(NONE)
RDEFINE RODMMGR RODM2 UACC(NONE)
RDEFINE RODMMGR RODM3 UACC(NONE)
RDEFINE RODMMGR RODM4 UACC(NONE)
RDEFINE RODMMGR RODM5 UACC(NONE)
RDEFINE RODMMGR RODM6 UACC(NONE)
```

If you have your own user-defined *class_name*, replace RODMMGR with the security class name on the RDEFINE commands. The resource names used are an example.

The RODM resource names consist of a prefix and a suffix. The suffix must have values of 1 through 6 for the different levels of security. The default resource name prefix is the RODM name specified in the RODM startup JCL. For example, the RODM name is ZZRODM using either of the following start commands:

```
S EKGXRODM,NAME=ZZRODM
S EKGXRODM.ZZRODM
```

It is recommended that your resources use the name of your RODM.

If you specify your own RODM resource names, the resource name prefix must be specified in EKGCUST on the SEC_RNAME statement if the resource name prefix is not the name of your RODM.

RODM restricts the length of resource names by requiring that the resource name be one less than the number you specify for MAXLNTH. For RODMMGR, MAXLNTH is 44, so the resource name must be 43 characters or fewer. If you define your own security class, MAXLNTH is specified when you define the RACF class descriptor table. See [“Defining the Resource Class to the RACF Class Descriptor Table”](#) on page 114 for information about defining a security class to the RACF class descriptor table.

2. To set the system-wide RACF options, enter:

```
SETROPTS CLASSACT(RODMMGR)
```

If you have your own user-defined *class_name*, replace RODMMGR with the security class name on the SETROPTS command. The resource class name used is an example.

Authorizing User IDs to RACF Resource Names

To access RODM, enter the following command from your authorized TSO ID for each *userid* that requires access:

```
PERMIT resource_name CLASS(RODMMGR) ID(userid)
```

Where:

userid

Specifies the RACF user ID. You can use the PERMIT command to authorize a group, instead of specific users, to the authority level resources. You can then connect or remove user IDs from the group as their need for RODM capabilities changes.

resource_name

Specifies the name of the RODM resource (such as RODM1 through RODM6) that has the appropriate security level for the function that the *userid* needs to be able to perform. Indicate the highest level RODM resource name the *userid* needs to access. If you indicate a user is authorized for RODM3, that user also has authorization for security level 1 (RODM1) and security level 2 (RODM2) capabilities.

For example:

```
PERMIT RODM3 CLASS(RODMMGR) ID(USER1)
```

indicates that USER1 is authorized to perform the capabilities of RODM security levels 1, 2, and 3. [Table 9 on page 116](#) describes the RODM security levels.

Note: If you have a user-defined *class_name*, replace RODMMGR with the security class name on the PERMIT commands.

Table 9. RODM Access Security Levels

Resource Name	Security Level	Capabilities
<i>rodm1</i>	1	Connect and disconnect to RODM
<i>rodm2</i>	2	Query and list of functions (queries only)
<i>rodm3</i>	3	Action and list of functions (queries or actions) including triggering methods and change methods
<i>rodm4</i>	4	Checkpointing
<i>rodm5</i>	5	Administrative functions (adding or deleting from the RODM data cache) and adding managerial objects
<i>rodm6</i>	6	Stopping RODM

NetView operators require RODM security level 2 or higher to use the QRS command to query whether they have span of control over resources defined.

Ensure the following authorizations:

- The RODM load function requires a minimum of RODM security level 3. If your RODM loader job is run as a started procedure, you can define it to the STARTED class in the SAF product to enable it to run as a trusted user. You can define the task in the started procedure table, ICHRIN03; however, using the STARTED class is preferred.

- The Graphic Monitor Facility host subsystem (GMFHS) procedure requires a minimum of RODM security level 5.
- The NetView procedure, if NetView user code accesses RODM.
- The SNA topology manager requires a minimum of RODM security level 5. The user ID to authorize is APPNTM.
- The DSIQTSK subtask requires RODM security level 6. Define user ID DSIQTSK or the value of the ID keyword of the REP statement in the DSIQTSKI initialization member.
- Any user who submits one of the preceding procedures.
- Any user who manipulates RODM from the NetView program by using RODMView panels or the RODMView command processors.
- A minimum of RODM security level 2 for the NetView domain name, if span of control is being applied to views and resources.

Connecting to RODM

When connecting to RODM, a user ID and password or password phrase are part of the API request. A password or password phrase is required, except when the program making the request is running in an APF-authorized library. The user ID can be specified on the connection request, or RODM can extract it from the SAF product.

You can connect to RODM with a blank user ID if the system on which RODM is installed has active RODM security. In this case, RODM extracts the user ID from the SAF product. Connecting to RODM with a blank user ID is not allowed when you are running without RODM security active.

If you have RODM security active, the user ID that is associated with the connection request must be defined to your SAF security product.

For started procedures, you can define the started procedure name to the STARTED class of the SAF product. In RACF, this can also be accomplished by defining the task in the started procedure table, ICHRIN03; however, using the STARTED class is recommended.

To activate RODM security:

- Install an SAF product.
- Activate a security class for RODM (RODMMGR or user defined).
- Identify the security class with the SEC_CLASS keyword in EKGCUST.

Chapter 13. Scenarios for Converting Types of Security

If you are installing the NetView program for the first time, use this chapter with [Chapter 1, “Overview of NetView Security,”](#) on [page 1](#) to plan your security implementation and determine how you can adapt your security in the future.

If you already have security defined for the NetView product, use this chapter to ensure your security continues to work as you intended.

Note:

1. CNMSCAT2 is a sample command authorization table supplied with the NetView program. This table contains recommended commands, keywords, and values to protect. Review this table and make changes necessary for your installation.
2. Sample CNMSAF2 contains RACF definitions for NetView operators and commands. Review this sample and modify it as necessary for your installation.

Use the NetView LIST SECOPTS command to display your current security settings as shown in the following example:

```

NetView V6R3 - 7E      NetView   NTV7E TOM      06/07/19 18:27:41
* NTV7E      list secopts
  NTV7E
BNH228I OPTION          VALUE          LAST UPDATED          UPDATE ID
BNH229I -----
BNH229I OPERSEC         NETVPW         06/07/19 09:59:30     INITIALIZATION
BNH229I OPSPAN          NETV           06/07/19 09:59:30     INITIALIZATION
BNH229I CMDAUTH         TABLE         06/07/19 09:59:30     INITIALIZATION
BNH229I TBLNAME         CNMSCAT2     06/07/19 09:59:30     INITIALIZATION
BNH229I AUTHCHK         SOURCEID       06/07/19 09:59:30     INITIALIZATION
BNH229I SPANAUTH        NONE          06/07/19 09:59:07     INITIALIZATION
BNH229I SPANCHK         SOURCEID       06/07/19 09:59:30     INITIALIZATION
BNH229I CATAUDIT        NONE          06/07/19 09:59:07     INITIALIZATION
BNH229I AUTOSEC         BYPASS          06/07/19 09:59:34     INITIALIZATION
BNH229I SURROGAT        NO            06/07/19 09:59:30     INITIALIZATION
BNH229I RMTSEC          NONE          06/07/19 09:59:40     INITIALIZATION
BNH229I RMTAUTH         SENDER        06/07/19 09:59:30     INITIALIZATION
BNH229I WEBAUTH         PASS          06/07/19 09:59:30     INITIALIZATION
BNH230I END OF LIST SECOPTS INFORMATION
-----

```

This example shows the following security settings:

OPERSEC

NETVPW indicates that operator IDs are defined in DSIPARM member DSIOPF.

OPSPAN

NETV indicates that valid span of control names for operators are defined by operator profiles located in the DSIPRF data set.

CMDAUTH

TABLE indicates that command authorization definitions are defined in DSIPARM by a command authorization table.

TBLNAME

CNMSCAT2 indicates the name of the command authorization table.

AUTHCHK

SOURCEID indicates that cross-task command authorization checking is performed against the authority of the original issuer of the command. When this information is not available, the identity of the task where the command first entered the NetView program is checked.

SPANAUTH

TABLE indicates that resources and views that are contained in the spans of control are defined in DSIPARM by a NetView span table.

SPANTBL

MAINSPAN indicates the name of the span of control authorization table.

SPANCHK

TARGETID indicates span checking is performed against the authority of the operator who issued the command.

CATAUDIT

NONE indicates that auditing of command authorization checking is not active.

AUTOSEC

CHECK indicates that command authorization checking is active for commands originating in the NetView automation table.

SURROGAT

NO indicates that no SAF surrogate support is available for NetView TSO servers.

RMTSEC

TABLE indicates that a remote security table in DSIPARM is used to contain the list of remote NetView domains that are allowed to send commands to this NetView domain.

TBLNAME

DSISECUR indicates that the name of the remote security table is DSISECUR.

WEBAUTH

CHECK indicates that authorization checking for access to the NetView Web server is performed as well as command authorization checking for any subsequent commands entered by the operator.

The NetView LIST command will display operator attribute values. The values of the attributes are set when the operator logs on. Use LIST ' ' to display your own values, or LIST *operid* command to display the values of other operator tasks.

```
- N5173 STATION: OPER1 TERM: NTD9L703
- N5173 HCOPI: NOT ACTIVE PROFILE: PROFAUTH
- N5173 STATUS: ACTIVE IDLE MINUTES: 0
- N5173 ATTENDED: YES CURRENT COMMAND: LIST
- N5173 AUTHRCVR: NO CONTROL: GLOBAL
- N5173 NGMFADMN: YES DEFAULT MVS CONSOLE NAME: NONE
- N5173 NGMFCMD: YES AUTOTASK: NO
- N5173 IP ADDRESS: N/A
- N5173 OP CLASS LIST: NONE
- N5173 DOMAIN LIST: N5173 (I) N5172 (I) N5171 (I) CNM99 (I) CNM02 (I)
- N5173 CNM01 (I)
- N5173 ACTIVE SPAN LIST: NONE
- N5173 Task Serial: 23741
- N5173 Messages Pending: 0 Held: 0
- N5173 WLM Service Class: Not Available
- N5173 END OF STATUS DISPLAY
```

Refer to the NetView online help for information about the NetView LIST command.

If you have not used security in the past, but now want to use security, use all the security chapters in this book to understand how security works; then see [“Scenario 1: Migrating from a System with No Security” on page 121](#) to use the default values that are supplied with the NetView product for security.

When security is established, you can use the information in this chapter to change how you define operator passwords and attributes, span of control, and command authorization security from the NetView program to an SAF product, such as RACF.

Because some types of security have prerequisites, it can be beneficial to review the steps in the order listed. You can arrange the parts of your security in any sequence that fulfills the prerequisites listed in each scenario. The scenarios describe how to:

1. Enable security, if you have disabled it.
2. Migrate your existing security settings to this release.

3. Change logon passwords to use an SAF product.
4. Define authorization for individual tasks, rather than at a global level for all NetView tasks.
5. Define the operator attributes in the NETVIEW segment of the SAF product instead of in NetView profiles in the DSIPRF data set.
6. If you use span of control, define the span names in an SAF product.
7. If you use span of control, convert existing span definitions in DSISPN and VTAMLST to NetView span table statements.
8. Change the type of command security from scope of command authorization to the NetView command authorization table.
9. Change the type of command security from the NetView command authorization table to an SAF product with NetView command authorization table as backup.

Although other security setups can be desirable for some purposes, this chapter describes the security tasks listed previously.

Many scenarios contain brief examples showing how to make security changes manually or using the NetView SECMIGR command. You need to understand the differences between the types of security methods whether you convert manually or by using the SECMIGR command, because any conversion must be carefully verified before being implemented on your system.

The SECMIGR command can be used through a panel interface, or by issuing the command with keywords through a command line interface. The following example illustrates the SECMIGR panel interface:

```

CNMKSER1          NetView Security Migration Tool

      To perform one of the following migration options, place
      any character next to the desired option and press enter.

      From                      To                      Keyword Name
      -----
      _ NetView Operator Definitions  RACF                      OPS2RACF
      _ NetView Span Definitions      RACF                      SPN2RACF
      _ NetView Scope Settings        NetView Command          SCP2TBL
      _ NetView Table Statements      Authorization Statements  TBL2RACF
      _ NetView Scope Settings        RACF                      SCP2RACF
      _ VTAM Span Definitions          NetView Span Definition   SPN2TBL
      Statements

CMD==>

```

Figure 8. SECMIGR Main Panel

For more information about the SECMIGR command, refer to the NetView online help.

Before changing any type of security, verify that your security is working as you expect with this release of the NetView program. To solve problems with the security function, see [Chapter 14, “Checklist for Debugging Security Problems,”](#) on page 139.

For more information about Version 2 of the RACF product, refer to the RACF library.

Scenario 1: Migrating from a System with No Security

Examine the SECOPTS statements in the CNMSTYLE member. Decide which definitions you plan to maintain in the NetView program and in an SAF product. For command authorization checking, the NetView program provides samples CNMSCAT2 and CNMSAF2.

Scenario 2: Converting Operator Passwords

This scenario assumes you currently define operator logon passwords in the NetView DSIOPF member. If you have an SAF product installed, and if you are already using it for NetView password security, skip this step. When you complete this scenario, your system will use the SAF product to validate NetView operator passwords.

The advantages for using an SAF product for passwords include:

- All NetView operator passwords can be maintained in a secure place.
- Enforcement of consistent rules for operator passwords across multiple products on your system.
- Passwords can be changed by operators.
- Password phrases can be used as a substitute for a password. A password phrase can include phrases that are 9 - 100 characters in length, without character restrictions.

If you are also going to convert operator logon attributes, skip to [“Scenario 3: Converting to Task-Level Checking”](#) on page 123, which also contains information about converting operator passwords. For an overview of the conversion process, see [“Example of Migrating an Operator Password and Logon Attributes”](#) on page 21.

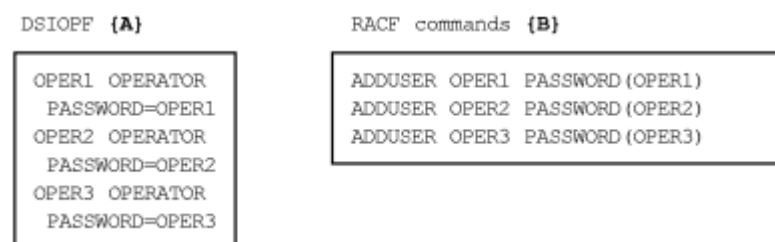


Figure 9. Converting Operator Passwords to RACF

You can convert NetView passwords to use the RACF product as shown in [Figure 9 on page 122](#).

{A}

In this example, three operators and their passwords are defined using the NetView DSIOPF member.

{B}

Use the RACF ADDUSER command to define the operators in RACF with an initial password.

RACF is configured so that all initial passwords expire when the operator logs on. RACF can also have rules enforced for passwords which might not be compatible with the NetView passwords, so an operator is forced to change the password after logging on the first time.

To test whether these definitions work as you expect, dynamically change your security so that an SAF product checks logon passwords by using the NetView REFRESH command:

```
REFRESH OPERSEC=SAFDEF
```

If operators cannot log on as you expect, issue `REFRESH OPERSEC=NETVPW` to go back to using the NetView program for passwords and see [Chapter 14, “Checklist for Debugging Security Problems,”](#) on [page 139](#).

When you are sure that your security is working correctly, modify the SECOPTS statement in the CNMSTUSR or CxxSTGEN member in the following way

```
SECOPTS.OPERSEC = SAFDEF
```

The next time the NetView program is initialized, it uses an SAF product to verify logon passwords. See [“Defining Operator Password Security”](#) on [page 10](#) for information about the commands used to define passwords in the SAF product.

Scenario 3: Converting to Task-Level Checking

This scenario assumes you currently use OPERSEC=SAFPW and want to migrate to OPERSEC=SAFCHECK. With OPERSEC=SAFCHECK, the security product calls made by MVS on the behalf of NetView tasks use the individual task authority, rather than the NetView program authority. The SAF product is invoked for security checking on behalf of NetView tasks for:

- Data set authorization
- MVS system commands that are protected in the OPERCMDS class of the security product.

Before changing to OPERSEC=SAFCHECK, you must define at least the PPT task to the security product. Check DSIOPF to ensure that all existing operators and autotasks are defined to the security product. See [“Scenario 6: Converting Operator Logon Attributes” on page 129](#) for more information. In RACF, issue the following ADDUSER commands if the domain name was CNM01:

```
ADDUSER CNM01PPT
```

When OPERSEC=SAFPW, command authorization and data set access were checked against NetView authority. After migrating to OPERSEC=SAFCHECK, these authority checks are made against the individual NetView operator tasks. Therefore, it is important to ensure that the NetView operator tasks that perform data set access, such as with EXECIO from REXX command lists, are authorized to the proper data sets in the DATASET class of the security product. You might also want to authorize some NetView operators to protected MVS system commands in the OPERCMDS class of the security product. To authorize OPER1 to update a protected data set called SYSPROG.LOGDATA, issue the following RACF PERMIT command:

```
PERMIT 'SYSPROG.LOGDATA' ID(OPER1) CLASS(DATASET) ACCESS(UPDATE)
```

Similarly, if the SET MPF command is protected in the OPERCMDS class, and OPER1 wants to issue it from the NetView program, issue the following PERMIT command in RACF to authorize OPER1 to the SET MPF command:

```
PERMIT MVS.SET.MPF ID(OPER1) CLASS(OPERCMDS) ACCESS(READ)
```

Giving operators access to the MVS.MCSOPER.*console_name* profile in the OPERCMDS class, gives those operators permission to obtain MCS consoles.

```
PERMIT MVS.MCSOPER.console_name ID (OPER1) CLASS (OPERCMDS) ACCESS (READ)
```

If no MVS.MCSOPER.*console_name* profile exists, you can define access generically as follows:

```
RDEFINE OPERCMDS MVS.MCSOPER.*VACC (READ)
```

For additional information see, [“Protecting EMCS Console Names Using an SAF Product” on page 27](#)

MVS commands can also be protected with the NetView command authorization table or the NETCMDS class of the security product.

When you are confident that all necessary authorizations to the OPERCMDS class and the DATASET class are complete, you can dynamically change to task-level checking by issuing:

```
REFRESH OPERSEC=SAFCHECK
```

After updating OPERCMDS class definition, refresh the OPERMDS class by entering:

```
SETROPTS RACLIST (OPERCMD) REFRESH
```

The next time each task issues a command, the task-level checking will take effect for that task.

If the task-level checking is working properly, you can include the following SECOPTS.OPERSEC statement in the CNMSTUSR or CxxSTGEN member:

```
SECOPTS.OPERSEC = SAFCHECK
```

The next time the NetView program is initialized, it uses task-level checking.

Scenario 4: Converting Operator Access to Span of Control

This scenario assumes that you already use an SAF product for operator passwords or password phrases and that you are using task-level checking (OPERSEC=SAFCHECK). It also assumes that you already used the NetView program to define span of control.

When spans are defined using the NetView program (OPSPAN=NETV), the SPAN or ISPAN statements are used in operator profiles to enable or restrict access to spans.

Whether access to spans is defined using the NetView program or in the NETSPAN class of the security product, the resources and views contained in the spans are defined in a NetView span table.

You cannot define span of control in the NETSPAN class of the security product (OPSPAN=SAF) unless the OPERSEC keyword has a value of SAFCHECK or is being concurrently migrated to a value of SAFDEF. When you complete this scenario, your span of control will be defined using an SAF product.

You must collect span names from SPAN and ISPAN statements in the operator profiles in the DSIPRF data set and define the span names in the NETSPAN class of an SAF product. For necessary background, read the overview in Chapter 4, “Using Spans to Protect Resources and Views,” on page 67 and refer to the example operator attributes described in “Using an SAF Product Exclusively” on page 19 to understand the desired results. The CTL setting, which affects whether span of control is used for individual operators, can be set in the NetView operator profile or in the NETVIEW segment, whichever you use to define operator attributes.

When migrating, source information for existing or default span of control is found in DSIPRF and various profile members in the DSIPRF data set. In this scenario, DSIPROFY and DSIPROFZ are used as example member names to demonstrate how to gather span-of-control information for operators OPERY and OPERZ, and use this information to create span-of-control definitions in an SAF product.

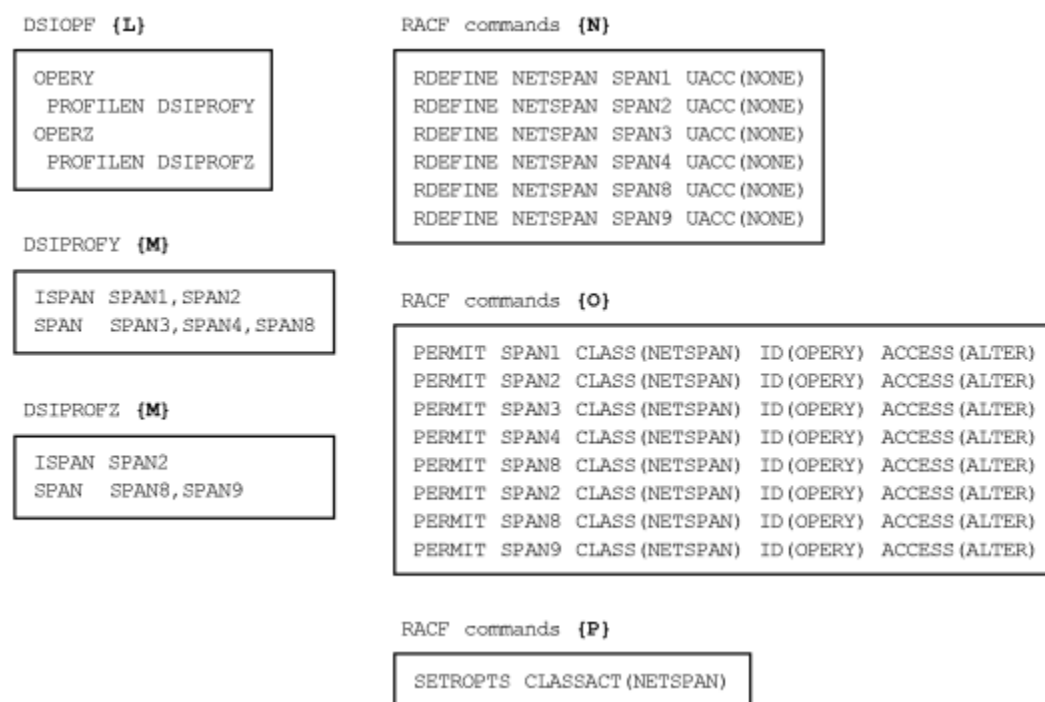


Figure 10. Converting Operator Passwords to RACF

Note: This example shows only one profile for each operator. If you specify multiple profiles for operators, it might be necessary to extract span information from additional profiles or members.

{L}

In your existing NetView operator definitions, the DSIOPF member defines operator task names and the names of the profiles used by each operator.

In this example, DSIOPF defines operator names OPERY and OPERZ, and specifies that OPERY uses profile DSIPROFY, and OPERZ uses DSIPROFZ.

{M}

These excerpts of profiles DSIPROFY and DSIPROFZ show which spans OPERY and OPERZ are allowed to start. OPERY is allowed to start five spans, including SPAN3, SPAN4, and SPAN8, and will start up with SPAN1 and SPAN2 already activated. OPERZ is allowed to start three spans, including SPAN8 and SPAN9, and will start up with SPAN2 already activated.

{N}

Each unique span is protected by defining it as UACC(NONE) in the NETSPAN class, for example using the RACF RDEFINE command. Each span only needs to be defined once, so span names which are duplicated across multiple operators do not require more than one RDEFINE.

{O}

When the span names are protected, each operator must be permitted to use every span in their corresponding NetView operator profile, as shown in **{M}**. There must be a PERMIT statement for each span name for each operator.

{P}

The NETSPAN class must be activated before attempting to activate the span-of-control definitions, using the RACF SETROPTS command. You can activate OPSPAN=SAF using the NetView REFRESH command, or by including the following statement in the CNMSTUSR or CxxSTGEN member and recycling the NetView program:

```
SECOPTS.OPSPAN = SAF
```

Use an access level of alter, ACCESS(ALTER), to provide equivalent access to resources as when span names are defined using OPSPAN=NETV.

To provide the same function as an ISPAN statement using an SAF product, add one START SPAN command to the operator's initial command list for each of the spans contained in the ISPAN statements that were in the NetView profile. For example, add these commands to the initial command list for OPERY:

```
START SPAN=SPAN1
START SPAN=SPAN2
```

Add the command START SPAN=SPAN2 to the initial command list for OPERZ.

This is an example of using the NetView REFRESH command to use the NETSPAN class of an SAF product:

```
REFRESH OPSPAN=SAF
```

The NetView LIST SECOPTS command will display security settings, including the OPSPAN setting as follows:

NetView V6R3 - 7E	NetView	NTV7E TOM	06/07/19 18:27:41
* NTV7E list secopts			
' NTV7E			
BNH228I OPTION	VALUE	LAST UPDATED	UPDATE ID
BNH229I -----	-----	-----	-----
BNH229I OPERSEC	NETVPW	06/07/19 09:59:30	INITIALIZATION
BNH229I OPSPAN	NETV	06/07/19 09:59:30	INITIALIZATION
BNH229I CMDAUTH	TABLE	06/07/19 09:59:30	INITIALIZATION
BNH229I TBLNAME	CNMSCAT2	06/07/19 09:59:30	INITIALIZATION
BNH229I AUTHCHK	SOURCEID	06/07/19 09:59:30	INITIALIZATION
BNH229I SPANAUTH	NONE	06/07/19 09:59:07	INITIALIZATION
BNH229I SPANCHK	SOURCEID	06/07/19 09:59:30	INITIALIZATION
BNH229I CATAUDIT	NONE	06/07/19 09:59:07	INITIALIZATION
BNH229I AUTOSEC	BYPASS	06/07/19 09:59:34	INITIALIZATION

```
BNH229I SURROGAT      NO      06/07/19 09:59:30      INITIALIZATION
BNH229I RMTSEC        NONE     06/07/19 09:59:40      INITIALIZATION
BNH229I RMTAUTH       SENDER   06/07/19 09:59:30      INITIALIZATION
BNH229I WEBAUTH       PASS     06/07/19 09:59:30      INITIALIZATION
BNH230I END OF LIST SECOPTS INFORMATION
-----
```

Note: The OPERSEC value must be either SAFCHECK or SAFDEF to be able to use OPSPAN=SAF.

The NetView QRS command can help you manage your span-of-control security. To verify that a resource is currently in an operator's active span, use the NetView QRS command. These are examples for both NetView and SAF span of control:

```
QRS OP=OPERY RESOURCE=A01CDRSC
DW0841I OPERY IS ALLOWED ACCESS TO A01CDRSC

QRS OP=OPERY RESOURCE=A01CMRSC
DW0842I OPERY IS NOT ALLOWED ACCESS TO A01CMRSC
```

Figure 11. Output from the QRS Command When Using the NetView Program for Span of Control

Note that when OPSPAN=SAF the QRS command produces different messages, which include the access level.

```
QRS OP=OPERY RESOURCE=A01CDRSC ACCLVL=READ
BNH224I OPERY IS ALLOWED ACCESS TO A01CDRSC AT ACCESS LEVEL READ

QRS OP=OPERY RESOURCE=A01CDRSC ACCLVL=CONTROL
BNH224I OPERY IS ALLOWED ACCESS TO A01CDRSC AT ACCESS LEVEL CONTROL

QRS OP=OPERY RESOURCE=A01CDRSC
BNH224I OPERY IS ALLOWED ACCESS TO A01CDRSC AT ACCESS LEVEL ALTER

QRS OP=OPERY RESOURCE=A01CMRSC ACCLVL=UPDATE
BNH225I OPERY IS NOT ALLOWED ACCESS TO A01CMRSC AT ACCESS LEVEL UPDATE
```

Figure 12. Output from the QRS Command When Using an SAF Product for Span of Control

To display the resources and views within a span of control, use the NetView LIST SPAN command. For example, if you know that OPERY has access to SPAN1, but you are uncertain if the span is active or which resources and views this span controls, enter LIST ' ' to display a list of active spans. Then, enter LIST SPAN=SPAN1 to display the following output at your console:

```
SPAN NAME:

SPECIFIC RESOURCES:  A01CDRSC, A01APPLS, A01CDRM, A01LOCAL, A01PATH,
                     A02CDRSC, A02APPLS, A02CDRM

GENERIC RESOURCES:   A01NET* (OMIT: A01NETA.*, A01NETB.*), A01SWNET.*

SPECIFIC VIEWS:      A01_Network, A02_RESOURCES

GENERIC VIEWS:       NONE
```

Figure 13. Output from a LIST SPAN Command

To display the span names that an operator can control, use the NetView LIST command. For example, if you do not know which span names OPERY has active, or what access level is active for the spans, enter LIST OPERY to display this output at your console:


```

* N5173 LIST OPERY
- N5173 STATION: OPERY TERM: NTD9L704
- N5173 HCOPIY: NOT ACTIVE PROFILE: DSIPROFA
- N5173 STATUS: ACTIVE IDLE MINUTES: 0
- N5173 ATTENDED: YES CURRENT COMMAND: LIST
- N5173 AUTHRCVR: NO CONTROL: SPECIFIC
- N5173 NGMFADMN: NO DEFAULT MVS CONSOLE NAME: NONE
- N5173 NGMFVSPN: NNNN (NO SPAN CHECKING ON NMC VIEWS)
- N5173 NGMFCMDS: YES AUTOTASK: NO
- N5173 IP ADDRESS: N/A
- N5173 OP CLASS LIST: NONE
- N5173 DOMAIN LIST: NONE
- N5173 ACTIVE SPAN LIST: SPAN1 (A) SPAN2 (A)
- N5173 R=READ U=UPDATE C=CONTROL A=ALTER
- N5173 Task Serial: 28318
- N5173 Messages Pending: 0 Held: 0
- N5173 WLM Service Class: Not Available
- N5173 END OF STATUS DISPLAY

```

Figure 14. Output from a LIST Command for an Operator Value

The spans of control defined by the NetView product (OPSPAN=NETV) are always started using the highest access level (ALTER). You can define various access levels for spans of control in the NETSPAN class (OPSPAN=SAF). The various access levels introduce increased granularity in the ability to control access to resources.

If you want to restrict or grant access to a specific VTAM resource, but are uncertain of which span names the resource belongs to, use the NetView LIST command to display VTAM spans. To find the span names which affect an LU named A01APPLS, issue LIST RESOURCE=A01APPLS which displays output that is similar to the following output:

```

A01APPLS SPECIFICALLY DEFINED TO SPANS:
SPAN1, SPAN9, SPAN10

A01APPLS GENERICALLY DEFINED TO SPANS:
SPAN7

```

Figure 15. Output from a LIST RESOURCE Command

For more information about defining span of control, see [Chapter 4, “Using Spans to Protect Resources and Views,”](#) on page 67.

When your span-of-control definitions meet your security objectives, add the following SECOPTS statement to the CNMSTUSR or CxxSTGEN member to keep this type of security in effect the next time that the NetView program initializes:

```
SECOPTS.OPSPAN = SAF
```

Scenario 5: Converting from DSISPN and VTAMLST to the NetView Span Table

This scenario assumes that you are currently using a span-of-control definition in the DSISPN and VTAMLST statements and are converting to use the NetView span table. DSISPN and VTAMLST statements are supported in migration mode only. The NetView span table offers the following improvements:

- Consolidation of span definitions into one member
- Dynamic updates without recycling the NetView program
- Powerful syntax that allows pattern-matching characters, such as the asterisk (*)

Creating a NetView Span Table

To create a NetView span table, you need the existing span names defined for your system. If you are using NetView profiles in the DSIPRF data set to define operator access to spans of control, the SPAN and

Scenarios for Converting Types of Security

ISPAN statements define valid span names. If you are using an SAF product for defining operator access to spans of control, the resources defined under the NETSPAN resource class define which span names can be active. This list of span names contains the spans that you need to define in the NetView span table.

You can determine the resources that are controlled under a span name by issuing the LIST SPAN command. For example, the following command:

```
LIST SPAN=SPAN1
```

produces output that is similar to the following output:

```
SPAN NAME: SPAN1

SPECIFIC RESOURCES:  A01CDRSC, A01APPLS, A01CDRM, A01LOCAL, A01PATH,
                     A02CDRSC, A02APPLS, A02CDRM

GENERIC RESOURCES:   NONE

SPECIFIC VIEWS:      NONE

GENERIC VIEWS:       NONE
```

Use the LIST SPAN command for each span name that can be made active. Depending on your resource naming conventions, you can create NetView span table statements to define the span contents. Use wild card characters to define patterns of resource names.

Use the SEC MIGR command to convert existing span-of-control definitions to the NetView span table. You can also code SEC OPTS.SPANAUTH = VTAMLST.*tblname* in the CNMSTUSR or CxxSTGEN member to create and put into effect this span table at NetView initialization.

Testing the NetView Span Table

Use the REFRESH command with the TEST keyword to test the syntax of your newly created NetView span table:

```
REFRESH SPANAUTH=TABLE, SPANTBL=span_table, TEST
```

Error messages will be generated for any syntax errors that are encountered.

Activating the NetView Span Table

The last step in converting your span-of-control definitions to the NetView span table is to change the NetView program settings. Activate the NetView span table by using the following NetView REFRESH command:

```
REFRESH SPANAUTH=TABLE, SPANTBL=span_table
```

Where *span_table* is the name of your NetView span table.

Use the LIST SEC OPTS command to display the following updated security information:

```
NetView V6R3 - 7E          NetView  NTV7E TOM          06/07/19 18:27:41
* NTV7E list secopts
' NTV7E
BNH228I OPTION          VALUE          LAST UPDATED          UPDATE ID
BNH229I -----
BNH229I OPERSEC         NETVPW         06/07/19 09:59:30     INITIALIZATION
BNH229I OPSPAN          NETV          06/07/19 09:59:30     INITIALIZATION
BNH229I CMDAUTH         TABLE        06/07/19 09:59:30     INITIALIZATION
BNH229I TBLNAME         CNMSCAT2     06/07/19 09:59:30     INITIALIZATION
BNH229I AUTHCHK         SOURCEID     06/07/19 09:59:30     INITIALIZATION
BNH229I SPANAUTH        TABLE        06/07/19 09:59:07     OPER1
BNH229I SPANTBL         MAINSPAN     06/07/19 09:59:07     OPER1
BNH229I SPANCHK         SOURCEID     06/07/19 09:59:30     INITIALIZATION
BNH229I CATAUDIT        NONE          06/07/19 09:59:07     INITIALIZATION
BNH229I AUTOSEC         BYPASS       06/07/19 09:59:34     INITIALIZATION
BNH229I SURROGAT        NO           06/07/19 09:59:30     INITIALIZATION
```

```

BNH229I RMTSEC      NONE      06/07/19 09:59:40      INITIALIZATION
BNH229I RMTAUTH     SENDER    06/07/19 09:59:30      INITIALIZATION
BNH229I WEBAUTH     PASS      06/07/19 09:59:30      INITIALIZATION
BNH230I END OF LIST SECOPTS INFORMATION
-----

```

Scenario 6: Converting Operator Logon Attributes

This scenario assumes you currently use the NetView program to define operator attributes and have OPERSEC set to a value other than MINIMAL.

This scenario also contains information about converting operator passwords, so it is a super set of “Scenario 2: Converting Operator Passwords” on page 122.

Source information for existing or default operator logon information is found in various profile members in the DSIPRF data set. In this scenario, we use DSIPROFX and DSIPROFZ as example member names which are converted from the NetView product to SAF product format.

When you convert operator logon attributes, you stop using the operator attribute definitions from profile members in the DSIPRF data set, and start using the NETVIEW segment of the SAF product. When using the NETVIEW segment (SECOPTS.OPERSEC=SAFDEF in the CNMSTYLE member, or OPERSEC=SAFDEF on the REFRESH command), you must use SECOPTS.OPSPAN=SAF in the CNMSTYLE member, or OPSPAN=SAF on the REFRESH command. Even if you do not use span of control, you need to activate the NETSPAN class. Otherwise an error message is issued.

Here is an example of NetView operator logon attributes as defined in members in the DSIPRF data set, and functionally equivalent definitions in RACF.

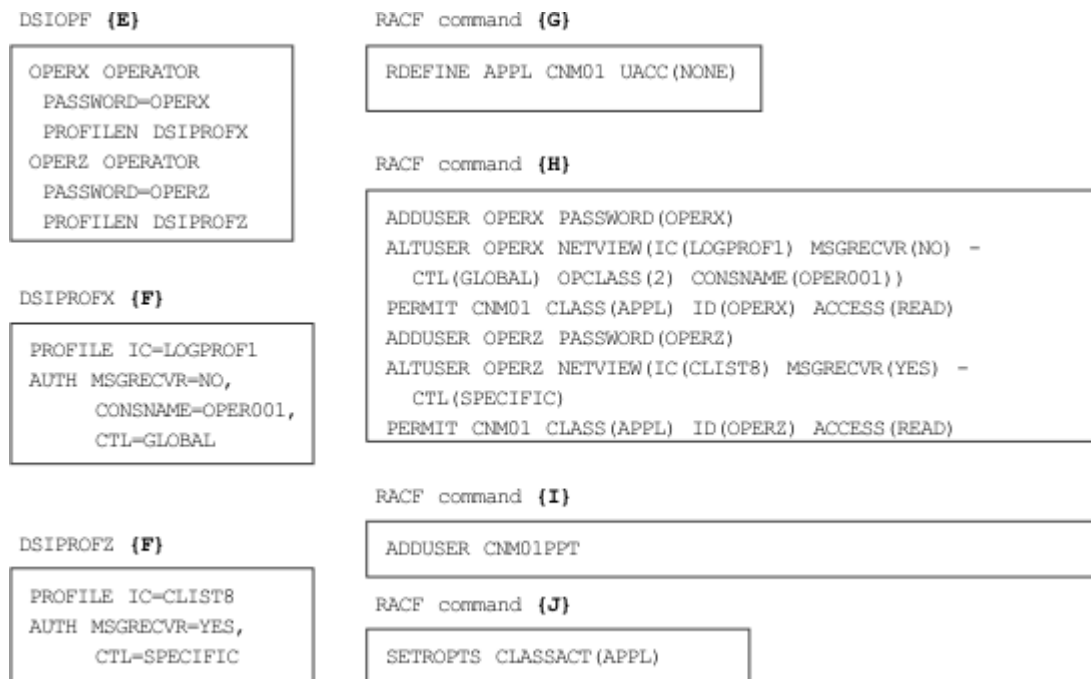


Figure 16. Converting Operator Attributes to an SAF Product

{E}

In your existing NetView operator definitions, the DSIOPF member defines operator task names and the names of the profiles used by each operator.

In this example, DSIOPF defines operator names OPERX and OPERZ, and specifies that OPERX uses profile DSIPROFX, and OPERZ uses DSIPROFZ.

{F}

These two NetView profiles contain operator attributes. The IC, MSGRECV, CONSNAME, and CTL attributes and their values can be defined in the NETVIEW segment of the user profile in an SAF product.

{G}

Each unique copy of the NetView product is protected by defining it to the APPL class, for example using the RACF RDEFINE command to protect the NetView domain name, CNM01. When the NetView program is defined to the RACF APPL class, operators cannot log on unless they are permitted to use the resource name in the APPL class that represents a particular NetView program.

{H}

Before an operator can be permitted to use an application, the operator must be created and any logon attributes must be defined. In this example, the RACF ADDUSER command is used to create the operator and set an initial password. The RACF ALTUSER command sets the operator logon attributes. You can define the attributes on the ADDUSER command, but if the operator already exists, the ADDUSER command will fail, all attributes will be ignored, and the password will not be reset.

It is more reliable to define the operator attributes on an ALTUSER command separate from the ADDUSER.

This example uses a TSO command list continuation character, which is required to enter such a long command if you used a TSO batch job. From an authorized TSO task, the text automatically wraps, and does not require a continuation character.

{I}

The PPT task must be defined to the SAF product when OPERSEC=SAFCHECK or OPERSEC=SAFDEF.

{J}

The APPL class must be activated to enable logon protection using the RACF SETROPTS CLASSACT (APPL) command. If the APPL class is not activated, or the NetView domain is not defined to the APPL class, any operator with a valid password is allowed to log on to the NetView program.

Global task definitions to an SAF product, such as defining the NetView domain name to APPL class and defining the PPT task, only need to be done once per domain. All operators who are defined in the NetView DSIOPF member must have the attributes from their profiles in the DSIPRF data set added to the NETVIEW segment in the SAF product.

Unlike the NetView PROFILE statement, which allowed operators to choose from multiple profiles (such as DSIPROFA and DSIPROFB), the NETVIEW segment of an SAF product only allows you to define one set of operator attributes for each operator.

Unlike NetView profiles, which can be used by multiple operators (such as OPER1 and OPER2), the NETVIEW segment requires that attributes are separately defined for each operator. For instance, if you have 6 NetView profiles which are shared between 100 operators, an SAF product requires 100 sets of operator attributes using the NETVIEW segment.

To dynamically change the NetView program to use an SAF product for logon password checking, the NETVIEW segment for operator attributes, and the NETSPAN class for span checking, enter use this NetView REFRESH command:

```
REFRESH OPERSEC=SAFDEF,OPSPAN=SAF
```

For recovery and backup purposes, keep the DSIOPF and profile members in the DSIPRF data set in case you need to use the REFRESH command to use OPERSEC=SAFCHECK.

Use the NetView LIST SECOPTS command to display your current security settings as illustrated in the following example:

```
NetView V6R3 - 7E      NetView   NTV7E TOM      06/07/19 18:27:41
* NTV7E      list seopts
' NTV7E
BNH228I OPTION      VALUE      LAST UPDATED      UPDATE ID
BNH229I -----
BNH229I OPERSEC     SAFDEF     06/07/19 09:59:30  OPER1
BNH229I OPSPAN      NETV      06/07/19 09:59:30  OPER1
```

```

BNH229I CMDAUTH      TABLE      06/07/19 09:59:30      INITIALIZATION
BNH229I TBLNAME      CNMSCAT2    06/07/19 09:59:30      INITIALIZATION
BNH229I AUTHCHK      SOURCEID    06/07/19 09:59:30      INITIALIZATION
BNH229I SPANAUTH     NONE        06/07/19 09:59:07      INITIALIZATION
BNH229I SPANCHK      SOURCEID    06/07/19 09:59:30      INITIALIZATION
BNH229I CATAUDIT     NONE        06/07/19 09:59:07      INITIALIZATION
BNH229I AUTOSEC      BYPASS     06/07/19 09:59:34      INITIALIZATION
BNH229I SURROGAT     NO         06/07/19 09:59:30      INITIALIZATION
BNH229I RMTSEC       NONE        06/07/19 09:59:40      INITIALIZATION
BNH229I RMTAUTH      SENDER     06/07/19 09:59:30      INITIALIZATION
BNH229I WEBAUTH      PASS        06/07/19 09:59:30      INITIALIZATION
BNH230I END OF LIST SECOPTS INFORMATION
-----

```

When your definitions are suitable for the OPERSEC=SAFDEF option, you can add the SECOPTS.OPERSEC statement in the CNMSTUSR or CxxSTGEN member for your permanent setup.

Scenario 7: Converting from Scope to the NetView Command Authorization Table

This scenario assumes you are now using scope of command authorization for command authorization. You must convert it to use either the NetView command authorization table or an SAF product.

The command authorization table offers the following improvements:

- Consolidation of command security definitions into one member, instead of having definitions in several places as with scope:
 - CMDCLASS, KEYCLASS, and VALCLASS statements in CNMCMD
 - PROFILE names in DSIOPF
- Dynamic updates without recycling the NetView program
- Powerful syntax that is similar to RACF definitions
 - Use of pattern-matching characters, such as the asterisk (*)
 - Command identifiers are the same format as resource names in the NETCMDs class for CMDAUTH=SAF

Even if you want to change your type of command security to an SAF product, there are advantages to migrating from scope of command authorization to the NetView command authorization table first. Having a NetView command authorization table is helpful because it is:

- An intermediate conversion step
- A backup in case the SAF product cannot make a security decision
- A backup if you want to change your type of command security
- A way to protect immediate commands when using CMDAUTH=SAF

Because the command identifiers used in the NetView command authorization table (as described in [“Command Identifiers”](#) on page 39) are similar to the resource identifiers used in RACF or an SAF product, it might be simpler to convert from scope checking to the NetView command authorization table, as an intermediate step. See [Chapter 3, “Controlling Access to Commands,”](#) on page 29 for information about the different types of command authorization and comparisons of command protection.

Creating a NetView Command Authorization Table

For a description of the syntax, the statements, and how to create and load the NetView command authorization table, see [“Using the NetView Command Authorization Table”](#) on page 39.

You can create a NetView command authorization table by using the SECMIGR command on your existing NetView system. Alternatively, the NetView program does this automatically when you specify SECOPTS.CMDAUTH=SCOPE.tbl_name in the CNMSTUSR or CxxSTGEN member. Review the NetView command authorization table produced by either of these processes.

Note: CNMCMC members do not contain scope definitions.

When you create the NetView command authorization table, test whether the syntax is valid using this NetView command:

```
REFRESH CMDAUTH=TABLE,TBLNAME=sec_table,TEST
```

Finding Data for Command Identifiers

To create command identifiers, you need the *netid* and *luname* values for your system. Use the NetView LISTVAR command to get these values, or else use pattern-matching characters, such as the asterisk (*) instead. You can also find these values using the REXX functions NETID() and DOMAIN() in a command list. In the example in [Figure 17 on page 132](#), the current values are NETA for *netid* and the domain, CNM01, for *luname*.

```
C NTV6D CNM353I LISTVAR : OPSYSTEM = MVS/ESA
C NTV6D CNM353I LISTVAR : MVSLEVEL = SP7.2.2
C NTV6D CNM353I LISTVAR : CURSYS = NMPIPL22
C NTV6D CNM353I LISTVAR : VTAMLVL = V622
C NTV6D CNM353I LISTVAR : VTCOMPID = 5695-11701-220
C NTV6D CNM353I LISTVAR : NetView = IBM Z NetView V6R3
C NTV6D CNM353I LISTVAR : NETID = NETA
C NTV6D CNM353I LISTVAR : DOMAIN = CNM01
C NTV6D CNM353I LISTVAR : APPLID = NTV6D010
C NTV6D CNM353I LISTVAR : OPID = OPER3
C NTV6D CNM353I LISTVAR : LU = NT6DL701
C NTV6D CNM353I LISTVAR : TASK = OST
C NTV6D CNM353I LISTVAR : NCCFCNT = 6
C NTV6D CNM353I LISTVAR : HCOPI = 
C NTV6D CNM353I LISTVAR : TYPE = ENT
C NTV6D CNM353I LISTVAR : CURCONID = KENNIE1
```

Figure 17. Example of NetView LISTVAR Command Output

[Figure 18 on page 134](#) shows source information for existing or default command authorization information that is found in CNMCMC (**{T}** and **{U}**), DSIOPF (**{R}**), and various profile members in the DSIPRF data set. In this scenario, DSIPROFA, DSIPROFB, DSIPROFC, and DSIPROFR (all identified with **{S}**) are used as example member names.

Defining Operator Groups from Scope Information

You can define command authorization for all your operators individually or in groups. In the NetView command authorization table, use GROUP statements to deal with multiple operators as a single unit. If you define all operators individually, it might be more difficult to manage changes.

One method of defining groups of operators for command security is to make a group for each existing OPCLASS value.

For example, if you currently use scope of command authorization, you might have defined your own OPCLASS values. The example in [Figure 18 on page 134](#) shows how to create groups which correspond to each OPCLASS value. All the operators who have the same OPCLASS value, such as OPCLASS 2, are defined to the same group: GRP2.

If you currently use scope-of-command authorization and have operators defined without OPCLASS values, these operators have access to all commands, keywords, and values. If you change the type of command security to use the NetView command authorization table or an SAF product, they will not automatically have unrestricted access to all commands and command lists. To provide equivalent function, you must create a group of all operators who have unrestricted command access, and this group must be permitted to use all protected commands.

In [Figure 18 on page 134](#), DSIPROFR defines operators without any OPCLASS values, who can issue any NetView commands, keywords, or values. This example uses GRPALL for the name of the new group. All the operators who are defined without OPCLASS values are defined to GRPALL.

If you want to give GRPALL unrestricted access to all commands, command lists, keywords, and values, each resource with a PROTECT statement must have PERMIT statement for GRPALL. This level of access is not recommended for most operators.

Creating Command Identifiers from Scope Information

After your groups are defined, create command identifiers to protect commands, keywords, and values using the NetView command authorization table. To convert CMDCLASS, KEYCLASS, and VALCLASS statements into command identifiers with corresponding PROTECT statements, you will need:

- The scope-of-command authorization definitions as source material
- Chapter 3, “Controlling Access to Commands,” on page 29 for the syntax of the NetView command authorization table
- Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151 as a reference

You can also use the NetView SECMIGR command to perform much of the conversion between security types.

As shown in box **{T}** in Figure 18 on page 134, the NetView AUTOTBL command has a CMDCLASS statement in a CMDMDL definition. Then, as shown in box **{W}**, a corresponding PROTECT command identifier statement restricts access to operators. Then there are PERMIT statements allowing GRPALL, GRP1, and GRP2 to issue the AUTOTBL command. Note that AUTOMSG is a synonym for the AUTOTBL command, so AUTOMSG is protected using the AUTOTBL command identifier, as shown in Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151. This command identifier typically is the value on the CMDMDL label, rather than the synonym (CMD SYN) values which operators can issue.

Not all commands, keywords, or values can be protected. Because of synonyms, abbreviations, and duplicate values, the command identifiers which you specify for commands, keywords, or values might not match the values on CMDMDL statements or values you enter. To find which command identifiers are used, see Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151. For usage instructions, see “Command Identifiers” on page 39.

Keywords use the same process of creating PROTECT and PERMIT statements on command identifiers which correspond to KEYCLASS values. As shown in **{Y}**, a PROTECT statement is required to limit access to the OFF keyword. The SECMIGR command will also generate a second generic PROTECT statement for values of the OFF keyword, even though this keyword does not use values. When using the SECMIGR command, manually delete any such unnecessary statements in order to save storage and improve performance.

If the keyword can have a corresponding value, you can protect each keyword-value pair with a command identifier. For example, the AUTOTBL command has a MEMBER=*member_name* pair, which is protected as shown in box **{X}** in Figure 18 on page 134. In this example, all values for *member_name* are protected by specifying a generic pattern matching character, the asterisk (*), in the value position.

In addition, the DSITBL01 automation table is specifically protected as a value of the MEMBER keyword on the AUTOTBL command. The example shows how the VALCLASS statement is converted to a PROTECT statement.

DSIOPF {R} OPER1 OPERATOR PROFILE DSIPROFA NETOP2 OPERATOR PROFILE DSIPROFB AUTO1 OPERATOR PROFILE DSIPROFC OPERR OPERATOR PROFILE DSIPROFR	NetView Command Authorization Table {V} GROUP GRPALL OPERR GROUP GRP1 NETOP2,AUTO1 GROUP GRP2 OPER1,NETOP2,AUTO1
DSIPROFA {S} OPCLASS 2	NetView Command Authorization Table {W} PROTECT NETA.CNM01.AUTOTBL PERMIT GRPALL NETA.CNM01.AUTOTBL PERMIT GRP1 NETA.CNM01.AUTOTBL PERMIT GRP2 NETA.CNM01.AUTOTBL
DSIPROFB {S} OPCLASS 1,2	NetView Command Authorization Table {X} PROTECT NETA.CNM01.AUTOTBL.MEMBER.DSITBL01 PERMIT GRPALL NETA.CNM01.AUTOTBL.MEMBER.DSITBL01 PERMIT GRP1 NETA.CNM01.AUTOTBL.MEMBER.DSITBL01 PERMIT GRP2 NETA.CNM01.AUTOTBL.MEMBER.DSITBL01 PROTECT NETA.CNM01.AUTOTBL.MEMBER.* PERMIT GRPALL NETA.CNM01.AUTOTBL.MEMBER.* PERMIT GRP1 NETA.CNM01.AUTOTBL.MEMBER.*
DSIPROFC {S} OPCLASS 1,2	NetView Command Authorization Table {Y} PROTECT NETA.CNM01.AUTOTBL.OFF PERMIT GRPALL NETA.CNM01.AUTOTBL.OFF PERMIT GRP1 NETA.CNM01.AUTOTBL.OFF
DSIPROFR {S} (no opclass specified)	NetView Command Authorization Table {Z} PROTECT NETA.CNM01.ALLOCATE PERMIT GRPALL NETA.CNM01.ALLOCATE PERMIT GRP1 NETA.CNM01.ALLOCATE PERMIT GRP2 NETA.CNM01.ALLOCATE PROTECT NETA.CNM01.ALLOCATE.DATASET PERMIT GRPALL NETA.CNM01.ALLOCATE.DATASET PERMIT GRP1 NETA.CNM01.ALLOCATE.DATASET
DSICMD {T} AUTOTBL CMDMDL ... CMD SYN AUTOMSG CMDCLASS 1,2 MEMBER KEYCLASS 1,2 DSITBL01 VALCLASS 1,2 =OTHER VALCLASS 1 OFF KEYCLASS 1	
DSICMD {U} ALLOCATE CMDMDL ... CMD SYN ALLOC CMDCLASS 1,2 DATASET KEYCLASS 1	

Figure 18. Converting from Scope to the NetView Command Authorization Table

{R} and {S}

In your existing NetView operator definitions, the DSIOPF member defines operator task names and the names of the profiles used by each operator, and the profiles specify the OPCLASS values. In this example,

- OPER1 uses profile DSIPROFA, which specifies an OPCLASS value of 2
- NETOP2 uses profile DSIPROFB, which specifies OPCLASS values of 1 and 2
- AUTO1 uses profile DSIPROFC, which specifies OPCLASS values of 1 and 2
- OPERR uses profile DSIPROFR which does not specify any OPCLASS values

{T}

Each NetView command that was protected using scope of command authorization was defined with a CMDMDL statement in the NetView CNMCMD member. This is where the CMDCLASS, KEYCLASS, and VALCLASS were set to protect commands, keywords, and values.

In this example, the NetView AUTOTBL command has a synonym (defined with the CMDSYN statement) of AUTOMSG. The command, its synonym, and the MEMBER keyword can only be issued by operators with an OPCLASS of 1 or 2, or operators without an OPCLASS. The OFF keyword can only be issued by operators with an OPCLASS of 1, or operators without an OPCLASS. The DSITBL01 value can only be issued by operators with an OPCLASS of 2, or operators without an OPCLASS.

{U}

In the next CMDMDL example, the NetView ALLOCATE command has a synonym of ALLOC. The command and its synonym can only be issued by operators with an OPCLASS of 1 or 2, or operators without an OPCLASS. The DATASET keyword can only be issued by operators with an OPCLASS of 1, or operators without an OPCLASS.

{V}

In this example, operator group definitions are based on their OPCLASS levels. For example, since NETOP2 and AUTO1 were defined with OPCLASS values of both 1 and 2, they are defined in both GRP1 and GRP2. Since OPER1 was defined with OPCLASS 2, this operator task is defined only in GRP2. Since OPERR was defined without any OPCLASS values, this operator task is defined in GRPALL.

{W}

To protect the AUTOTBL command (or its synonym, AUTOMSG) from unauthorized usage, code a NetView command authorization table PROTECT statement with the command identifier found in [Appendix A, "NetView Commands, Keywords, and Values that Can Be Protected," on page 151.](#)

Then, to allow groups of operators to issue the AUTOTBL command, correlate the group names with PERMIT statements. For instance, code a NetView command authorization table PERMIT statement to allow access to the operators in GRPALL, GRP1, and GRP2.

{X}

To protect the MEMBER keyword and the DSITBL01 value on the AUTOTBL command from unauthorized usage, code a PROTECT statement with the command identifier found in [Appendix A, "NetView Commands, Keywords, and Values that Can Be Protected," on page 151.](#) To allow authorized operators to activate DSITBL01, code a PERMIT statement for GRPALL, GRP1, and GRP2.

To mimic the function of the =OTHER statement, use a generic character in the value field of the command identifier.

{Y}

To protect the OFF keyword on the AUTOTBL command from unauthorized use, code a PROTECT statement with the command identifier found in [Appendix A, "NetView Commands, Keywords, and Values that Can Be Protected," on page 151.](#) To enable authorized operators to use the keyword, code a PERMIT statement for GRPALL and GRP1.

The OFF keyword on the AUTOTBL command is never issued with a corresponding value; therefore, it does not require a PROTECT NETA.CNM01.AUTOTBL.OFF.* statement. However, if you use the NetView SECMIGR command to convert your scope of command authorization definitions, the NetView command authorization table will contain this extra statement. Having unnecessary statements requires extra storage and processing.

{Z}

To protect the ALLOCATE command (or its abbreviated synonym, ALLOC) from unauthorized use, code a PROTECT statement with the command identifier found in [Appendix A, "NetView Commands, Keywords, and Values that Can Be Protected," on page 151.](#)

To protect the DATASET keyword on the ALLOCATE command from unauthorized use, code a PROTECT statement with the command identifier. Then, to allow authorized operators to issue the DATASET keyword of the ALLOCATE command, code a PERMIT statement for GRPALL and GRP1.

Testing the NetView Command Authorization Table

Use the REFRESH command with the TEST keyword to test the syntax of your newly created NetView command authorization table.

```
REFRESH CMDAUTH=TABLE,TBLNAME=sec_table,TEST
```

Error messages will be generated for any syntax errors that are encountered.

Activating the NetView Command Authorization Table

The last step in converting your command security to the NetView command authorization table is to change the NetView program settings.

The NetView command authorization table can be activated by specifying SECOPTS.CMDAUTH=TABLE.*sec_table* in the CNMSTYLE member and recycling the NetView program, or by issuing the following command:

```
REFRESH CMDAUTH=TABLE TBLNAME=sec_table
```

Use the LIST SECOPTS command to display the updated security information.

Scenario 8: Converting from NetView Command Authorization Table to SAF Command Authorization

This scenario assumes you are using the NetView command authorization table (CMDAUTH=TABLE), and want to define command authorization using an SAF product (CMDAUTH=SAF). It is recommended that you use the NetView command authorization table as a backup for SAF command authorization (BACKTBL=*sec_table*), because it enables command security in the following circumstances:

- The SAF product is not functioning
- The SAF product cannot make a decision about command authorization
- A command is an immediate command

Although it is possible to convert command authorization directly from scope checking to an SAF product, it is safer to have a NetView command authorization table as a backup security mechanism, in case your SAF product cannot make a command security decision. However, if you want to always make the same command security decision when the SAF product does not yield a decision, specify SECOPTS.CMDAUTH=SAF.PASS or SECOPTS.CMDAUTH=SAF.FAIL in the CNMSTYLE member, or specify the CMDAUTH=SAFNODEC keyword on the REFRESH command.

To convert your NetView command authorization table statements to RACF definitions, take the following actions:

- Only once, ensure the NETCMDS class is active by entering:

```
SETOPTS CLASSACT(NETCMDS) GRPLIST
```

- Define the NETCMDS class as a GENERIC class to allow the use of generic characters, since use of generic characters is recommended:

```
SETOPTS GENERIC(NETCMDS)
```

- For each GROUP statement, use the RACF ADDGROUP and CONNECT commands to establish the groups.

```
ADDGROUP group_name  
CONNECT oper_id GROUP(group_name) UACC(READ)
```

- For each PROTECT statement, use the RACF RDEFINE command to define the resource (command identifier) in the NETCMDS class with a universal access of NONE:

```
RDEFINE NETCMDS cmd_identifier UACC(NONE)
```

- For each EXEMPT statement, use the RACF RDEFINE command to define the resource (command identifier) in the NETCMDS class with a universal access of READ:

```
RDEFINE NETCMDS cmd_identifier UACC(READ)
```

- For each PERMIT statement, use the RACF PERMIT command to allow the operator or group to access the protected resource. The access level required to issue commands that are protected in the NETCMDS class is READ:

```
PERMIT cmd_identifier CLASS(NETCMDS) ID(operator_id) ACCESS(READ)
```

- Refresh the NETCMDS class definitions by entering:

```
SETROPTS RACLIST(NETCMDS) REFRESH
```

If you use the SECMIGR tool, it understands how to convert the NetView command authorization table statements enclosed by <BEGIN> and <END> labels into RACF commands. The SECMIGR tool also supports automatic translation of the NetView SETVAR statements into hard-coded RACF commands.

If you create a NetView command authorization table that duplicates the protection defined in your SAF product, ensure both the SAF product and the NetView command authorization table allow at least one operator to issue the NetView REFRESH command. If you do not, you will not be able to use the REFRESH command to activate another method of security, if necessary.

Activating the SAF Command Authorization

Activate command security defined in the NETCMDS class in your SAF product by entering:

```
REFRESH CMDAUTH=SAF, BACKTBL=SECTABLE
```

In this example, a NetView command authorization table named SECTABLE is used to protect immediate commands, and to backup the SAF product.

When you are certain that you do not want to use the NetView command authorization table as the primary type of command security, add the following statement to the CNMSTUSR or CxxSTGEN member:

```
SECOPTS.CMDAUTH = SAF.sec_table
```

To check your command authorization settings, use the NetView LIST SECOPTS commands to display the messages indicating that you are using an SAF product with a backup NetView command authorization table as shown in the following example:

```
NetView V6R3 - 7E      NetView  NTV7E TOM      06/07/19 18:27:41
* NTV7E      list secopts
+ NTV7E
BNH228I OPTION      VALUE      LAST UPDATED      UPDATE ID
BNH229I -----
BNH229I OPERSEC     NETVPW     06/07/19 09:59:30  INITIALIZATION
BNH229I OPSPAN      SAF       06/07/19 09:59:30  OPER1
BNH229I CMDAUTH     SAF       06/07/19 09:59:30  OPER1
BNH229I BACKTBL     CNMSCAT2  06/07/19 09:59:30  INITIALIZATION
BNH229I AUTHCHK     SOURCEID  06/07/19 09:59:30  INITIALIZATION
BNH229I SPANAUTH    NONE      06/07/19 09:59:07  INITIALIZATION
BNH229I SPANCHK     SOURCEID  06/07/19 09:59:30  INITIALIZATION
BNH229I CATAUDIT    NONE      06/07/19 09:59:07  INITIALIZATION
BNH229I AUTOSEC     BYPASS   06/07/19 09:59:34  INITIALIZATION
BNH229I SURROGAT    NO       06/07/19 09:59:30  INITIALIZATION
BNH229I RMTSEC      NONE      06/07/19 09:59:40  INITIALIZATION
BNH229I RMTAUTH     SENDER   06/07/19 09:59:30  INITIALIZATION
BNH229I WEBAUTH     PASS      06/07/19 09:59:30  INITIALIZATION
BNH230I END OF LIST SECOPTS INFORMATION
-----
```

Scenarios for Converting Types of Security

After successfully completing these steps, the command authorization in an SAF product provides function equivalent to the NetView command authorization table in the NetView product.

Chapter 14. Checklist for Debugging Security Problems

This checklist is a quick reference to help you avoid potential security problems and debugging problems when you have defined security using the following methods:

- Operator definitions and profiles
- The NetView command authorization table
- A system authorization facility product such as RACF
- Span-of-control authorization
- The NetView span table

Review “Check These Things First” on [page 139](#) for general points to check before you do anything else. Subsequent sections address common security problems and possible solutions for each security method. The section “If You Cannot Isolate the Problem” on [page 148](#) gives additional ways to collect data if you are unable to determine the problem.

Check These Things First

1. If you are using an SAF product for security, does the product have the capabilities you need?

The ICH520I message on the MVS system log shows you which level of RACF you are using. Use the RACF SETROPTS LIST command to see which classes are active. Ensure that the NETCMDS class is listed as a generic class.

2. How are your security options currently set?

- Issue the NetView LIST SECOPTS command to list the security options that are currently in use.
- Browse the NetView logs to see possible error messages and command echoes. See “[If Your Specified Initial Security Settings Were Not Taken](#)” on [page 147](#) for more information.

3. What are error messages telling you about the problem?

Error messages can help you determine whether your security is working as you intend.

When a task has authorization for a command, but cannot issue the command, you might have protected a command that you did not intend to protect. If you are using the NetView command authorization table, an error message tells you where in your security settings that command is being protected. If you are using an SAF product and you are auditing the command identifier, it is identified in error messages. See “[If a Command Cannot Be Accessed by an Authorized Operator](#)” on [page 142](#).

4. If you are using the SAF product or the NetView command authorization table, is a generic statement taking effect when you did not expect it to?

Error messages can indicate that a generic statement protected commands, keywords, or values that you did not intend.

Generic characters are used to generalize command identifiers. Both the asterisk (*) and percent sign (%) are pattern-matching (wild card) characters. See “[Command Identifiers](#)” on [page 39](#) for an explanation of pattern-matching characters.

5. If operators cannot log on, does your OPERSEC setting match the operator definitions?

If the SECOPTS.OPERSEC statement is set to SAFPW, SAFCHECK or SAFDEF in the CNMSTYLE member (or the equivalent on the REFRESH command), ensure that the operator is defined to the SAF product. Issue the LISTUSER *userid* from an authorized TSO operator to see if the operator is defined to RACF.

If OPERSEC is set to NETVPW, ensure that the operator is defined in DSIOPF. If OPERSEC is set to SAFPW or SAFCHECK, ensure that the operator is defined in DSIOPF or that a default logon profile is in effect by entering LIST DEFAULTS and checking what is specified for LOGPROF.

See [“If an Operator Cannot Log On” on page 145](#) for more to check.

6. How is your NGMFVSPN attribute currently set?

Issue this NetView command to list the NGMFVSPN value currently in use for the operator.

```
LIST opername
```

Substitute the operator ID for *opername* in this command.

See [“Defining Operator Attributes” on page 14](#) and the *IBM Z NetView Administration Reference* for information about how to code the NGMFVSPN attribute.

If a Command Can Be Accessed by an Unauthorized Operator

If you protected a command, keyword, or value, yet it was able to be used by an unauthorized operator, check these steps:

1. Is the command one that can be protected?

Because some commands, keywords and values cannot be protected, check [Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151](#) to ensure your statements match what can be protected.

2. Is there a SEC=BY coded on the CMDDEF statement in CNMCMD?

If so, security checking is unconditionally bypassed for this command for all types of command authorization. All users can issue this command until the SEC=BY is changed and the NetView program is recycled.

3. Is there an exception which prevents the command from being protected?

For a list of reasons why a command might not be protected, see [“Exceptions to Command Authorization Checking” on page 31](#).

If You Are Using the NetView Command Authorization Table

1. Is there a PROTECT statement which provides security for this command?
2. Does the PROTECT statement have the correct command, keyword, and value?
3. Is there an EXEMPT statement with a generic that is more specific than the PROTECT statement?

For example, in the following scenario, the RESET keyword on the NetView AUTOCNT command is restricted, but passed authorization checking. Initially, a system programmer used the generic AUTOCNT.R* on the PROTECT statement in the table to disallow both the REPORT and RESET keywords on AUTOCNT:

```
PROTECT *.*.AUTOCNT.R*
```

Later, another system programmer wanted to allow the REPORT keyword, and added an EXEMPT statement after the PROTECT statement in the table:

```
PROTECT *.*.AUTOCNT.R*  
EXEMPT *.*.AUTOCNT.RE*
```

Because the RE* on the EXEMPT is more specific than the R* on the PROTECT, the effect is that not only is the REPORT keyword allowed, but the RESET keyword is now also allowed.

4. Is the SECOPTS.AUTHCHK statement in the CNMSTYLE member, or AUTHCHK on the REFRESH command set as you intended?

For example, AUTHCHK might be set to TARGETID when you thought it was set to SOURCEID. Issue a NetView LIST SECOPTS to see how AUTHCHK is currently set.

5. Is there a PERMIT statement for this command either for the operator ID or for a GROUP that the operator is in?
6. Is there a generic PERMIT that the command might have matched?

For example, if a PERMIT statement exists for ABC*, then ABCD is allowed if the ABC* statement is the most specific match.

7. Are you using DEFAULTS.AUTOSEC=BYPASS in the CNMSTYLE member, or AUTOSEC=BYPASS on the DEFAULTS command?

If the command is issued from the automation table, and you are using AUTOSEC=BYPASS, security checking is bypassed for that command or command list, as well as for any commands or command lists nested within it. To protect a command that is issued inside another command or command list, use SEC=CH on the CMDDEF statement in CNMCMDU for the command that you want to protect.

8. Are you using special characters and translating them correctly in the command identifier?

See [“Protecting Commands Containing Special Characters” on page 38](#) for the correct format.

9. Did you incorrectly protect a synonym for a command or keyword rather than the command or keyword itself? See Appendix A, [“NetView Commands, Keywords, and Values that Can Be Protected,” on page 151](#) for a table of commands, keywords and values that can be protected.

10. Are NetView component identifiers specified correctly?

Check to see whether the command identifiers for NPDA and NLDM commands are specified correctly. See Appendix A, [“NetView Commands, Keywords, and Values that Can Be Protected,” on page 151](#) for the list of command identifiers.

11. Are you using pattern-matching characters for the *luname* and *netid*? If not, does the *luname* on the command identifier match the system where the command was processed? See [“Using the NetView Command Authorization Table” on page 39](#) for more information.

If You Are Using an SAF Product

1. Is the SAF NETCMDS class active in the SAF product?

Use the RACF SETROPTS LIST command from an authorized TSO user to check the active classes in RACF.

2. Is the NETCMDS class set up to handle generic characters? For example, the asterisk (*) and percent sign (%) are pattern-matching generic characters (wild cards).

Use the RACF SETROPTS LIST command from an authorized TSO user to check whether the NETCMDS class is defined as GENERIC. If the NETCMDS class was not defined as GENERIC:

- a. Delete all existing NETCMDS resource names which contain generic characters.
- b. Enter SETROPTS GENERIC(NETCMDS) .
- c. Redefine the resource names using generic characters.

3. Is the command an immediate command?

Immediate commands cannot be protected in the NETCMDS class. If you want to protect this command, you can protect it in a backup NetView command authorization table.

4. Are you using DEFAULTS.AUTOSEC=BYPASS in the CNMSTYLE member, or AUTOSEC=BYPASS on the DEFAULTS command?

If the command is issued from the automation table, and you are using AUTOSEC=BYPASS, security checking is bypassed for that command or command list, as well as for any commands or command lists nested within it. To protect a nested command, use SEC=CH on the CMDDEF statement for the command in CNMCMDU.

5. What are your current NETCMDS class settings?

For RACF, use SEARCH CLASS(NETCMDS) to show everything that is defined to NETCMDS.

6. Are you using SAF.PASS on the SECOPTS statement in the CNMSTYLE member, or SAFNODEC=PASS with the REFRESH command?

If the SAF product was unable to make a decision on this command, and there was no backup table restricting this command, SAF.PASS or SAFNODEC=PASS allows all operators to run any command.

7. Are NetView component identifiers specified correctly?

Check to see whether the command identifiers for NPDA and NLDM commands are specified correctly. See Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151 for a list of command identifiers.

8. Did you incorrectly protect a synonym for a command or keyword rather than the command or keyword itself? See Appendix A, “NetView Commands, Keywords, and Values that Can Be Protected,” on page 151 for a table of commands, keywords and values that can be protected.

9. Are you using special characters and translating them correctly in the command identifier?

See “Protecting Commands Containing Special Characters” on page 38 for the correct format.

10. Is the operator, or a group the operator is in, allowed to access the protected command?

11. Do the *luname* and *netid* on the command identifier match the system where the command was processed?

12. Is the SECOPTS.AUTHCHK statement in the CNMSTYLE member, or AUTHCHK on the REFRESH command set as you intended?

For example, AUTHCHK might be set to TARGETID when you thought it was set to SOURCEID. Enter the NetView LIST SECOPTS command to see how AUTHCHK is currently set.

13. Are you using a backup table that does not protect this command?

See “Using the NETCMDS Class in an SAF Product for Command Authorization” on page 49 for more information.

If a Command Cannot Be Accessed by an Authorized Operator

1. What are the error messages telling you?

If a user is not authorized to issue a particular command, keyword, or value, messages might indicate which user ID was rejected, the command identifier, and the type of command authorization in effect. If you are using the NetView command authorization table or an SAF product, you get messages BNH232E and BNH233E. Here are example messages:

```
BNH232E 'userid' IS NOT AUTHORIZED TO ISSUE COMMAND 'command'  
BNH233E THE COMMAND 'command' IS PROTECTED BY COMMAND IDENTIFIER  
'commandid' IN 'auth_method'
```

If the SECOPTS.CMDAUTH statement is set to SAF in the CNMSTYLE member, or CMDAUTH=SAF on the REFRESH command, message BNH233E does not contain the *commandid* unless auditing is in effect for that command identifier in the NETCMDS class.

If the SECOPTS.AUTHCHK statement is set to SOURCEID in the CNMSTYLE member, or AUTHCHK=SOURCEID on the REFRESH command, this user ID might be different from the task that is running the command. Ensure you are protecting the correct user ID. If this user ID is supposed to have command authorization for the command, look at the command identifier in the error message. Ensure you are protecting the correct command, keyword, or value.

2. Is the command that failed nested within a command or command list?

If a task can issue a command or command list, the task also needs to be authorized to run any commands issued from within the command or command list. Determine whether it is appropriate for tasks to issue the nested commands or command lists. For instance, if an automation task is failing:

- You might want to use `DEFAULTS.AUTOSEC=BYPASS` in the `CNMSTYLE` member (or `AUTOSEC=BYPASS` on the `DEFAULTS` command) to allow automation to continue, if the command came from an automation table. See [Chapter 8, “Security Considerations for Automation,”](#) on page 99 for more information about automation security.
- Check which task's authority is being used for the command security check, if authorization is failing based on the issuing task or the destination task. It might not be the task that is running if `SECOPTS.AUTHCHK=SOURCEID` is in effect. For more information about `AUTHCHK`, see [“Authority Checking Commands against the Command Source”](#) on page 34.

If You Are Using NetView Command Authorization Table

1. Is an error message identifying the failing command identifier?

For example, operator `OPER1` might be authorized to issue an `AUTOTBL` command but restricted from issuing an `AUTOTBL` command with a keyword of `OFF`. When `OPER1` issues:

```
AUTOTBL OFF
```

`OPER1` gets the following message:

```
BNH234E 'OPER1' IS NOT AUTHORIZED TO USE KEYWORD 'OFF'
BNH235E THE KEYWORD 'OFF' IS PROTECTED BY COMMAND IDENTIFIER
'commandid' IN 'auth_method'
```

If operator `OPER1` is authorized to issue `AUTOTBL` with `MEMBER` equal to some values, but restricted from issuing `AUTOTBL MEMBER` with a value of `MEMBER1`, when `OPER1` issues:

```
AUTOTBL MEMBER=MEMBER1
```

Then `OPER1` gets the following message:

```
BNH236E 'OPER1' IS NOT AUTHORIZED TO USE THE KEYWORD 'MEMBER' AND
VALUE 'MEMBER1' COMBINATION
BNH237E THE KEYWORD 'MEMBER' AND VALUE 'MEMBER1' ARE PROTECTED BY
COMMAND IDENTIFIER 'commandid' IN 'auth_method'
```

2. Does the operator have command authorization?

Whether you are using command authorization for the `SOURCEID` or `TARGETID`, ensure that the operator who is checked has command authorization. If `SECOPTS.AUTHCHK=SOURCEID` is in effect, a different task might need to be authorized.

For more information about `AUTHCHK`, see [“Authority Checking Commands against the Command Source”](#) on page 34.

3. Is the *luname* specified incorrectly, so the command identifier mistakenly matches the system where the command was processed?
4. Is there a `PROTECT` statement specific to this command?
If so, either ensure a `PERMIT` statement exists to allow the task to issue the command or add an `EXEMPT` statement and delete the `PROTECT` statement to ensure that everyone is allowed to issue the command.
5. Did you activate changes to the NetView command authorization table?
If the NetView command authorization table has changed, be sure you activated your changes by issuing the NetView `REFRESH` command.
6. Is there a generic `PROTECT` statement that the command matched?

For example, if a `PROTECT` statement exists for `ABC*`, then `ABCD` is protected if the `ABC*` statement is the most specific match. If this is the case, add `PROTECT` and `PERMIT` statements for the specific command you want to authorize. If all operators must have access, you might want to consider adding an `EXEMPT` statement for the specified command.

7. Is the operator authorized to issue the command by itself, in addition to the *command.keyword* or *command.keyword.value* combination in effect?

Even if the operator is allowed to issue a keyword or keyword and value combination, the operator must also be authorized to issue the command by itself, if it is protected. Commands with protected keywords are first authority checked for just the command, then for the command with each of the protected keywords.

If You Are Using an SAF Product

1. Have any changes been made to the definitions in the NETCMDS class that alters SAF command authorization?

Use the command identifier in the error message to help you identify the operators authorized to issue this command. For example, a failed attempt to run the NetView DEFAULTS commands generates the following message:

```
BNH233E THE COMMAND 'DEFAULTS' IS PROTECTED BY COMMAND IDENTIFIER  
'NETA.CNM01.DEFAULTS' IN 'SAF'
```

If the command identifier value is NOT AVAILABLE, turn on auditing in the SAF product for resources in the NETCMDS class and try to recreate the failure.

From an authorized TSO user ID, you can find out which users are authorized for the DEFAULTS command by issuing the RACF RLIST command using the command identifier from the error message:

```
RLIST NETCMDS (NETA.CNM01.DEFAULTS) AUTHUSER
```

2. Is there a generic statement in the NETCMDS class that is protecting this command?
3. What are your current NETCMDS class settings?

For RACF, use SEARCH CLASS(NETCMDS) to show everything that is defined to NETCMDS.

4. Should the operator task be permitted to use the command, either as an individual task or as part of a group, but currently is not?
5. Are you using a backup table to protect this command? If so, check messages BNH233, BNH235, or BNH237 for the *auth_method* which can indicate a backup NetView command authorization table.

```
BNH233E THE COMMAND 'command' IS PROTECTED BY COMMAND IDENTIFIER  
'commandid' IN 'auth_method'  
BNH235E THE KEYWORD 'keyword' IS PROTECTED BY COMMAND IDENTIFIER  
'commandid' IN 'auth_method'  
BNH237E THE KEYWORD 'kywd1' AND VALUE 'value1' ARE PROTECTED BY  
COMMAND IDENTIFIER 'commandid' IN 'auth_method'
```

6. Are you using the SAF.FAIL operand on the SECOPTS statement in the CNMSTYLE member, or the SAFNODEC=FAIL operand with the REFRESH command?

If the command was prevented because of SAF.FAIL or SAFNODEC=FAIL, the following message is displayed:

```
BNH274E A COMMAND AUTHORIZATION DECISION COULD NOT BE MADE BY THE  
SECURITY PRODUCT. RACROUTE MACRO RC IS X'racroute_rc', REQUEST TYPE IS  
'request', SECURITY PRODUCT RC IS X'security_rc', SECURITY PRODUCT  
REASON CODE IS X'security_rsn', COMMAND IDENTIFIER IS 'identifier'
```

If you specify SAF.FAIL or SAFNODEC=FAIL and a problem occurs in the SAF product, no operator can issue any command except immediate commands and commands with SEC=BY specified on the CMDDEF statement.

7. Is the operator authorized to issue the command? Is the *command.keyword* or *command.keyword.value* combination in effect?

Even if the operator can issue a keyword or keyword and value combination, the operator must also be authorized to issue the protected command by itself. Commands with protected keywords are first

authority checked for just the command, then for the command with each of the keywords which can be protected.

If an Operator Cannot Log On

1. What are the error messages telling you?

Issue a NetView LIST SECOPTS command to list security options which are currently in use.

2. Are there spelling errors in the operator definition?
3. Was the password or password phrase reset?

See “Defining Operator Password Security” on page 10 for more information.

If You Are Using SECOPTS.OPERSEC=NETVPW, SAFPW, or SAFCHECK

1. Is the operator listed in DSIOPF?
2. Is the syntax of the operator's logon profile correct?
3. If you have recently added the operator, have you issued a REFRESH OPERS command since the change to dynamically add the operator definitions to DSIOPF?
4. Is a default logon profile in effect? Determine this by entering LIST DEFAULTS and checking what is specified for LOGPROF. If a default logon profile is in effect, the following conditions are true:
 - For SAFPW or SAFCHECK, no DSIOPF definition is required.
 - For NETVPW, no profile is required for the DSIOPF entry for the operator.

If You Are Using SECOPTS.OPERSEC=SAFDEF

1. Is the operator defined in the security product?

If not, define the operator, using a RACF ADDUSER command, for example.
2. Is the APPL class active?

For RACF, use the SETROPTS LIST command to see which classes are active.
3. Is the operator permitted to the resource in the APPL class of the security product which represents this NetView program?

For example, if the NetView domain name is CNM01, ensure you have a RACF PERMIT command for this operator to CNM01.
4. Is the time of day, the day of the week, or the terminal restricted?

If a Resource or View Can Be Accessed by an Unauthorized Operator

1. What are the active security definitions?

Issue a NetView LIST SECOPTS command to determine the current specification for SPANAUTH. If SPANAUTH=TABLE, find the name of the span authorization table.

Issue a NetView LIST RESOURCE or LIST VIEW command to list the spans to which the resource or view is defined.

Issue a NetView LIST *operid* command to list the active spans for an operator. If appropriate, also check the NGMFVSPN setting. For more information about the LIST command, see the NetView online help.

2. If the NetView LIST *operid* command shows GLOBAL for the CTL setting, no protection is provided.
3. If the command is a D NET,EE command, no span checking is performed, and all resources specified in the D NET,EE command are accessible to all NetView operators who have authority to issue the D NET,EE command, regardless of their span-of-control authorization settings.

See [Chapter 4, “Using Spans to Protect Resources and Views,”](#) on page 67 for more information.

When using the SECOPTS.SPANAUTH statement in the CNMSTYLE member, or the SPANAUTH keyword on the REFRESH command, consider the following questions:

1. Has the operator been unintentionally granted access to a span that contains the resource or view?
2. Has the resource been incorrectly included in a span?
3. Is the resource or view included in a specific or generic definition in the span table?

If the NetView LIST *operid* command shows N (none) in the first position (*span_level*) of the NGMFVSPN attribute, no span checking is done for NetView management console.

4. Is the span that contains the resource active for the operator?

Issue the QRS command to see if the resource is in an active span for the operator.

5. Are there spelling errors? For example:

- Has the resource name been changed accidentally?
- Has a keyword been misspelled (for example, SPAN)?
- Has the span name been misspelled?
- Has the resource or view name been misspelled?

6. If you have recently updated the NetView span table, have you issued a REFRESH command to make the updated table active?
7. Does the table contain a generic definition that is inadvertently too generic which permits access to more resources and views than was intended? If so, make the generic definition more specific or use OMIT values to narrow its definition.
8. Does the protected view name in the table exceed the allowable length for a NetView management console view name? See the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for the maximum length allowed.
9. If you used the SECMIGR migration tool to create the NetView span table, was the correct VTAMLST used?
10. If you used the SECMIGR migration tool to create the NetView span table, and either explicitly or implicitly specified that double-asterisk identifiers were to be generated, you might consider running the tool specifying not to create double-asterisk identifiers to make the identifiers more specific.

If a Resource or View Cannot Be Accessed by an Authorized Operator

1. What are the active security definitions?

Issue a NetView LIST SECOPTS command to determine the current specification for SPANAUTH. If SPANAUTH=TABLE, find the name of the span table.

Issue a NetView LIST RESOURCE or LIST VIEW command to list the spans to which the resource or view is defined.

Issue a NetView LIST *operid* command to list the active spans for an operator. If appropriate, also check the NGMFVSPN setting. If the first position (*span_level*) of the NGMFVSPN attribute is shown as A (all), V (views), or R (resources), span checking might have restricted the operator's access to the resource or view. For more information about the LIST command, see the NetView online help.

2. If the NetView LIST *operid* command shows SPECIFIC for the CTL setting, the operator can only access resources included in a span that is active for the operator.

See [Chapter 4, “Using Spans to Protect Resources and Views,”](#) on page 67 for more information.

If you are using SECOPTS.SPANAUTH=TABLE statement in the CNMSTYLE member, or SPANAUTH=TABLE on the REFRESH command, consider the following items:

1. Is the span that contains the resource or view active for the operator?

2. If the NetView LIST *operid* command shows any value except N in the first position (*span_level*) of the NGMFVSPN attribute, some span checking is being done that might have restricted the operator from seeing that resource or view.
3. Is CTL=GENERAL specified for the operator? If so, and the resource or view is included in any defined span, the operator must have access to one of these spans.
4. Issue the QRS command to see if the resource or view is in a span that is active for the operator.
5. Are there spelling errors? For example:
 - Has a keyword been misspelled that caused a table statement to be ignored?
 - Has the span name been misspelled?
 - Has the resource or view name been misspelled?
6. If you have recently updated the NetView span table, have you issued a REFRESH command to make the updated table active?
7. Does the protected view name in the table exceed the allowable length for a view name?
8. If you used the SECMIGR migration tool to create the NetView span table, was the correct VTAMLST used?
9. If the resource is correctly included in a span and also matches an omit string specification, the resource does not match and access is not granted.

If Your Specified Initial Security Settings Were Not Taken

1. What level of RACF or other SAF product are you using?

You need RACF Version 2 or later, or an SAF product with equivalent function to use the NETCMDS class for command authorization and the NETSPAN class for span authorization. Only Version 2 or a later release of the RACF product, or an SAF product with equivalent function, has the NETVIEW segment to store operator information.

The ICH520I message on the MVS system log shows you which level of RACF you are using.

2. Do the security options in your CNMSTYLE statements match the options shown when you enter a NetView LIST SECOPTS command?

If not, check the following items:

- If the NetView REFRESH command has changed the initial settings, output does not show INITIALIZATION for the updated ID value. Check for a REFRESH command in the log.
- If the settings in CNMSTYLE statements are in conflict, default values are used that might be different from your intended values. See the *IBM Z NetView Administration Reference* for valid combinations and check for error messages in the MVS system log.

If Performance Is Degraded When Using the NGMFVSPN Attribute

1. Are you checking the NetView span table for operator authorization to both view names and resources when only view names need to be span checked?

You can improve performance by having only view names span checked. For any operator profiles that allow the operator to see all the resources in some views or no resources in other views, you can use the NGMFVSPN option that only span checks view names (NGMFVSPN=Vxxx). This uses less system processing time than checking all resources (NGMFVSPN=Rxxx), or checking all resources and view names (NGMFVSPN=Axxx).

Note: The placeholder xxx is used to represent the 2nd, 3rd, and 4th positions of the NGMFVSPN attribute in the above examples because the values in these positions do not apply here. Only the first position (*span_level*) of the NGMFVSPN attribute determines the level of span checking.

2. Are you checking the NetView span table for authorization to both view names and resources for operators and system administrators who are authorized to see all views and all resources in views?
You might be having views and resources in views span checked unnecessarily.

If Performance Is Degraded When Using SAF Security

1. Are you protecting commands or command lists that anyone should be allowed to use?

You might be protecting commands or command lists that do not need to be protected. For example, the NetView HELP command is useful to anyone logged on to the NetView program, and must not be protected. Look at the NetView commands in the NetView member CNMCMD. For those commands that you consider harmless or safe, add a CMDDEF statement for the command in CNMCMDU specifying SEC=BY . This improves performance by eliminating security checking for those commands no matter what method you are using for command authorization.

2. Have you set up automation security to perform unnecessary checks of commands and command lists?

If you already secured automation table members and command lists as described in [Chapter 8, "Security Considerations for Automation,"](#) on page 99, you might not need to do any further checking of automated commands and command lists. Use the NetView DEFAULTS command to set AUTOSEC=BYPASS in order to bypass command authorization checking for all commands originating from the automation table.

3. Are you monitoring the SAF product unnecessarily?

Using RACF auditing for all resources can degrade system performance. See *RACF Auditor's Guide* for details on RACF auditing. Setting RACF auditing to NONE for resources in the NETCMDS class can improve performance.

4. Are you using unnecessary traces on calls to external security products?

Use the NetView LIST TRACE command to see how you are tracing security calls. The display of TRACE settings is similar to the following example:

```
LIST TRACE
STATUS:    ACTIVE
MODE:      INT
SIZE:      250 PAGES (1000K)
OPTIONS:    DISP, MOD, PSS, QUE, STOR, UEXIT, SAF
TASKTYPES: OST
SAF TRACE:  FAILURES
SAF TYPES:  AUTH, EXTRACT, FASTAUTH, LIST, STAT, TOKENMAP, TOKENXTR,
            VERIFY
END OF LIST TRACE DISPLAY
```

Look at:

- The OPTIONS line of the LIST TRACE display to see which trace options are specified on the TRACE command. If the SAF option is indicated, you might be tracing more NetView calls to the SAF product than you want. Use TRACE for SAF only when you need to debug a problem between the NetView program and the SAF product or to provide service information.
- The SAF TYPES line to see which types of SAF requests are being traced from the NetView program to the SAF product. Limit the number of SAF TYPES to include only those which are necessary for problem determination.

See the *IBM Z NetView Troubleshooting Guide* for more information about using the NetView traces.

If You Cannot Isolate the Problem

If you cannot isolate a problem, you can capture additional data in one of several ways.

Capturing Data by Auditing the NetView Command Authorization Table

1. Add an AUDIT keyword to PROTECT and EXEMPT statements in the NetView command authorization table to record which tasks attempt to use commands, keywords, or values for which they do not have command authorization. For example, the following command traces all failing command authorization calls for this command identifier:

```
PROTECT (AUDIT=FAILURES) neta.cnm01.allocate.space
```

For more information about the AUDIT parameter, see [Chapter 3, “Controlling Access to Commands,” on page 29](#).

2. Use the NetView DEFAULTS command with the CATAUDIT keyword to globally change the auditing of the NetView command authorization table.

For example, to begin tracing all failing command authorization calls at once, enter DEFAULTS CATAUDIT=FAILURES. For more information about the DEFAULTS command, see the NetView online help.

The AUDIT keyword and the DEFAULTS CATAUDIT command both write records to SMF. The XITXL exit can write the records to another external log. The audit records are in SMF format, record type 38.

Capturing Data Using RACF Auditing

Use the auditing facility provided by RACF in order to monitor your security setup. Using RACF AUDIT provides you with an audit trail of attempts to issue unauthorized commands or command lists.

Refer to the RACF library for detailed information.

Capturing Data by Tracing SAF Calls From the NetView Program

Use the NetView TRACE command as a cross-product serviceability aid to help you isolate problems with RACROUTE calls from the NetView program to the SAF product.

For more information about the TRACE command, see the *IBM Z NetView Troubleshooting Guide* or the NetView online help.

Note: The TRACE command does not trace calls made to the DATASET or OPERCMDS classes since these calls are not made directly from the NetView program, but by MVS on behalf of the NetView program.

Appendix A. NetView Commands, Keywords, and Values that Can Be Protected

“Protecting NetView Command Names, Keywords, and Values” on page 152 lists the NetView product commands that can be protected using the NetView command authorization table or an SAF product such as RACF. Keywords that can be protected and values associated with those keywords are listed.

You can restrict the following items with command authorization:

- Commands, keywords, and values listed in this appendix.
- Command lists, user-written commands, VTAM commands and operands defined in CNMCMD, and some of the commands that are listed in the NetView online help (displayed using `HELP COMMANDS`). Attempting to use command authorization for commands other than these can cause the NetView program to issue an error message.

Note:

1. When VTAM commands are entered from a NetView command line, data within quotation marks is treated as one operand (value) for authorization checking.
2. Authority checking defaults for keywords and values does not occur unless the keywords and values are explicitly stated and unless they can be protected.

For example, if you protect `STATS=SUMMARY` of the `AUTOCNT` command, you cannot enter: `AUTOCNT REPORT=BOTH, STATS=SUMMARY` but you can enter: `AUTOCNT REPORT=BOTH` even though the `AUTOCNT` command uses `STATS=SUMMARY` as a default value.

The exceptions to this rule, when authority checking applies even to unstated default values, are footnoted in “Protecting NetView Command Names, Keywords, and Values” on page 152. The explanations for the footnotes are shown at the end of the table.

3. Keywords that allow more than one value at a time, each value must be protected by a separate `netid.luname.command.keyword.value` command identifier, even though the command is issued in the form `keyword=(value1,value2)`.

To restrict command authorization, ensure that the `SECOPTS` statement in the `CNMSTUSR` member is not coded using `SECOPTS.OPERSEC = MINIMAL`. Unless `OPERSEC` is a value other than `MINIMAL`, you cannot restrict command authorization for operators.

For more information about defining command authorization, see [Chapter 3, “Controlling Access to Commands,”](#) on page 29.

Protecting NetView Management Console Commands

Some NetView management console functions and commands issued against real resources managed by GMFHS and MSM can be protected at the host. The NetView management console functions and commands that have corresponding NetView commands that can be protected are listed in [Table 10 on page 152](#). The resource identifiers for these NetView commands are listed in “Protecting NetView Command Names, Keywords, and Values” on page 152.

The NetView management console functions and commands in [Table 10 on page 152](#) are accessed by selecting a resource and then right-clicking. If additional pull-downs are necessary before the NetView management console function or command can be selected, they are indicated in [Table 10 on page 152](#).

Table 10. Protecting NetView Management Console Functions and Commands

NetView Management Console Function or Command	Additional Pull-down Necessary?	NetView Command
Activate	No	DUIFSACT
Change Status	No	DUIFSSET
Current® Status	Yes, Resource services	DUIFSDIS
Inactivate	No	DUIFSINA
Service Point Command Line	Yes, Network	DUIFSNTV
Recycle	No	DUIFSRCY

Although the NetView commands in [Table 10 on page 152](#) are not available to resources managed by a manager other than GMFHS, they can be protected with the Command Profile Editor. Because objects displayed by NetView management console come from many sources, many types of commands can be added to the Command Profile Editor (shipped as the ihsscpe.en_US.rsp file). The defaults that are supplied with the NetView program include examples of NetView, VTAM, and SNA Topology Manager commands. To prevent a command from being included in an operator's **Command** menu for a particular resource, remove the command from the operator's command profile.

For information about...	See...
Restricting commands processed by host tasks	Chapter 3, “Controlling Access to Commands,” on page 29 and “Protecting NetView Command Names, Keywords, and Values” on page 152
How to implement NetView management console security	Chapter 10, “Security for the NetView Management Console,” on page 109

Protecting NetView Command Names, Keywords, and Values

This section contains the table of NetView command names, keywords, and values that can be protected along with their associated SAF resource or command authorization table identifier.

Table 11. NetView Command Identifiers		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
ACQ	CNME0001	netid.luname.CNME0001
ACT	CNME0002	netid.luname.CNME0002
ACTION	CNME3001	netid.luname.CNME3001

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
ADDCMD CMDSYN ECHO MOD NAME PARMSYN PARSE REPLACE RES SEC TYPE		netid.luname.ADDCMD netid.luname.ADDCMD.CMDSYN.name netid.luname.ADDCMD.ECHO.value netid.luname.ADDCMD.MOD.name netid.luname.ADDCMD.NAME.name netid.luname.ADDCMD.PARMSYN netid.luname.ADDCMD.PARSE.value netid.luname.ADDCMD.REPLACE.value netid.luname.ADDCMD.RES.value netid.luname.ADDCMD.SEC.value netid.luname.ADDCMD.TYPE.type
ADDLINE	CNME0040	netid.luname.CNME0040
AFTER PPT ROUTE SAVE		netid.luname.AFTER netid.luname.AFTER.PPT netid.luname.AFTER.ROUTE.OPERID ¹ netid.luname.AFTER.SAVE
AGTBRW	CNME8205	netid.luname.CNME8205
AGTSESMG	CNME8221	netid.luname.CNME8221
AGTTCPC	CNME8225	netid.luname.CNME8225
AINQ ORIGNET TARGNET		netid.luname.AINQ netid.luname.AINQ.ORIGNET.name ² netid.luname.AINQ.TARGNET.name
ALLOC		See ALLOCATE command

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
ALLOCATE		netid.luname.ALLOCATE
ALX		netid.luname.ALLOCATE.ALX
BLK		netid.luname.ALLOCATE.BLOCK
BLKS		netid.luname.ALLOCATE.BLOCK
BLKSIZE		netid.luname.ALLOCATE.BLKSIZE
BLOCK		netid.luname.ALLOCATE.BLOCK
BUFNO		netid.luname.ALLOCATE.BUFNO
CATALOG		netid.luname.ALLOCATE.CATALOG
CATLG		netid.luname.ALLOCATE.CATALOG
CONTIG		netid.luname.ALLOCATE.CONTIG
COPIES		netid.luname.ALLOCATE.COPIES
CYL		netid.luname.ALLOCATE.CYLINDER
CYLINDERS		netid.luname.ALLOCATE.CYLINDER
DA		netid.luname.ALLOCATE.DATASET
DATACLAS		netid.luname.ALLOCATE.DATACLAS.value
DATASET		netid.luname.ALLOCATE.DATASET
DD		netid.luname.ALLOCATE.FILE.ddname
DDN		netid.luname.ALLOCATE.FILE.ddname
DDNAME		netid.luname.ALLOCATE.FILE.ddname
DEFER		netid.luname.ALLOCATE.DEFER
DELETE		netid.luname.ALLOCATE.DELETE
DEN		netid.luname.ALLOCATE.DEN.density
DEST		netid.luname.ALLOCATE.DEST
DIR		netid.luname.ALLOCATE.DIR
DS		netid.luname.ALLOCATE.DATASET
DSN		netid.luname.ALLOCATE.DATASET
DSNAME		netid.luname.ALLOCATE.DATASET
DSORG		netid.luname.ALLOCATE.DSORG.organization
DUMMY		netid.luname.ALLOCATE.DUMMY
EROPT		netid.luname.ALLOCATE.EROPT.option
EXPDT		netid.luname.ALLOCATE.EXPDT
F		netid.luname.ALLOCATE.FILE.ddname
FI		netid.luname.ALLOCATE.FILE.ddname
FILE		netid.luname.ALLOCATE.FILE.ddname
FORMS		netid.luname.ALLOCATE.FORMS.number
FREE		netid.luname.ALLOCATE.FREE
HOLD		netid.luname.ALLOCATE.HOLD
INPUT		netid.luname.ALLOCATE.INPUT
KEEP		netid.luname.ALLOCATE.KEEP
KEYLEN		netid.luname.ALLOCATE.KEYLEN
LABEL		netid.luname.ALLOCATE.LABEL.type
LIKE		netid.luname.ALLOCATE.LIKE.dsname
LRECL		netid.luname.ALLOCATE.LRECL
MGMTCLAS		netid.luname.ALLOCATE.MGMTCLAS.value
MOD		netid.luname.ALLOCATE.MOD

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
ALLOCATE (continued) MSVGP MXIG NEW NOHOLD OLD OUTLIM OUTPUT PASSWORD POS POSITION PROTECT RECFM RELEASE RETPD RLSE RND ROUND SHR SPACE STORCLAS SYSOUT TRACKS TRK TRKS UCS UNCATALOG UNCATLG UNIT VOLUME VSEQ WRITER		netid.luname.ALLOCATE.MSVGP.volgroup netid.luname.ALLOCATE.MXIG netid.luname.ALLOCATE.NEW netid.luname.ALLOCATE.NOHOLD netid.luname.ALLOCATE.OLD netid.luname.ALLOCATE.OUTLIM netid.luname.ALLOCATE.OUTPUT netid.luname.ALLOCATE.PASSWORD netid.luname.ALLOCATE.POSITION netid.luname.ALLOCATE.POSITION netid.luname.ALLOCATE.PROTECT netid.luname.ALLOCATE.RECFM netid.luname.ALLOCATE.RELEASE netid.luname.ALLOCATE.RETPD netid.luname.ALLOCATE.RELEASE netid.luname.ALLOCATE.ROUND netid.luname.ALLOCATE.ROUND netid.luname.ALLOCATE.SHR netid.luname.ALLOCATE.SPACE netid.luname.ALLOCATE.STORCLAS.value netid.luname.ALLOCATE.SYSOUT.class netid.luname.ALLOCATE.TRACKS netid.luname.ALLOCATE.TRACKS netid.luname.ALLOCATE.TRACKS netid.luname.ALLOCATE.UCS.charset netid.luname.ALLOCATE.UNCATALO netid.luname.ALLOCATE.UNCATALO netid.luname.ALLOCATE.UNIT.type netid.luname.ALLOCATE.VOLUME.serial netid.luname.ALLOCATE.VSEQ netid.luname.ALLOCATE.WRITER.extwriter
AONCMD		See EXCMD
APERSIST	CNME7022	netid.luname.CNME7022
APPLS	CNME0003	netid.luname.CNME0003
APPLSPEN	CNME0005	netid.luname.CNME0005
APSERV	CNMEAPCS	netid.luname.CNMEAPCS
ASSIGN		netid.luname.ASSIGN
ASSISCMD	CNME6220	netid.luname.CNME6220

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
AT PPT ROUTE SAVE		netid.luname.AT netid.luname.AT.PPT netid.luname.AT.ROUTE. <i>operid</i> ¹ netid.luname.AT.SAVE
ATTACH DUMP		netid.luname.ATTACH netid.luname.ATTACH.DUMP
AUPD ORIGNET		netid.luname.AUPD netid.luname.AUPD.ORIGNET. <i>name</i> ²
AUTBNABL	CNMEAUTB	netid.luname.CNMEAUTB
AUTOB146	CNME7007	netid.luname.CNME7007
AUTOCMD		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
AUTOCHFP	CNME7010	netid.luname.CNME7010
AUTOCHK	CNME7004	netid.luname.CNME7004
AUTOCNT RESET REPORT NAME STATS DISPLAY FILE REPLACE TEST		netid.luname.AUTOCNT netid.luname.AUTOCNT.RESET netid.luname.AUTOCNT.REPORT. <i>value</i> netid.luname.AUTOCNT.NAME. <i>tblname</i> netid.luname.AUTOCNT.STATS. <i>value</i> netid.luname.AUTOCNT.DISPLAY netid.luname.AUTOCNT.FILE. <i>membername</i> ³ netid.luname.AUTOCNT.REPLACE netid.luname.AUTOCNT.TEST
AUTOCOLL	CNME2001	netid.luname.CNME2001
AUTODMSG	CNME7008	netid.luname.CNME7008
AUTODROP	CNMS8003	netid.luname.CNMS8003
AUTOMAN		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199
AUTOMSG		See AUTOTBL command
AUTONEWF	CNME7011	netid.luname.CNME7011
AUTONTFY	CNME7006	netid.luname.CNME7006
AUTORECD	CNME2005	netid.luname.CNME2005
AUTOSEND	CNME7005	netid.luname.CNME7005
AUTOSTDN	CNME7003	netid.luname.CNME7003
AUTOSTUN	CNME7002	netid.luname.CNME7002

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
AUTOTBL OFF REMOVE MEMBER TEST SWAP INSERT AT BEFORE AFTER FIRST LAST LISTING REPLACE DISABLE ENABLE NAME SEQUENCE LABEL ENDLABEL BLOCK GROUP STATUS		netid.luname.AUTOTBL netid.luname.AUTOTBL.OFF netid.luname.AUTOTBL.REMOVE. <i>tblname</i> netid.luname.AUTOTBL.MEMBER. <i>membername</i> netid.luname.AUTOTBL.TEST netid.luname.AUTOTBL.SWAP.# netid.luname.AUTOTBL.INSERT netid.luname.AUTOTBL.AT.# netid.luname.AUTOTBL.BEFORE.# netid.luname.AUTOTBL.AFTER.# netid.luname.AUTOTBL.FIRST netid.luname.AUTOTBL.LAST netid.luname.AUTOTBL.LISTING. <i>listname</i> ³ netid.luname.AUTOTBL.REPLACE netid.luname.AUTOTBL.DISABLE netid.luname.AUTOTBL.ENABLE netid.luname.AUTOTBL.NAME. <i>name</i> netid.luname.AUTOTBL.SEQUENCE. <i>seq</i> netid.luname.AUTOTBL.LABEL. <i>labelname</i> netid.luname.AUTOTBL.ENDLABEL. <i>lablename</i> netid.luname.AUTOTBL.BLOCK. <i>labelname</i> netid.luname.AUTOTBL.GROUP. <i>groupname</i> netid.luname.AUTOTBL.STATUS
AUTOTASK OPID CONSOLE DROP		netid.luname.AUTOTASK netid.luname.AUTOTASK.OPID. <i>operid</i> netid.luname.AUTOTASK.CONSOLE. <i>value</i> netid.luname.AUTOTASK.DROP
AUTOTEST CZID CZRECORD OFF STATUS MEMBER DD LISTING REPLACE SOURCE TASK REPORT LOGREPL RECORD RPTREPL		netid.luname.AUTOTEST netid.luname.AUTOTEST.CZID. <i>czid</i> netid.luname.AUTOTEST.CZRECORD. <i>recname</i> netid.luname.AUTOTEST.OFF netid.luname.AUTOTEST.STATUS netid.luname.AUTOTEST.MEMBER. <i>membername</i> netid.luname.AUTOTEST.DD. <i>ddname</i> netid.luname.AUTOTEST.LISTING. <i>lname</i> ³ netid.luname.AUTOTEST.REPLACE netid.luname.AUTOTEST.SOURCE. <i>sname</i> ³ netid.luname.AUTOTEST.TASK. <i>taskname</i> netid.luname.AUTOTEST.REPORT. <i>repname</i> ³ netid.luname.AUTOTEST.LOGREPL netid.luname.AUTOTEST.RECORD. <i>recname</i> netid.luname.AUTOTEST.RPTREPL
AUTOTR	CNME0006	netid.luname.CNME0006
AUTOWRAP		netid.luname.AUTOWRAP ⁴

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
BFRUSE	CNME0007	netid.luname.CNME0007
BFSESS	CNME1001	netid.luname.CNME1001
BGNSESS APPLID SRCLU		netid.luname.BGNSESS netid.luname.BGNSESS.APPLID. <i>applid</i> netid.luname.BGNSESS.SRCLU. <i>srclu</i>
BLOG	CNME1099	netid.luname.CNME1099
BOOKS	CNMEHTSP	See CNMEHTSP command
BOSESS	CNME1002	netid.luname.CNME1002
BR		See BROWSE command
BROWSE <i>ddname.member</i>	CNME5001	netid.luname.CNME5001 netid.luname.READSEC. <i>ddname.member</i>
C		See COMMAND command
CALC	CNME8001	netid.luname.CNME8001
CANCEL		See RESET command
CANCMD TAG ID SP		netid.luname.CANCMD netid.luname.CANCMD.TAG. <i>tag</i> netid.luname.CANCMD.ID. <i>name</i> netid.luname.CANCMD.SP. <i>spname</i>
CANZLOG	CNMECANZ	netid.luname.CNMECANZ.common
CCDEF	CNME1504	netid.luname.CNME1504
CCPDR		netid.luname.CCPDR. <i>name</i>
CCLOADF		netid.luname.CCPLOADF. <i>name</i>
CCLOADI		netid.luname.CCPLOADI. <i>name</i>
CCLOADT		netid.luname.CCPLOADT. <i>name</i>
CDRMS	CNME0008	netid.luname.CNME0008
CDRSCS	CNME0009	netid.luname.CNME0009
CDRSCS	CNME0009	netid.luname.CNME0009
CGED		netid.luname.EZLE840A
CHANGEFP	CNME7009	netid.luname.CNME7009
CHNGFP		netid.luname.CHNGFP

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
CHRON • AFTER • AT • COMMAND • DEBUG • EVERY • INTERVAL • FOR • MXREPEAT • OFF • REMAFTER • REMOVE • DAYSWEEK • DAYSMON • CALENDAR • GMT • ID • LOCAL • NOTIFY IGNORE PURGE REMOVE RUN • RECOVERY • REFRESH • REM • ROUTE • NOSAVE • SAVE • TEST		netid.luname.CHRON • netid.luname.CHRON.AFTER.value • netid.luname.CHRON.AT.value • netid.luname.CHRON.COMMAND.value • netid.luname.CHRON.DEBUG • netid.luname.CHRON.EVERY.value • netid.luname.CHRON.EVERY(INTERVAL.value • netid.luname.CHRON.EVERY(INTERVAL(FOR.value • netid.luname.CHRON.EVERY(INTERVAL(MXREPEAT.value • netid.luname.CHRON.EVERY(INTERVAL(OFF.value • netid.luname.CHRON.EVERY(REMAFTER.value • netid.luname.CHRON.EVERY(REMOVE.value • netid.luname.CHRON.EVERY(DAYSWEEK.value • netid.luname.CHRON.EVERY(DAYSMON.value • netid.luname.CHRON.EVERY(CALENDAR.value • netid.luname.CHRON.GMT • netid.luname.CHRON.ID • netid.luname.CHRON.LOCAL • netid.luname.CHRON.NOTIFY.value netid.luname.CHRON.NOTIFY(IGNORE.value netid.luname.CHRON.NOTIFY(PURGE.value netid.luname.CHRON.NOTIFY(REMOVE.value netid.luname.CHRON.NOTIFY(RUN.value • netid.luname.CHRON.RECOVERY.value • netid.luname.CHRON.REFRESH • netid.luname.CHRON.REM.value • netid.luname.CHRON.ROUTE.value • netid.luname.CHRON.NOSAVE • netid.luname.CHRON.SAVE • netid.luname.CHRON.TEST
CLEAR		netid.luname.CLEAR
CLOSE		netid.luname.CLOSE ⁴
CLRSTATS		netid.luname.CLRSTATS
CLSTRS	CNME0010	netid.luname.CLSTRS
CMD		netid.luname.CMD ⁴

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
CMDMON • STATS • CAPTURE • LOGSMF		netid.luname.CMDMON netid.luname.CMDMON.STATS.value netid.luname.CMDMON.CAPTURE netid.luname.CMDMON.LOGSMF.value
CMDSERV AUTHSNDR NAME		netid.luname.CMDSERV netid.luname.CMDSERV.AUTHSNDR.value netid.luname.CMDSERV.NAME.value
CNMDLAR	CNMDLAR	netid.luname.CNMDLAR
CNME8204	CNME8204	netid.luname.CNME8204
CNME8240	CNME8240	netid.luname.CNME8240
CNMEDRCL	CNMEDRCL	netid.luname.CNMEDRCL
CNMEDUPS	CNMEDUPS	netid.luname.CNMEDUPS
CNMEERSC	CNMEERSC	netid.luname.CNMEERSC
CNMEHTSP	CNMEHTSP	netid.luname.CNMEHTSP netid.luname.CNMEHTSP.TYPE.value
CNMEMCXX	CNMEMCXX	netid.luname.CNMEMCXX
CNMEMCXY	CNMEMCXY	netid.luname.CNMEMCXY
CNMEMIBB	CNMEMIBB	netid.luname.CNMEMIBB
CNMEMIBR	CNMEMIBR	netid.luname.CNMEMIBR
CNMEMIBW	CNMEMIBW	netid.luname.CNMEMIBW
CNMENV39	CNMENV39	netid.luname.CNMENV39
CNMESNMP	CNMESNMP	netid.luname.CNMESNMP
CNME0044	CNME0044	netid.luname.CNME0044
CNME1087	CNME1087	netid.luname.CNME1087
CNME1103	CNME1103	netid.luname.CNME1103 ⁵
CNME8601 INITNRM	CNME8601	netid.luname.CNME8601 netid.luname.CNME8601.INITNRM
CNME8602 SUSPNRM	CNME8602	netid.luname.CNME8602 netid.luname.CNME8602.SUSPNRM
CNVOSI	CNME2103	netid.luname.CNME2103
COLLCTL	CNME8250 START STOP	netid.luname.CNME8250 netid.luname.CNME8250.START netid.luname.CNME8250.STOP
COLLECT		See NLDM command

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
COMMAND	CNME1036	netid.luname.CNME1036
CONNSEC COUNT DEFINE JOBNAME LADDR LPORT MAXRECS OPID PROTOCOL QUERY RADDR RPORT START STARTTIME STOP USERID		netid.luname.CONNSEC netid.luname.CONNSEC.COUNT.value netid.luname.CONNSEC.DEFINE netid.luname.CONNSEC.JOBNAME.value netid.luname.CONNSEC.LADDR.value netid.luname.CONNSEC.LPORT.value netid.luname.CONNSEC.MAXRECS.value netid.luname.CONNSEC.OPID.value netid.luname.CONNSEC.PROTOCOL.value netid.luname.CONNSEC.QUERY netid.luname.CONNSEC.RADDR.value netid.luname.CONNSEC.RPORT.value netid.luname.CONNSEC.START netid.luname.CONNSEC.STARTTIME.value netid.luname.CONNSEC.STOP netid.luname.CONNSEC.USERID.value
COS	CNME0045	netid.luname.CNME0045
CPTBL MEMBER TEST		netid.luname.CPTBL netid.luname.CPTBL.MEMBER.membername netid.luname.CPTBL.TEST
CSCF OP PU PURGE		netid.luname.CSCF netid.luname.CSCF.OP.value netid.luname.CSCF.PU.puname netid.luname.CSCF.PURGE
CURTIME	CNME7021	netid.luname.CNME7021
D		See DISPLAY command
DBAUTO	CNME2008	netid.luname.CNME2008
DBFULL	CNME2010	netid.luname.CNME2010
DBINIT	CNME2009	netid.luname.CNME2009
DCL		See DROPCL command

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
DEFAULTS AUTOLOGN AUTOSEC AVLMAX AVLSLOW BEEP BRUNLOCK CATAUDIT CMD CNM493I COSTIME DISPLAY EMCSPARM EVERYCON HELPTXT HOLD HCYLOG LOGSPNCF LOGSPNCP LOGSPNVF LOGSPNVP LOGTSTAT LONGDATE LONGTIME MAXABEND MAXCPU MAXCSSIR MAXIO MAXLOGON MAXMQIN MAXMQOUT MAXREPLY MAXSTG MDCFGTIM MSGMODID MSGTOUT NETLOG NETVASIS NOREPLY RCVREPLY REACQPRI REXXENV REXXENVL REXXSLMT REXXSTOR		netid.luname.DEFAULTS netid.luname.DEFAULTS.AUTOLOGN.value netid.luname.DEFAULTS.AUTOSEC.value netid.luname.DEFAULTS.AVLMAX.value netid.luname.DEFAULTS.AVLSLOW.value netid.luname.DEFAULTS.BEEP.value netid.luname.DEFAULTS.BRUNLOCK.value netid.luname.DEFAULTS.CATAUDIT.value netid.luname.DEFAULTS.CMD.value netid.luname.DEFAULTS.CNM493I.value netid.luname.DEFAULTS.COSTIME.costime netid.luname.DEFAULTS.DISPLAY.value netid.luname.DEFAULTS.EMCSPARM.value netid.luname.DEFAULTS.EVERYCON.value netid.luname.DEFAULTS.HELPTXT.value netid.luname.DEFAULTS.HOLD.value netid.luname.DEFAULTS.HCYLOG.value netid.luname.DEFAULTS.LOGSPNCF.value netid.luname.DEFAULTS.LOGSPNCP.value netid.luname.DEFAULTS.LOGSPNVF.value netid.luname.DEFAULTS.LOGSPNVP.value netid.luname.DEFAULTS.LOGTSTAT.value netid.luname.DEFAULTS.LONGDATE.value netid.luname.DEFAULTS.LONGTIME.value netid.luname.DEFAULTS.MAXABEND.n netid.luname.DEFAULTS.MAXCPU.value netid.luname.DEFAULTS.MAXCSSIR.value netid.luname.DEFAULTS.MAXIO.value netid.luname.DEFAULTS.MAXLOGON.n netid.luname.DEFAULTS.MAXMQIN.value netid.luname.DEFAULTS.MAXMQOUT.value netid.luname.DEFAULTS.MAXREPLY.nnnnn netid.luname.DEFAULTS.MAXSTG.value netid.luname.DEFAULTS.MDCFGTIM.mdcfgtime netid.luname.DEFAULTS.MSGMODID.value netid.luname.DEFAULTS.MSGTOUT.msgtonumber netid.luname.DEFAULTS.NETLOG.value netid.luname.DEFAULTS.NETVASIS.value netid.luname.DEFAULTS.NOREPLY.nnn netid.luname.DEFAULTS.RCVREPLY.nnn netid.luname.DEFAULTS.REACQPRI.value netid.luname.DEFAULTS.REXXENV.value netid.luname.DEFAULTS.REXXENVL.value netid.luname.DEFAULTS.REXXSLMT.value netid.luname.DEFAULTS.REXXSTOR.value

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
DEFAULTS (continued) REXXSTRF RMTMAXL SCRNFMT SCROLL SENDMSG SESINACT SHORTDAT SHORTTIM SLOGCMDR SLOWSTG STARTCOL STORDUMP STRTSERV SUPZDATE SUPZTIME SYSLOG TAFPREFX TAFRECLN TIMEFMSG WRNCPU WRNIO WRNMQIN WRNMQOUT WRNMSGCT WRNSTG		netid.luname.DEFAULTS.REXXSTRF.value netid.luname.DEFAULTS.RMTMAXL.value netid.luname.DEFAULTS.SCRNFMT.member netid.luname.DEFAULTS.SCROLL.value netid.luname.DEFAULTS.SENDMSG.value netid.luname.DEFAULTS.SESINACT.nnn netid.luname.DEFAULTS.SHORTDAT.value netid.luname.DEFAULTS.SHORTTIM.value netid.luname.DEFAULTS.SLOGCMDR.value netid.luname.DEFAULTS.SLOWSTG.value netid.luname.DEFAULTS.STARTCOL.value netid.luname.DEFAULTS.STORDUMP.n netid.luname.DEFAULTS.STRTSERV.value netid.luname.DEFAULTS.SUPZDATE.value netid.luname.DEFAULTS.SUPZTIME.value netid.luname.DEFAULTS.SYSLOG.value netid.luname.DEFAULTS.TAFPREFX.value netid.luname.DEFAULTS.TAFRECLN.value netid.luname.DEFAULTS.TIMEFMSG.value netid.luname.DEFAULTS.WRNCPU.value netid.luname.DEFAULTS.WRNIO.value netid.luname.DEFAULTS.WRNMQIN.value netid.luname.DEFAULTS.WRNMQOUT.value netid.luname.DEFAULTS.WRNMSGCT.value netid.luname.DEFAULTS.WRNSTG.value
DEL		See NPDA command
DELCMD FREE NAME		netid.luname.DELCMD netid.luname.DELCMD.FREEvalue netid.luname.DELCMD.NAME.name
DELAY	CNME1021	netid.luname.CNME1021
DELAY2	CNME1022	netid.luname.CNME1022
DELDS	CNME1055	netid.luname.CNME1055
DELHMSG	EZLEDMSG	netid.luname.EZLEDMSG
DETACH		netid.luname.DETACH
DIALCDRM	CNME7023	netid.luname.CNME7023
DIS	CNME1023	netid.luname.CNME1023
DISABLE		See NLDM command

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
DISBQL ALL <i>recvr</i>		netid.luname.DISBQL netid.luname.DISBQL.ALL netid.luname.DISBQL. <i>recvr</i>
DISC		netid.luname.DISC
DISCONID		netid.luname.DISCONID
DISG	CNME1070	netid.luname.CNME1070
DISK	CNME0046	netid.luname.CNME0046
DISKEEP		See NLDM command
DISPCMD A ALL ID OP PPT SP		netid.luname.DISPCMD netid.luname.DISPCMD.ALL netid.luname.DISPCMD.ALL netid.luname.DISPCMD.ID. <i>name</i> netid.luname.DISPCMD.OP. <i>operid</i> netid.luname.DISPCMD.PPT netid.luname.DISPCMD.SP. <i>sname</i>
DISPFK	CNME1048	netid.luname.CNME1048
DISPLAY <i>keyword</i>		netid.luname.DISPLAY ⁶ netid.luname.DISPLAY. <i>keyword.value</i> ⁶
DISPMOD		netid.luname.DISPMOD
DISPPI RCVRID BUFQ ALL TRACE TABLE		netid.luname.DISPPI netid.luname.DISPPI.RCVRID. <i>receiverid</i> netid.luname.DISPPI.BUFQ netid.luname.DISPPI.ALL netid.luname.DISPPI.TRACE netid.luname.DISPPI.TABLE
DISPREG		netid.luname.DISPREG
DISPTOPO		netid.luname.DISPTOPO
DM	EZLEDMSG	netid.luname.EZLEDMSG
DMCS		netid.luname.DMCS
DNIC	CNME7015	netid.luname.CNME7015

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
DOM MSG TOKEN SYSID SCOPE NVDELID CURMSG ONESYS ALLSYS		netid.luname.DOM netid.luname.DOM.MSG netid.luname.DOM.TOKEN netid.luname.DOM.SYSID netid.luname.DOM.SCOPE netid.luname.DOM.NVDELID netid.luname.DOM.CURMSG netid.luname.DOM.ONESYS netid.luname.DOM.ALLSYS
DRDS	CNME0011	netid.luname.CNME0011
DROPCL		netid.luname.DROPCL
DROUTE	CNME0012	netid.luname.CNME0012
DSILCMD DATE MOD NAME OPID USAGE		netid.luname.DSILCMD netid.luname.DSILCMD.DATE netid.luname.DSILCMD.MOD.name netid.luname.DSILCMD.NAME.name netid.luname.DSILCMD.OPID.name netid.luname.DSILCMD.USAGE
DSILMEXP		netid.luname.DSILMEXP
DSILSSIR STATS ALLSTATS SHOWMSG		netid.luname.DSILSSIR netid.luname.DSILSSIR.STATS netid.luname.DSILSSIR.ALLSTATS netid.luname.DSILSSIR.SHOWMSG ⁵
DSIMCAP	DSIMCAP	netid.luname.DSIMCAP
DSISAPDR	CNME1085	netid.luname.CNME1085
DSITSTAT		netid.luname.DSITSTAT
DSIVSMX IDCAMS OPEN PUT GET GETREV INQUIRE DEL CLOSE (all data set names) DD (all data set names) DD		netid.luname.DSIVSMX netid.luname.DSIVSMX.IDCAMS netid.luname.DSIVSMX.OPEN.ddname ⁷ netid.luname.DSIVSMX.PUT.ddname ⁸ netid.luname.DSIVSMX.GET.ddname ⁷ netid.luname.DSIVSMX.GETREV.ddname ⁷ netid.luname.DSIVSMX.INQUIRE.ddname ⁷ netid.luname.DSIVSMX.DEL.ddname ⁸ netid.luname.DSIVSMX.CLOSE.ddname ⁷ netid.luname.READSEC.(ALLDSN) netid.luname.READSEC.fully_qualified_dsname netid.luname.WRITESEC.(ALLDSN) netid.luname.WRITESEC.fully_qualified_dsname

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
DSIVSAM PUT GET GETREV INQUIRE DEL		netid.luname.DSIVSAM netid.luname.DSIVSAM.PUT.taskname netid.luname.DSIVSAM.GET.taskname netid.luname.DSIVSAM.GETREV.taskname netid.luname.DSIVSAM.INQUIRE.taskname netid.luname.DSIVSAM.DEL.taskname
DSIZKNYJ		netid.luname.DSIZKNYJ
DSRBS		netid.luname.DSRBS
DUIFSACT		netid.luname.DUIFSACT
DUIFSDIS		netid.luname.DUIFSDIS
DUIFSINA		netid.luname.DUIFSINA
DUIFSNTV		netid.luname.DUIFSNTV
DUIFSRCY		netid.luname.DUIFSRCY
DUIFSRSC		netid.luname.DUIFSRSC
DUIFSSET		netid.luname.DUIFSSET
DUILOGON NMC		netid.luname.DUILOGON.NMC
DVIPALOG	CNME8260	netid.luname.CNME8260
DVIPCONN	CNME8213	netid.luname.CNME8213
DVIPDDCR	CNME8215	netid.luname.CNME8215
DVIPHLTH	CNME8214	netid.luname.CNME8214
DVIPPLEX	CNME8211	netid.luname.CNME8211
DVIPSTAT	CNME8210	netid.luname.CNME8210
DVIPTARG	CNME8212	netid.luname.CNME8212
EKGVACTL		netid.luname.EKGVACTL
EKGVACTM CHECKPNT CONNECT DISCON STOP UCONNECT UPDATE		netid.luname.EKGVACTM netid.luname.EKGVACTM.CHECKPNT netid.luname.EKGVACTM.CONNECT netid.luname.EKGVACTM.DISCON netid.luname.EKGVACTM.STOP netid.luname.EKGVACTM.UCONNECT netid.luname.EKGVACTM.UPDATE
EKGVCREM		netid.luname.EKGVCREM
EKGVDELM		netid.luname.EKGVDELM
EKGVLNKM		netid.luname.EKGVLNKM
EKGVLOCM		netid.luname.EKGVLOCM

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
EKGVMETM		netid.luname.EKGVMETM
EKGVQUEM		netid.luname.EKGVQUEM
EKGVSUBM		netid.luname.EKGVSUBM
ENABLE		See NLDM command
ENDSESS		netid.luname.ENDSESS
ENDTASK ALL DISCONN DOMAIN FORCE IP IPV6 LU NETID OPERID STOP		netid.luname.ENDTASK netid.luname.ENDTASK.ALL netid.luname.ENDTASK.DISCONN netid.luname.ENDTASK.DOMAIN.value netid.luname.ENDTASK.FORCE netid.luname.ENDTASK.IP netid.luname.ENDTASK.IPV6 netid.luname.ENDTASK.LU.luname netid.luname.ENDTASK.NETID.value netid.luname.ENDTASK.OPERID.value netid.luname.ENDTASK.STOP
ERST	CNME0014	netid.luname.CNME0014
ESESS	CNME1004	netid.luname.CNME1004
EVENTS	CNME3003	netid.luname.CNME3003
EVERY PPT ROUTE SAVE		netid.luname.EVERY netid.luname.EVERY.PPT netid.luname.EVERY.ROUTE.operid ¹ netid.luname.EVERY.SAVE
EXCMD opid/cmd		netid.luname.EXCMD netid.luname.EXCMD.opid.cmd ⁹
EZLEACNA		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLEASLN		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLEDAPI HANDLE TASKID	EZLEDAPI	netid.luname.EZLEDAPI netid.luname.EZLEDAPI.HANDLE.value netid.luname.EZLEDAPI.TASKID.value
EZLEF000	EZLEF000	netid.luname.EZLEF000
EZLEF002	EZLEF002	netid.luname.EZLEF002
EZLEF005	EZLEF005	netid.luname.EZLEF005

<i>Table 11. NetView Command Identifiers (continued)</i>		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEF007	EZLEF007	netid.luname.EZLEF007
EZLEF008	EZLEF008	netid.luname.EZLEF008
EZLEINFU		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLEMSG		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLENETF		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLENFRM		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLEIPC		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLEQAPI HANDLE TASKID	EZLEQAPI	netid.luname.EZLEQAPI netid.luname.EZLEQAPI.HANDLE.value netid.luname.EZLEQAPI.TASKID.value
EZLEQCAL	EZLEQCAL	netid.luname.EZLEQCAL
EZLERTVE		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLETAPI ACTION HANDLE TASKID TYPE	EZLETAPI	netid.luname.EZLETAPI netid.luname.EZLETAPI.ACTION.value netid.luname.EZLETAPI.HANDLE.value netid.luname.EZLETAPI.TASKID.value netid.luname.EZLETAPI.TYPE.value
EZLEXIST		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLE1900		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLE5500	EZLE5500	netid.luname.EZLE5500 • netid.luname.EZLE5500.REQUEST.value • netid.luname.EZLE5500.SAFE.value
EZLE8500		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLMSG		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLSATDF		netid.luname.EZLSATDF
EZLSPIPC		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLSRTVE		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLSTRAC		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
EZLTRACE		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
F		See MODIFY command
FINDNCP	CNME0042	netid.luname.CNME0042
FKXESCMD -h STACK	FKXESCMD	netid.luname.FKXESCMD netid.luname.FKXESCMD._H.hostname netid.luname.FKXESCMD.STACK.stackname
FMTPACKT	CNMEFPTC	netid.luname.CNMEFPTC Note: Permission must also be granted for the PKTS QUERY command and associated parameters. The PKTS QUERY command parameters are only checked by the PKTS command, not by the FMTPACKT command.
FOC		See FOCALPT command

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
FOCALPT		netid.luname.FOCALPT
A		netid.luname.FOCALPT.ACQUIRE
AB		netid.luname.FOCALPT.ADDBKUP
ABK		netid.luname.FOCALPT.ADDBKUP
ACQ		netid.luname.FOCALPT.ACQUIRE
ACQUIRE		netid.luname.FOCALPT.ACQUIRE
ACTIVE		netid.luname.FOCALPT.ACTIVE
ADD		netid.luname.FOCALPT.ADDBKUP
ADDBKUP		netid.luname.FOCALPT.ADDBKUP
ADDBU		netid.luname.FOCALPT.ADDBKUP
ADDPEND		netid.luname.FOCALPT.ADDPEND
ALL		netid.luname.FOCALPT.ALL
BACKUP		netid.luname.FOCALPT.BACKUP.value
BACKLIST		netid.luname.FOCALPT.BACKLIST.backup_focalpt
BACNET		netid.luname.FOCALPT.BACNET.value
BL		netid.luname.FOCALPT.BACKLIST.backup_focalpt
BN		netid.luname.FOCALPT.BACNET.value
BU		netid.luname.FOCALPT.BACKUP.value
C		netid.luname.FOCALPT.CHANGE
CAT		netid.luname.FOCALPT.FPCAT.value
CHANGE		netid.luname.FOCALPT.CHANGE
CHG		netid.luname.FOCALPT.CHANGE
D		netid.luname.FOCALPT.DISPSOC
DEFFOCPT		netid.luname.FOCALPT.DEFFOCPT.value
DELAPND		netid.luname.FOCALPT.DELAPND
DELPEND		netid.luname.FOCALPT.DELPEND
DELETE		netid.luname.FOCALPT.DELETE
DF		netid.luname.FOCALPT.DEFFOCPT.value
DISPSOC		netid.luname.FOCALPT.DISPSOC

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
FOCALPT (continued) DROP DRP FPCAT INACTIVE OVERRIDE PN PRI PRIMARY PRINET Q QRY QUERY REFRESH TARGET TARGLIST TARGNET TG TN TYP TYPE UNKNOWN		netid.luname.FOCALPT.DROP netid.luname.FOCALPT.DROP netid.luname.FOCALPT.FPCAT.value netid.luname.FOCALPT.INACTIVE netid.luname.FOCALPT.OVERRIDE netid.luname.FOCALPT.PRINET.value netid.luname.FOCALPT.PRIMARY.value netid.luname.FOCALPT.PRIMARY.value netid.luname.FOCALPT.PRINET.value netid.luname.FOCALPT.QUERY netid.luname.FOCALPT.QUERY netid.luname.FOCALPT.QUERY netid.luname.FOCALPT.REFRESH netid.luname.FOCALPT.TARGET.value netid.luname.FOCALPT.TARGLIST.target_name netid.luname.FOCALPT.TARGNET.value netid.luname.FOCALPT.TARGET.value netid.luname.FOCALPT.TARGNET.value netid.luname.FOCALPT.FPCAT.value netid.luname.FOCALPT.FPCAT.value netid.luname.FOCALPT.UNKNOWN
FORCE E/AS		See NLDM command See Modify command
FPDLCMD		netid.luname.FPDLCMD
FPIC	CNME7013	netid.luname.CNME7013
FPT		See FOCALPT command
FREE ¹⁰ CATALOG DATASET DD DDN DDNAME DELETE F FI FILE HOLD KEEP NOHOLD SYSOUT UNCATALOG		netid.luname.FREE netid.luname.FREE.CATALOG netid.luname.FREE.DATASET netid.luname.FREE.FILE.ddname netid.luname.FREE.FILE.ddname netid.luname.FREE.FILE.ddname netid.luname.FREE.DELETE netid.luname.FREE.FILE.ddname netid.luname.FREE.FILE.ddname netid.luname.FREE.FILE.ddname netid.luname.FREE.HOLD netid.luname.FREE.KEEP netid.luname.FREE.NOHOLD netid.luname.FREE.SYSOUT.class netid.luname.FREE.UNCATALO
FTRACE	CNME0015	netid.luname.CNME0015

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
GEMDCR	CNME9535	netid.luname.CNME9535
GEMSETTH	CNME9506	netid.luname.CNME9506
GEMDEREG	CNME9512	netid.luname.CNME9512
GEMDREGC	CNME9533	netid.luname.CNME9533
GEMGETPI	CNME9525	netid.luname.CNME9525
GEMPPIRQ	CNME9524	netid.luname.CNME9524
GEMQRYCO	CNME9534	netid.luname.CNME9534
GEMQRYCN	CNME9515	netid.luname.CNME9515
GEMQRYL	CNME9509	netid.luname.CNME9509
GEMQRYPI	CNME9517	netid.luname.CNME9517
GEMQRYST	CNME9526	netid.luname.CNME9526
GEMQRYTH	CNME9516	netid.luname.CNME9516
GEMQRYTR	CNME9523	netid.luname.CNME9523
GEMQRYV	CNME9508	netid.luname.CNME9508
GEMREG	CNME9511	netid.luname.CNME9511
GEMREGC	CNME9532	netid.luname.CNME9532
GEMRNC	CNME9521	netid.luname.CNME9521
GEMRPTCO	CNME9522	netid.luname.CNME9522
GEMSETP	CNME9513	netid.luname.CNME9513
GEMSETPI	CNME9507	netid.luname.CNME9507
GEMSNDCC	CNME9520	netid.luname.CNME9520
GEMSNDHB	CNME9518	netid.luname.CNME9518
GEMSNDTH	CNME9519	netid.luname.CNME9519
GEMTHRSH	CNME9510	netid.luname.CNME9510
GENALERT		netid.luname.GENALERT
GETCONID ALERTPCT AUTH CONSOLE MIGRATE QLIMIT QRESUME STORAGE		netid.luname.GETCONID netid.luname.GETCONID.ALERTPCT.alertpct netid.luname.GETCONID.AUTH.value netid.luname.GETCONID.CONSOLE.console_name netid.luname.GETCONID.MIGRATE.value netid.luname.GETCONID.QLIMIT.qlimit netid.luname.GETCONID.QRESUME.qresume netid.luname.GETCONID.STORAGE.storage
GETMLINE		netid.luname.GETMLINE
GETMPRES		netid.luname.GETMPRES

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
GETMTFLG		netid.luname.GETMTFLG
GETMTYPE		netid.luname.GETMTYPE
GETMSIZE		netid.luname.GETMSIZE
GETPW		netid.luname.GETPW
GETTOPO		netid.luname.GETTOPO
GLOBALV * AUTOC AUTOT DEFC DEFT GETC GETT PURGEC PURGET PUTC PUTT RESTC RESTOREC RESTORET RESTT SAVEC SAVET TRACEC TRACET		netid.luname.GLOBALV netid.luname.GLOBALV.ASTERISK netid.luname.GLOBALV.AUTOC netid.luname.GLOBALV.AUTOT netid.luname.GLOBALV.DEFC netid.luname.GLOBALV.DEFT netid.luname.GLOBALV.GETC netid.luname.GLOBALV.GETT netid.luname.GLOBALV.PURGEC netid.luname.GLOBALV.PURGET netid.luname.GLOBALV.PUTC netid.luname.GLOBALV.PUTT netid.luname.GLOBALV.RESTOREC netid.luname.GLOBALV.RESTOREC netid.luname.GLOBALV.RESTORET netid.luname.GLOBALV.RESTRET netid.luname.GLOBALV.SAVEC netid.luname.GLOBALV.SAVET netid.luname.GLOBALV.TRACEC netid.luname.GLOBALV.TRACET
GMFHS	CNME2101	netid.luname.CNME2101
GO		netid.luname.GO ⁴
GROUPS	CNME0047	netid.luname.CNME0047
H		See HELP command
HD		See HELPDESK command
HELP		netid.luname.HELP
HELPDESK	CNME1026	netid.luname.CNME1026
HELPMMSG		See HELP command
HEXDEC	CNME1027	netid.luname.CNME1027
HIPERSOC	CNME8237	netid.luname.CNME8237

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
HLEENV ALLOC CHANGE CRITENVS DEFAULT LIST NORESET PHEAP PSTACK REGENVS RESET STATS TYPE		netid.luname.HLEENV netid.luname.HLEENV.ALLOC netid.luname.HLEENV.CHANGE netid.luname.HLEENV.CRITENVS.critenvs netid.luname.HLEENV.DEFAULT.PREINIT netid.luname.HLEENV.LIST netid.luname.HLEENV.NORESET netid.luname.HLEENV.PHEAP.value netid.luname.HLEENV.PSTACK.value netid.luname.HLEENV.REGENVS.regenvs netid.luname.HLEENV.RESET netid.luname.HLEENV.STATS netid.luname.HLEENV.TYPE.value
HM		See HELP command
HOLD		netid.luname.HOLD ⁴
IDCAMS		netid.luname.IDCAMS
IDLEOFF	CNME1057	netid.luname.CNME1057
IFSTAT	CNME8231	netid.luname.CNME8231
ILOG®		See page Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
IMR	CNME0016	netid.luname.CNME0016
INACT	CNME0017	netid.luname.CNME0017
INACTF	CNME0018	netid.luname.CNME0018
INDEX	CNME1024	netid.luname.CNME1024
INFORM		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
INFORMLG		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
INFORMTB		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
INFRMTBL		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
INIC	CNME7014	netid.luname.CNME7014
INITAPI	CNME9501	netid.luname.INME9501
INITCNFG	CNME0039	netid.luname.CNME0039
INITTOPO		netid.luname.INITTOPO
INPUT		netid.luname.INPUT

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
INSTORE <i>ddname/member</i>		netid.luname.DSIPIINS netid.luname.DSIPIINS.COMMON. <i>ddname/member</i>
IPCMD		See page Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199
IPTRACE	FKXE2A0A	netid.luname.FKXE2A0A
IOPD	CNME0019	netid.luname.CNME0019
JAVAACC TCPADDR		netid.luname.JAVAACC.TCPADDR.value ²⁵
KEEP		See NLDM command
LCL		See LOADCL command
LINEMAP		See NLDM command
LINES	CNME0020	netid.luname.CNME0020
LIST (NCCF) • CLIST <i>ddname.member</i> • CONNSEC • DST • OP • PROFILE <i>ddname.member</i> • SAFOP ¹¹ • SEGMENT ¹¹ • TCPCONN		netid.luname.LIST • netid.luname.LIST.CLIST netid.luname.READSEC. <i>ddname.member</i> • netid.luname.LIST.CONNSEC • netid.luname.LIST.DST. <i>dsname</i> • netid.luname.LIST.OP. <i>value</i> • netid.luname.LIST.PROFILE netid.luname.READSEC. <i>ddname.member</i> • netid.luname.LIST.SAFOP. <i>value</i> • netid.luname.LIST.SEGMENT. <i>value</i> • netid.luname.LIST.TCPCONN
LIST (NLDM)		See NLDM command
LISTA		netid.luname.LISTA
LISTALC		See LISTA command
LISTCAT		netid.luname.LISTCAT
LISTCMD	CNME1104	netid.luname.CNME1104
LISTSESS		netid.luname.LISTSESS
LISTVAR	CNME1006	netid.luname.CNME1006
LL2	CNME0021	netid.luname.CNME0021
LOADCL		netid.luname.LOADCL
LOG		See LOGOFF command
LOGAUTOD	CNME7018	netid.luname.CNME7018
LOGAUTOF	CNME7016	netid.luname.CNME7016

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
LOGAUTOI	CNME7017	netid.luname.CNME7017
LOGOFF		netid.luname.LOGOFF ⁴
LOGPROF1	CNME1049	netid.luname.CNME1049
LOGPROF2	CNME1032	netid.luname.CNME1032
LOGPROF3	CNME1033	netid.luname.CNME1033
LOGTSTAT		netid.luname.LOGTSTAT
LSESS	CNME1007	netid.luname.CNME1007
M		See MSG command
MAINMENU	CNME1066	netid.luname.CNME1066
MAJNODES	CNME0022	netid.luname.CNME0022
MAPCL		netid.luname.MAPCL
MCL		See MAPCL command
MEMLIST	CNME1058	netid.luname.CNME1058
MEMSTORE	CNME1054	netid.luname.CNME1054
MEMSTOUT	CNMEOUTS	netid.luname.CNMEOUTS
MIGRATE	CNME1084	netid.luname.CNME1084
MODIFY keyword		netid.luname.MODIFY netid.luname.MODIFY.keyword.value ⁶
MONIT ALL ID START STOP		netid.luname.MONIT netid.luname.MONIT.ALL netid.luname.MONIT.ID.nodename netid.luname.MONIT.START netid.luname.MONIT.STOP
MONOFF	CNME9001	netid.luname.CNME9001
MONON	CNME9002	netid.luname.CNME9002
MONOPERP	CNME7024	netid.luname.CNME7024
MSG		netid.luname.MSG
MSGROUTE BEEP HCYLOG HOLD NETLOG operatorid SYSLOG		netid.luname.MSGROUTE netid.luname.MSGROUTE.BEEP.value netid.luname.MSGROUTE.HCYLOG.value netid.luname.MSGROUTE.HOLD.value netid.luname.MSGROUTE.NETLOG.value netid.luname.MSGROUTE.operatorid netid.luname.MSGROUTE.SYSLOG.value

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
MVS ^{12 13 14} <i>token1</i> <i>token1.token2</i>		netid.luname.MVS netid.luname.MVS. <i>token1</i> netid.luname.MVS. <i>token1.token2</i>
MVSPING		See page Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
NACMD	CNME8200	netid.luname.CNME8200
NACTL LISTCONN LISTINFO LISTOPID START STOP	CNME7204	netid.luname.CNME7204 - netid.luname.CNME7204.LISTCONN - netid.luname.CNME7204.LISTINFO - netid.luname.CNME7204.LISTOPID - netid.luname.CNME7204.START - netid.luname.CNME7204.STOP
NCCF		netid.luname.NCCF
NCPDUMP	CNME0013	netid.luname.CNME0013
NCPSTOR	CNME0023	netid.luname.CNME0023
NETCONV ACTION IP LU OPID PORT STARTCON ²⁴ STOPCON ²⁴		netid.luname.NETCONV netid.luname.NETCONV.ACTION. <i>value</i> netid.luname.NETCONV.IP. <i>pcipid</i> netid.luname.NETCONV.LU. <i>pcluname</i> netid.luname.NETCONV.OPID. <i>value</i> netid.luname.NETCONV.PORT. <i>portnum</i> netid.luname.NETCONV.STARTCON. <i>sysdef</i> netid.luname.NETCONV.STOPCON. <i>sysdef</i>
NETWORK	CNME1060	netid.luname.CNME1060
NEWS	CNME1008	netid.luname.CNME1008

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
NLDM • AR • COLLECT • DISABLE • DISKEEP • ENABLE • ER • FORCE • KEEP • LINEMAP • LIST • P • PG • PT • PURGE • RECORD • RELOAD • RT • RTREND • RTSUM • SD • SDOMAIN • SENSE • SESS • SET DOMAIN • SG • SMDR • ST • TRACE • VR	CNME2003 CNME2004	netid.luname.NLDM • netid.luname.NLDM.AR • netid.luname.NLDM.COLLECT • netid.luname.NLDM.DISABLE • netid.luname.NLDM.DISKEEP • netid.luname.NLDM.ENABLE • netid.luname.NLDM.ER • netid.luname.NLDM.FORCE • netid.luname.NLDM.KEEP • netid.luname.NLDM.LINEMAP • netid.luname.NLDM.LIST • netid.luname.NLDM.P • netid.luname.NLDM.PG • netid.luname.NLDM.PT • netid.luname.NLDM.PURGE • netid.luname.NLDM.RECORD • netid.luname.NLDM.RELOAD • netid.luname.NLDM.RT • netid.luname.NLDM.RTREND • netid.luname.NLDM.RTSUM • netid.luname.NLDM.DOMAIN.value ¹⁵ • netid.luname.NLDM.DOMAIN.value ¹⁵ • netid.luname.NLDM.SENSE • netid.luname.NLDM.SESS • netid.luname.NLDM.DOMAIN.value ¹⁵ • netid.luname.NLDM.SG • netid.luname.NLDM.SMDR • netid.luname.NLDM.ST • netid.luname.NLDM.TRACE • netid.luname.NLDM.VR
NODE	CNME0024	netid.luname.CNME0024
NOSTAT	CNME0025	netid.luname.CNME0025

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
NPDA DEL P PRG PURGE PRGATT REPORTS SD SDOMAIN SR SRATIO SRF SRFILTER SW SWRAP		netid.luname.NPDA netid.luname.NPDA.DEL netid.luname.NPDA.P netid.luname.NPDA.PURGE netid.luname.NPDA.PURGE netid.luname.NPDA.PRATT netid.luname.NPDA.REPORTS netid.luname.NPDA.SDOMAIN netid.luname.NPDA.SDOMAIN netid.luname.NPDA.SRATIO netid.luname.NPDA.SRATIO netid.luname.NPDA.SRFILTER netid.luname.NPDA.SRFILTER netid.luname.NPDA.SWRAP.value netid.luname.NPDA.SWRAP.value
NRMCTL Listmon Listparm XCLDOM XCLTASK XCLTYPE		netid.luname.NRMCTL netid.luname.NRMCTL.Listmon netid.luname.NRMCTL.Listparm netid.luname.NRMCTL.XCLDOM netid.luname.NRMCTL.XCLTASK.value netid.luname.NRMCTL.XCLTYPE.value
NUMVERFY	CNME1031	netid.luname.CNME1031
NVSNMP		See page Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
NVSTAT	CNME8235	netid.luname.CNME8235
ORC		See ORCNTL command
ORCNTL CHKPT CHNG CONN DISC IMMED LIST NOAOREP OR REPOSIT TASK		netid.luname.ORCNTL netid.luname.ORCNTL.CHKPT netid.luname.ORCNTL.CHNG netid.luname.ORCNTL.CONN netid.luname.ORCNTL.DISC netid.luname.ORCNTL.IMMED netid.luname.ORCNTL.LIST netid.luname.ORCNTL.NOAOREP netid.luname.ORCNTL.OR.name netid.luname.ORCNTL.REPOSIT netid.luname.ORCNTL.TASK

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
ORCONV CLASS DATA ERROR FIELD ID MSGFIELD MSGPARMS OBJECT OBJINDEP OR PARM PARMTYPE SUBFIELD TYPE WAITF WAITT		netid.luname.ORCONV netid.luname.ORCONV.CLASS. <i>class_name</i> netid.luname.ORCONV.DATA. <i>field_data</i> netid.luname.ORCONV.ERROR. <i>err_routine</i> netid.luname.ORCONV.FIELD. <i>field_name</i> netid.luname.ORCONV.ID. <i>identifier</i> netid.luname.ORCONV.MSGFIELD. <i>message_field</i> netid.luname.ORCONV.MSGPARMS netid.luname.ORCONV.OBJECT. <i>object_name</i> netid.luname.ORCONV.OBJINDEP. <i>method</i> netid.luname.ORCONV.OR. <i>name</i> netid.luname.ORCONV.PARM. <i>method_parm</i> netid.luname.ORCONV.PARMTYPE. <i>method-parm_data_type</i> netid.luname.ORCONV.SUBFIELD. <i>value</i> netid.luname.ORCONV.TYPE. <i>value</i> netid.luname.ORCONV.WAITF. <i>value</i> netid.luname.ORCONV.WAITT. <i>integer</i>
OSAPORT	CNME8236	netid.luname.CNME8236
OSNMP BULKWALK GET GETBULK GETNEXT SET TRAP WALK		netid.luname.OSNMP netid.luname.OSNMP.BULKWALK. <i>mib_value</i> netid.luname.OSNMP.GET. <i>mib_value</i> netid.luname.OSNMP.GETBULK. <i>mib_value</i> netid.luname.OSNMP.GETNEXT. <i>mib_value</i> netid.luname.OSNMP.SET. <i>mib_value</i> netid.luname.OSNMP.TRAP. <i>mib_value</i> netid.luname.OSNMP.WALK. <i>mib_value</i>

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
OVERRIDE BEEP BNJPNL1 BNJPNL2 BRUNLOCK CMD CNMPNL1 CNM493I DISPLAY DSIARPT DSIASRC DSICLD DSILIST DSIMSG DSIOPEN DSIPARM DSIPRF DSIVTAM EMCSPARM EVERYCON HCYLOG HELPTXT HOLD LOGSPNCF LOGSPNCP LOGSPNVF LOGSPNVP LOGTSTAT LONGDATE LONGTIME MAXCPU MAXIO MAXMQIN MAXMQOUT MAXSTG MDCFGTIM MSGTOUT		netid.luname.OVERRIDE netid.luname.OVERRIDE.BEEP.value netid.luname.OVERRIDE.BNJPNL1.value netid.luname.OVERRIDE.BNJPNL2.value netid.luname.DEFAULTS.BRUNLOCK.value netid.luname.OVERRIDE.CMD.value netid.luname.DEFAULTS.CNMPNL1.value netid.luname.OVERRIDE.CNM493I.value netid.luname.OVERRIDE.DISPLAY.value netid.luname.OVERRIDE.DSIARPT.value netid.luname.OVERRIDE.DSIASRC.value netid.luname.OVERRIDE.DSICLD.value netid.luname.OVERRIDE.DSILIST.value netid.luname.OVERRIDE.DSIMSG.value netid.luname.OVERRIDE.DSIOPEN.value netid.luname.OVERRIDE.DSIPARM.value netid.luname.OVERRIDE.DSIPRF.value netid.luname.OVERRIDE.DSIVTAM.value netid.luname.OVERRIDE.EMCSPARM.value netid.luname.OVERRIDE.EVERYCON.value netid.luname.OVERRIDE.HCYLOG.value netid.luname.OVERRIDE.HELPTXT.value netid.luname.OVERRIDE.HOLD.value netid.luname.OVERRIDE.LOGSPNCF.value netid.luname.OVERRIDE.LOGSPNCP.value netid.luname.OVERRIDE.LOGSPNVF.value netid.luname.OVERRIDE.LOGSPNVP.value netid.luname.OVERRIDE.LOGTSTAT.value netid.luname.OVERRIDE.LONGDATE.value netid.luname.OVERRIDE.LONGTIME.value netid.luname.OVERRIDE.MAXCPU.value netid.luname.OVERRIDE.MAXIO.value netid.luname.OVERRIDE.MAXMQIN.value netid.luname.OVERRIDE.MAXMQOUT.value netid.luname.OVERRIDE.MAXSTG.value netid.luname.OVERRIDE.MDCFGTIM.mdcfgtime netid.luname.OVERRIDE.MSGTOUT.msgtonumber

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
OVERRIDE (continued) NETLOG NETVASIS REXXENV REXXENVL REXXSLMT REXXSTOR REXXSTRF RMTMAXL SCNFLOW SCRNFMT SCROLL SHORTDAT SHORTTIM SLOGCMDR SLOWSTG STARTCOL STMREFR SUPZDATE SUPZTIME SYSLOG TAFRECLN TASK TIMEFMSG WRNCPU WRNIO WRNMQIN WRNMQOUT WRNMSGCT WRNSTG		netid.luname.OVERRIDE.NETLOG.value netid.luname.OVERRIDE.NETVASIS.value netid.luname.OVERRIDE.REXXENV.value netid.luname.OVERRIDE.REXXENVL.value netid.luname.OVERRIDE.REXXSLMT.value netid.luname.OVERRIDE.REXXSTOR.value netid.luname.OVERRIDE.REXXSTRF.value netid.luname.OVERRIDE.RMTMAXL.value netid.luname.OVERRIDE.SCNFLOW.value netid.luname.OVERRIDE.SCRNFMT.member netid.luname.OVERRIDE.SCROLL.value netid.luname.OVERRIDE.SHORTDAT.value netid.luname.OVERRIDE.SHORTTIM.value netid.luname.OVERRIDE.SLOGCMDR.value netid.luname.OVERRIDE.SLOWSTG.value netid.luname.OVERRIDE.STARTCOL.value netid.luname.OVERRIDE.STMREFR.value netid.luname.OVERRIDE.SUPZDATE.value netid.luname.OVERRIDE.SUPZTIME.value netid.luname.OVERRIDE.SYSLOG.value netid.luname.OVERRIDE.TAFRECLN.value netid.luname.OVERRIDE.TASK.value netid.luname.OVERRIDE.TIMEFMSG.value netid.luname.DEFAULTS.WRNCPU.value netid.luname.DEFAULTS.WRNIO.value netid.luname.DEFAULTS.WRNMQIN.value netid.luname.DEFAULTS.WRNMQOUT.value netid.luname.DEFAULTS.WRNMSGCT.value netid.luname.DEFAULTS.WRNSTG.value
P		See NPDA command
PARSE		netid.luname.PARSE
PARSEL2R		netid.luname.PARSEL2R
PATHS	CNME0026	netid.luname.CNME0026
PDFILTER	CNME3004	netid.luname.CNME3004
PENDING	CNME0028	netid.luname.CNME0028
PING	CNMEPING	netid.luname.CNMEPING
PFKDEF	CNME1010	netid.luname.CNME1010

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
PKTS • COUNT • DEFINE • INTFNAME • LADDR • LPORT • MAXRECS • OPID • PORTNUM • PROTOCOL • PSOURCE • PURGE • QUERY • RADDR • RPORT • START • STOP • STOPCOLL • STORAGE • TCPNAME • TIME • TRUNCATE • WRITE		netid.luname.PKTS netid.luname.PKTS.COUNT.value netid.luname.PKTS.DEFINE netid.luname.PKTS.INTFNAME.value netid.luname.PKTS.LADDR.value netid.luname.PKTS.LPORT.value netid.luname.PKTS.MAXRECS.value netid.luname.PKTS.OPID.value netid.luname.PKTS.PORTNUM.value netid.luname.PKTS.PROTOCOL.value netid.luname.PKTS.PSOURCE.value netid.luname.PKTS.PURGE netid.luname.PKTS.QUERY netid.luname.PKTS.RADDR.value netid.luname.PKTS.RPORT.value netid.luname.PKTS.START netid.luname.PKTS.STOP netid.luname.PKTS.STOPCOLL netid.luname.PKTS.STORAGE.value netid.luname.PKTS.TCPNAME.value netid.luname.PKTS.TIME.value netid.luname.PKTS.TRUNCATE.value netid.luname.PKTS.WRITE
PLEXCTL • RANK • TAKEOVER		netid.luname.PLEXCTL netid.luname.PLEXCTL.RANK.new_rank netid.luname.PLEXCTL.TAKEOVER.value
POLICY REQ ENTRY TYPE SAFE Keyword	EZLEPOLY	netid.luname.EZLEPOLY netid.luname.EZLEPOLY.REQ.value netid.luname.EZLEPOLY.ENTRY.value netid.luname.EZLEPOLY.TYPE.value netid.luname.EZLEPOLY.SAFE.value netid.luname.EZLEPOLY.Keyword.value
PPI SEND RECEIVE		netid.luname.DSIPIPPI netid.luname.DSIPIPPI.SEND.rcvrname netid.luname.DSIPIPPI.RECEIVE.name
PPTUPDCG	CNME1082	netid.luname.CNME1082
PPTSETCG	CNME1083	netid.luname.CNME1083
PRG		See NPDA command

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
PRGATT		See NPDA command
PRINT	CNMECZPR	netid.luname.CNMECZPR
PURGE (NCCF) COSCONF DST OP REQ TIMER		netid.luname.PURGE netid.luname.PURGE.COSCONF netid.luname.PURGE.DST. <i>dstname</i> netid.luname.PURGE.OP. <i>value</i> netid.luname.PURGE.REQ. <i>value</i> netid.luname.PURGE.TIMER. <i>value</i>
PURGE (NLDM)		See NLDM command
PURGE (NPDA)		See NPDA command
PURGEDB	CNME2007	netid.luname.CNME2007
PWDSEC		netid.luname.PWDSEC.resourcename
Q		See QUEUE command
QHCL	CNME1011	netid.luname.CNME1011
QOS OP		netid.luname.QOS netid.luname.QOS.OP. <i>operid</i>
QREXX	CNME8002	netid.luname.CNME8002
QRS ACCLVL OP RESOURCE RODMOBID VIEW		netid.luname.QRS netid.luname.QRS.ACCLVL. <i>acclvl</i> netid.luname.QRS.OP. <i>operid</i> netid.luname.QRS.RESOURCE. <i>resource</i> netid.luname.QRS.RODMOBID. <i>value</i> netid.luname.QRS.VIEW. <i>value</i>
QRYGLOBL AUTOCNT BOTH COMMON FILE MODE REPLACE TASK VARS		netid.luname.QRYGLOBL netid.luname.QRYGLOBL.AUTOCNT netid.luname.QRYGLOBL.BOTH ² netid.luname.QRYGLOBL.COMMON netid.luname.QRYGLOBL.FILE. <i>filename</i> ³ netid.luname.QRYGLOBL.MODE. <i>modename</i> netid.luname.QRYGLOBL.REPLACE netid.luname.QRYGLOBL.TASK netid.luname.QRYGLOBL.VARS. <i>varspec</i>
QSAM stage DD DSN		netid.luname.READSEC.(ALLDSN) netid.luname.READSEC. <i>fully_qualified_dsname.member</i> ¹⁶ netid.luname.READSEC. <i>fully_qualified_dsname.member</i> ¹⁶
QUEUE		netid.luname.QUEUE
R		See REPLY command
RCFB	CNME0029	netid.luname.CNME0029

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
READSEC <i>ddname</i> <i>ddname.member</i> (all data set names) <i>dsname</i> <i>dsname.member</i>		netid.luname.READSEC netid.luname.READSEC. <i>ddname</i> netid.luname.READSEC. <i>ddname.member</i> netid.luname.READSEC.(ALLDSN) ¹⁷ netid.luname.READSEC. <i>fully_qualified_dsname</i> ¹⁸ netid.luname.READSEC. <i>fully_qualified_dsname.member</i> ¹⁶
RECORD		See NLDM command
RECYCLE	CNME0030	netid.luname.CNME0030
RECYCLET	CNME1089	netid.luname.CNME1089
REDIAL	CNME0031	netid.luname.CNME0031
REFRESH AUTHCHK BACKTBL CMDAUTH DELVTMLS OPERS OPERSEC OPSPAN RMTAUTH RMTSEC SAFNODEC SPANAUTH SPANCHK SPANTBL SURROGAT TBLNAME TEST WEBAUTH		netid.luname.REFRESH netid.luname.REFRESH.AUTHCHK. <i>value</i> netid.luname.REFRESH.BACKTBL. <i>backtbl_name</i> netid.luname.REFRESH.CMDAUTH. <i>value</i> netid.luname.REFRESH.DELVTMLS. <i>value</i> netid.luname.REFRESH.OPERS netid.luname.REFRESH.OPERSEC. <i>value</i> netid.luname.REFRESH.OPSPAN. <i>value</i> netid.luname.REFRESH.RMTAUTH. <i>value</i> netid.luname.REFRESH.RMTSEC. <i>value</i> netid.luname.REFRESH.SAFNODEC. <i>value</i> netid.luname.REFRESH.SPANAUTH. <i>value</i> netid.luname.REFRESH.SPANCHK. <i>value</i> netid.luname.REFRESH.SPANTBL. <i>value</i> netid.luname.REFRESH.SURROGAT. <i>value</i> netid.luname.REFRESH.TBLNAME. <i>name</i> netid.luname.REFRESH.TBLNAME.DSISECUR ² netid.luname.REFRESH.TEST netid.luname.REFRESH.WEBAUTH. <i>value</i>
REGISTER APPL COMMAND FOCALPT FPCAT LOGMODE NOTIFY PRI QUERY REPLACE TYPE		netid.luname.REGISTER netid.luname.REGISTER.APPL. <i>applname</i> netid.luname.REGISTER.COMMAND. <i>cmdname</i> netid.luname.REGISTER.FOCALPT. <i>value</i> netid.luname.REGISTER.FPCAT. <i>fp_cat</i> netid.luname.REGISTER.LOGMODE. <i>logmode_name</i> netid.luname.REGISTER.NOTIFY. <i>value</i> netid.luname.REGISTER.PRI. <i>value</i> netid.luname.REGISTER.QUERY. <i>value</i> netid.luname.REGISTER.REPLACE. <i>value</i> netid.luname.REGISTER.TYPE. <i>value</i>
REL	CNME0032	netid.luname.CNME0032

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
RELCONID SWITCH		netid.luname.RELCONID netid.luname.RELCONID.SWITCH
RELOAD		See NLDM command
REMOTEBR		netid.luname.REMOTEBR
REMOVBJ		netid.luname.REMOVBJ
REPLY		netid.luname.REPLY
REPORTS		See NPDA command
REQMS		netid.luname.REQMS
RES		See RESOURCE command
RESET		netid.luname.RESET ⁴
RESETDB		netid.luname.RESETDB
RESOURCE		netid.luname.RESOURCE
RESTOPO		netid.luname.RESTOPO
RESTORE DELETE TIMER		netid.luname.RESTORE netid.luname.RESTORE.DELETE netid.luname.RESTORE.TIMER
RESTYLE		netid.luname.RESTYLE
RETURN		netid.luname.RETURN
REVISE ALL CMD MEMBER MSG OFF REPORT STATUS		netid.luname.REVISE netid.luname.REVISE.ALL netid.luname.REVISE.CMD netid.luname.REVISE.MEMBER.value netid.luname.REVISE.MSG netid.luname.REVISE.OFF netid.luname.REVISE.REPORT netid.luname.REVISE.STATUS
REVISMSG MEMBER OFF REPORT STATUS		netid.luname.REVISMSG netid.luname.REVISMSG.MEMBER.value netid.luname.REVISMSG.OFF netid.luname.REVISMSG.REPORT netid.luname.REVISMSG.STATUS
RID		netid.luname.RID
RGWY		See Appendix B, “AON Commands, Keywords, and Values that Can Be Protected,” on page 199.
RMTACC TCPADDR NETIDDOM		netid.luname.RMTACC.TCPADDR.value netid.luname.RMTACC.NETIDDOM.value

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
RMTCMD DOMAIN IP (for query) IP (for send) IPV6 LCLAUTOS LU NETID OPERID PORT QUERY RMTAUTOS RMTDOMS RMTLUS SEND TASKID		netid.luname.DSIUSNDM netid.luname.DSIUSNDM.DOMAIN. <i>value</i> netid.luname.DSIUSNDM.IP netid.luname.DSIUSNDM.IP. <i>value</i> netid.luname.DSIUSNDM.IPV6 netid.luname.DSIUSNDM.LCLAUTOS netid.luname.DSIUSNDM.LU. <i>domname</i> netid.luname.DSIUSNDM.NETID. <i>netid</i> netid.luname.DSIUSNDM.OPERID. <i>operid</i> ¹⁹ netid.luname.DSIUSNDM.PORT. <i>value</i> netid.luname.DSIUSNDM.QUERY netid.luname.DSIUSNDM.RMTAUTOS netid.luname.DSIUSNDM.RMTDOMS netid.luname.DSIUSNDM.RMTLUS netid.luname.DSIUSNDM.SEND ² netid.luname.DSIUSNDM.TASKID. <i>taskid</i>
RMTSEND		See RMTCMD
RMTSESS	CNME1092	netid.luname.CNME1092
RODM	CNME1098	netid.luname.CNME1098
ROLL		netid.luname.ROLL
ROUTE		netid.luname.ROUTE
RSESS	CNME1012	netid.luname.CNME1012
RT		See ROUTE command
RTREND		See NLDM command
RTRINIT		netid.luname.RTRINIT
RTRNSESS		See BGNSESS command
RTRQUEUE		netid.luname.RTRQUEUE
RTSUM		See NLDM command
RTTBL		netid.luname.RTTBL
RUNCMD APPL CLISTVAR NETID SP		netid.luname.RUNCMD netid.luname.RUNCMD.APPL. <i>application-name</i> netid.luname.RUNCMD.CLISTVAR. <i>value</i> netid.luname.RUNCMD.NETID. <i>netid</i> netid.luname.RUNCMD.SP. <i>service-point-name</i>
SAVECMD	CNME6221	netid.luname.CNME6221
SD		See NPDA command
SDOMAIN		See NPDA command
SECMIGR	CNME8004	netid.luname.CNME8004

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
SEEOPCTL		See DSILMEXP command
SEEOPTCL		See DSILMEXP command
SENDSSESS		netid.luname.SENDSSESS
SENSE	CNME2003	Issued from NLDM, see NLDM command netid.luname.CNME2003 (issued from NCCF)
SEQUENT MONITOR MONOPER		netid.luname.SEQUENT netid.luname.SEQUENT.MONITOR. <i>name</i> netid.luname.SEQUENT.MONOPER. <i>name</i>
SESMGET	CNME2011	netid.luname.CNME2011
SESS	CNME2004	Issued from NLDM, see NLDM command netid.luname.CNME2004 (issued from NCCF)
SESSC	CNME2012	netid.luname.CNME2012
SESSIONS	CNME0048	netid.luname.CNME0048
SESSMDIS		netid.luname.SESSMDIS
SET		netid.luname.SET
SETBQL <i>receiver_id</i>		netid.luname.SETBQL netid.luname.SETBQL. <i>receiver_id</i>
SETADIAL	CNME7012	netid.luname.CNME7012
SETCONID CONSOLE		netid.luname.SETCONID netid.luname.SETCONID.CONSOLE. <i>console_name</i>
SETRVAR		netid.luname.SETRVAR
SETTINGS E/AS		See Modify command
SHOWCODE	CNME5002	netid.luname.CNME5002
SMDR		See NLDM command
SNMP BULKWALK GET GETBULK GETNEXT INFORM SET TRAP WALK	CNMESNMP	netid.luname.CNMESNMP.BULKWALK netid.luname.CNMESNMP.GET netid.luname.CNMESNMP.GETBULK netid.luname.CNMESNMP.GETNEXT netid.luname.CNMESNMP.INFORM netid.luname.CNMESNMP.SET netid.luname.CNMESNMP.TRAP netid.luname.CNMESNMP.WALK
SNMPVIEW	FKXE280A	netid.luname.FKXE280A

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
SOCACC TCPADDR		netid.luname.SOCACC.TCPADDR. <i>value</i>
SOCKET FAMILY SOCKTYPE TYPE		netid.luname.SOCKET netid.luname.SOCKET.FAMILY. <i>value</i> netid.luname.SOCKET.SOCKTYPE. <i>value</i> netid.luname.SOCKET.TYPE. <i>value</i> Note: 1. The SOCKTYPE keyword is checked only when TYPE=SOCKET is being used. 2. The FAMILY keyword is checked only when TYPE=SOCKET or TYPE=TAKESOCKET is being used.
SOLICIT		netid.luname.SOLICIT
SPLOOKUP	CNME0041	netid.luname.CNME0041
SQSELECT	CNMSQSEL	netid.luname.CNMSQSEL
SRVRNV	CNME0054	netid.luname.CNME0054
STACK		netid.luname.STACK
STACSTAT	CNME8230	netid.luname.CNME8230
START DOMAIN FILE HCL LOGMODE MEM MEMBER MOD OP PRI SPAN TASK TSOSERV UNIXSERV XCFGROUP		netid.luname.START netid.luname.START.DOMAIN. <i>dom</i> netid.luname.START.MEM. <i>member</i> netid.luname.START.HCL. <i>hclname</i> netid.luname.START.LOGMODE. <i>modename</i> netid.luname.START.MEM. <i>member</i> netid.luname.START.MEM. <i>member</i> netid.luname.START.MOD. <i>module</i> netid.luname.START.OP. <i>opname</i> netid.luname.START.PRI. <i>taskpri</i> netid.luname.START.SPAN. <i>spanname</i> netid.luname.START.TASK. <i>taskname</i> netid.luname.START.TSOSERV. <i>tsouserid</i> netid.luname.START.UNIXSERV.+ netid.luname.START.XCFGROUP. <i>groupname</i>
STARTCNM	CNME1015	netid.luname.CNME1015
STARTDOM	CNME7001	netid.luname.CNME7001
STATIONS	CNME0033	netid.luname.CNME0033
STATMON		netid.luname.STATMON
STATS	CNME3005	netid.luname.CNME3005
STATUS(NCCF)	CNME0034	netid.luname.CNME0034

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
STOP FORCE TASK UNCOND TSOSERV XCFGROUP		netid.luname.STOP netid.luname.STOP.FORCE. <i>value</i> netid.luname.STOP.TASK. <i>taskname</i> netid.luname.STOP.UNCOND. <i>value</i> netid.luname.STOP.TSOSERV. <i>tsouserid</i> netid.luname.STOP.XCFGROUP. <i>groupname</i>
STOPCNM	CNME1016	netid.luname.CNME1016
STOPNA	CNME7201	netid.luname.CNME7201
STOPSESS	CNME7020	netid.luname.CNME7020
SUBMIT for DSIPARM <i>dsname</i> <i>dsname.member</i> for other data sets		netid.luname.SUBMIT netid.luname.SUBMIT.DSIPARM. <i>jobname</i> netid.luname.SUBMIT. <i>fully_qualified_dsname</i> ¹⁸ netid.luname.SUBMIT. <i>fully_qualified_dsname.member</i> ¹⁶ netid.luname.SUBMIT.DATASET
SUSPTOPO		netid.luname.SUSPTOPO
SW		See NPDA command
SWITCH DSILOG DSITRACE <i>taskname</i>		netid.luname.SWITCH netid.luname.SWITCH.DSILOG netid.luname.SWITCH.DSITRACE netid.luname.SWITCH. <i>taskname</i>
SWLD	CNME2002	netid.luname.CNME2002
SWPD	CNME3006	netid.luname.CNME3006
SWRAP		See NPDA command
TASKMON	CNME1100	netid.luname.CNME1100
TASKURPT	CNME1101	netid.luname.CNME1101
TASKUTIL DURATION SORT <i>taskname</i> TYPE		netid.luname.TASKUTIL netid.luname.TASKUTIL.DURATION netid.luname.TASKUTIL.SORT. <i>sortfield</i> netid.luname.TASKUTIL. <i>taskname</i> netid.luname.TASKUTIL.TYPE. <i>value</i>

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
TCPCONN APPLNAME COUNT DEFINE ENDTIME FORCE GTF JOBNAME LADDR LPORT LUNAME MAXRECS OPID PURGE QUERY QUERYACT RADDR RPORT SELECT STACK START STARTIME STOP		netid.luname.TCPCONN netid.luname.TCPCONN.APPLNAME.value netid.luname.TCPCONN.COUNT.value netid.luname.TCPCONN.DEFINE netid.luname.TCPCONN.ENDTIME.value netid.luname.TCPCONN.FORCE.value netid.luname.TCPCONN.GTF.value netid.luname.TCPCONN.JOBNAME.value netid.luname.TCPCONN.LADDR.value netid.luname.TCPCONN.LPORT.value netid.luname.TCPCONN.LUNAME.value netid.luname.TCPCONN.MAXRECS.value netid.luname.TCPCONN.OPID.value netid.luname.TCPCONN.PURGE netid.luname.TCPCONN.QUERY netid.luname.TCPCONN.QUERYACT netid.luname.TCPCONN.RADDR.value netid.luname.TCPCONN.RPORT.value netid.luname.TCPCONN.SELECT.value netid.luname.TCPCONN.STACK.value netid.luname.TCPCONN.START netid.luname.TCPCONN.STARTIME.value netid.luname.TCPCONN.STOP
TE		netid.luname.TE ⁴
TELSTAT	CNME8232	netid.luname.CNME8232
TERMAPI	CNME9502	netid.luname.CNME9502
TERMS	CNME0035	netid.luname.CNME0035
TEST		See NPDA command
TESTPORT	CNMETSTL	netid.luname.CNMETSTL
TESTRCMD	CNME0049	netid.luname.CNME0049
TESTSP	CNME0043	netid.luname.CNME0043
TIMEP		netid.luname.TIMEP
TIMER TARGET	EZLE600A	netid.luname.EZLE600A netid.luname.EZLE600A.target.value
TIMERS	EZLE600A	netid.luname.EZLE600A
TIMR	EZLE600A	netid.luname.EZLE600A
TNPTSTAT	CNME8233	netid.luname.CNME8233
TNSTAT	CNME0036	netid.luname.CNME0036

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
TOPOSNA • ACTIVATE LCLNAME NODE OBJECTID RESTYPE • CRITICAL LIST STARTMON STOPMON TYPE • INACT LCLNAME NODE OBJECTID RESTYPE WAIT • LISTPOOL • LISTREQS • LISTRODM • LISTSTOR • MONITOR – LOCAL NODE OBJECTID • LUCOL LCLNAME NODE OBJECTID • NETWORK NODE OBJECTID • MONTIME • PURGE PURGDAYS • QUERYDEF • RECYCLE LCLNAME NODE OBJECTID RESTYPE		netid.luname.TOPOSNA • netid.luname.TOPOSNA.ACTIVATE netid.luname.TOPOSNA.LCLNAME.value ²⁰ netid.luname.TOPOSNA.NODE.value ^{20 21} netid.luname.TOPOSNA.OBJECTID.value, ^{22 20} netid.luname.TOPOSNA.RESTYPE.value ²⁰ • netid.luname.TOPOSNA.CRITICAL netid.luname.TOPOSNA.LIST netid.luname.TOPOSNA.STARTMON.value netid.luname.TOPOSNA.STOPMON.value netid.luname.TOPOSNA.TYPE.value • netid.luname.TOPOSNA.INACT netid.luname.TOPOSNA.LCLNAME.value ²⁰ netid.luname.TOPOSNA.NODE.value ^{20 21} netid.luname.TOPOSNA.OBJECTID.value ^{20 22} netid.luname.TOPOSNA.RESTYPE.value ²⁰ netid.luname.TOPOSNA.WAIT.value • netid.luname.TOPOSNA.LISTPOOL • netid.luname.TOPOSNA.LISTREQS • netid.luname.TOPOSNA.LISTRODM • netid.luname.TOPOSNA.LISTSTOR • • netid.luname.TOPOSNA.MONITOR • netid.luname.TOPOSNA.LOCAL netid.luname.TOPOSNA.NODE.value ^{20 21} netid.luname.TOPOSNA.OBJECTID.value ^{20 22} • netid.luname.TOPOSNA.LUCOL ²⁰ netid.luname.TOPOSNA.LCLNAME.value ²⁰ netid.luname.TOPOSNA.NODE.value ^{20 21} netid.luname.TOPOSNA.OBJECTID.value ^{20 22} • netid.luname.TOPOSNA.NETWORK netid.luname.TOPOSNA.NODE.value ^{20 21} netid.luname.TOPOSNA.OBJECTID.value ^{20 22} • netid.luname.TOPOSNA.MONTIME.value • netid.luname.TOPOSNA.PURGE netid.luname.TOPOSNA.PURGDAYS.value • netid.luname.TOPOSNA.QUERYDEF • netid.luname.TOPOSNA.RECYCLE netid.luname.TOPOSNA.LCLNAME.value ²⁰ netid.luname.TOPOSNA.NODE.value ^{20 21} netid.luname.TOPOSNA.OBJECTID.value ^{20 22} netid.luname.TOPOSNA.RESTYPE.value ²⁰

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
TOPOSNA (continued) • REFRESH ALLTABLS CLASS EXVIEW OSIDISP RESOLVE • SETDEFS – AUTOMON ALL value ENLOCAL value NNLOCAL value SALocal value SANET value – ERRLIMIT – CMPRETRY – LCLRETRY – LURETRY – NETRETRY – RDMRETRY • STOP – LOCAL NODE OBJECTID – LUCOL LCLNAME NODE OBJECTID – NETWORK NODE OBJECTID • STOPMGR		• netid.luname.TOPOSNA.REFRESH - netid.luname.TOPOSNA.ALLTABLS - netid.luname.TOPOSNA.CLASS.value - netid.luname.TOPOSNA.EXVIEW - netid.luname.TOPOSNA.OSIDISP - netid.luname.TOPOSNA.RESOLVE • netid.luname.TOPOSNA.SETDEFS – netid.luname.TOPOSNA.AUTOMON.value - netid.luname.TOPOSNA.AUTOMON.ALL - netid.luname.TOPOSNA.ALL.value - netid.luname.TOPOSNA.AUTOMON.ENLOCAL - netid.luname.TOPOSNA.ENLOCAL.value - netid.luname.TOPOSNA.AUTOMON.NNLOCAL - netid.luname.TOPOSNA.NNLOCAL.value - netid.luname.TOPOSNA.AUTOMON.SALocal - netid.luname.TOPOSNA.SALocal.value - netid.luname.TOPOSNA.AUTOMON.SANET - netid.luname.TOPOSNA.SANET.value – netid.luname.TOPOSNA.ERRLIMIT.value – netid.luname.TOPOSNA.CMPRETRY.value – netid.luname.TOPOSNA.LCLRETRY.value – netid.luname.TOPOSNA.LURETRY.value – netid.luname.TOPOSNA.NETRETRY.value – netid.luname.TOPOSNA.RDMRETRY.value • netid.luname.TOPOSNA.STOP – netid.luname.TOPOSNA.LOCAL - netid.luname.TOPOSNA.NODE.value^{20 21} - netid.luname.TOPOSNA.OBJECTID.value^{20 22} – netid.luname.TOPOSNA.LUCOL - netid.luname.TOPOSNA.LCLNAME.value²⁰ - netid.luname.TOPOSNA.NODE.value^{20 21} - netid.luname.TOPOSNA.OBJECTID.value^{20 22} – netid.luname.TOPOSNA.NETWORK - netid.luname.TOPOSNA.NODE.value^{20 21} - netid.luname.TOPOSNA.OBJECTID.value^{20 22} • netid.luname.TOPOSNA.STOPMGR - netid.luname.TOPOSNA.NODE.value^{20 21} - netid.luname.TOPOSNA.OBJECTID.value^{20 22}

Table 11. NetView Command Identifiers (continued)

Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
TOPOSNA (continued) • TRACE CLASS MODE OFF ON QUERY SIZE		• netid.luname.TOPOSNA.TRACE netid.luname.TOPOSNA.CLASS. <i>value</i> netid.luname.TOPOSNA.MODE. <i>value</i> netid.luname.TOPOSNA.OFF. <i>value</i> netid.luname.TOPOSNA.ON. <i>value</i> netid.luname.TOPOSNA.QUERY netid.luname.TOPOSNA.SIZE. <i>value</i>
TRACE (NCCF)		netid.luname.TRACE
TRACE E/AS		See NLDM command See Modify command
TRACERTE	CNMETRTE	netid.luname.CNMETRTE
TRACERT	CNMETRTE	netid.luname.CNMETRTE
TRANRCV		netid.luname.TRANRCV
TRANMSG MEMBER		netid.luname.TRANMSG netid.luname.TRANMSG.MEMBER. <i>member-name</i>
TRANSND		netid.luname.TRANSND
TRAP		netid.luname.TRAP
TRCROUTE	CNMETRTE	netid.luname.CNMETRTE
TRCRTE	CNMETRTE	netid.luname.CNMETRTE
TRCRT	CNMETRTE	netid.luname.CNMETRTE
TS		netid.luname.TS ⁴
TSO stage to protect servers to protect TSO commands ²³		netid.luname.DSIPITSO netid.luname.DSIPITSO.TSOSERV.userid/job netid.luname.DSIPITSO.VERB.tso_command
TSOUSER	CNME0037	netid.luname.CNME0037
TUTOR	CNME5003	netid.luname.CNME5003
UNALLOC		See FREE command
UNIQUE		netid.luname.UNIQUE
UNIX		netid.luname.DSIPIUNX
UNSTACK		netid.luname.UNSTACK
UPPER		netid.luname.UPPER
V		See VARY command

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
VARY keyword		netid.luname.VARY ⁶ netid.luname.VARY.keyword.value ⁶
VBSERV		netid.luname.EKGBSERV
VIPAROUT	CNME8216	netid.luname.CNME8216
VPDALL ADD CLIST CONFIG CREATE ERROR EXEC EXECUTE NOERROR OPER OPERATOR REPLACE		netid.luname.VPDALL netid.luname.VPDALL.ADD ² netid.luname.VPDALL.CLIST.clist name netid.luname.VPDALL.CONFIG.vtam configuration member netid.luname.VPDALL.CREATE ² netid.luname.VPDALL.ERROR netid.luname.VPDALL.EXEC netid.luname.VPDALL.EXECUTE netid.luname.VPDALL.NOERROR netid.luname.VPDALL.OPER.operid netid.luname.VPDALL.OPERATOR.operid netid.luname.VPDALL.REPLACE
VPDCMD		netid.luname.VPDCMD
VPDDCE	CNME0052	netid.luname.CNME0052
VPDLOG		netid.luname.VPDLOG
VPDLOGC	CNME0050	netid.luname.CNME0050
VPDPU	CNME0051	netid.luname.CNME0051
VPDXDOM	CNME0053	netid.luname.CNME0053
VRST	CNME0038	netid.luname.CNME0038
VSAMPOOL		netid.luname.VSAMPOOL
WEBACC TCPADDR		netid.luname.WEBACC.TCPADDR.value
WEBCMD		netid.luname.WEBCMD
WEBSITES	CNMEHTSP	See CNMEHTSP command
WHO	CNME1019	netid.luname.CNME1019
WINDOW	CNME1505	netid.luname.CNME1505
WRAP	CNME1020	netid.luname.CNME1020
WRITE	CNME7019	netid.luname.CNME7019

Table 11. NetView Command Identifiers (continued)		
Commands and Keywords	Command List Name	SAF Resource or Command Authorization Table Identifier
WRITESEC <i>ddname</i> (all data set names) <i>dsname</i> <i>dsname.member</i>		netid.luname.WRITESEC netid.luname.WRITESEC. <i>ddname</i> netid.luname.WRITESEC.(ALLDSN) ¹⁷ netid.luname.WRITESEC. <i>fully_qualified_dsname</i> ¹⁸ netid.luname.WRITESEC. <i>fully_qualified_dsname.member</i> ¹⁶
WTO		netid.luname.WTO
WTOR		netid.luname.WTOR
XCFMSG stage		netid.luname.DSIPIXCF
XCFTABLE stage		netid.luname.DSIPIXTB
< (From Disk) stage <i>ddname</i>		(not protectable) netid.luname.READSEC. <i>ddname</i>

Table notes:

1. If ROUTE=* or ROUTE=*myoperid* is specified, no security checking is performed for the ROUTE keyword.
2. Defaulted keyword always checked.
3. In addition to command authorization, the task needs write access to the data set. For information, see Chapter 5, “Controlling Access to Data Sets and Members,” on page 87.
4. When issued as an immediate command, only the NetView command authorization table can provide command security, as described in [“Protecting Immediate Commands When CMDAUTH=SAF”](#) on page 52.
5. DSILSSIR SHOWMSG is used in CNME1103 which runs on the PPT task. CNME1103 is used by the DSITBL01 sample automation table to automate message BNH535A.
6. The keyword/value can be any VTAM command keyword/value pair or just the keyword, if the keyword does not accept any value.

If the keyword is ID, SLU, PLU, LU1, LU2, IPADDR (excluding START command), or DVIPA, the value is a VTAM resource name or IP address. If the VTAM resource name is qualified by the network ID, the network ID, and resource name must be protected separately (for example, in separate PROTECT statements). The network ID and resource names can be up to 8 characters long.

If IDTYPE=IPADDR is entered with the DISPLAY command, the value of the ID keyword is an IP address. The value of IPADDR (MODIFY/DISPLAY VTAMOPTS command) or DVIPA(DISPLAY STATS command) is also an IP address. The IP address can have more than 8 characters. Refer to [Appendix F, “How to Protect IP Addresses for Command Security and Span of Control,”](#) on page 231 for examples of protecting IP addresses.

7. This keyword uses READSEC only.
8. This keyword uses WRITESEC only.
9. The *cmd* is treated as a value of the *opid* keyword. When protecting the target verb of EXCMD, specify the command verb and not any command synonym. Unless otherwise documented, the verb is either (1) the command name specified on the CMDDEF statement or (2) the value of the NAME keyword of your ADDCMD command.

Note: The verb for RMTCMD and for remote labeled commands is DSIUSNDM. The verb for labeled commands beginning with a slash is EXCMD.

10. The list of FREE keywords might not be complete. Check DSIUNALL for the complete synonym list.
11. For an operator to see more than their own base segment information, the permission must be authorized in the field class. For additional information about field class refer to *Resource Access Control Facility Administrator's Guide (RACF)*.
12. If you are using CMDDEF statements with MOD=CNMCMJC to enter MVS commands, you need to create different to protect those commands when using the command authorization table or the NETCMDS class. Entering MVS commands without the MVS prefix is illustrated in the CNMS6401, CNMS6402, and CNMS6403 samples.

For example, to protect the NetView command MVS D T, the command identifier is "netid.luname.MVS.D.T". However, if D is defined with a CMDDEF with MOD=CNMCMJC, to protect D T when issued without the MVS prefix, the command identifier is "netid.luname.D.T".
13. If the MVS command ROUTE is issued from a NetView task, the originating source ID is always passed to the SAF product for authorization checks in the OPERCMDS class. This occurs for all settings of AUTHCHK and CMDAUTH.
14. Because extended MCS consoles are used for issuing MVS commands, operators must have authority to an mvs.MCSOPER.console_name profile in the OPERCMDS class. See ["Protecting EMCS Console Names Using an SAF Product"](#) on page 27.
15. If the CP keyword is specified with the SD, SDOMAIN, or SET DOMAIN subcommands of the NLDM command, the authorization check is performed against the domain name for that CP specification, not against the value specified with the CP keyword.
16. You can protect fully qualified data set names only when using an SAF product such as RACF or the NetView command authorization table. The format of the command identifier must include the fully qualified data set name with an optional member name, for example:

netid.luname.READSEC.hi_qualif/mid_qualif/lo_qualif.membername
17. ALLDSN is an all-or-nothing switch that was intended for use with CMDAUTH=SCOPE. As scope of commands is no longer supported, use generics for the data set names rather than ALLDSN.
18. You can protect fully qualified data set names only when using an SAF product such as RACF or the NetView command authorization table. The format of the command identifier must include the fully qualified data set name, for example:

netid.luname.WRITESEC.hi_qualif/mid_qualif/lo_qualif
19. If OPERID=* is specified, no security checking is performed for the OPERID keyword.
20. The command identifier is identical to other command identifiers for other keywords on this command. Use caution when defining security with this command identifier because it applies to multiple, unrelated keywords.
21. For a network qualified name supplied as a value, up to the first eight characters of each field are checked.
22. For OBJECTID values greater than eight characters, only the first eight characters are checked.
23. For additional information refer to ["Defining TSO Stage Authorization"](#) on page 53.
24. To use this NETCONV keyword, the operator must also have the authority to issue the EXCMD command.
25. The JAVAACC command is used by the REXEC server and the NetView correlation service.

Appendix B. AON Commands, Keywords, and Values that Can Be Protected

This section lists the AON base commands, command synonyms, and command lists that can be protected using the NetView command authorization table or a system authorization facility (SAF) product such as Resource Access Control Facility (RACF).

The following table contains the commands for base AON:

For more information about defining command authorization, refer to the *IBM Z NetView Administration Reference*.

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected		
Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
ACTMON	EZLE450A	netid.luname.EZLE450A
AHED	EZLE100A	netid.luname.EZLE100A
ANO	EZLE000	netid.luname.EZLE0000
ANOMENU	EZLE000	netid.luname.EZLE0000
AON	EZLE000	netid.luname.EZLE0000
AONAIP	EZLESAIP	netid.luname.EZLESAIP
AONCTRL	EZLE400A	netid.luname.EZLE400A
AONENABL	EZLE830A	netid.luname.EZLE830A
AONGW	EZLE520A	netid.luname.EZLE520A
AONHD	EZLE100A	netid.luname.EZLE100A
AONINFO	EZLT0000	netid.luname.EZLT0000
AONINIT	EZLE820A	netid.luname.EZLE820A
AONMAINT	EZLE700A	netid.luname.EZLE700A
AONOIV	EZLEOIVS	netid.luname.EZLEOVIS
AONTAF	EZLE540A	netid.luname.EZLE540A
AONTASK	EZLE770A	netid.luname.EZLE770A

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
AONTRACE	EZLE810A	netid.luname.EZLE810A
AUTOAIP	EZLESAIP	netid.luname.EZLESAIP
AUROCMD	EZLEF002	netid.luname.EZLEF002
AUTOMAN	EZLEAMAN	netid.luname.EZLEAMAN
AUTOOIV	EZLEOIVS	netid.luname.EZLEOVIS
AUTOSET	EZLE400A	netid.luname.EZLE400A
AUTOVIEW	EZLE200A	netid.luname.EZLE200A
CDLOG	EZLE500B	netid.luname.EZLE500B
CGED	EZLE840A	netid.luname.EZLE840A
CGLOBAL		netid.luname.EZLSVLST
CLEARSTS	EZLE750A	netid.luname.EZLE750A
DBMAINT	EZLE750A	netid.luname.EZLE750A
DDF		netid.luname.EZLODDF
DDFADD		netid.luname.EZLADDF
DDFASGN	EZLEBASN	netid.luname.EZLBASN
DDFCLEAR	EZLEADCR	netid.luname.EZLEADCR
DDFDEL		netid.luname.EZLDDDF
DDFPANEL		netid.luname.EZLPDDF
DDFPANL		netid.luname.EZLPDDF
DDFQRY		netid.luname.EZLQDDF
DDFTREE		netid.luname.EZLPDDF

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
DDFUNAS	EZLEBUAS	netid.luname.EZLEBUAS
DDFUPD	EZLEAXST	netid.luname.EZLEAXST
DELAUTO	EZLE412A	netid.luname.EZLE412A
DELMONIT	EZLE441A	netid.luname.EZLE441A
DELNTFY	EZLE421A	netid.luname.EZLE421A
DELTHRES	EZLE431A	netid.luname.EZLE431A
DETAIL	EZLEADET	netid.luname.EZLEADET
DISAUTO	EZLE411A	netid.luname.EZLE411A
DISCFG	EZLE710A	netid.luname.EZLE710A
DISNODE	EZLE200A	netid.luname.EZLE200A
DISNTFY	EZLE420A	netid.luname.EZLE420A
DISPCGBL	EZLEAGBL	netid.luname.EZLEAGBL
DISSTS	EZLE720A	netid.luname.EZLE720A
DISTHRES	EZLE430A	netid.luname.EZLE430A
DSPCFG	EZLE710A	netid.luname.EZLE710A
DSPSTS	EZLE720A	netid.luname.EZLE720A
EMAIL	EZLEMAIL	netid.luname.EZLEMAIL
EZLACFG		netid.luname.EZLSMCFG
EZLADDF		netid.luname.EZLADDF
EZLALBF		netid.luname.EZLALBF
EZLALOG		netid.luname.EZLALOG

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLASTS		netid.luname.EZLASTS
EZLAUST		netid.luname.EZLAUST
EZLCALL	EZLECALL	netid.luname.EZLECALL
EZLCFG		netid.luname.EZLCFG
EZLDDDF		netid.luname.EZLDDDF
EZLEAAGD	EZLEAAGD	netid.luname.EZLEAAGD
EZLEAAIC	EZLEAAIC	netid.luname.EZLEAAIC
EZLEAANV	EZLEAANV	netid.luname.EZLEAANV
EZLEAATR	EZLEAATR	netid.luname.EZLEAATR
EZLEAATS	EZLEAATS	netid.luname.EZLEAATS
EZLEAAT1	EZLEAAT1	netid.luname.EZLEAAT1
EZLEAAT2	EZLEAAT2	netid.luname.EZLEAAT2
EZLEAAT3	EZLEAAT3	netid.luname.EZLEAAT3
EZLEAAT4	EZLEAAT4	netid.luname.EZLEAAT4
EZLEAAT5	EZLEAAT5	netid.luname.EZLEAAT5
EZLEAAT6	EZLEAAT6	netid.luname.EZLEAAT6
EZLEAAT8	EZLEAAT8	netid.luname.EZLEAAT8
EZLEAAT9	EZLEAAT9	netid.luname.EZLEAAT9
EZLEACAF		netid.luname.EZLSACAF
EZLEACGA	EZLEACGA	netid.luname.EZLEACGA
EZLEACGL	EZLEACGL	netid.luname.EZLEACGL

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEACGT	EZLEACGT	netid.luname.EZLEACGT
EZLEACG0	EZLEACG0	netid.luname.EZLEACG0
EZLEACG1	EZLEACG1	netid.luname.EZLEACG1
EZLEACG2	EZLEACG2	netid.luname.EZLEACG2
EZLEACG3	EZLEACG3	netid.luname.EZLEACG3
EZLEACG4	EZLEACG4	netid.luname.EZLEACG4
EZLEACG5	EZLEACG5	netid.luname.EZLEACG5
EZLEACG6	EZLEACG6	netid.luname.EZLEACG6
EZLEACG7	EZLEACG7	netid.luname.EZLEACG7
EZLEACG8	EZLEACG8	netid.luname.EZLEACG8
EZLEACG9	EZLEACG9	netid.luname.EZLEACG9
EZLEACKT	EZLEACKT	netid.luname.EZLEACKT
EZLEACNA	EZLEACNA	netid.luname.EZLEACNA
EZLEACNT	EZLEACNT	netid.luname.EZLEACNT
EZLEACST	EZLEACST	netid.luname.EZLEACST
EZLEACSX	EZLEACSX	netid.luname.EZLEACSX
EZLEACT1	EZLEACT1	netid.luname.EZLEACT1
EZLEACT2	EZLEACT2	netid.luname.EZLEACT2
EZLEAC10	EZLEAC10	netid.luname.EZLEAC10
EZLEAC11	EZLEAC11	netid.luname.EZLEAC11
EZLEADLY	EZLEADLY	netid.luname.EZLEADLY

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEAEXI	EZLEAEXI	netid.luname.EZLEAEXI
EZLEAFST	EZLEAFST	netid.luname.EZLEAFST
EZLEAGEN	EZLEAGEN	netid.luname.EZLEAGEN
EZLEAGN1	EZLEAGN1	netid.luname.EZLEAGN1
EZLEAGRN		netid.luname.EZLAGRN
EZLEAHED	EZLEAHED	netid.luname.EZLEAHED
EZLEAINL	EZLEAINL	netid.luname.EZLEAINL
EZLEAINT	EZLEAINT	netid.luname.EZLEAINT
EZLEAIOP	EZLEAIOP	netid.luname.EZLEAIOP
EZLEAIPL	EZLEAIPL	netid.luname.EZLEAIPL
EZLEAIRP	EZLEAIRP	netid.luname.EZLEAIRP
EZLEAISM	EZLEAISM	netid.luname.EZLEAISM
EZLEAJUL	EZLEAJUL	netid.luname.EZLEAJUL
EZLEALCL	EZLEALCL	netid.luname.EZLEALCL
EZLEALDR	EZLEALDR	netid.luname.EZLEALDR
EZLEALD1	EZLEALD1	netid.luname.EZLEALD1
EZLEALFL	EZLEALFL	netid.luname.EZLEALFL
EZLEALIC	EZLEALIC	netid.luname.EZLEALIC
EZLEALRS	EZLEALRS	netid.luname.EZLEALRS
EZLEALSW	EZLEALSW	netid.luname.EZLEALSW
EZLEANTL	EZLEANTL	netid.luname.EZLEANTL

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEARCY	EZLEARCY	netid.luname.EZLEARCY
EZLEARFR	EZLEARFR	netid.luname.EZLEARFR
EZLEARST	EZLEARST	netid.luname.EZLEARST
EZLEASAO	EZLEASAO	netid.luname.EZLEASAO
EZLEASCD	EZLEASCD	netid.luname.EZLEASCD
EZLEASCN	EZLEASCN	netid.luname.EZLEASCN
EZLEASLN	EZLEASLN	netid.luname.EZLEASLN
EZLEASTK	EZLEASTK	netid.luname.EZLEASTK
EZLEASTM	EZLEASTM	netid.luname.EZLEASTM
EZLEASTP	EZLEASTP	netid.luname.EZLEASTP
EZLEATDF		netid.luname.EZLSATDF
EZLEATDS	EZLEATDS	netid.luname.EZLEATDS
EZLEATHR		netid.luname.EZLSATHR
EZLEATRC	EZLEATRC	netid.luname.EZLEATRC
EZLEATST	EZLEATST	netid.luname.EZLEATST
EZLEAUCG	EZLEAUCG	netid.luname.EZLEAUCG
EZLEAUCL	EZLEAUCL	netid.luname.EZLEAUCL
EZLEAUSF	EZLEAUSF	netid.luname.EZLEAUSF
EZLEAUST	EZLEAUST	netid.luname.EZLEAUST
EZLEAUS1	EZLEAUS1	netid.luname.EZLEAUS1
EZLEAU01	EZLEAU01	netid.luname.EZLEAU01

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEAU02	EZLEAU02	netid.luname.EZLEAU02
EZLEAU03	EZLEAU03	netid.luname.EZLEAU03
EZLEAU07		netid.luname.EZLSAU07
EZLEAX00	EZLEAX00	netid.luname.EZLEAX00
EZLEAX01	EZLEAX01	netid.luname.EZLEAX01
EZLEBELG	EZLEBELG	netid.luname.EZLEBELG
EZLECATV	EZLECATV	netid.luname.EZLECATV
EZLECAUT		netid.luname.EZLECAUT
EZLECHAU	EZLECHAU	netid.luname.EZLECHAU
EZLECHGF	EZLECHGF	netid.luname.EZLECHGF
EZLECMOD		netid.luname.EZLSCMOD
EZLECTHR	EZLECTHR	netid.luname.EZLECTHR
EZLEDAN1	EZLEDAN1	netid.luname.EZLEDAN1
EZLEDTSK	EZLEDTSK	netid.luname.EZLEDTSK
EZLEDUTL	EZLEDUTL	netid.luname.EZLEDUTL
EZLEFAIL	EZLEFAIL	netid.luname.EZLEFAIL
EZLEF00B	EZLEF00B	netid.luname.EZLEF00B
EZLEF00D	EZLEF00D	netid.luname.EZLEF00D
EZLEF001	EZLEF001	netid.luname.EZLEF001
EZLEF003	EZLEF003	netid.luname.EZLEF003
EZLEF004	EZLEF004	netid.luname.EZLEF004

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEF009	EZLEF009	netid.luname.EZLEF009
EZLEGTID	EZLEGTID	netid.luname.EZLEGTID
EZLEHBLD	EZLEHBLD	netid.luname.EZLEHBLD
EZLEHNDE		netid.luname.EZLSHNDE
EZLEHRCY	EZLEHRCY	netid.luname.EZLEHRCY
EZLEICGS	EZLEICGS	netid.luname.EZLEICGS
EZLEICGV	EZLEICGV	netid.luname.EZLEICGV
EZLEIDNT	EZLEIDNT	netid.luname.EZLEIDNT
EZLEINFU	EZLEINFU	netid.luname.EZLEINFU
EZLEITWR	EZLEITWR	netid.luname.EZLEITWR
EZLELSTH	EZLELSTH	netid.luname.EZLELSTH
EZLEMCOL	EZLEMCOL	netid.luname.EZLEMCOL
EZLEMSG	EZLEMSG	netid.luname.EZLEMSG
EZLEMSU	EZLEMSU	netid.luname.EZLEMSU
EZLENDET	EZLENDET	netid.luname.EZLENDET
EZLENETF	EZLENETF	netid.luname.EZLENETF
EZLENFRM	EZLENFRM	netid.luname.EZLENFRM
EZLENPS2	EZLENPS2	netid.luname.EZLENPS2
EZLEOIVT	EZLEOIVT	netid.luname.EZLEOIVT
EZLEOPER	EZLEOPER	netid.luname.EZLEOPER
EZLEPAR	EZLEPAR	netid.luname.EZLEPAR

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEPDEL	EZLEPDEL	netid.luname.EZLEPDEL
EZLEPDIS	EZLEPDIS	netid.luname.EZLEPDIS
EZLEIPC		netid.luname.EZLSIPC
EZLEPRCY	EZLEPRCY	netid.luname.EZLEPRCY
EZLERAIP	EZLERAIP	netid.luname.EZLERAIP
EZLERCMD	EZLERCMD	netid.luname.EZLERCMD
EZLERGWY	EZLERGWY	netid.luname.EZLERGWY
EZLERECV	EZLERECV	netid.luname.EZLERECV
EZLERMSU	EZLERMSU	netid.luname.EZLERMSU
EZLERNGE	EZLERNGE	netid.luname.EZLERNGE
EZLEROUT	EZLEROUT	netid.luname.EZLEROUT
EZLERTVE		netid.luname.EZLERTVE
EZLESLCT	EZLESLCT	netid.luname.EZLESLCT
EZLESNTX	EZLESNTX	netid.luname.EZLESNTX
EZLESRMD	EZLESRMD	netid.luname.EZLESRMD
EZLESTOP	EZLESTOP	netid.luname.EZLESTOP
EZLESTRT	EZLESTRT	netid.luname.EZLESTRT
EZLEVACT	EZLEVACT	netid.luname.EZLEVACT
EZLEVIEW	EZLEVIEW	netid.luname.EZLEVIEW
EZLEVINA	EZLEVINA	netid.luname.EZLEVINA
EZLEVMOV	EZLEVMOV	netid.luname.EZLEVMOV

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLEW001	EZLEW001	netid.luname.EZLEW001
EZLEW002	EZLEW002	netid.luname.EZLEW002
EZLEXIST		netid.luname.EZLSEXST
EZLEXIT7	EZLEXIT7	netid.luname.EZLEXIT7
EZLE0100	EZLE0100	netid.luname.EZLE0100
EZLE1CDL	EZLE1CDL	netid.luname.EZLE1CDL
EZLE1CNT	EZLE1CNT	netid.luname.EZLE1CNT
EZLE1DAL	EZLE1DAL	netid.luname.EZLE1DAL
EZLE1DOM	EZLE1DOM	netid.luname.EZLE1DOM
EZLE1FUL	EZLE1FUL	netid.luname.EZLE1FUL
EZLE1FWD	EZLE1FWD	netid.luname.EZLE1FWD
EZLE1GXC	EZLE1GXC	netid.luname.EZLE1GXC
EZLE1GXD	EZLE1GXD	netid.luname.EZLE1GXD
EZLE1GXE	EZLE1GXE	netid.luname.EZLE1GXE
EZLE1ICK	EZLE1ICK	netid.luname.EZLE1ICK
EZLE1IGT	EZLE1IGT	netid.luname.EZLE1IGT
EZLE1IMN	EZLE1IMN	netid.luname.EZLE1IMN
EZLE1INT	EZLE1INT	netid.luname.EZLE1INT
EZLE1ITF	EZLE1ITF	netid.luname.EZLE1ITF
EZLE1IXD	EZLE1IXD	netid.luname.EZLE1IXD
EZLE1IXL	EZLE1IXL	netid.luname.EZLE1IXL

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLE1I01	EZLE1I01	netid.luname.EZLE1I01
EZLE1I02	EZLE1I02	netid.luname.EZLE1I02
EZLE1I03	EZLE1I03	netid.luname.EZLE1I03
EZLE1I04	EZLE1I04	netid.luname.EZLE1I04
EZLE1I05	EZLE1I05	netid.luname.EZLE1I05
EZLE1I06	EZLE1I06	netid.luname.EZLE1I06
EZLE1I07	EZLE1I07	netid.luname.EZLE1I07
EZLE1I08	EZLE1I08	netid.luname.EZLE1I08
EZLE1NTF	EZLE1NTF	netid.luname.EZLE1NTF
EZLE1RGT	EZLE1RGT	netid.luname.EZLE1RGT
EZLE1RNT	EZLE1RNT	netid.luname.EZLE1RNT
EZLE1RSP	EZLE1RSP	netid.luname.EZLE1RSP
EZLE1RTN	EZLE1RTN	netid.luname.EZLE1RTN
EZLE1RUD	EZLE1RUD	netid.luname.EZLE1RUD
EZLE1RUR	EZLE1RUR	netid.luname.EZLE1RUR
EZLE1RUT	EZLE1RUT	netid.luname.EZLE1RUT
EZLE1RUU	EZLE1RUU	netid.luname.EZLE1RUU
EZLE1RUX	EZLE1RUX	netid.luname.EZLE1RUX
EZLE1TMX	EZLE1TMX	netid.luname.EZLE1TMX
EZLE1UFW	EZLE1UFW	netid.luname.EZLE1UFW
EZLE1XMN	EZLE1XMN	netid.luname.EZLE1XMN

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLE1XTF	EZLE1XTF	netid.luname.EZLE1XTF
EZLE1000	EZLE1000	netid.luname.EZLE1000
EZLE1900	EZLE1900	netid.luname.EZLE1900
EZLE2000	EZLE2000	netid.luname.EZLE2000
EZLE4000	EZLE4000	netid.luname.EZLE4000
EZLE4100	EZLE4100	netid.luname.EZLE4100
EZLE4110	EZLE4110	netid.luname.EZLE4110
EZLE4120	EZLE4120	netid.luname.EZLE4120
EZLE4200	EZLE4200	netid.luname.EZLE4200
EZLE4300	EZLE4300	netid.luname.EZLE4300
EZLE4400	EZLE4400	netid.luname.EZLE4400
EZLE4500	EZLE4500	netid.luname.EZLE4500
EZLE500A	EZLE500A	netid.luname.EZLE500A
EZLE5000	EZLE5000	netid.luname.EZLE5000
EZLE5200	EZLE5200	netid.luname.EZLE5200
EZLE5300	EZLE5300	netid.luname.EZLE5300
EZLE5400	EZLE5400	netid.luname.EZLE5400
EZLE6000	EZLE6000	netid.luname.EZLE6000
EZLE7000	EZLE7000	netid.luname.EZLE7000
EZLE7100	EZLE7100	netid.luname.EZLE7100
EZLE7110	EZLE7110	netid.luname.EZLE7110

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLE7200	EZLE7200	netid.luname.EZLE7200
EZLE7210	EZLE7210	netid.luname.EZLE7210
EZLE7500	EZLE7500	netid.luname.EZLE7500
EZLE7600	EZLE7600	netid.luname.EZLE7600
EZLE7700	EZLE7700	netid.luname.EZLE7700
EZLE8000	EZLE8000	netid.luname.EZLE8000
EZLE8100	EZLE8100	netid.luname.EZLE8100
EZLE8110	EZLE8110	netid.luname.EZLE8110
EZLE8120	EZLE8120	netid.luname.EZLE8120
EZLE8200	EZLE8200	netid.luname.EZLE8200
EZLE8300	EZLE8300	netid.luname.EZLE8300
EZLE8400	EZLE8400	netid.luname.EZLE8400
EZLE8410	EZLE8410	netid.luname.EZLE8410
EZLE8500	EZLE8500	netid.luname.EZLE8500
EZLE8600	EZLE8600	netid.luname.EZLE8600
EZLE8611	EZLE8611	netid.luname.EZLE8611
EZLE8612	EZLE8612	netid.luname.EZLE8612
EZLIPLDT		netid.luname.EZLIPLDT
EZLLOG		netid.luname.EZLLOG
EZLMSG		netid.luname.EZLMSG
EZLODDF		netid.luname.EZLODDF

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLPDDF		netid.luname.EZLPDDF
EZLQDDF		netid.luname.EZLQDDF
EZLSACAF		netid.luname.EZLSACAF
EZLSAGRN		netid.luname.EZLEAGRN
EZLSASND		netid.luname.EZLSASND
EZLSATHR		netid.luname.EZLSATHR
EZLSAU07		netid.luname.EZLSAU07
EZLSCAUT		netid.luname.EZLSCAUT
EZLSCMOD		netid.luname.EZLSCMOD
EZLSHNDE		netid.luname.EZLSHNDE
EZSLCFG		netid.luname.EZSLCFG
EZLSMSU		netid.luname.EZLSMSU
EZLSNHLP	EZLSNHLP	netid.luname.EZLSNHLP
EZLSPIPC		netid.luname.EZLSPIPC
EZLSPIPS		netid.luname.EZLSPIPS
EZLSRTVE		netid.luname.EZLSRTVE
EZLSTMEM		netid.luname.EZLSTMEM
EZLSTRAC		netid.luname.EZLSTRAC
EZLSTS		netid.luname.EZLSCSTS
EZLSVLST		netid.luname.EZLSVLST
EZLSX001		netid.luname.EZLSX001

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
EZLTRACE		netid.luname.EZLSTRAC
FKVEAGRN		netid.luname.EZLEAGRN
FKVECATV	EZLECATV	netid.luname.EZLECATV
FKVSPIPB		netid.luname.EZLSPIPS
GETPW		netid.luname.EZLSUPPW
ILOG	EZLEINFL	netid.luname.EZLEINFL
INFORM	EZLECALL	netid.luname.EZLECALL
INFORMLG	EZLEINFL	netid.luname.EZLEINFL
INFORMTB	EZLEITBL	netid.luname.EZLEITBL
INFRMTBL	EZLEITBL	netid.luname.EZLEITBL
LOAD	EZLE860A	netid.luname.EZLE860A
LOADTABL	EZLE860A	netid.luname.EZLE860A
LOADTBL	EZLE860A	netid.luname.EZLE860A
MARK	EZLEBASN	netid.luname.EZLEBASN
NLOG		netid.luname.EZLALBF
RELEASE	EZLEBUAS	netid.luname.EZLEBUAS
RGWY	EZLERGWY	netid.luname.EZLERGWY
SENDCMD	EZLE530A	netid.lunname.EZLE530A
SENDMSG		netid.luname.EZLSASND
SETALERT	EZLE420A	netid.luname.EZLE420A
SETAUTO	EZLE410A	netid.luname.EZLE410A

Table 12. AON Command Identifiers. AON Base Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
SETBEEP	EZLEABEP	netid.luname.EZLEABEP
SETMONIT	EZLE440A	netid.luname.EZLE440A
SETNTFY	EZLE420A	netid.luname.EZLE420A
SETTHRES	EZELE430A	netid.luname.EZLE430A
SIGNOUT	EZLEBASN	netid.luname.EZLEBASN
SM		netid.luname.EZLSASND
SNAVIEW	EZLEVIE1	netid.luname.EZLEVIE1
STARTEZL	EZLE760A	netid.luname.EZLE760A
STARTGWY	EZLE1SGW	netid.luname.EZLE1SGW
STOPEZL	EZLE761A	netid.luname.EZLE761A
TGLOBAL	EZLSVLST	netid.luname.EZLSVLST
UNMARK	EZLEBUAS	netid.luname.EZLEBUAS

Appendix C. AON/SNA Command Names and Synonyms that can be Protected

This section lists the AON/SNA commands and synonyms that can be protected.

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected		
Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
@N	FKVE500A	netid.luname.FKVE500A
AONSNA	FKVE000A	netid.luname.FKVE000A
AONX25	FKVEX00A	netid.luname.FKVEX00A
APPN	FKVEA00A	netid.luname.FKVEA00A
CHGSPEED	FKVEECGB5	netid.luname.FKVEECGB5
DELDOWN	FKVEECGB5	netid.luname.FKVEECGB5
DISPNET	FKVE400A	netid.luname.FKVE400A
DISPOOL	FKVEECGB3	netid.luname.FKVEECGB3
EZLASNB		netid.luname.FKVASNB
EZLENCH1		netid.luname.EZLENCH1
EZLENCH2		netid.luname.EZLENCH2
EZLENCH3		netid.luname.EZLENCH3
EZLENCH4		netid.luname.EZLENCH4
FKVASNB		netid.luname.FKVASNB
FKVEADMP	FKVEADMP	netid.luname.FKVEADMP
FKVEAIDA	FKXEAIDA	netid.luname.FKVEAIDA
FKVEAIDB	FKVEAIDB	netid.luname.FKVEAIDB
FKVEAIDC	FKVEAIDC	netid.luname.FKVEAIDC

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
FKVEAIDD	FKVEAIDD	netid.luname.FKVEAIDD
FKVEAIDE	FKVEAIDE	netid.luname.FKVEAIDE
FKVEAIDF	FKVEAIDF	netid.luname.FKVEAIDF
FKVEAIDG	FKVEAIDG	netid.luname.FKVEAIDG
FKVEAIDH	FKVEAIDH	netid.luname.FKVEAIDH
FKVEAIDI	FKVEAIDI	netid.luname.FKVEAIDI
FKVEAIDJ	FKVEAIDJ	netid.luname.FKVEAIDJ
FKVEAIDK	FKVEAIDK	netid.luname.FKVEAIDK
FKVEAID1	FKVEAID1	netid.luname.FKVEAID1
FKVEAID2	FKVEAID2	netid.luname.FKVEAID2
FKVEAID3	FKVEAID3	netid.luname.FKVEAID3
FKVEAID4	FKVEAID4	netid.luname.FKVEAID4
FKVEAID5	FKVEAID5	netid.luname.FKVEAID5
FKVEAID6	FKVEAID6	netid.luname.FKVEAID6
FKVEAID7	FKVEAID7	netid.luname.FKVEAID7
FKVEAID8	FKVEAID8	netid.luname.FKVEAID8
FKVEAID9	FKVEAID9	netid.luname.FKVEAID9
FKVEAMS1	FKVEAMS1	netid.luname.FKVEAMS1
FKVEARLD	FKVEARLD	netid.luname.FKVEARLD
FKVEA0IC	FKVEA0IC	netid.luname.FKVEA0IC
FKVEA000	FKVEA000	netid.luname.FKVEA000

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
FKVEA100	FKVEA100	netid.luname.FKVEA100
FKVEA200	FKVEA200	netid.luname.FKVEA200
FKVEA210	FKVEA210	netid.luname.FKVEA210
FKVEA300	FKVEA300	netid.luname.FKVEA300
FKVEA400	FKVEA400	netid.luname.FKVEA400
FKVEA410	FKVEA410	netid.luname.FKVEA410
FKVECAPL	FKVECAPL	netid.luname.FKVECAPL
FKVECGBA	FKVECGBA	netid.luname.FKVECGBA
FKVECGBB	FKVECGBB	netid.luname.FKVECGBB
FKVECGBC	FKVECGBC	netid.luname.FKVECGBC
FKVECGBD	FKVECGBD	netid.luname.FKVECGBD
FKVECGBE	FKVECGBE	netid.luname.FKVECGBE
FKVECGBF	FKVECGBF	netid.luname.FKVECGBF
FKVECGBG	FKVECGBG	netid.luname.FKVECGBG
FKVEGBH	FKVEGBH	netid.luname.FKVEGBH
FKVECGCA	FKVECGCA	netid.luname.FKVECGCA
FKVECGCC	FKVECGCC	netid.luname.FKVECGCC
FKVECGCD	FKVECGCD	netid.luname.FKVECGCD
FKVECGDA	FKVECGDA	netid.luname.FKVECGDA
FKVECGDB	FKVECGDB	netid.luname.FKVECGDB
FKVECGDC	FKVECGDC	netid.luname.FKVECGDC

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
FKVECGDD	FKVECGDD	netid.luname.FKVECGDD
FKVECGDE	FKVECGDE	netid.luname.FKVECGDE
FKVECGDF	FKVECGDF	netid.luname.FKVECGDF
FKVECGDG	FKVECGDG	netid.luname.FKVECGDG
FKVECGEA	FKVECGEA	netid.luname.FKVECGEA
FKVECGEB	FKVECGEB	netid.luname.FKVECGEB
FKVECGEC	FKVECGEC	netid.luname.FKVECGEC
FKVECGED	FKVECGED	netid.luname.FKVECGED
FKVECGFD	FKVECGFD	netid.luname.FKVECGFD
FKVECGFF	FKVECGFF	netid.luname.FKVECGFF
FKVECGFG	FKVECGFG	netid.luname.FKVECGFG
FKVECGFH	FKVECGFH	netid.luname.FKVECGFH
FKVECGHA	FKVECGHA	netid.luname.FKVECGHA
FKVECGHB	FKVECGHB	netid.luname.FKVECGHB
FKVECGHD	FKVECGHD	netid.luname.FKVECGHD
FKVECHCM	FKVECHCM	netid.luname.FKVECHCM
FKVECHIN	FKVECHIN	netid.luname.FKVECHIN
FKVECHRP	FKVECHRP	netid.luname.FKVECHRP
FKVECHSG	FKVECHSG	netid.luname.FKVECHSG
FKVECHSR	FKVECHSR	netid.luname.FKVECHSR
FKVECNCP	FKVECNCP	netid.luname.FKVECNCP

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
FKVEDETL	FKVEDETL	netid.luname.FKVEDETL
FKVEF005	FKVEF005	netid.luname.FKVEF005
FKVEINIT	FKVEINIT	netid.luname.FKVEINIT
FKVEOG01	FKVEOG01	netid.luname.FKVEOG01
FKVEOG02	FKVEOG02	netid.luname.FKVEOG02
FKVEOG03	FKVEOG03	netid.luname.FKVEOG03
FKVEOG04	FKVEOG04	netid.luname.FKVEOG04
FKVEOG05	FKVEOG05	netid.luname.FKVEOG05
FKVEOG06	FKVEOG06	netid.luname.FKVEOG06
FKVEOG07	FKVEOG07	netid.luname.FKVEOG07
FKVEOG08	FKVEOG08	netid.luname.FKVEOG08
FKVEOG09	FKVEOG09	netid.luname.FKVEOG09
FKVEOI00	FKVEOI00	netid.luname.FKVEOI00
FKVEOPFI	FKVEOPFI	netid.luname.FKVEOPFI
FKVEOSEC	FKVEOSEC	netid.luname.FKVEOSEC
FKVEPULT	FKVEPULT	netid.luname.FKVEPULT
FKVERDIS	FKVERDIS	netid.luname.FKVERDIS
FKVESN	FKVESN	netid.luname.FKVESN
FKVETGSW	FKVETGSW	netid.luname.FKVETGSW
FKVEXACT	FKVEXACT	netid.luname.FKVEXACT
FKVEXCDB	FKVEXCDB	netid.luname.FKVEXCDB

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
FKVEXCON	FKVEXCON	netid.luname.FKVEXCON
FKVEXDIS	FKVEXDIS	netid.luname.FKVEXDIS
FKVEXINA	FKVEXINA	netid.luname.FKVEXINA
FKVEXMCH	FKVEXMCH	netid.luname.FKVEXMCH
FKVEXRES	FKVEXRES	netid.luname.FKVEXRES
FKVEXTRK	FKVEXTRK	netid.luname.FKVEXTRK
FKVEX000	FKVEX000	netid.luname.FKVEX000
FKVEX100	FKVEX100	netid.luname.FKVEX100
FKVEX200	FKVEX200	netid.luname.FKVEX200
FKVEX74E	FKVEX74E	netid.luname.FKVEX74E
FKVEX74X	FKVEX74X	netid.luname.FKVEX74X
FKVE0000	FKVE0000	netid.luname.FKVE0000
FKVE095A	FKVE095A	netid.luname.FKVE095A
FKVE1000	FKVE1000	netid.luname.FKVE1000
FKVE1100	FKVE1100	netid.luname.FKVE1100
FKVE1101	FKVE1101	netid.luname.FKVE1101
FKVE1102	FKVE1102	netid.luname.FKVE1102
FKVE1103	FKVE1103	netid.luname.FKVE1103
FKVE1104	FKVE1104	netid.luname.FKVE1104
FKVE1110	FKVE1110	netid.luname.FKVE1110
FKVE1200	FKVE1200	netid.luname.FKVE1200

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
FKVE1300	FKVE1300	netid.luname.FKVE1300
FKVE1310	FKVE1310	netid.luname.FKVE1310
FKVE1320	FKVE1320	netid.luname.FKVE1320
FKVE1330	FKVE1330	netid.luname.FKVE1330
FKVE2000	FKVE2000	netid.luname.FKVE2000
FKVE2100	FKVE2100	netid.luname.FKVE2100
FKVE270I	FKVE270I	netid.luname.FKVE270I
FKVE284A	FKVE284A	netid.luname.FKVE284A
FKVE285I	FKVE285I	netid.luname.FKVE285I
FKVE3000	FKVE3000	netid.luname.FKVE3000
FKVE380I	FKVE380I	netid.luname.FKVE380I
FKVE4000	FKVE4000	netid.luname.FKVE4000
FKVE464I	FKVE464I	netid.luname.FKVE464I
FKVE5000	FKVE5000	netid.luname.FKVE5000
FKVE5100	FKVE5100	netid.luname.FKVE5100
FKVE530I	FKVE530I	netid.luname.FKVE530I
FKVE881I	FKVE881I	netid.luname.FKVE881I
FKVE897I	FKVE897I	netid.luname.FKVE897I
LUDRPOOL	FKVEX20A	netid.luname.FKVEX20A
LUDRSTAT	FKVEX20A	netid.luname.FKVEX20A
NETCHK	FKVE400A	netid.luname.FKVE400A

Table 13. AON Command Identifiers. AON/SNA Command Names and Synonyms That Can Be Protected
(continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
NETSTAT	FKVEX20A	netid.luname.FKVEX20A
SETPool	FKVEcGB4	netid.luname.FKVEcGB4
SNAHD	FKVE100A	netid.luname.FKVE100A
SNAHEAD	FKVE100A	netid.luname.FKVE100A
SNAHED	FKVE100A	netid.luname.FKVE100A
SNAMAP	FKVE200A	netid.luname.FKVE200A
SNAPULST	FKVEPULA	netid.luname.FKVEPULA
VTAMCMD	FKVE500A	netid.luname.FKVE500A
VTAMCMDS	FKVE500A	netid.luname.FKVE500A
VTAMOPT	FKVE300A	netid.luname.FKVE300A
VTAMOPTS	FKVE300A	netid.luname.FKVE300A
X25	FKVEX00A	netid.luname.FKVEX00A
X25INIT	FKVEXINI	netid.luname.FKVEXINI
X25MONIT	FKVEX10A	netid.luname.FKVEX10A

Appendix D. AON/TCP Command Names and Synonyms that Can Be Protected

This section lists the AON/TCP commands and synonyms that can be protected.

Table 14. AON Command Identifiers. AON/TCP Command Names and Synonyms That Can Be Protected		
Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
AONTCP	FKXE000A	netid.luname.FKXE000A
FKXEACTION2	FKXEACTION2	netid.luname.FKXEACTION2
FKXEAID1	FKXEAID1	netid.luname.FKXEAID1
FKXEAID2	FKXEAID2	netid.luname.FKXEAID2
FKXEALRT	FKXEALRT	netid.luname.FKXEALRT
FKXEAMS1	FKXEAMS1	netid.luname.FKXEAMS1
FKXECATV	FKXECATV	netid.luname.FKXECATV
FKXEDDFP	FKXEDDFP	netid.luname.FKXEDDFP
FKXEDROP	FKXEDROP	netid.luname.FKXEDROP
FKXEGTID	FKXEGTID	netid.luname.FKXEGTID
FKXEHNDE	FKXEHNDE	netid.luname.FKXEHNDE
FKXEICMD	FKXEICMD	netid.luname.FKXEICMD
FKXEINIT	FKXEINIT	netid.luname.FKXEINIT
FKXEIPMN	FKXEIPMN	netid.luname.FKXEIPMN
FKXEIPSM	FKXEIPSM	netid.luname.FKXEIPSM
FKXENSTH	FKXENSTH	netid.luname.FKXENSTH
FKXEOTHR	FKXEOTHR	netid.luname.FKXEOTHR
FKXEPING	FKXEPING	netid.luname.FKXEPING

Table 14. AON Command Identifiers. AON/TCP Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
FKXESSF	FKXESSF	netid.luname.FKXESSF
FKXESVPT	FKXESVPT	netid.luname.FKXESVPT
FKXETCMD	FKXETCMD	netid.luname.FKXETCMD
FKXEUCMD	FKXEUCMD	netid.luname.FKXEUCMD
FKXE0000	FKXE0000	netid.luname.FKXE0000
FKXE210A	FKXE210A	netid.luname.FKXE210A
FKXE230A	FKXE230A	netid.luname.FKXE230A
FKXE251A	FKXE251A	netid.luname.FKXE251A
FKXE1000	FKXE1000	netid.luname.FKXE1000
FKXE2000	FKXE2000	netid.luname.FKXE2000
FKXE3000	FKXE3000	netid.luname.FKXE3000
FKXE400A	FKXE400A	netid.luname.FKXE400A
FKXE4000	FKXE4000	netid.luname.FKXE4000
FKXE500A	FKXE500A	netid.luname.FKXE500A
FKXE5000	FKXE5000	netid.luname.FKXE5000
FKXWIND1	FKXWIND1	netid.luname.FKXWIND1
FKXWIND2	FKXWIND2	netid.luname.FKXWIND2
IPCMD	FKXEICMD	netid.luname.FKXEICMD
IPMAN	FKXEIPMN	netid.luname.FKXEIPMN
IPMANSSF	FKXEIPSS	netid.luname.FKXEIPSS
IPMNGR	FKXEIPMN	netid.luname.FKXEIPMN

Table 14. AON Command Identifiers. AON/TCP Command Names and Synonyms That Can Be Protected (continued)

Commands and Synonyms	Command List Name	SAF Resource or Command Authorization Table Identifier
IPSTAT	FKXE220A	netid.luname.FKXE220A
MVSPING	FKXE210A	netid.luname.FKXE210A
NVSNMP	FKXE251A	netid.luname.FKXE251A
NV6KCMD	FKXE120A	netid.luname.FKXE120A
NV6KLIST	FKXE300A	netid.luname.FKXE300A
NV6KPERF	FKXE140A	netid.luname.FKXE140A
NV6KPING	FKXE110A	netid.luname.FKXE110A
NV6KRPNG	FKXE130A	netid.luname.FKXE130A
NV6KVIEW	FKXEVIEW2	netid.luname.FKXEVIEW2
RPING	FKXE130A	netid.luname.FKXE130A
TCPLIST	FKXE300A	netid.luname.FKXE300A
TCPVIEW	FKXEVIEW2	netid.luname.FKXEVIEW2
T390LIST	FKXE300A	netid.luname.FKXE300A
T390VIEW	FKXEVIEW2	netid.luname.FKXEVIEW2
WEBIP	FKXEIPSM	netid.luname.FKXEPSM

Appendix E. VTAM Command Keyword Values that Can Be Protected

This section lists the VTAM keyword values that can be protected.

Table 15. VTAM Command Keyword Values. VTAM Command Keyword Values That Can Be Protected		
Keyword	Abbreviation	Notes
DVIPA		
ID		The value of the ID keyword is not span checked if TSOUSER (U) is specified.
IPADDR	IP	
HOSTNAME	HN	
LUNAME		
LU1		
LU2		
PLU		
SLU		

Appendix F. How to Protect IP Addresses for Command Security and Span of Control

The IP addresses specified on commands are converted to a standard format for security and span checking. The information below illustrates the format for protecting different types of IP addresses.

1. Regular IPv4 addresses are passed as IPv4 addresses.
2. Leading zeroes in IPv4 address are insignificant and are removed for command authorization and span checking.
3. IPv4-compatible IPv6 addresses are passed as IPv4 addresses.
4. IPv4-mapped IPv6 addresses are passed as IPv4 addresses.
5. IPv6 addresses are passed as a full-blown IPv6 addresses with leading zeroes dropped. There are always 8 octets of data separated by colons.

Table 16 on page 231 provides examples of IP addresses and indicates which rule (1, 2, 3, 4, or 5) applies to the format of the address.

Table 16. NetView Command Identifiers		
IPADDR Entered on Command	PROTECT Definition	Span Definition
9.67.50.44 (1)	9/67/50/44	9.67.50.44
009.067.050.044 (2)	9/67/50/44	9.67.50.44
0:0:0:0:0:9.67.50.44 (3)	9/67/50/44	9.67.50.44
::9.67.50.44 (3)	9/67/50/44	9.67.50.44
0:0:0:0:FFFF:9.67.50.44 (4)	9/67/50/44	9.67.50.44
1234:0000:5678::9ABC (5)	1234:0:5678:0:0:0:9ABC	1234:0:5678:0:0:0:9ABC
1234:BA98:5678:4321:3214:9ABC:3210:417A (5)	1234:BA98:5678:4321:3214:9ABC:3210:417A	1234:BA98:5678:4321:3214:9ABC:3210:417A

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